1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K62 MLB

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
				2011-02-08

LAST_MODIFIED=Tue Feb 8 15:20:30 2011

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=Tue Feb 8 15:20:30 2011

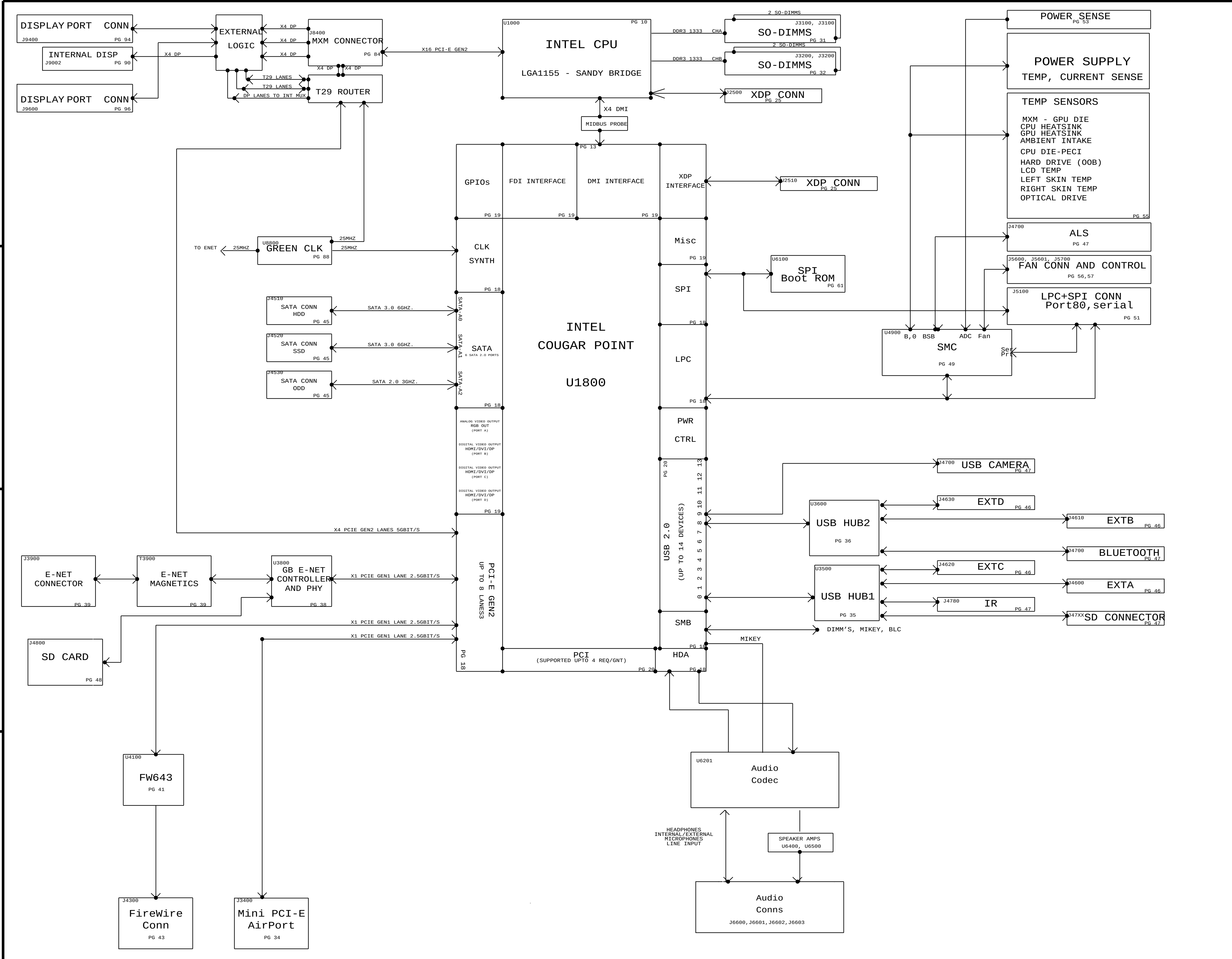
DRAWING TITLE: SCH,K62,MLB

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

BRANCH



SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

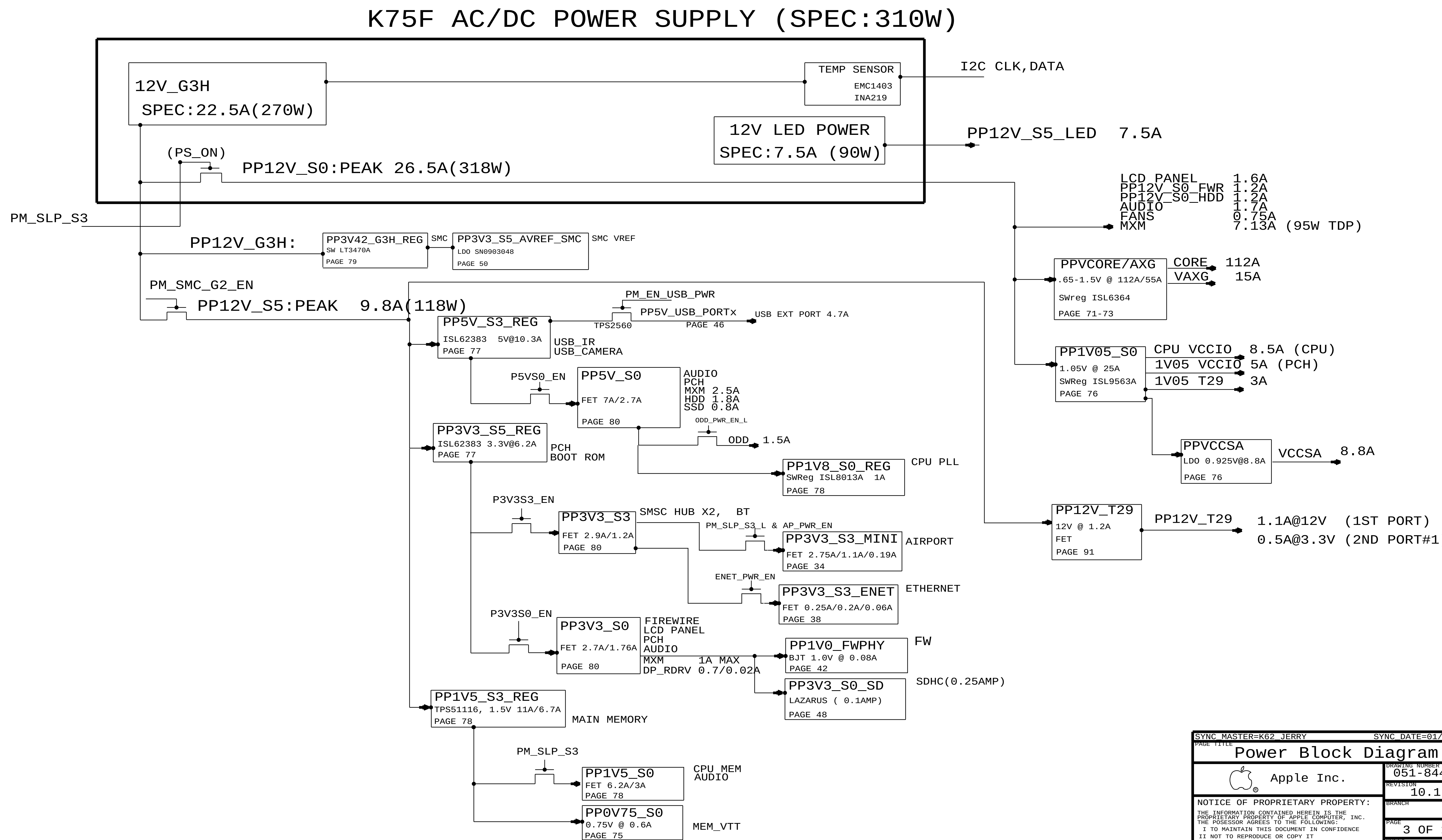
System Block Diagram

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: Power Block Diagram

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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D C B A

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-1226	PCBA,MLB,DEV,K62	DEVELOPMENT,DEV_GROUP
639-1769	PCBA,MLB,K62,2.8G,4C,PRQ,P2_ODD	K62,2P8GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,YES_DBG
639-1770	PCBA,MLB,K62,3.1G,4C,PRQ,P2_ODD	K62,3P1GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,YES_DBG
639-1771	PCBA,MLB,K62,3.4G,4C,PRQ,P2_ODD	K62,3P4GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,YES_DBG
639-2186	PCBA,MLB,K62,2.8G,4C,PRQ,P2_ODD,NO_DBG	K62,2P8GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,NO_DBG
639-2187	PCBA,MLB,K62,3.1G,4C,PRQ,P2_ODD,NO_DBG	K62,3P1GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,NO_DBG
639-2188	PCBA,MLB,K62,3.4G,4C,PRQ,P2_ODD,NO_DBG	K62,3P4GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P2,NO_DBG
639-2124	PCBA,MLB,K62,2.8G,4C,PRQ,P1_ODD	K62,2P8GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,YES_DBG
639-2121	PCBA,MLB,K62,3.1G,4C,PRQ,P1_ODD	K62,3P1GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,YES_DBG
639-2123	PCBA,MLB,K62,3.4G,4C,PRQ,P1_ODD	K62,3P4GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,YES_DBG
639-2129	PCBA,MLB,K62,2.8G,4C,PRQ,P1_ODD,NO_DBG	K62,2P8GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,NO_DBG
639-2131	PCBA,MLB,K62,3.1G,4C,PRQ,P1_ODD,NO_DBG	K62,3P1GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,NO_DBG
639-2130	PCBA,MLB,K62,3.4G,4C,PRQ,P1_ODD,NO_DBG	K62,3P4GHZ_SNB_CPU,BASIC1,BASIC2,CPUVCORE-4PH,ODD_SATA:P1,NO_DBG

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC1	COMMON,ALTERNATE,MXM,FCIM,CPU_1V5_SENSE,CPU_VCCSA_SENSE,1V05_PCH_SENSE
BASIC2	HUB_USX2061,AP,BT,IR,T29,VAXG,PRODUCTION
DEV_GROUP	VREFMRGN_A,VREFMRGN_B,DIMM_1V5_SENSE
YES_DBG	XDP,XDP_CONN,XDP_CPU_BPM,MOJOMUX:YES,LPCPLUS:YES
NO_DBG	MOJOMUX:NO,LPCPLUS:NO

CPU SOCKET & ILM SUB-BOMS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0071	1	SOCKET,LGA1155,CPU-LF	U1000	CRITICAL	TYCO_SOCKET
604-1474	1	ASSY,PURCHASED,ILM,TYCO	ILM	CRITICAL	TYCO_SOCKET
511S0073	1	SOCKET,LGA1155,CPU-LF	U1000	CRITICAL	MOLEX_SOCKET
604-1161	1	ASSY,PURCHASED,ILM,MOLEX	ILM	CRITICAL	MOLEX_SOCKET

BOM NUMBER	BOM NAME	BOM OPTIONS
085-2451	SUB ASSY,CPU SOCKET,K62,TYCO	TYCO_SOCKET
085-2450	SUB ASSY,CPU SOCKET,K62,MOLEX	MOLEX_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-2451	1	TYCO CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-2450	085-2451		SKT_ILM	MOLEX ALTERNATE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
	337S4088	1	IC,COUGAR POINT,SLJ4F,BD82Z68,PRQ,B3	U1800	CRITICAL	
	353S3055	2	IC,PI3VEDP212,X2 DP MUX,QFN	U9390,U9590	CRITICAL	
	338S0753	1	IC,FW643,1394B_PCIE,PHY/LINK	U4100	CRITICAL	
	825-7122	1	MLB LABEL,48.0X4.8	X14	CRITICAL	
	343S0534	1	IC,BCM57765,ENET&SD,8X8	U3900	CRITICAL	
RAW: 335S0807	341T0184	1	FLASH,EFI BOOTROM,K60/K62	U6100	CRITICAL	
RAW: 335S0539	341T0328	1	SFLASH ENET 2MBIT,CIV	U3990	CRITICAL	
	338S0945	1	T29 ROUTER, IC, ASSP	U9700	CRITICAL	T29
RAW: 335S0550	341T0329	1	IC,T29 SERIAL EEPROM	U9790	CRITICAL	T29
RAW: 337S3997	341T0327	2	IC,MCU,32B,LPC1112A,16KB/2KB,HVQFN25	U9330,U9530	CRITICAL	T29
RAW: 338S0878	341T0186	1	IC,SMC,K62	U4900	CRITICAL	

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4042	1	SNB,SR00D,PRQ,D2,2.8,95W,4+1,6M,LGA	CPU	CRITICAL	2P8GHZ_SNB_CPU
337S4040	1	SNB,SR00Q,PRQ,D2,3.1,95W,4+1,6M,LGA	CPU	CRITICAL	3P1GHZ_SNB_CPU
337S4041	1	SNB,SR00B,PRQ,D2,3.4,95W,4+1,8M,LGA	CPU	CRITICAL	3P4GHZ_SNB_CPU

K62 PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-8442	1	SCH,MLB,K62	SCH1		K62
820-2828	1	PCBF,MLB,K62	MLB1		K62

K62 ALTERNATE PARTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0298	128S0293		ALL	330UF
371S0679	371S0652		ALL	PIN DIODE
377S0107	377S0066		ALL	USB DIODE
376S0972	376S0612		ALL	ROHM TRA-BJT

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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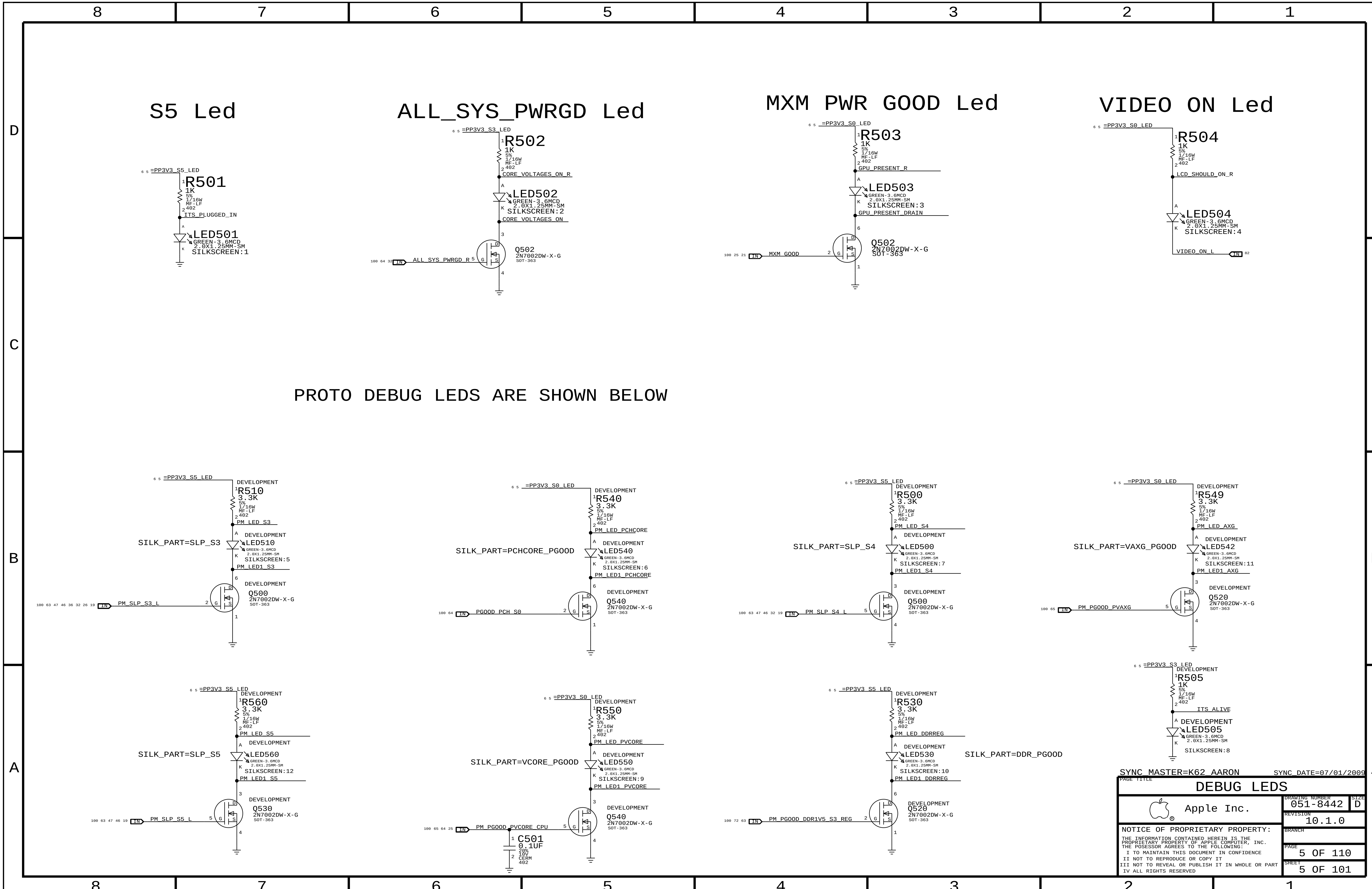
S5 Led
=PP3V3_S5_LED
R501
1K
5%
1/16W
MF-LF
402
ITS_PLUGGED_IN
LED501
GREEN-3.6MCD
2.0X1.25MM-SM
SILKSCREEN:1
ALL_SYS_PWRGD Led
=PP3V3_S3_LED
R502
1K
5%
1/16W
MF-LF
402
CORE_VOLTAGES_ON_R
LED502
GREEN-3.6MCD
2.0X1.25MM-SM
SILKSCREEN:2
CORE_VOLTAGES_ON
ALL_SYS_PWRGD_R
Q502
2N7002DW-X-G
SOT-363
MXM PWR GOOD Led
=PP3V3_S0_LED
R503
1K
5%
1/16W
MF-LF
402
GPU_PRESENT_R
LED503
GREEN-3.6MCD
2.0X1.25MM-SM
SILKSCREEN:3
GPU_PRESENT_DRAIN
MXM_GOOD
Q502
2N7002DW-X-G
SOT-363
VIDEO ON Led
=PP3V3_S0_LED
R504
1K
5%
1/16W
MF-LF
402
LCD_SHOULD_ON_R
LED504
GREEN-3.6MCD
2.0X1.25MM-SM
SILKSCREEN:4
VIDEO_ON_L
PROTO DEBUG LEDS ARE SHOWN BELOW
=PP3V3_S5_LED
DEVELOPMENT
R510
3.3K
PM_LED_S3
SILK_PART=SLP_S3
LED510
SILKSCREEN:5
PM_LED1_S3
Q500
2N7002DW-X-G
SOT-363
PM_SLP_S3_L
=PP3V3_S0_LED
R540
3.3K
PM_LED_PCHCORE
SILK_PART=PCHCORE_PGOOD
LED540
SILKSCREEN:6
PM_LED1_PCHCORE
Q540
PGOOD_PCH_S0
=PP3V3_S5_LED
R500
3.3K
PM_LED_S4
SILK_PART=SLP_S4
LED500
SILKSCREEN:7
PM_LED1_S4
Q500
PM_SLP_S4_L
=PP3V3_S0_LED
R549
3.3K
PM_LED_AXG
SILK_PART=VAXG_PGOOD
LED542
SILKSCREEN:11
PM_LED1_AXG
Q520
PM_PGOOD_PVAXG
=PP3V3_S3_LED
R505
1K
ITS_ALIVE
LED505
SILKSCREEN:8
=PP3V3_S5_LED
R560
3.3K
PM_LED_S5
SILK_PART=SLP_S5
LED560
SILKSCREEN:12
PM_LED1_S5
Q530
PM_SLP_S5_L
=PP3V3_S0_LED
R550
3.3K
PM_LED_PVCORE
SILK_PART=VCORE_PGOOD
LED550
SILKSCREEN:9
PM_LED1_PVCORE
Q540
PM_PGOOD_PVCORE_CPU
C501
0.1UF
20%
10V
CERM
402
=PP3V3_S5_LED
R530
3.3K
PM_LED_DDRREG
SILK_PART=DDR_PGOOD
LED530
SILKSCREEN:10
PM_LED1_DDRREG
Q520
PM_PGOOD_DDR1V5_S3_REG
SYNC_MASTER=K62_AARON
SYNC_DATE=07/01/2009
PAGE TITLE
DEBUG LEDS
Apple Inc.
DRAWING NUMBER
051-8442
SIZE
D
REVISION
10.1.0
BRANCH
PAGE
5 OF 110
SHEET
5 OF 101
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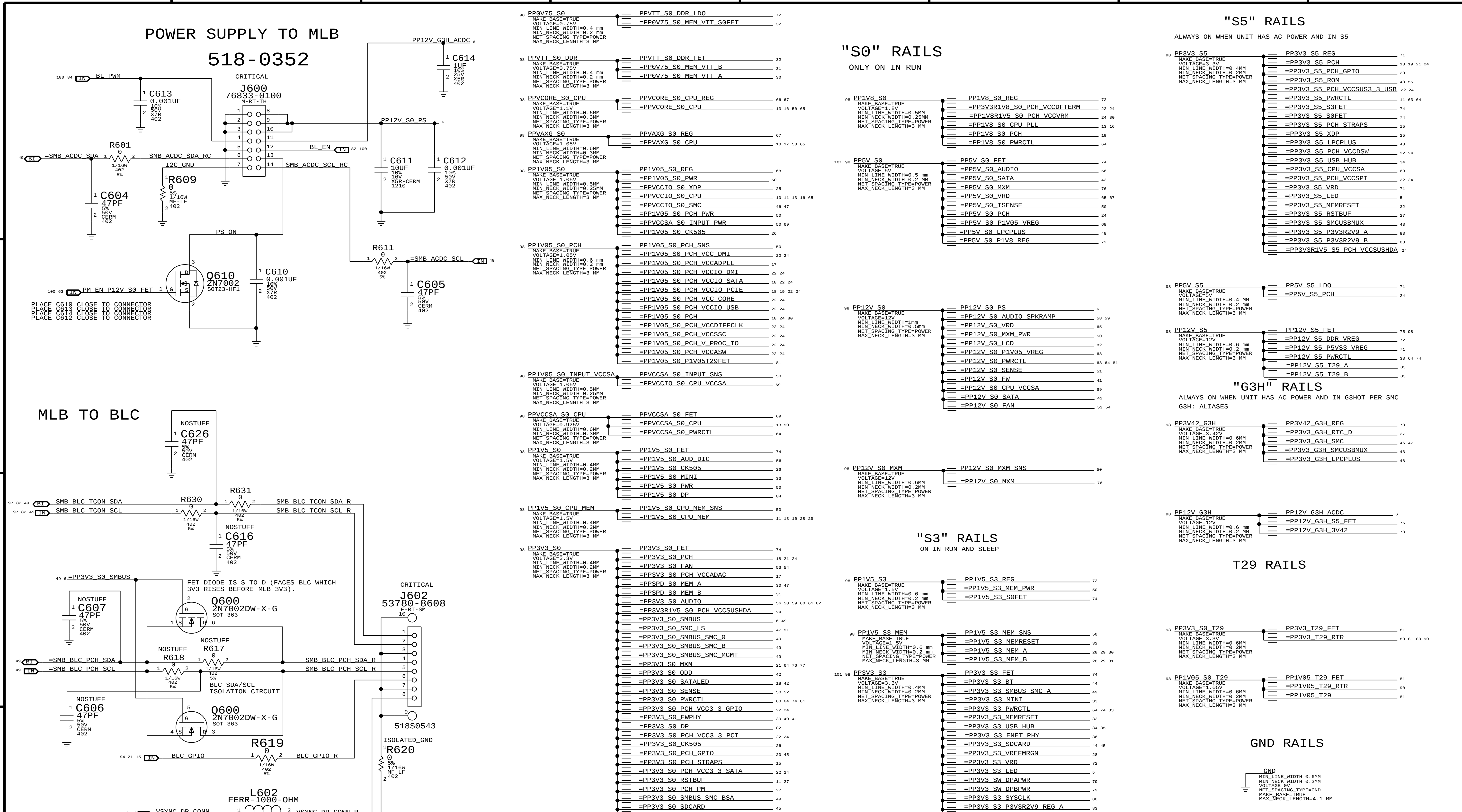

SYNC_MASTER=K62_AARON SYNC_DATE=12/30/2010

PAGE TITLE

Power Conn / Alias

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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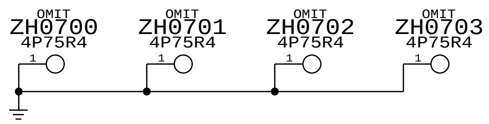
SHEET 6 OF 101

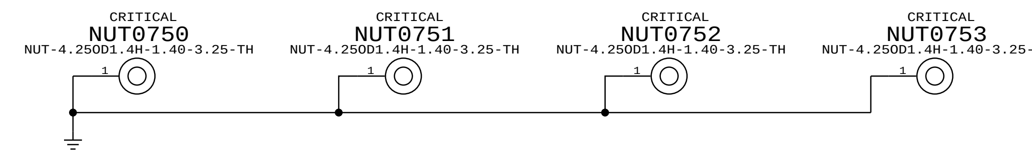
CPU Heatsink

4mm Plated Holes (998-0850)



DIMM CONNECTOR NUTS

Nuts (805-9582)



PCH HEATSINK MTG HOLES (NON-PLATED HOLE ON PCB ONLY)

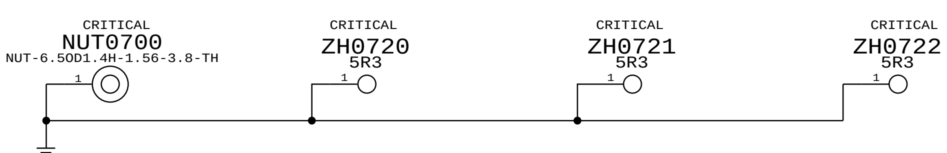
Rear Cover

STANDOFFS (WAS 860-1255 BUT NOW REPLACED WITH 860-1430)

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
860-1430	6	STANDOFF,MLB,K60/K62	SDF0713,SDF0714,SDF0715,SDF0717,SDF0718,SDF0719	COMMON

Backer Plate

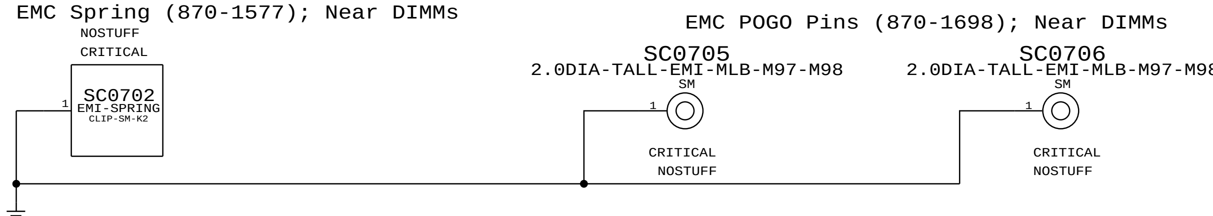
Nuts (835-0269)



For EMC

EMC Spring (870-1577); Near DIMMs

EMC POGO Pins (870-1698); Near DIMMs



SYNC_MASTER=K62_AARON SYNC_DATE=11/30/2009

PAGE TITLE: Holes

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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PAGE: 7 OF 110

SHEET: 7 OF 101

UNUSED CPU SIGNALS

TP_CPU_RSVD<16..1>	NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE
TP_CPU_RSVD<46..19>	NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCI ALIASES

TP_PCI_AD<31..0>	NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_C_BE_L<3..0>	NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_PAR	NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCI_RESET_L	NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE
TP_LPC_DREQ0_L	NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MEM ALIASES

TP_MEM_A_DQ_CB<7..0>	NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_MEM_A_DQS_N<8>	NC_MEM_A_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE
TP_MEM_A_DQS_P<8>	NC_MEM_A_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE
TP_MEM_B_DQ_CB<7..0>	NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_MEM_B_DQS_N<8>	NC_MEM_B_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE
TP_MEM_B_DQS_P<8>	NC_MEM_B_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MISC ALIASES

TP_HDA_SDIN1	NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE
TP_HDA_SDIN2	NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE
TP_HDA_SDIN3	NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_PWM0	NC_PCH_PWM0 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_PWM1	NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_PWM2	NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_PWM3	NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_SST	NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_CL_CLK1	NC_PCH_CL_CLK1 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_CL_DATA1	NC_PCH_CL_DATA1 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_CL_RST1	NC_PCH_CL_RST1 MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCIE ALIASES

TP_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5P MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5N MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE6P	NC_PCIE_CLK100M_PE6P MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE6N	NC_PCIE_CLK100M_PE6N MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE7P	NC_PCIE_CLK100M_PE7P MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE7N	NC_PCIE_CLK100M_PE7N MAKE_BASE=TRUE NO_TEST=TRUE
TP_PE_RX_N<3..0>	NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PE_RX_P<3..0>	NC_PE_RXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PE_TX_N<3..0>	NC_PE_TXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PE_TX_P<3..0>	NC_PE_TXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_D2R_PERN4	NC_PCIE_D2R_PERN4 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_D2R_PERP4	NC_PCIE_D2R_PERP4 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_R2D_PETN4	NC_PCIE_R2D_PETN4 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_R2D_PETP4	NC_PCIE_R2D_PETP4 MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4P MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4N MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED USB ALIASES

TP_USB_1N	NC_USB_1N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_1P	NC_USB_1P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_2N	NC_USB_2N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_2P	NC_USB_2P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_3N	NC_USB_3N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_3P	NC_USB_3P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_4N	NC_USB_4N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_4P	NC_USB_4P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_5N	NC_USB_5N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_5P	NC_USB_5P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_6N	NC_USB_6N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_6P	NC_USB_6P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_7N	NC_USB_7N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_7P	NC_USB_7P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_10N	NC_USB_10N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_10P	NC_USB_10P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_11N	NC_USB_11N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_11P	NC_USB_11P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_12N	NC_USB_12N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_12P	NC_USB_12P MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_13N	NC_USB_13N MAKE_BASE=TRUE NO_TEST=TRUE
TP_USB_13P	NC_USB_13P MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED DISPLAY ALIASES

TP_CRT_IG_DDC_CLK	NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_DDC_DATA	NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_RED	NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_GREEN	NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_BLUE	NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_HSYNC	NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE
TP_CRT_IG_VSYNC	NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_MLN<3..0>	NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_MLP<3..0>	NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_AUX_N	NC_DP_IG_B_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_AUX_P	NC_DP_IG_B_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_HPD	NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_DDC_CLK	NC_DP_IG_B_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_B_DDC_DATA	NC_DP_IG_B_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_MLN<3..0>	NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_MLP<3..0>	NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_AUX_N	NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_AUX_P	NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_HPD	NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_CTRL_CLK	NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_C_CTRL_DATA	NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_MLN<3..0>	NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_MLP<3..0>	NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_AUXN	NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_AUXP	NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_HPD	NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_CTRL_CLK	NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
TP_DP_IG_D_CTRL_DATA	NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_TVCLKINN	NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_TVCLKINP	NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_STALLN	NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_STALLP	NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_INTN	NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SDVO_INTP	NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_L_BKLTCTL	NC_PCH_L_BKLTCTL MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_L_BKLTEN	NC_PCH_L_BKLTEN MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_L_VDD_EN	NC_PCH_L_VDD_EN MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_CLKOUT_DPN	NC_PCH_CLKOUT_DPN MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_CLKOUT_DPP	NC_PCH_CLKOUT_DPP MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED FDI ALIASES

TP_CPU_FDI_TX_N<7..0>	NC_CPU_FDI_TXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_CPU_FDI_TX_P<7..0>	NC_CPU_FDI_TXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_FDI_RX_N<7..0>	NC_PCH_FDI_RXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
TP_PCH_FDI_RX_P<7..0>	NC_PCH_FDI_RXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED SATA ALIASES

TP_SATA_D_D2RN	NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_D_D2RP	NC_SATA_D_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_D_R2D_CN	NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_D_R2D_CP	NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_E_D2RN	NC_SATA_E_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_E_D2RP	NC_SATA_E_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_E_R2D_CN	NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_E_R2D_CP	NC_SATA_E_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_F_D2RN	NC_SATA_F_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_F_D2RP	NC_SATA_F_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_F_R2D_CN	NC_SATA_F_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
TP_SATA_F_R2D_CP	NC_SATA_F_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE
UNUSED SIGNAL ALIAS

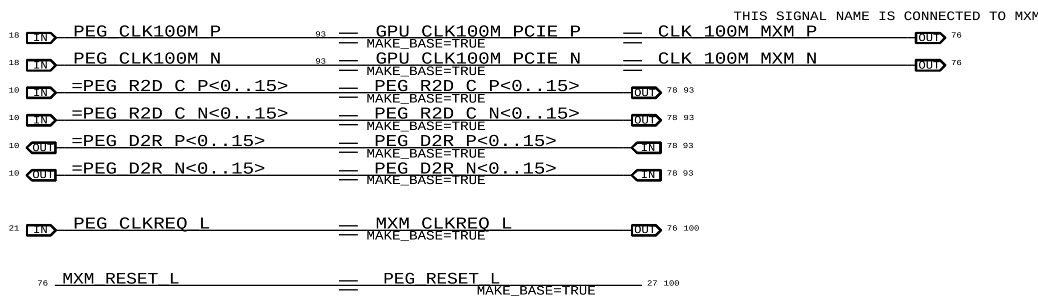
Apple Inc.

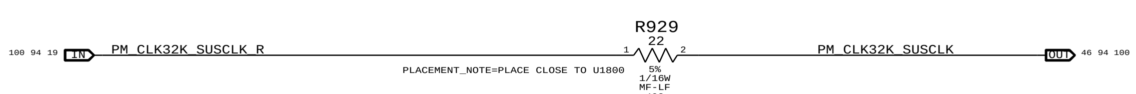
DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

PEG Slot Support





SYNC_MASTER=K62_SIJI SYNC_DATE=09/11/2010

PAGE TITLE

Signal Aliases

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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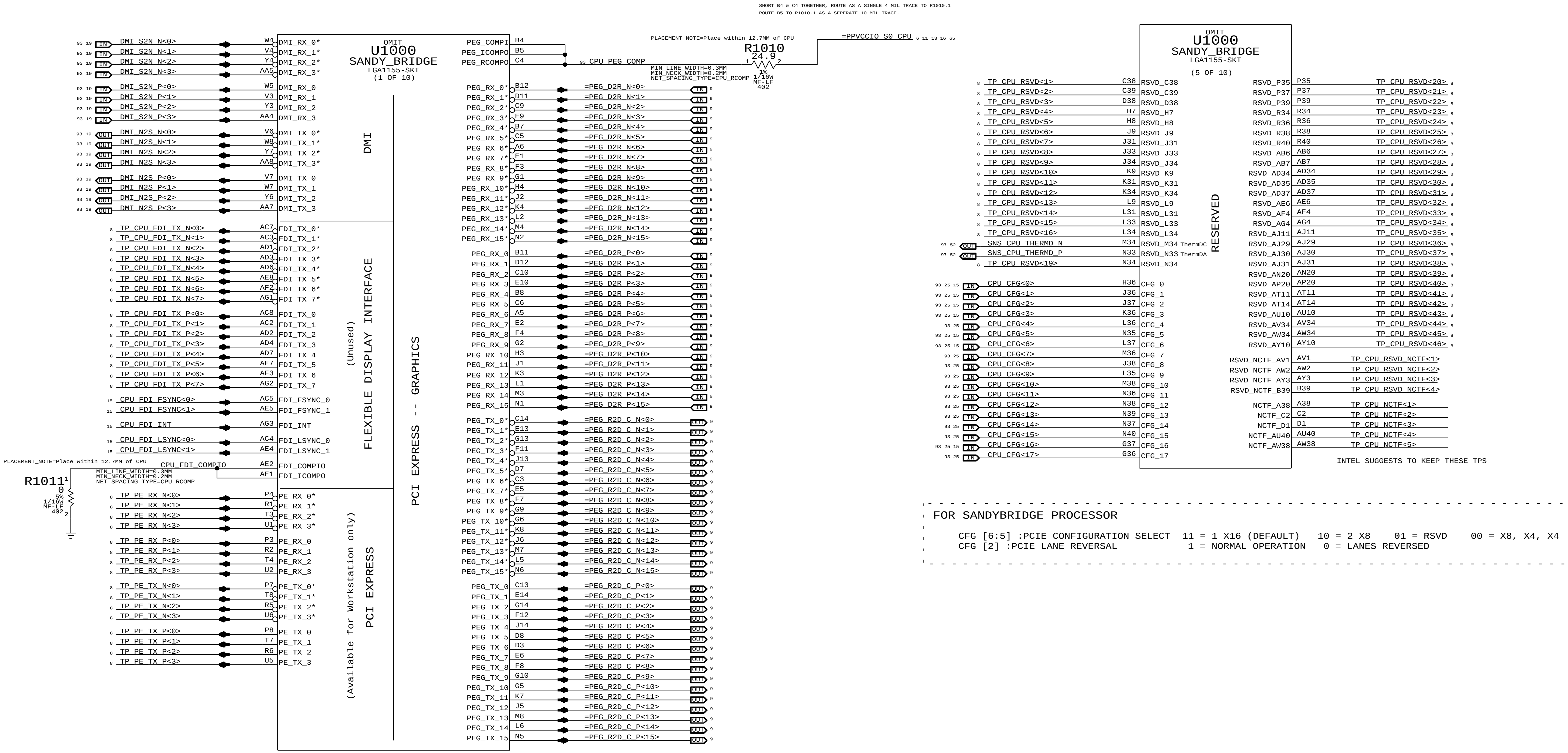

FOR SANDYBRIDGE PROCESSOR

CFG [6:5] :PCIE CONFIGURATION SELECT 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [2] :PCIE LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

PAGE TITLE

CPU DMI/PEG/FDI/RSVD

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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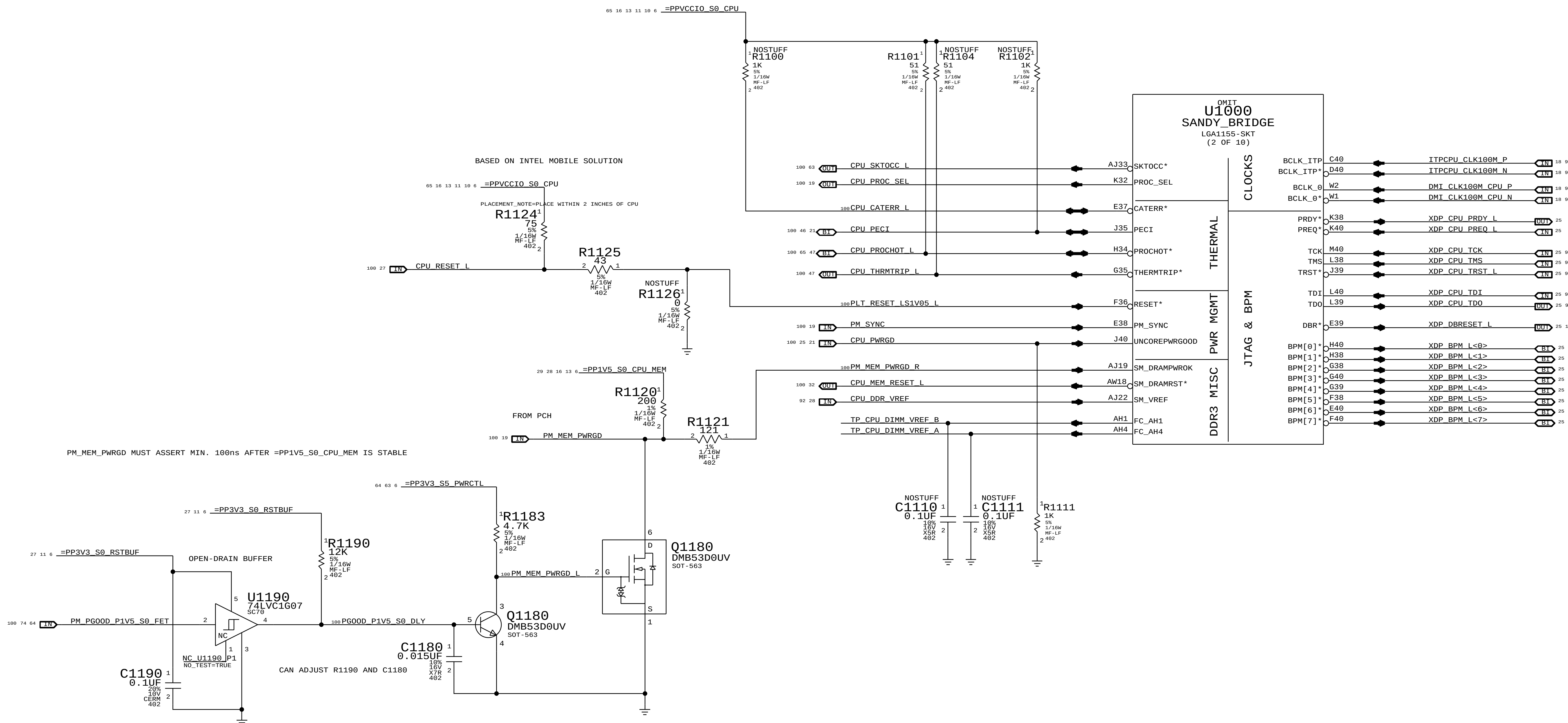


PAGE TITLE

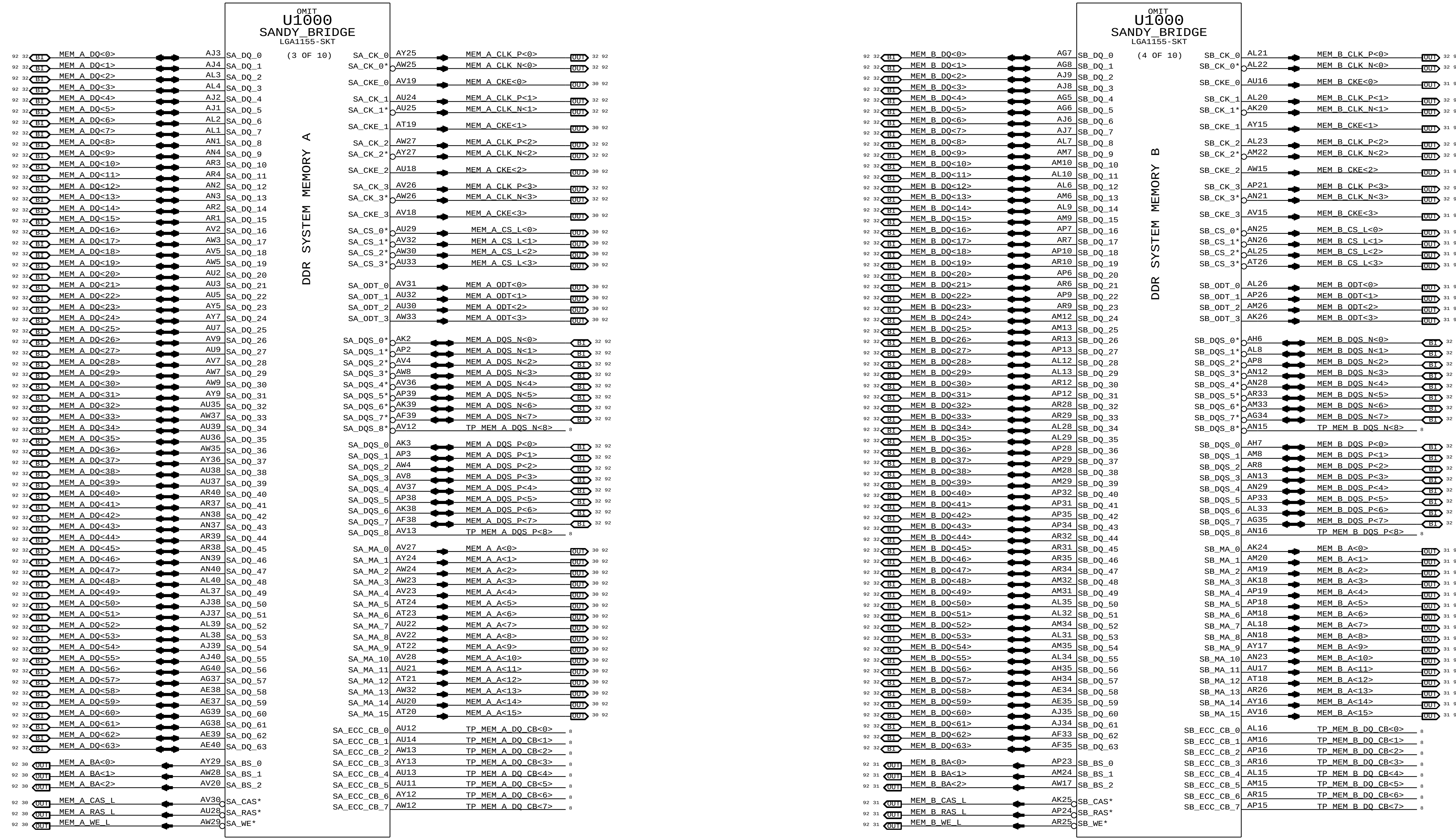
CPU CLOCK/MISC/JTAG

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



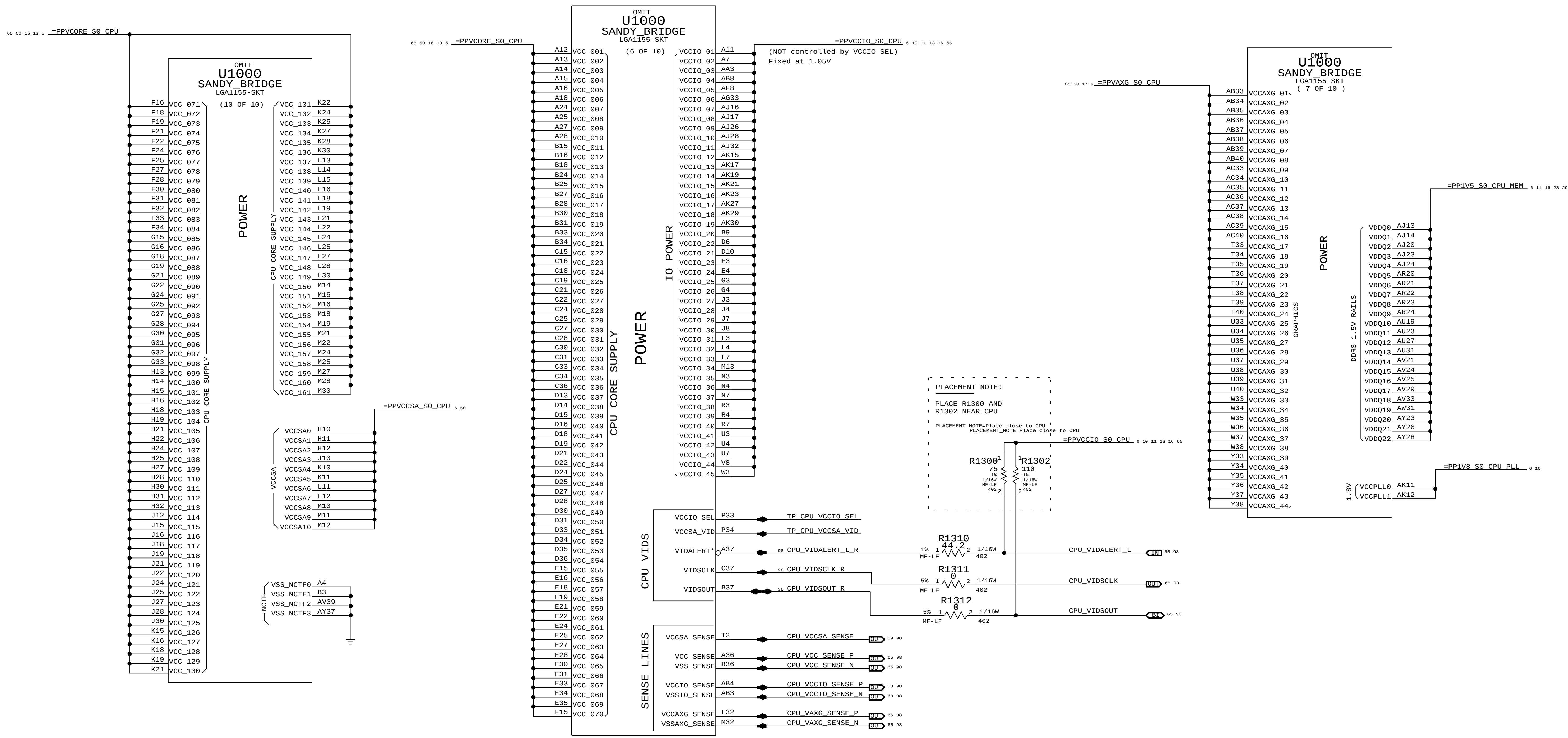
SYNC_DATE=01/09/2011

PAGE TITLE

CPU DDR3 INTERFACES

Apple Inc.

DRAWING NUMBER	051-8442
SIZE	D
REVISION	10.1.0
BRANCH	
PAGE	12 OF 110
SHEET	12 OF 101



SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

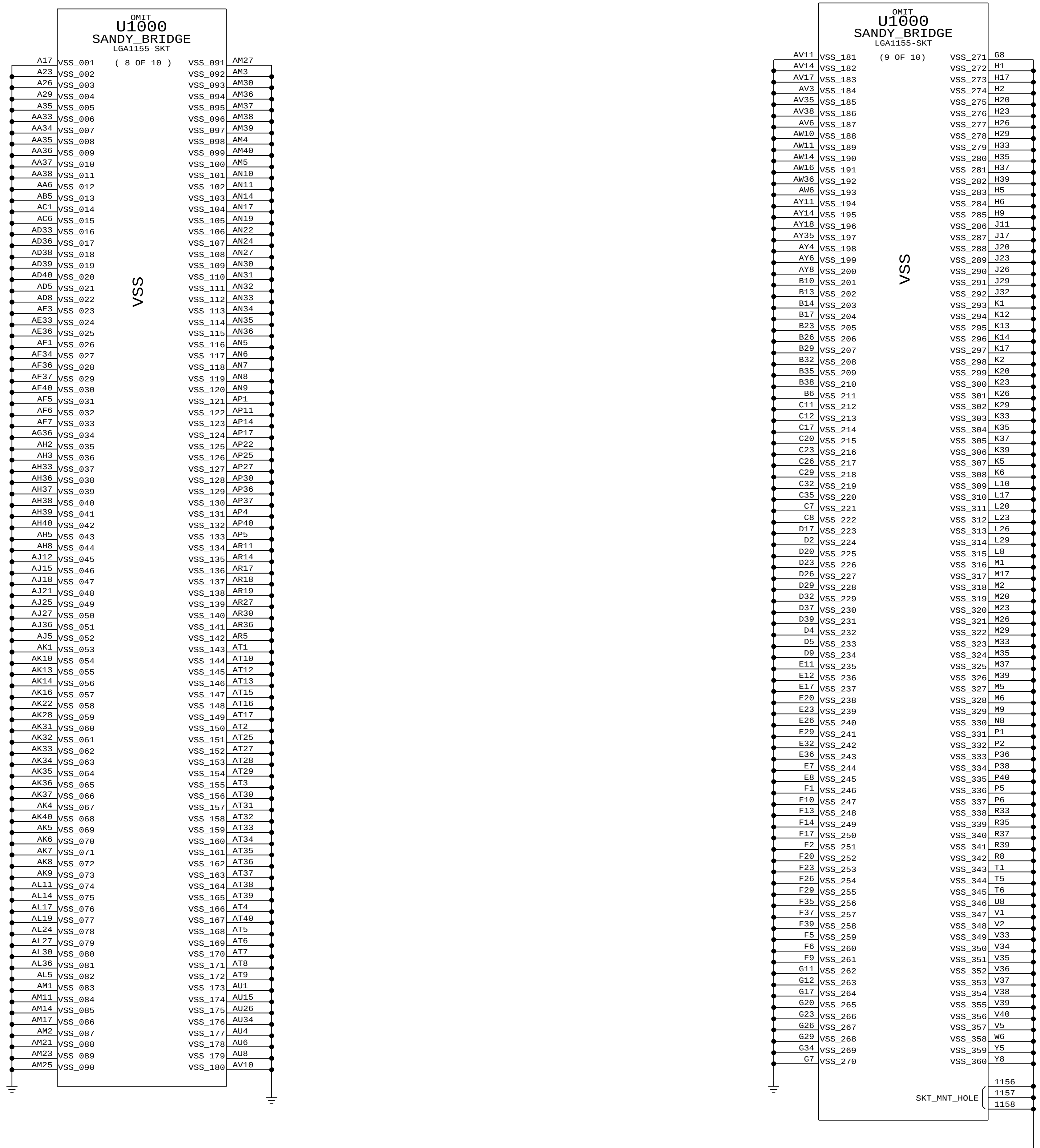
CPU POWER

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



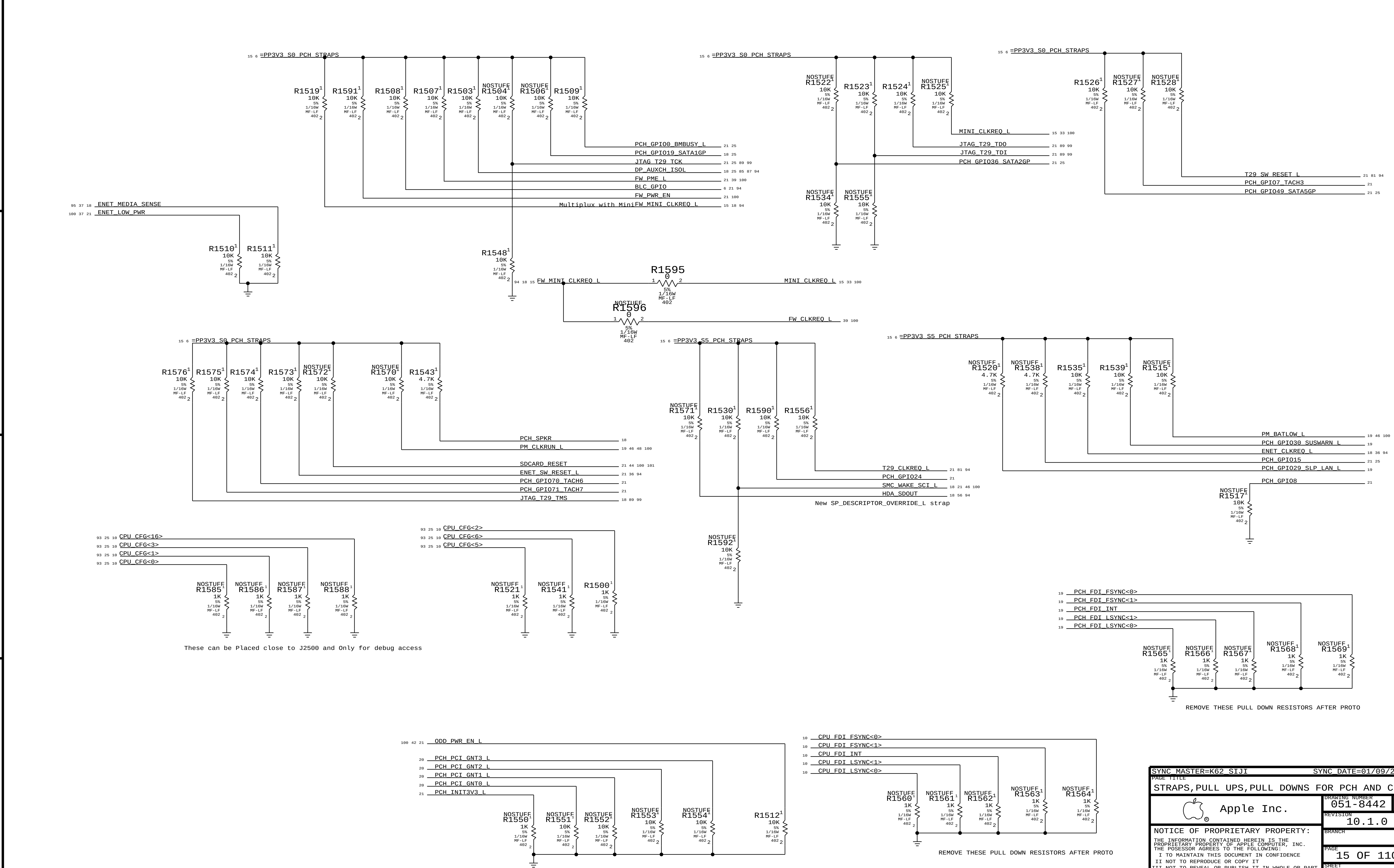
PAGE TITLE
CPU GROUNDS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

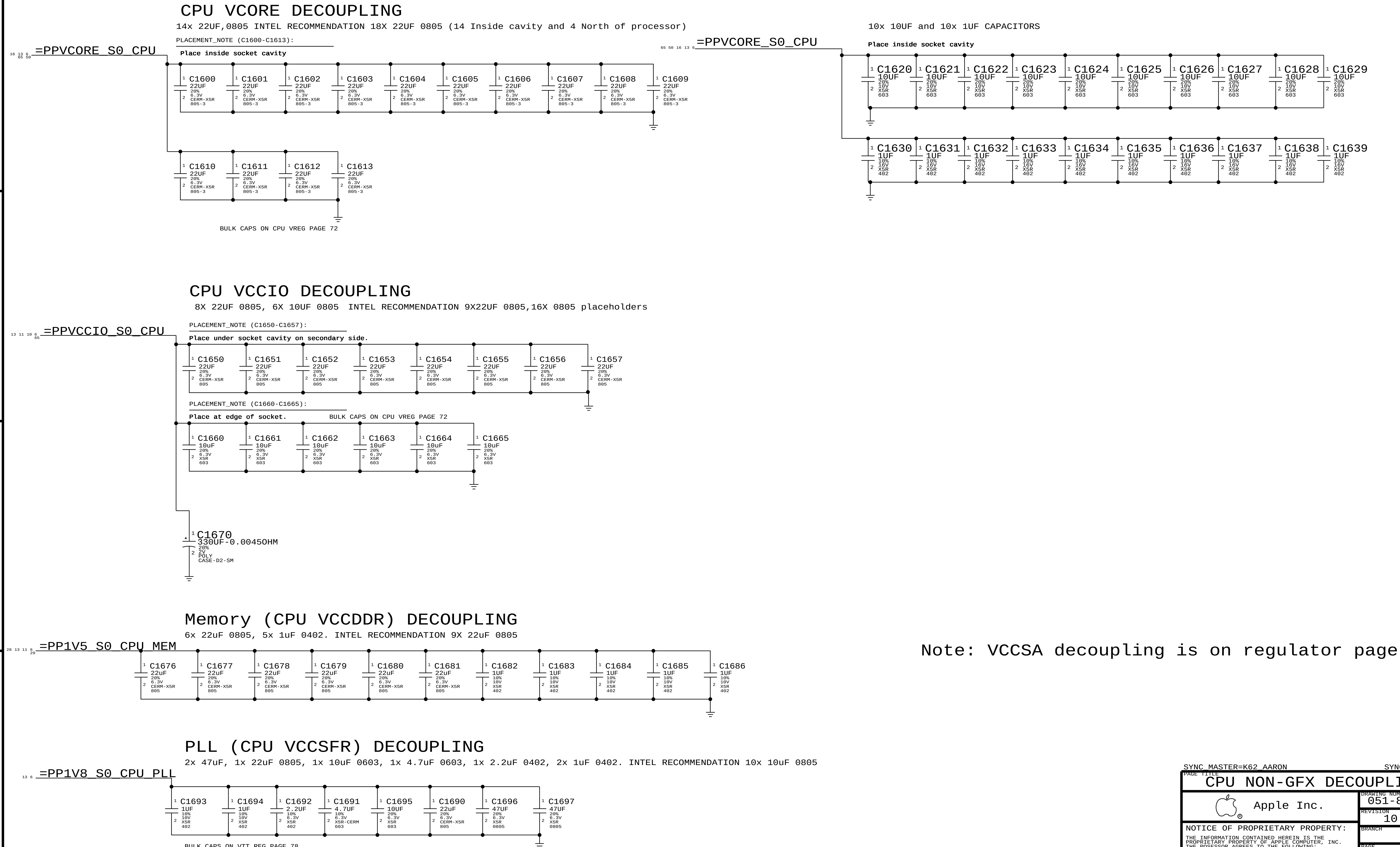
STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



Note: VCCSA decoupling is on regulator page

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

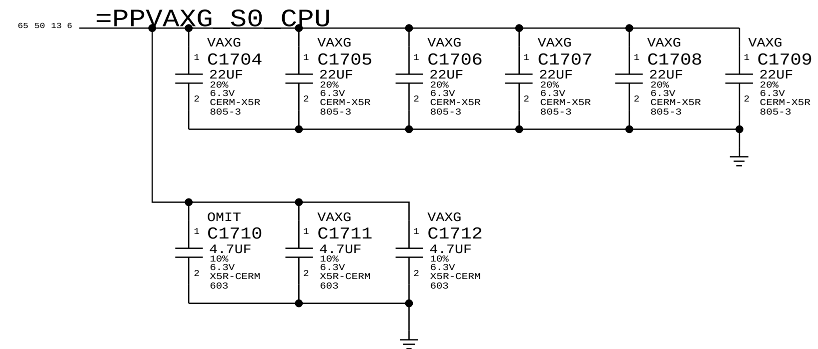
BRANCH

VAXG DECOUPLING

INTEL RECOMMENDATION 6X22UF 0805,3X 4.7UF

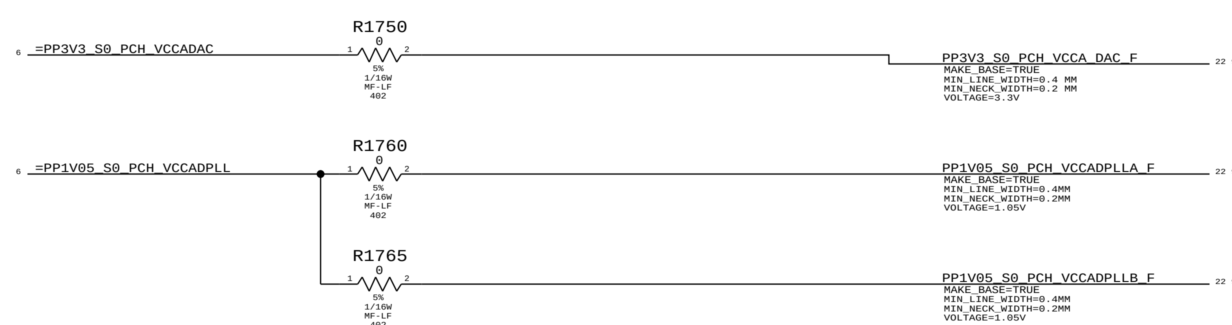
PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



BULK CAPS ON CPU VREG PAGE 73

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
138S0586	1	CAP,4.7UF,10%,6.3V,0603	C1710	VAXG
113S0022	1	RES,0 OHM,5%,0603	C1710	NO_VAXG



SYNC_MASTER=K62_AARON SYNC_DATE=11/30/2009

PAGE TITLE
GFX DECOUPLING & PCH PWR ALIAS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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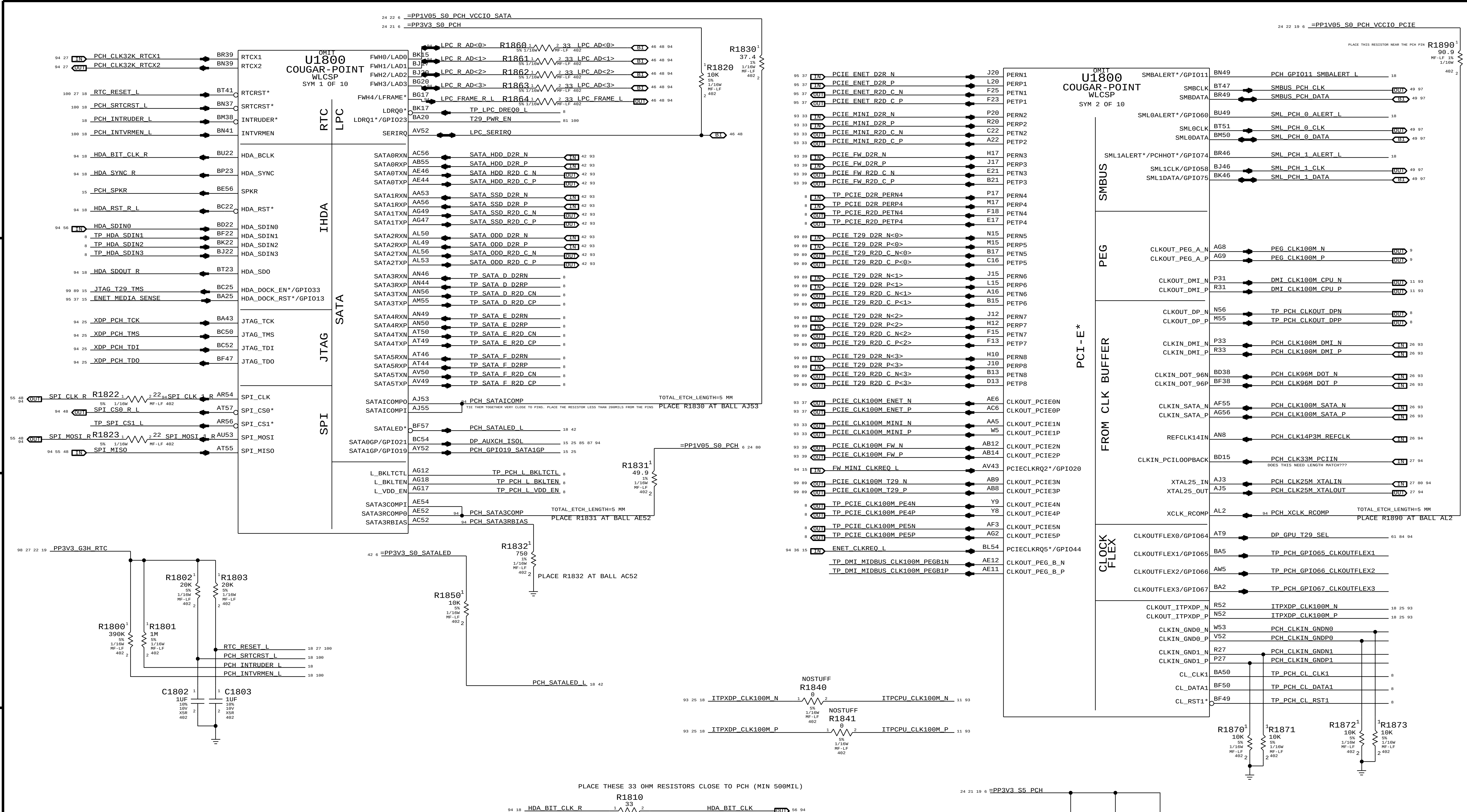


SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

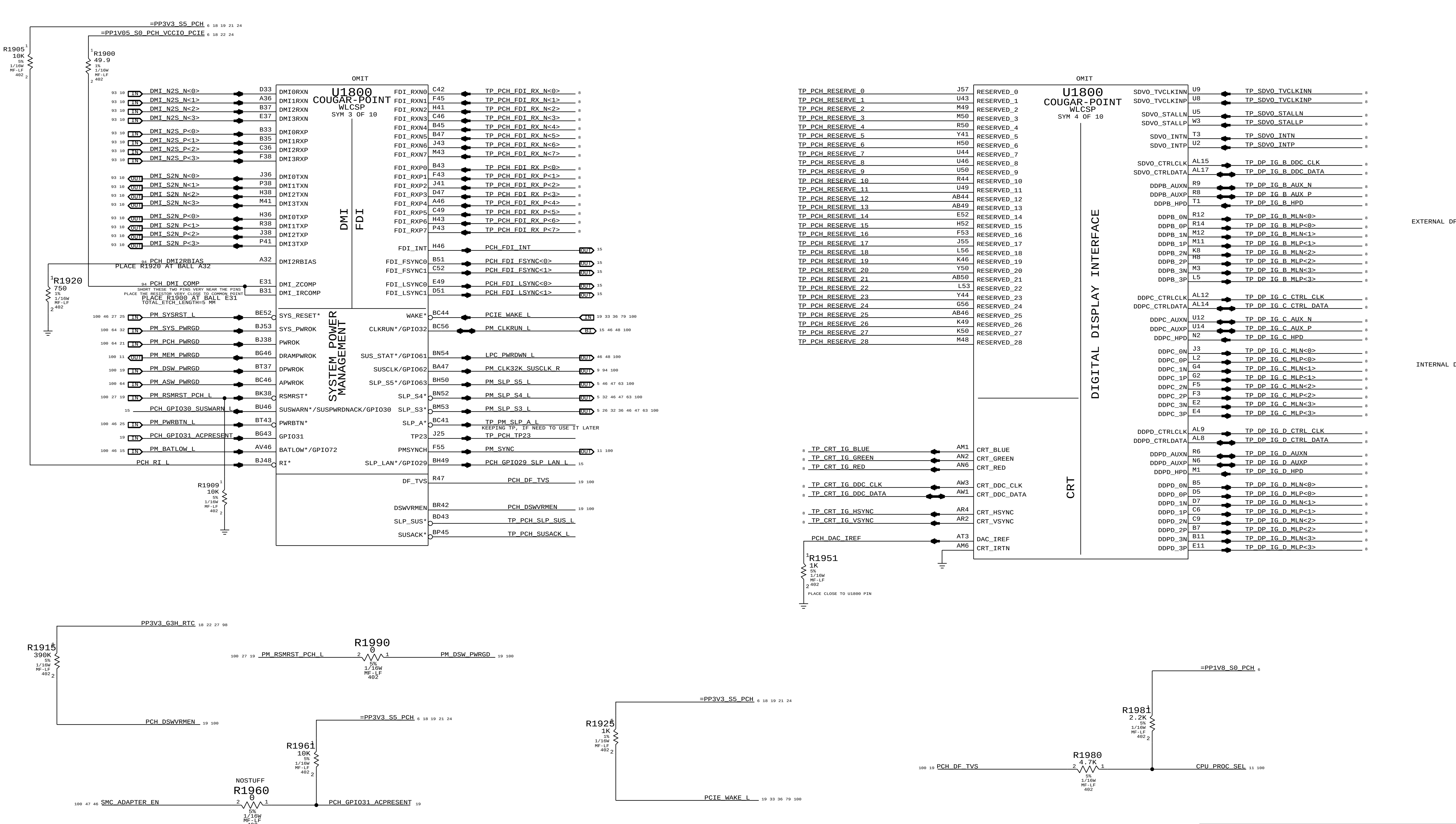
PAGE TITLE
PCH SATA/PCIE/CLK/LPC/SPI

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



PAGE TITLE

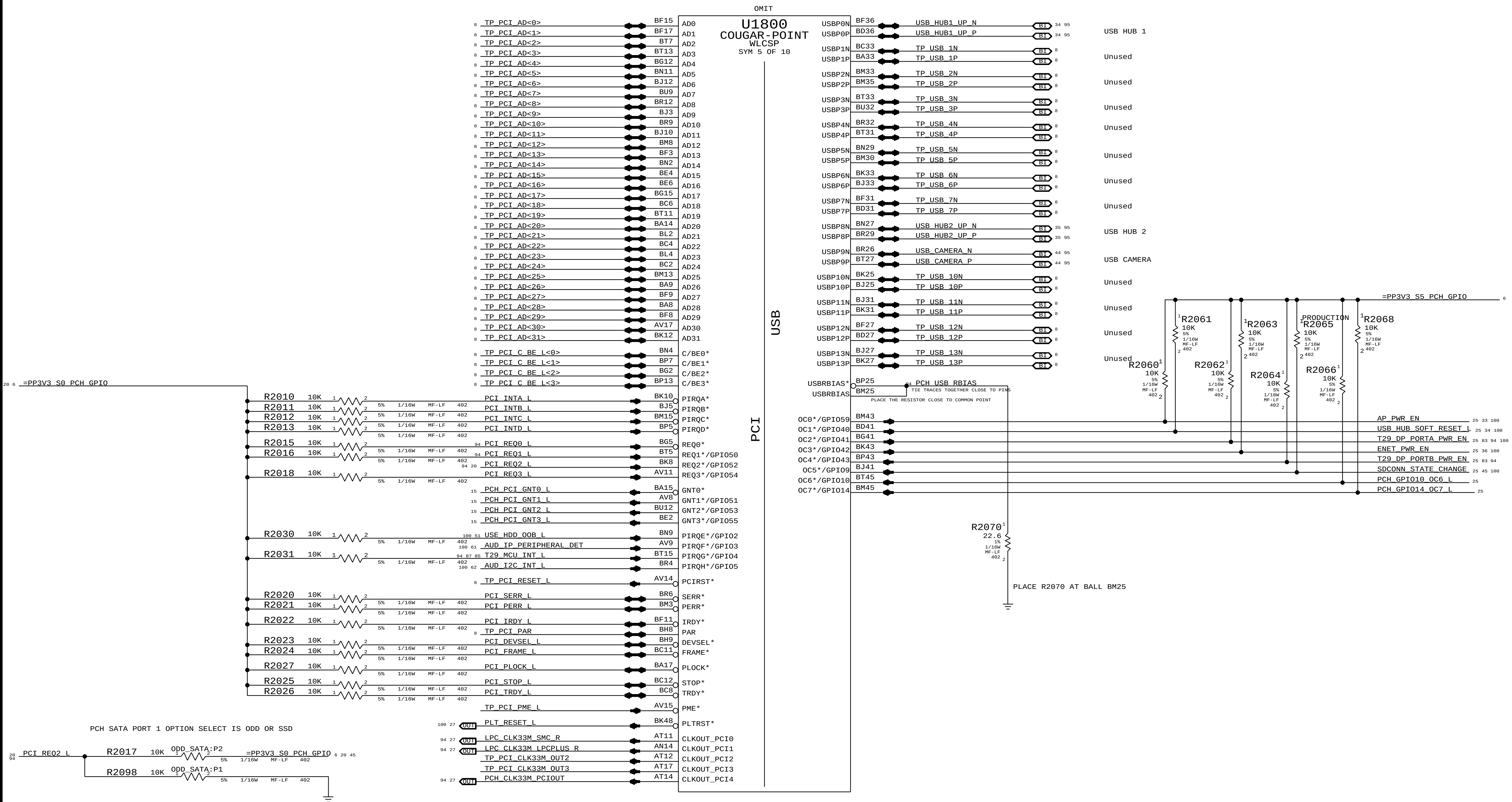
PCH DMI/FDI/GRAPHICS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH



SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

PCH PCI/FLASHCACHE/USB

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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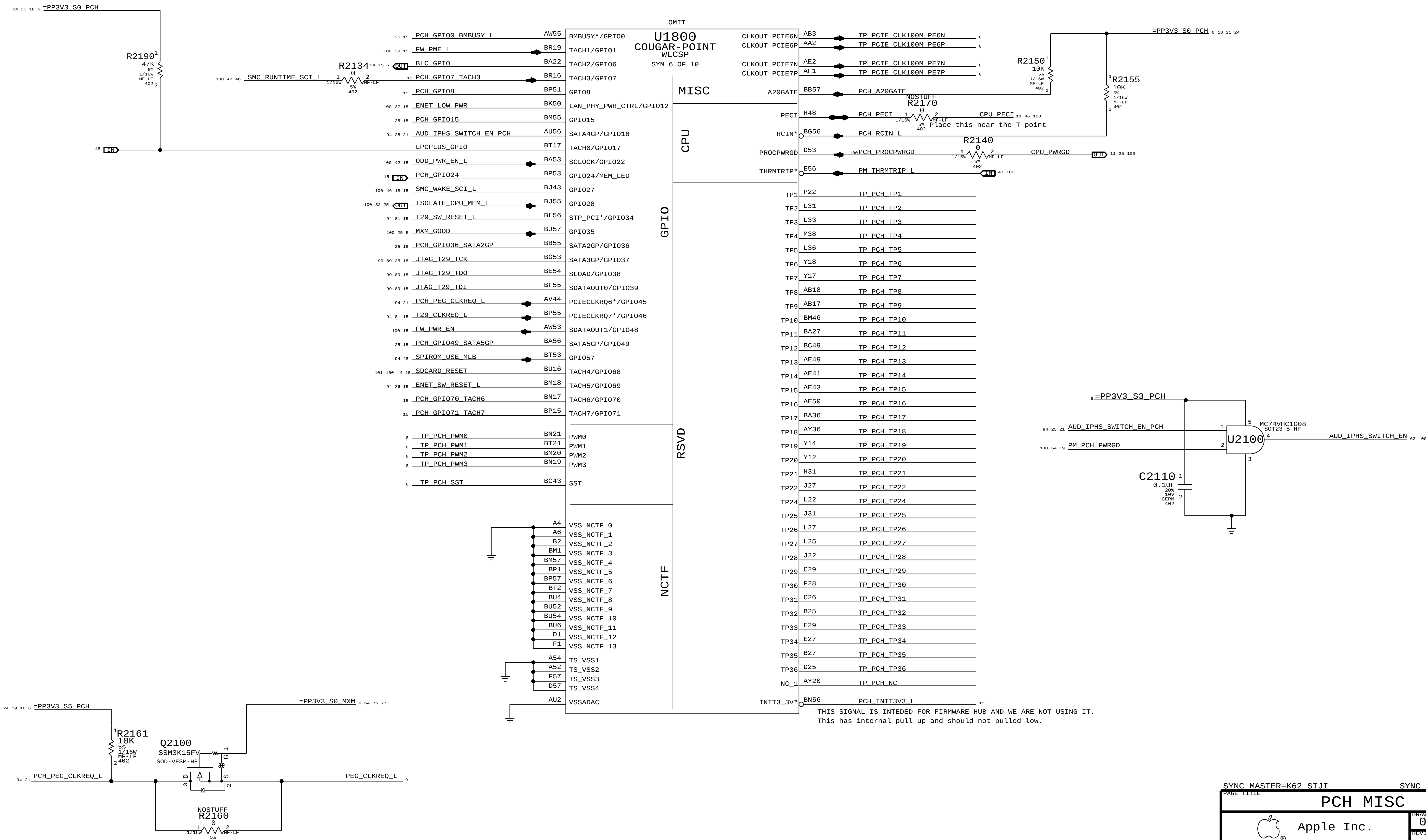

SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

PCH MISC

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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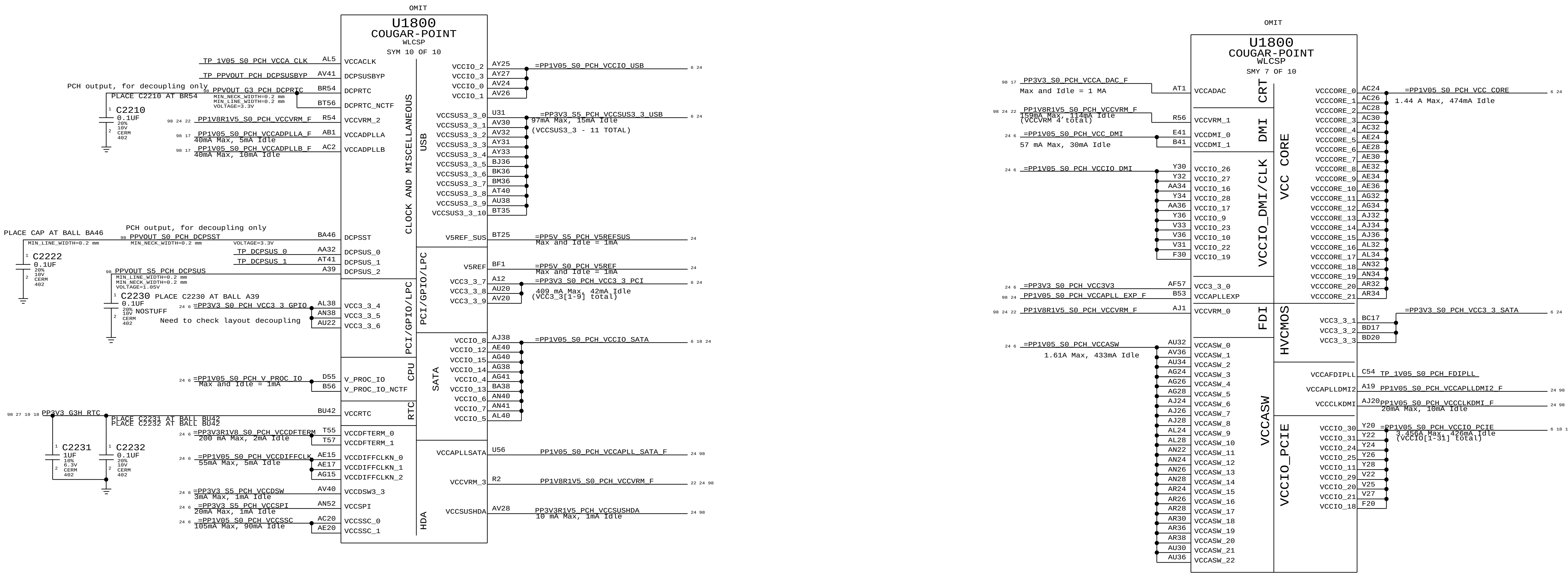


PAGE TITLE

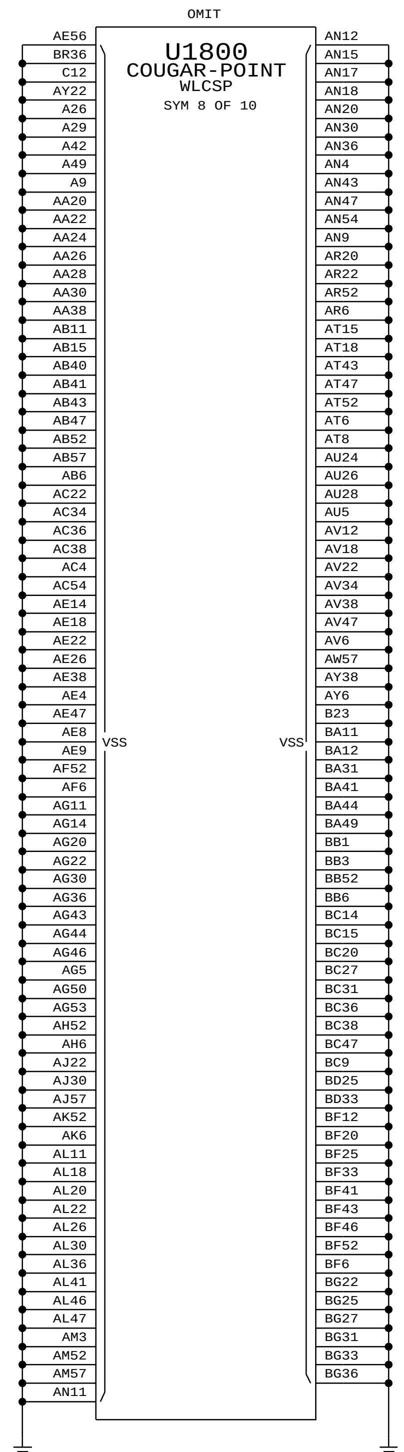
PCH POWER

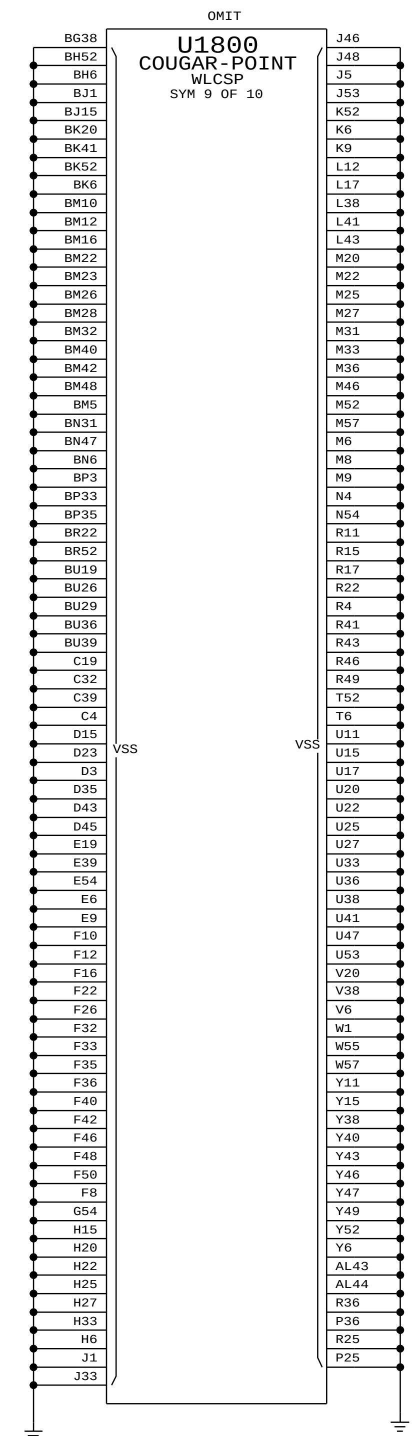
Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH





SYNC_MASTER=K62_AARON SYNC_DATE=07/01/2009

PAGE TITLE

PCH GROUNDS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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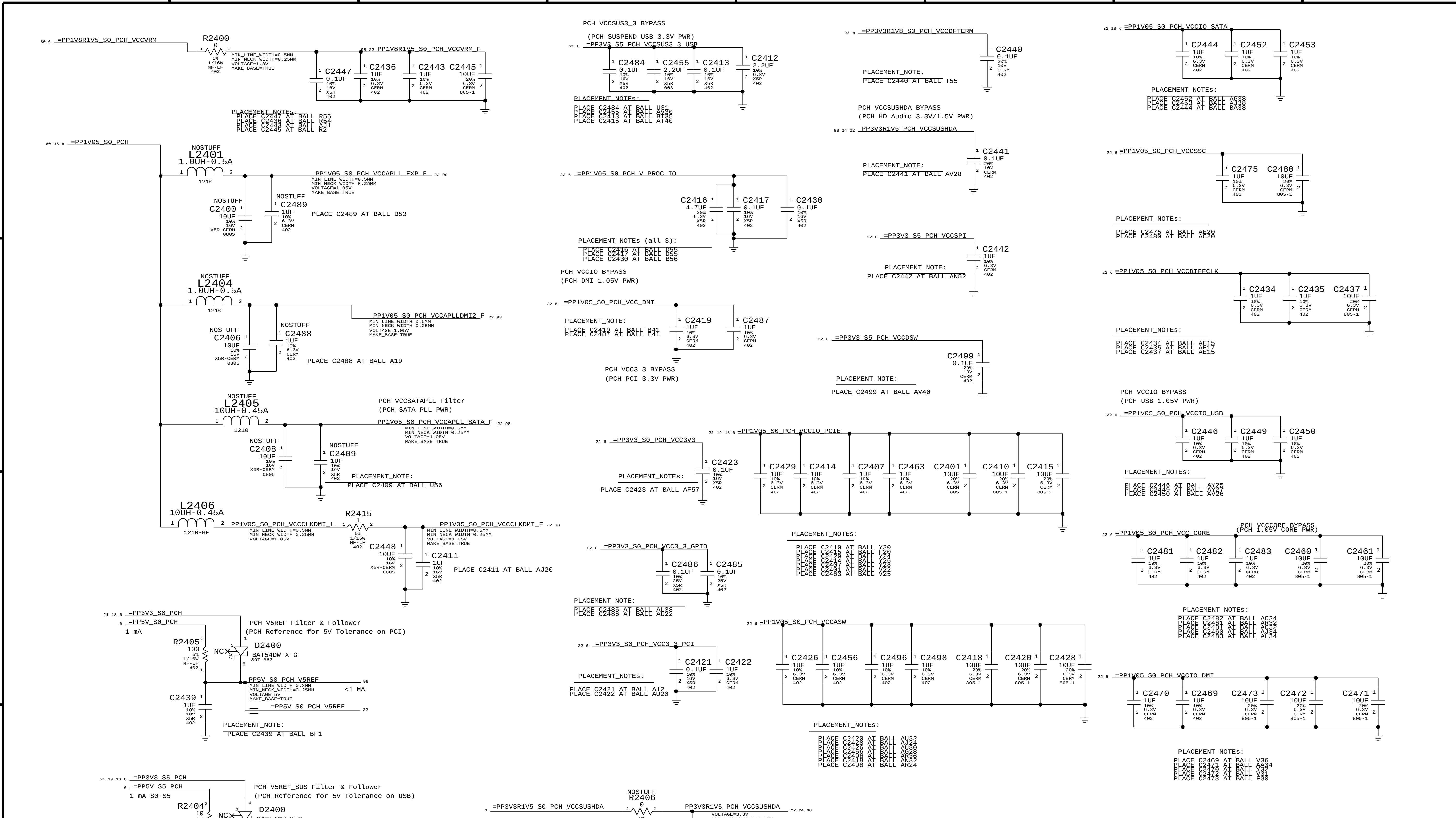


SYNC_MASTER=K62_AARON SYNC_DATE=07/01/2009

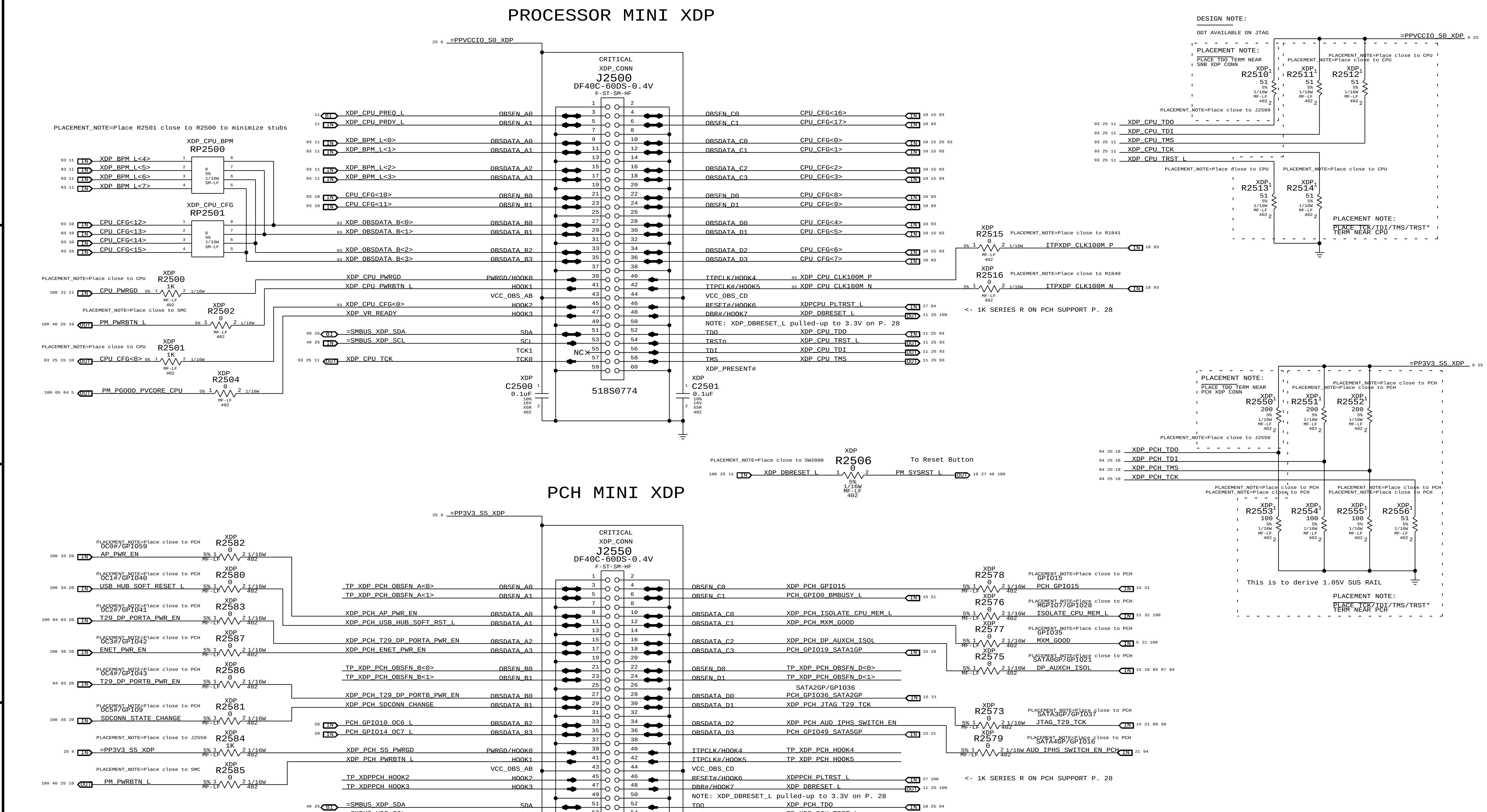
PAGE TITLE: PCH DECOUPLING

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

BRANCH



SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

CPU & PCH XDP

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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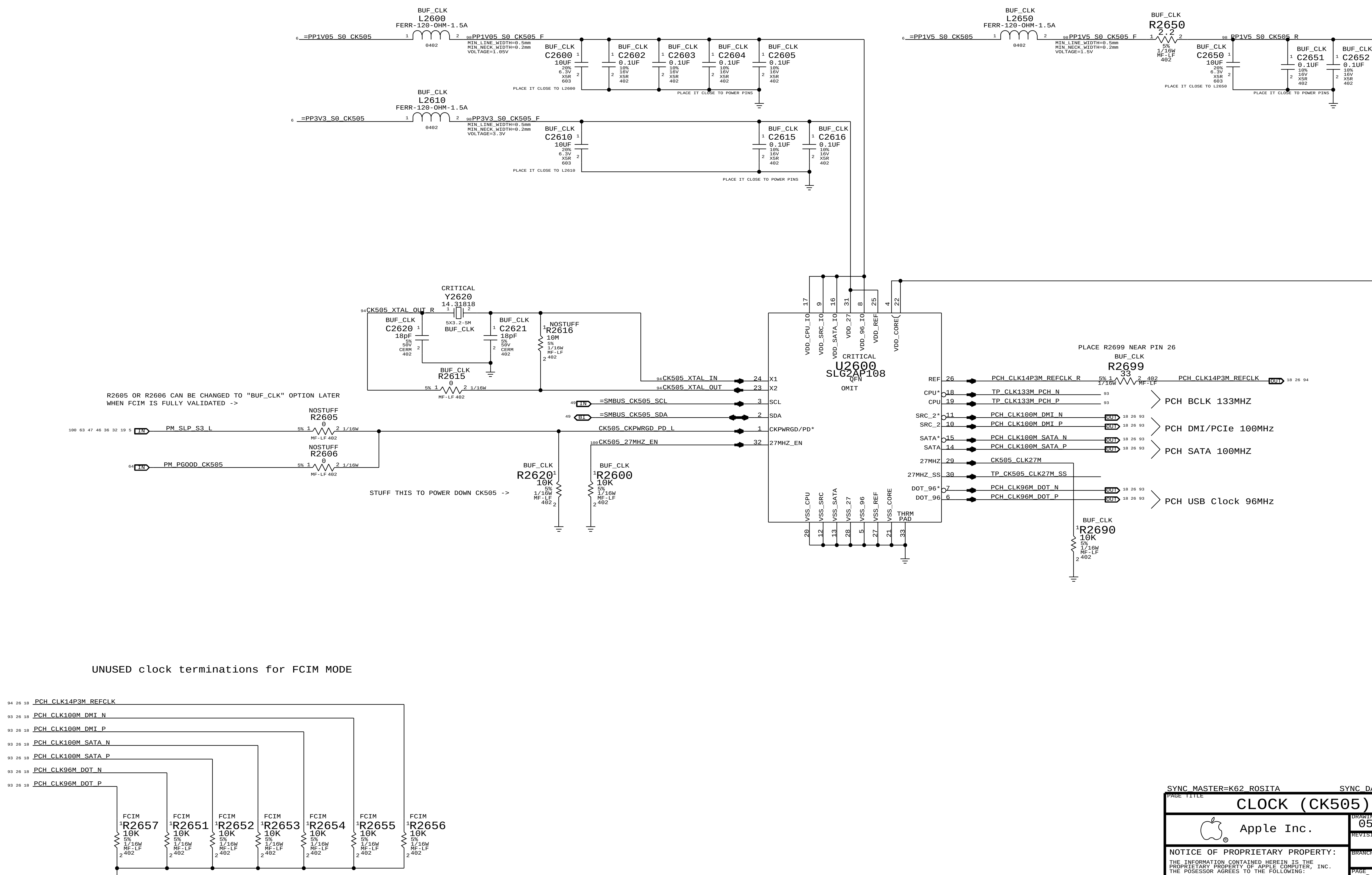


SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

CLOCK (CK505)

Apple Inc.

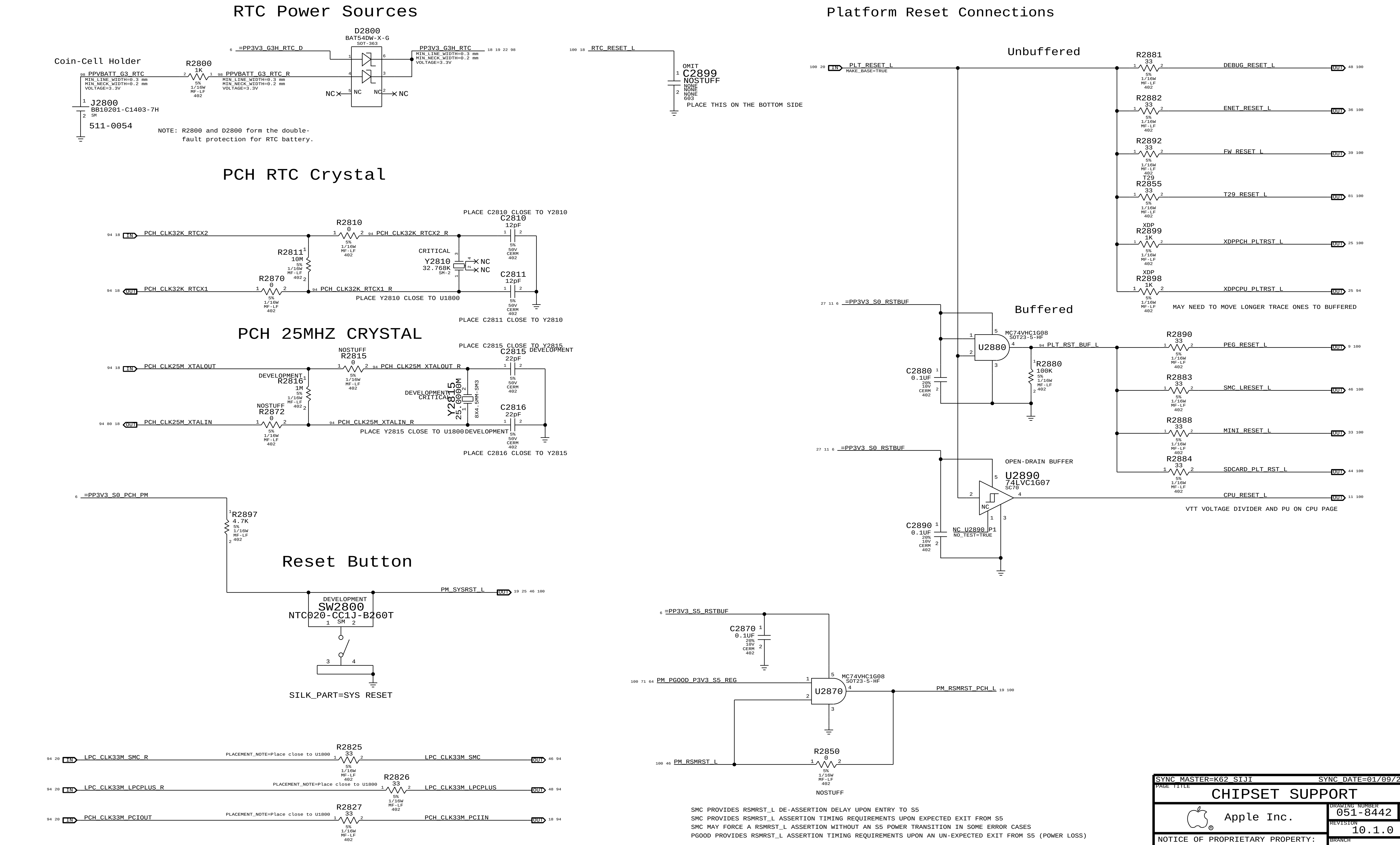
DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

PAGE 26 OF 110

SHEET 26 OF 101



SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

CHIPSET SUPPORT

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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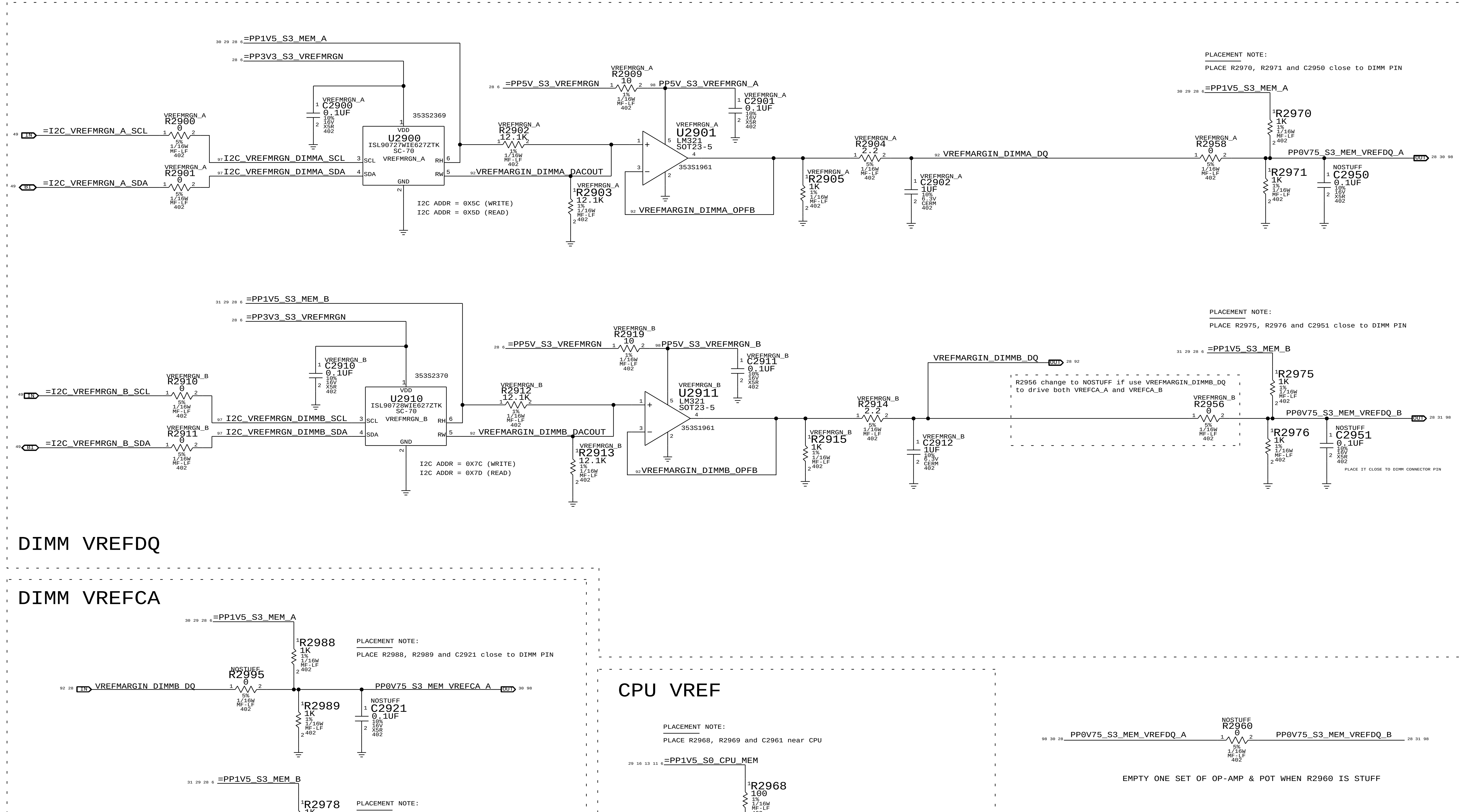


SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

DDR3 VREF MARGINING

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

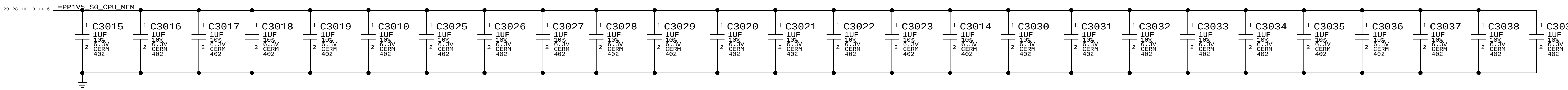
REVISION 10.1.0

BRANCH

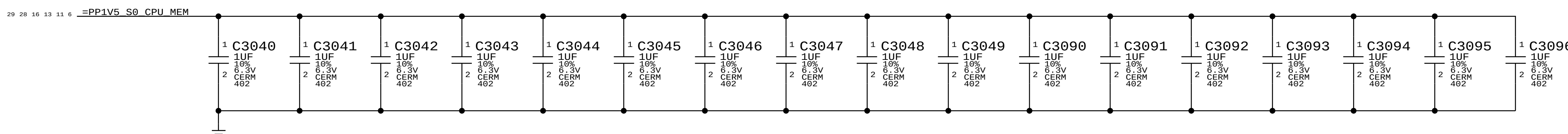
DIMM A (CLOSER TO CPU)

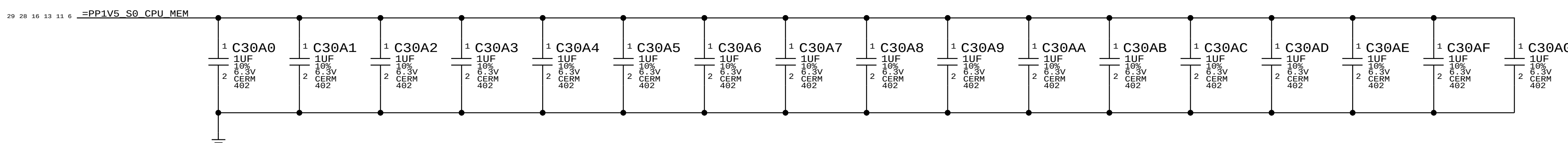
CAPS TO STITCH 1V5_CPU_MEM TO GND NEAR DIMM

DIMM B (FURTHER FROM CPU)

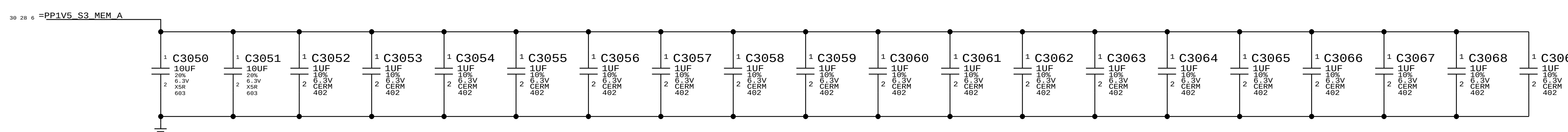


EXTRA DECOUPLING CAPS FOR 1V5_CPU_MEM RAIL

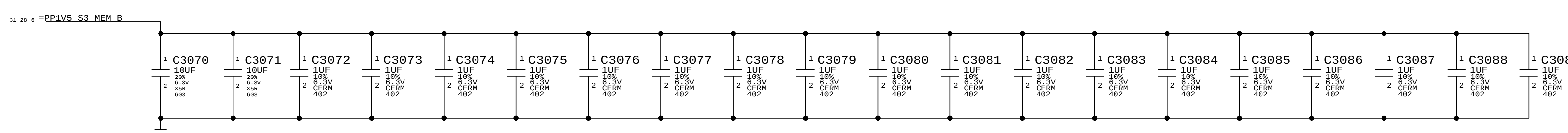




DECOUPLING CAPS FOR 1V5_S3_MEM AT CHANNEL A DIMM CONNECTOR



DECOUPLING CAPS FOR 1V5_S3_MEM AT CHANNEL B DIMM CONNECTOR



SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

MEMORY CAPS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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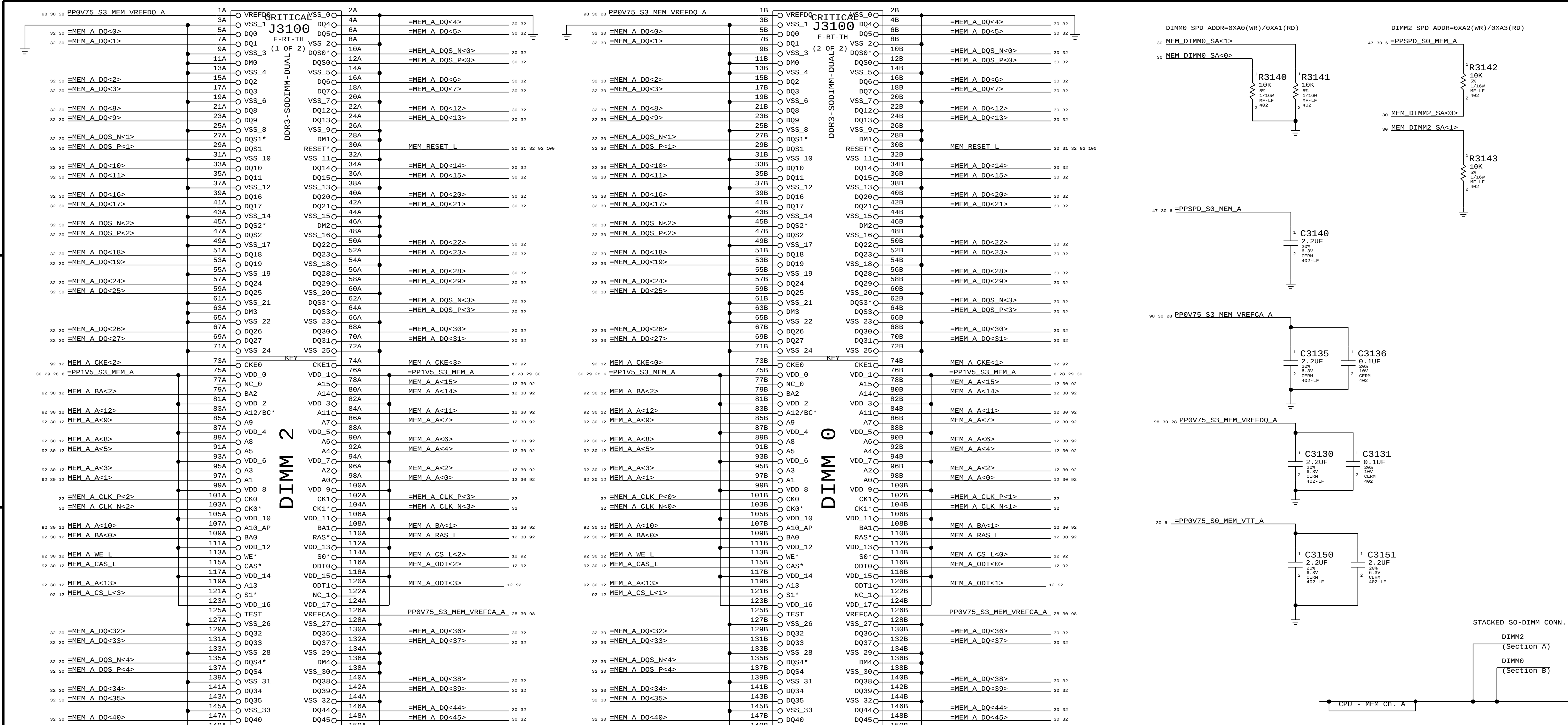


Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL — ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

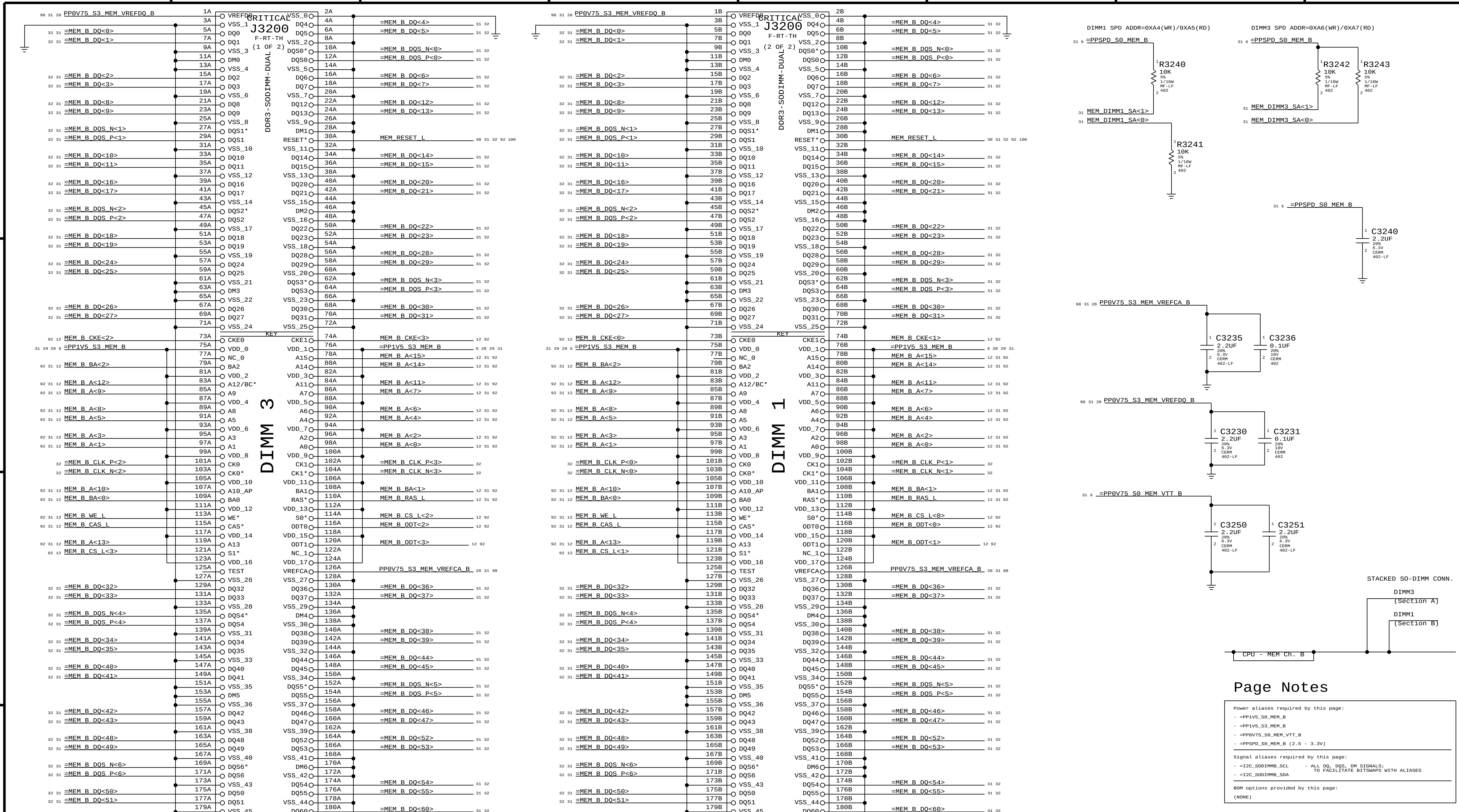
DDR3 SO-DIMM 0 & 2

Apple Inc.

DRAWING NUMBER	SIZE
051-8442	D

REVISION 10.1.0

BRANCH



Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMB_SCL
- =I2C_SODIMMB_SDA

- ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES

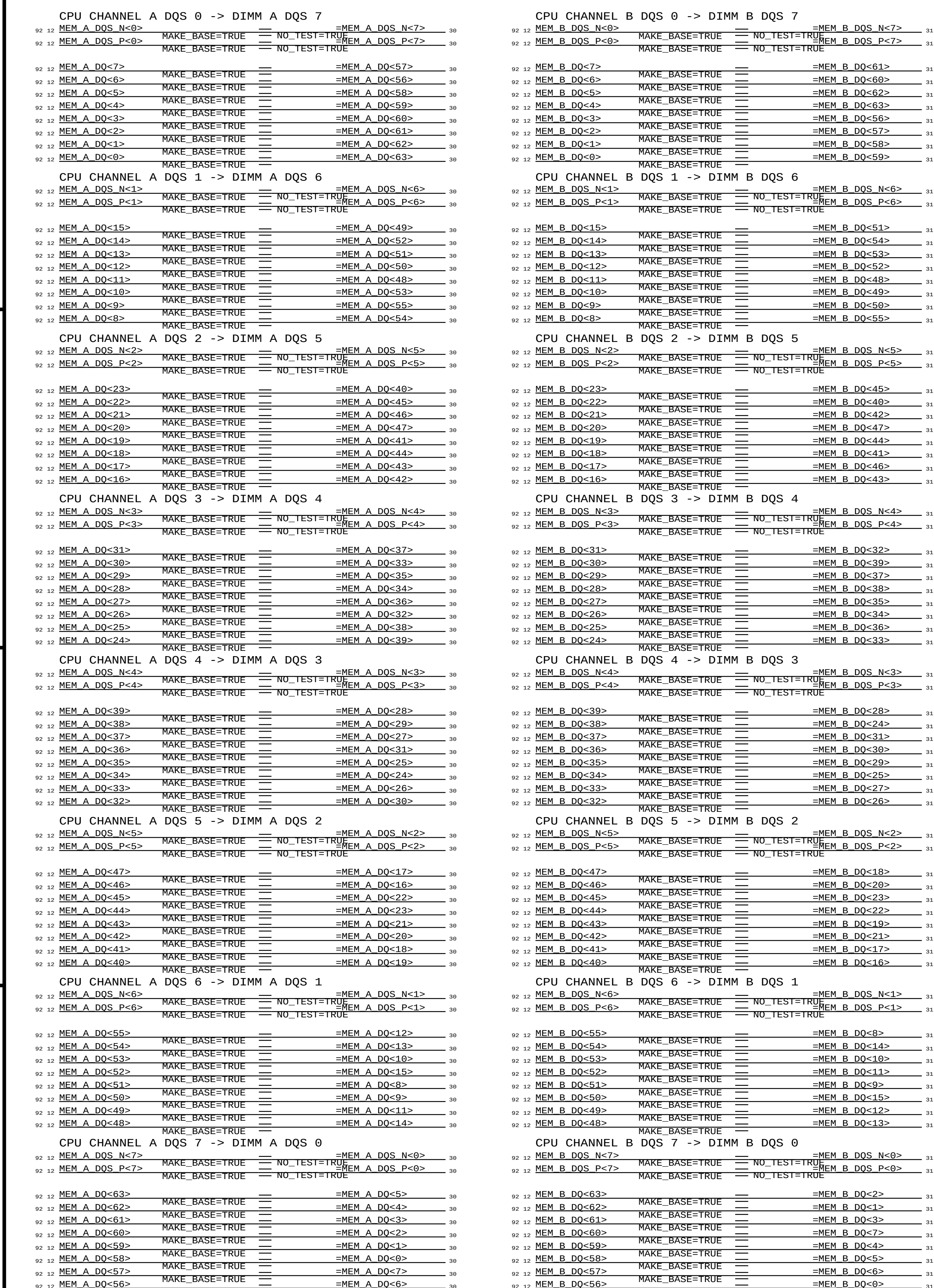
BOM options provided by this page:
(NONE)

SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE
DDR3 SO-DIMM 1 & 3

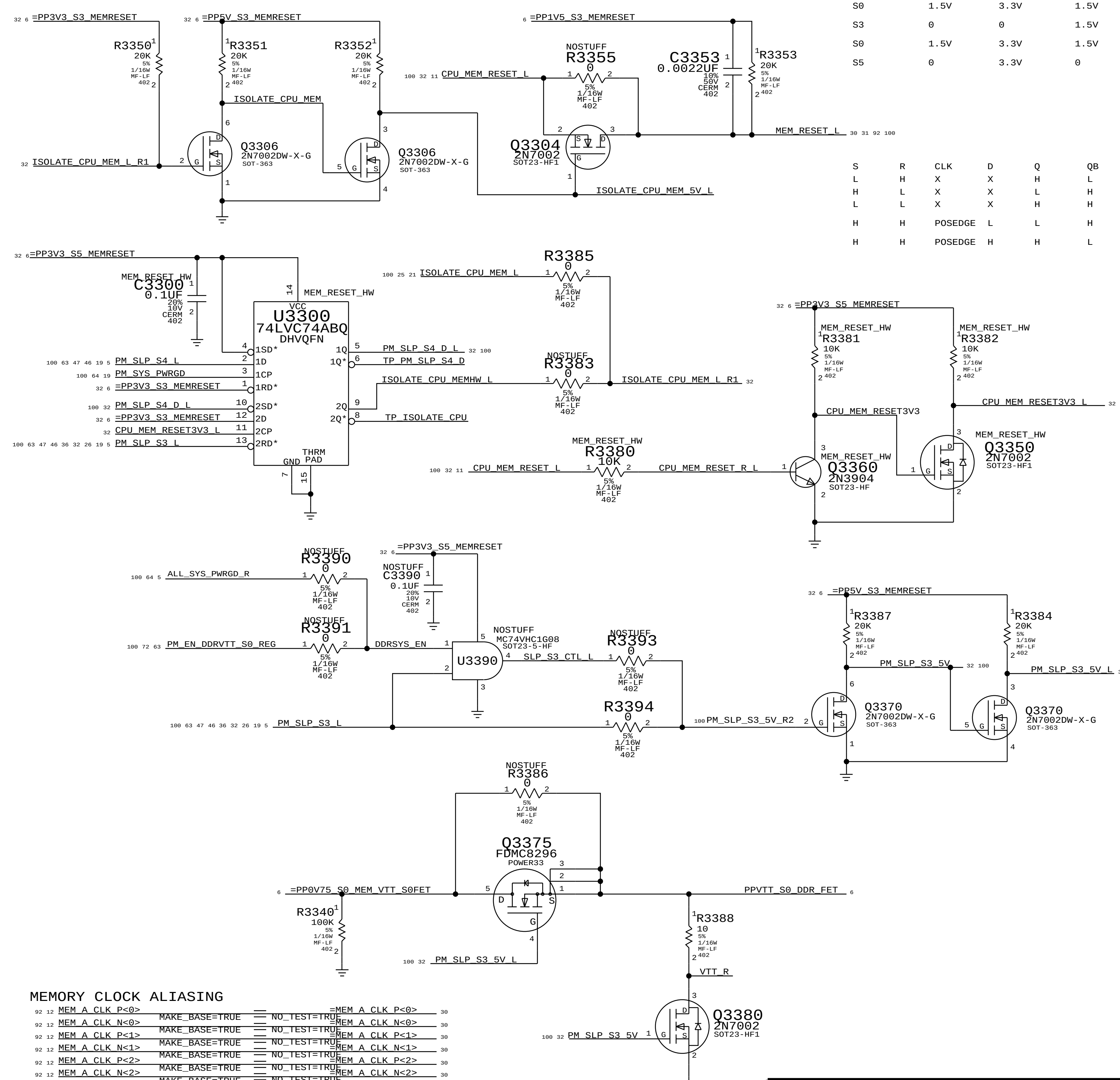
Apple Inc.

DRAWING NUMBER 051-8442 SIZE D
REVISION 10.1.0
BRANCH
PAGE 32 OF 110
SHEET 31 OF 101



DDR3 RESET SUPPORT

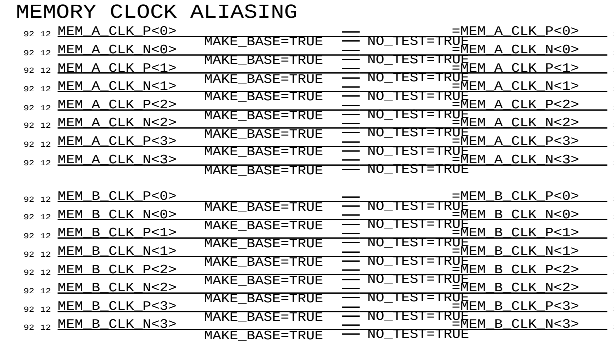
SNB? CANNOT CONTROL THIS SIGNAL DIRECTLY SINCE IT MUST BE HIGH IN SLEEP AND CPU MEM RAILS ARE NOT POWERED IN SLEEP.



	CPU_RESET_L	ISOLATE_L	MEM_RESET_L
S5	0	3.3V	0
S0	0	3.3V	0
S0	1.5V	3.3V	1.5V
S3	0	0	1.5V
S0	1.5V	3.3V	1.5V
S5	0	3.3V	0

S	R	CLK	D	Q	QB
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H	H
H	H	POSEDGE	L	L	H
H	H	POSEDGE	H	H	L

MEMORY CLOCK ALIASING



SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE: DDR3 SUPPORT AND BITSWAPS

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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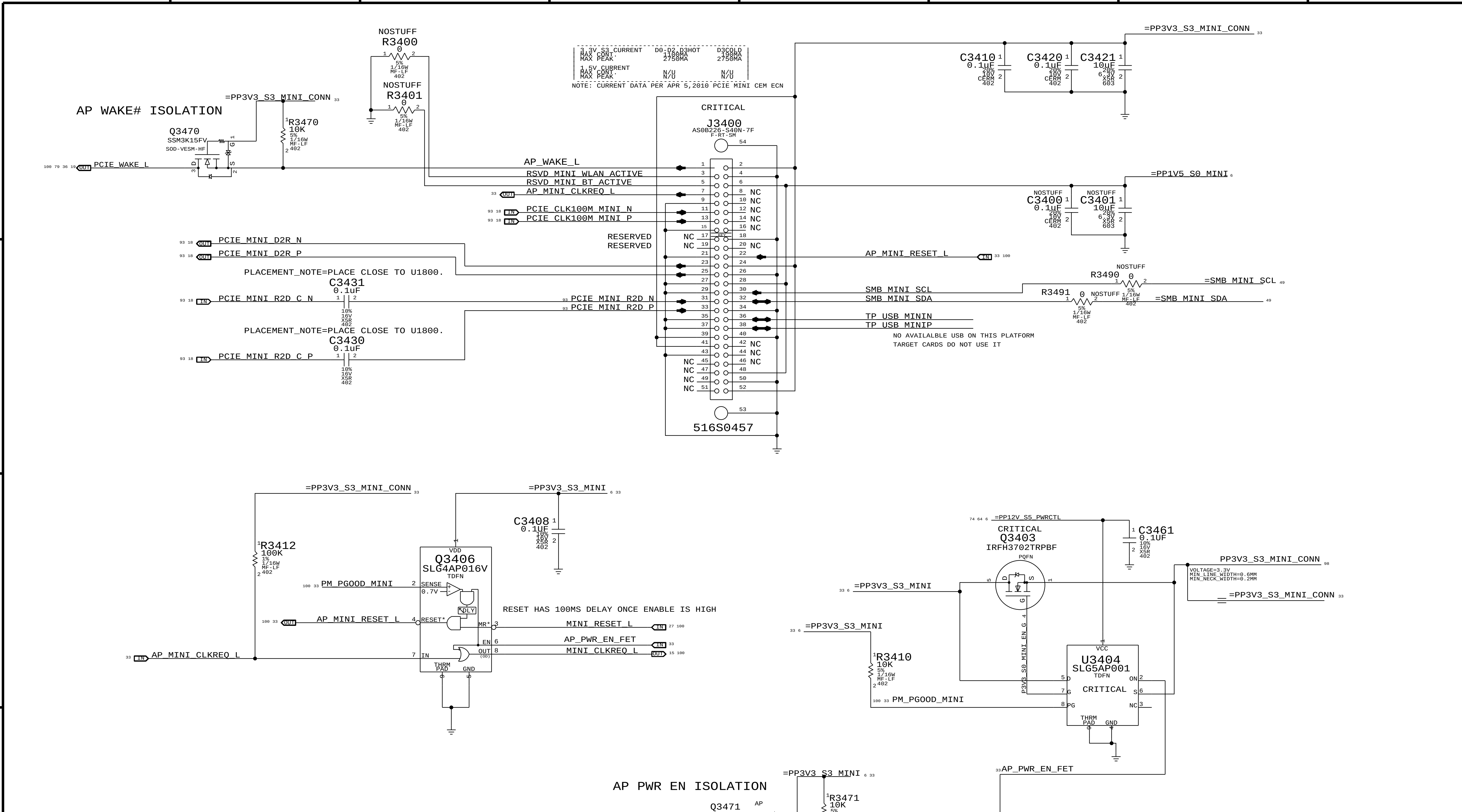

SYNC_MASTER=K62_AARON SYNC_DATE=07/16/2009

PAGE TITLE

PCI-E Wireless Connector

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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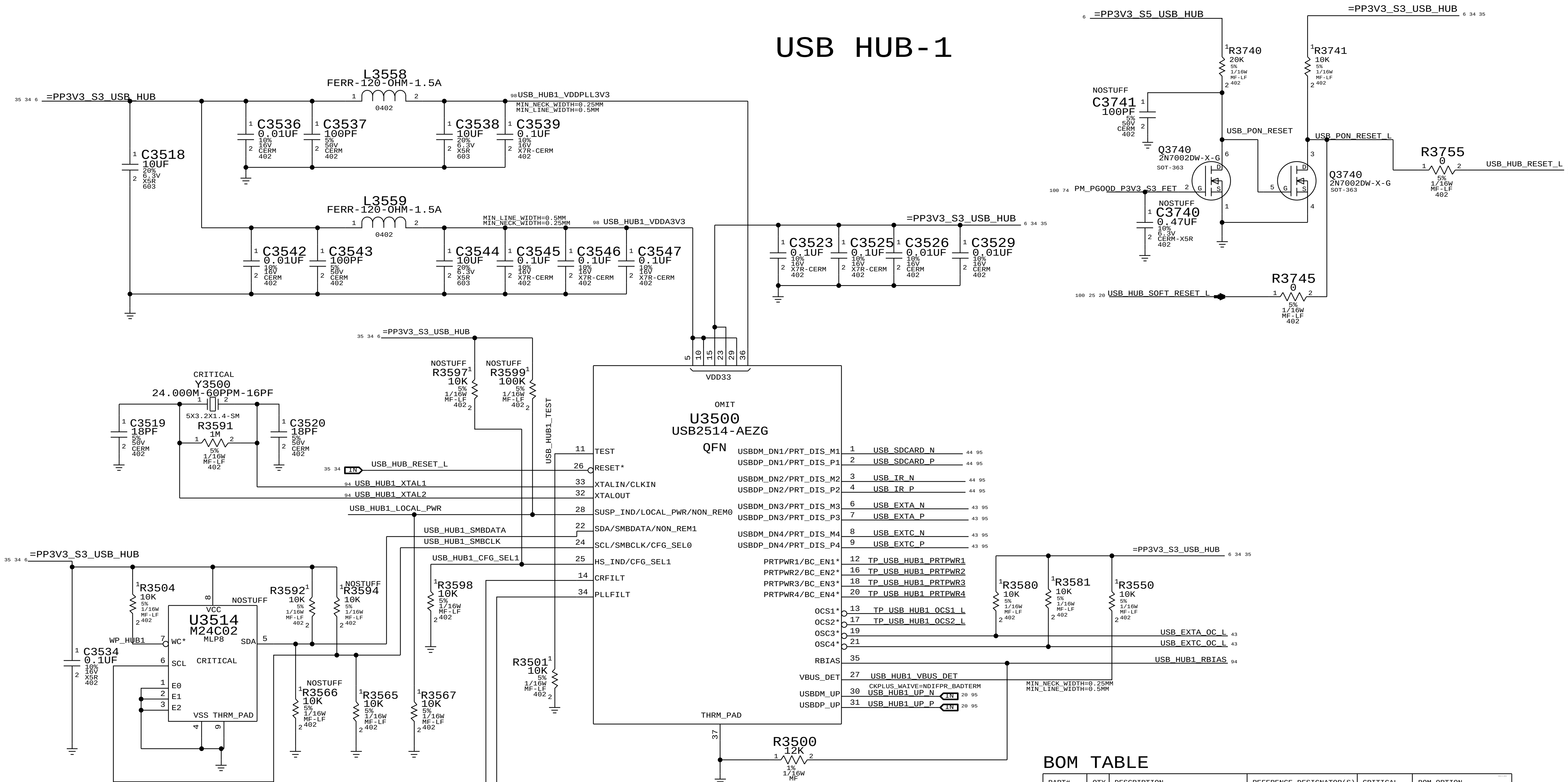
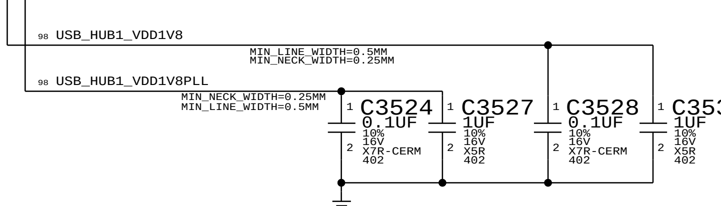
USB HUB-1



	SEL1	SEL0	DESCRIPTION
DEFAULT K23F ==>	0	0	Internal Default with Self powered Operation
	0	1	SMBUS Slave Config
	1	0	Internal Default with Bus powered Operation
	1	1	EEPROM Supported



	NON_REM1	NON_REM0	DESCRIPTION
	0	0	All ports are Non removable
	0	1	Port1 is non removable
DEFAULT K23F ==>	1	0	Port 1 and 2 are non removable
	1	1	Port1,2 and 3 are non Removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0721	2	SMSC USX2061-AEZG	U3500,U3600	CRITICAL	HUB_USX2061
338S0824	2	SMSC USB2514B	U3500,U3600	CRITICAL	HUB_USB2514B

SYNC_MASTER=K62_SIJI SYNC_DATE=11/14/2010

PAGE TITLE: USB HUB 1

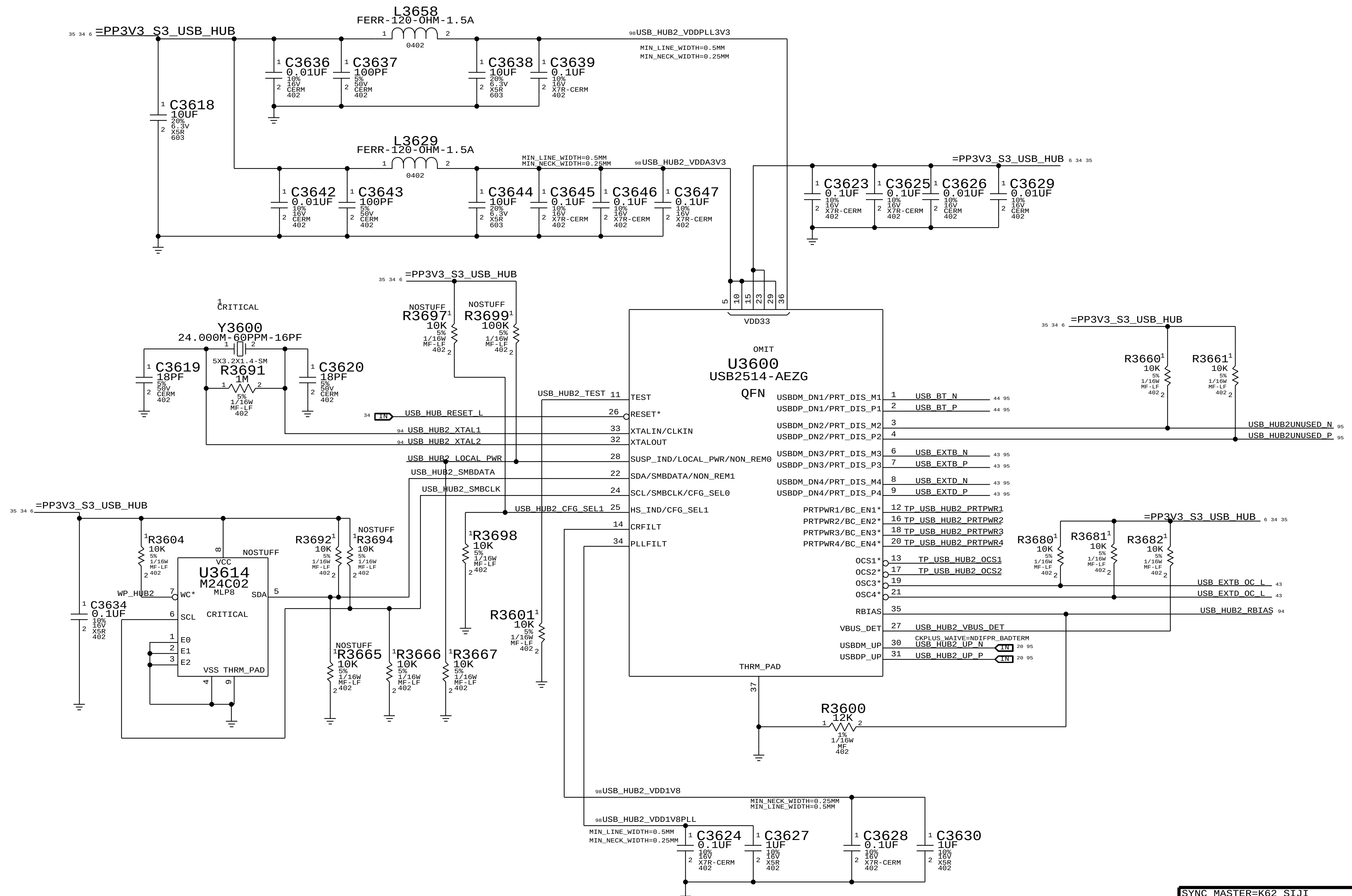
Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

BRANCH

USB HUB-2



SYNC_MASTER=K62_SIJI SYNC_DATE=11/14/2010

PAGE TITLE

USB HUB 2

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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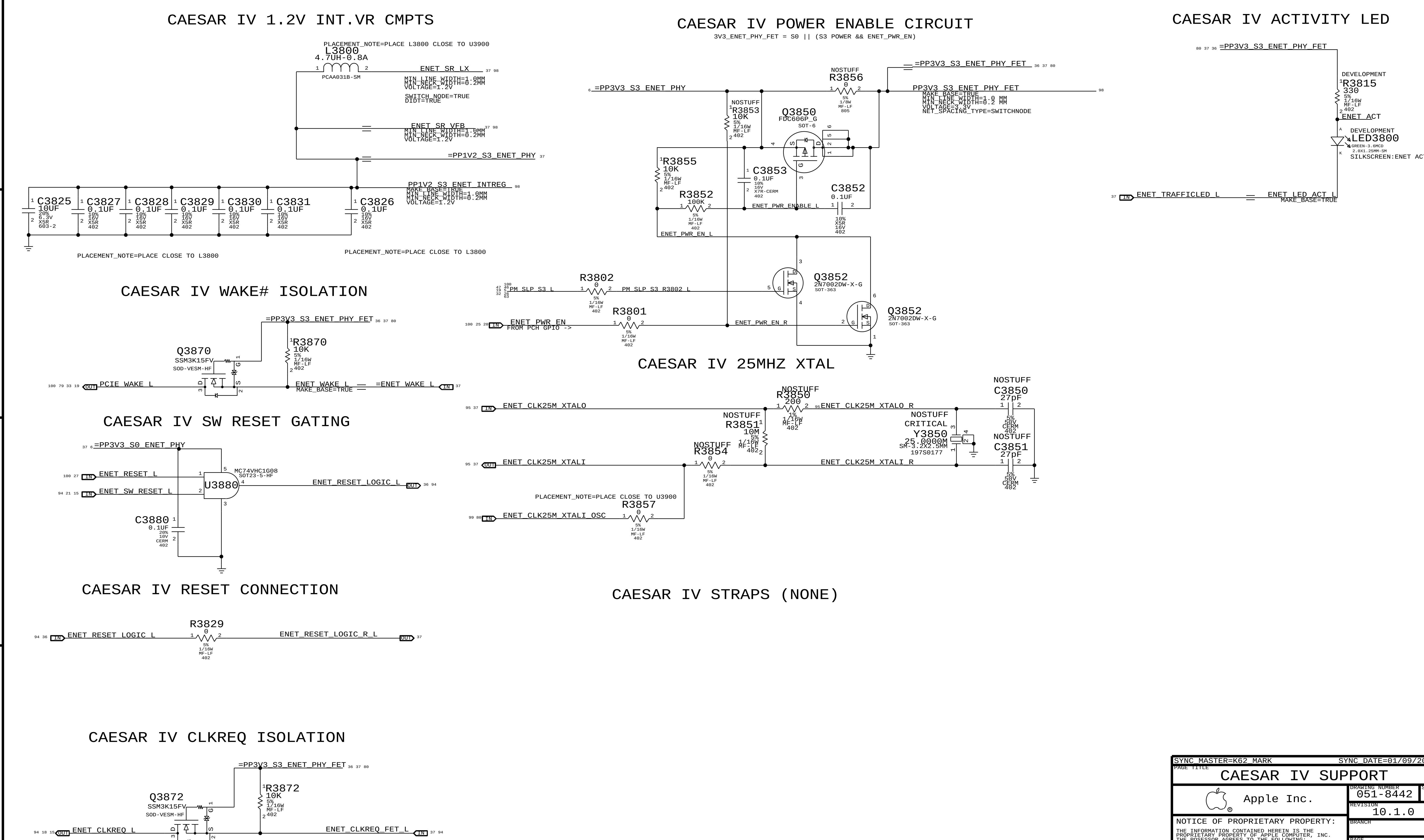

SYNC_MASTER=K62_MARK SYNC_DATE=01/09/2011

PAGE TITLE

CAESAR IV SUPPORT

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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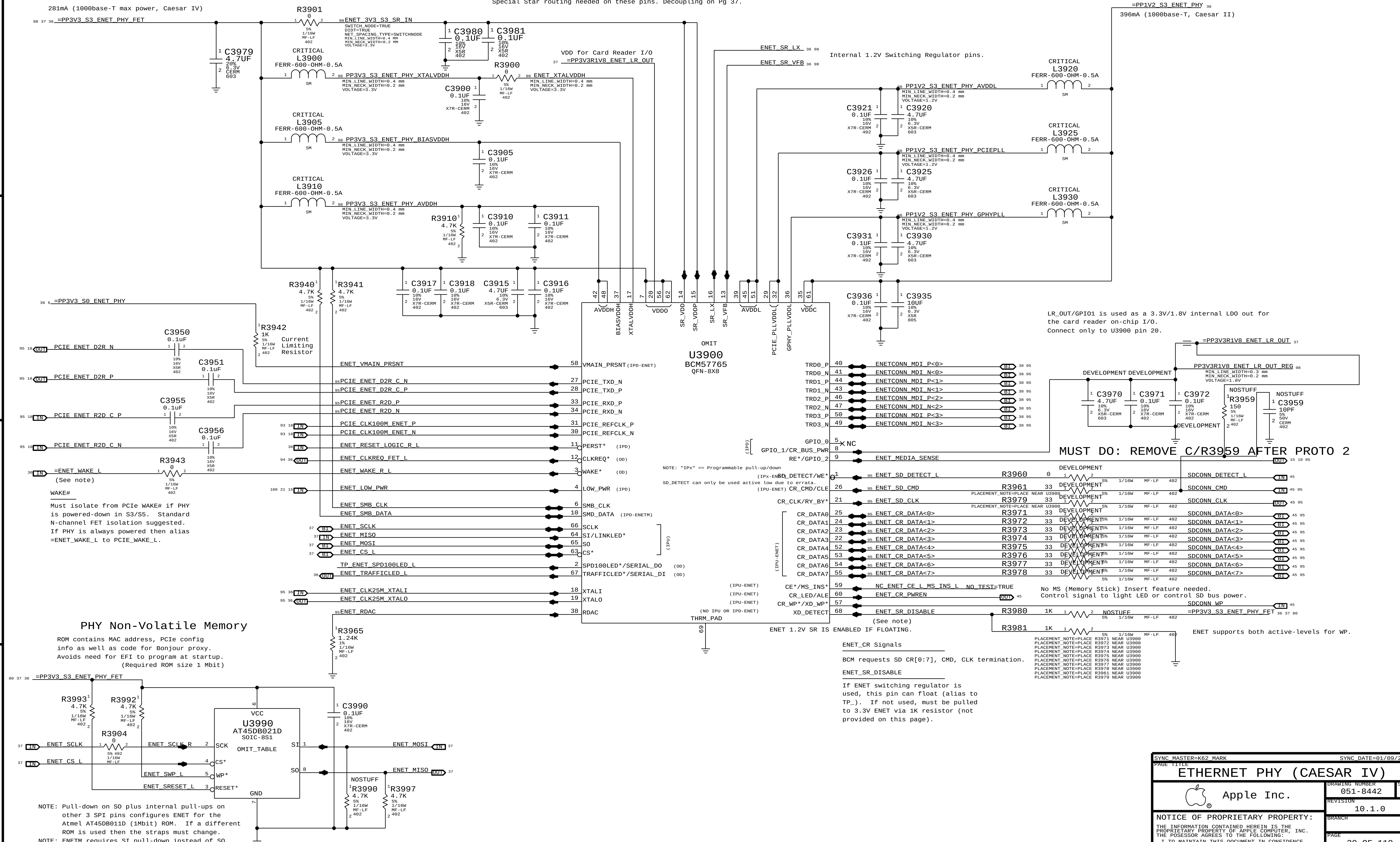
BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.



WAKE#

Must isolate from PCIe WAKE# if PHY is powered-down in S3/S5. Standard N-channel FET isolation suggested. If PHY is always powered then alias =ENET_WAKE_L to PCIE_WAKE_L.

PHY Non-Volatile Memory

ROM contains MAC address, PCIe config info as well as code for Bonjour proxy. Avoids need for EFI to program at startup. (Required ROM size 1 Mbit)

NOTE: Pull-down on SO plus internal pull-ups on other 3 SPI pins configures ENET for the Atmel AT45DB011D (1Mbit) ROM. If a different ROM is used then the straps must change.
NOTE: ENETM requires SI pull-down instead of SO.

ENET_CR Signals

BCM requests SD CR[0:7], CMD, CLK termination.

ENET_SR_DISABLE

If ENET switching regulator is used, this pin can float (alias to TP_). If not used, must be pulled to 3.3V ENET via 1K resistor (not provided on this page).

SYNC_MASTER=K62_MARK SYNC_DATE=01/09/2011

PAGE TITLE
ETHERNET PHY (CAESAR IV)

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D
REVISION 10.1.0
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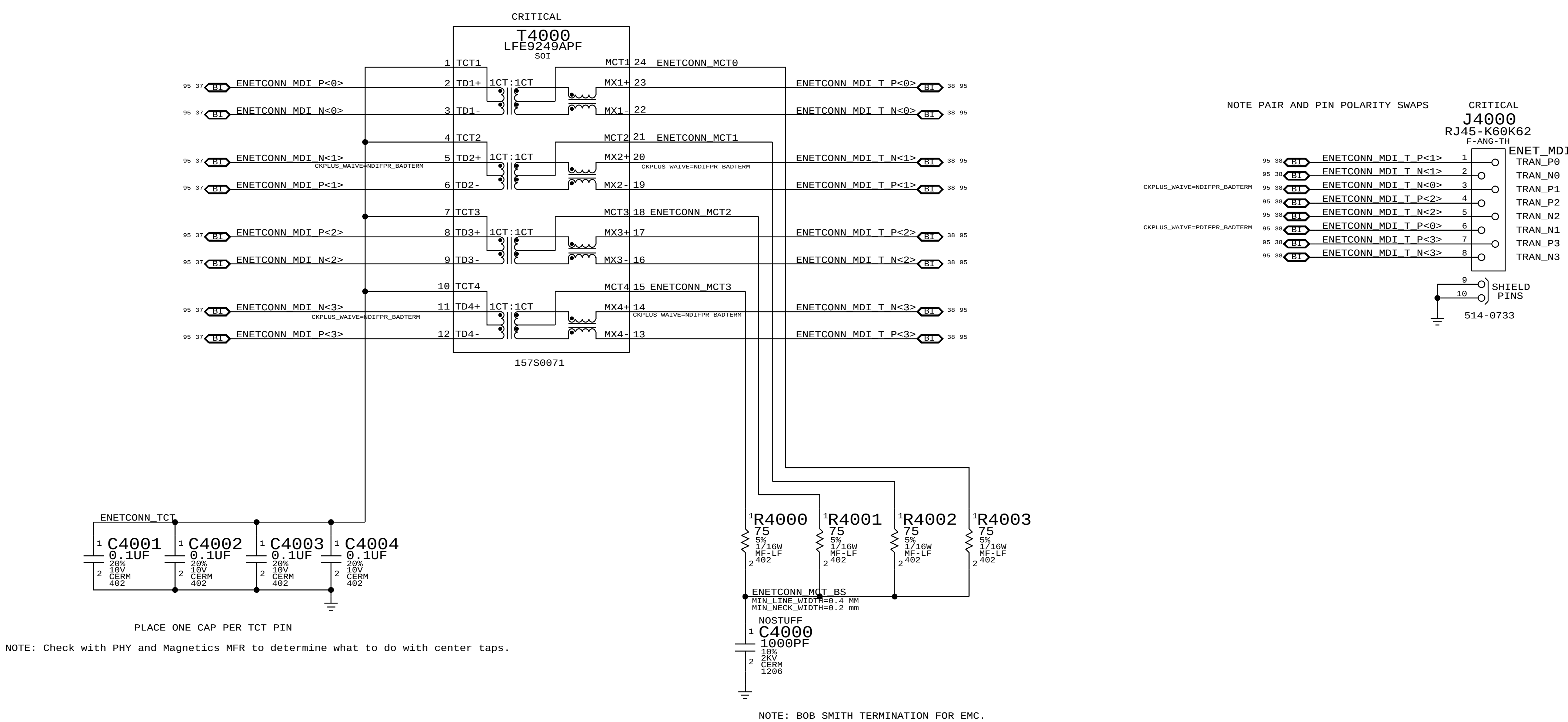
THIS PAGE IS DIFFERENT BETWEEN K60 AND K62.



SYNC_MASTER=K62_MARK	SYNC_DATE=01/09/2011
PAGE TITLE: Ethernet Connector	
Apple Inc.	DRAWING NUMBER: 051-8442 SIZE: D
	REVISION: 10.1.0
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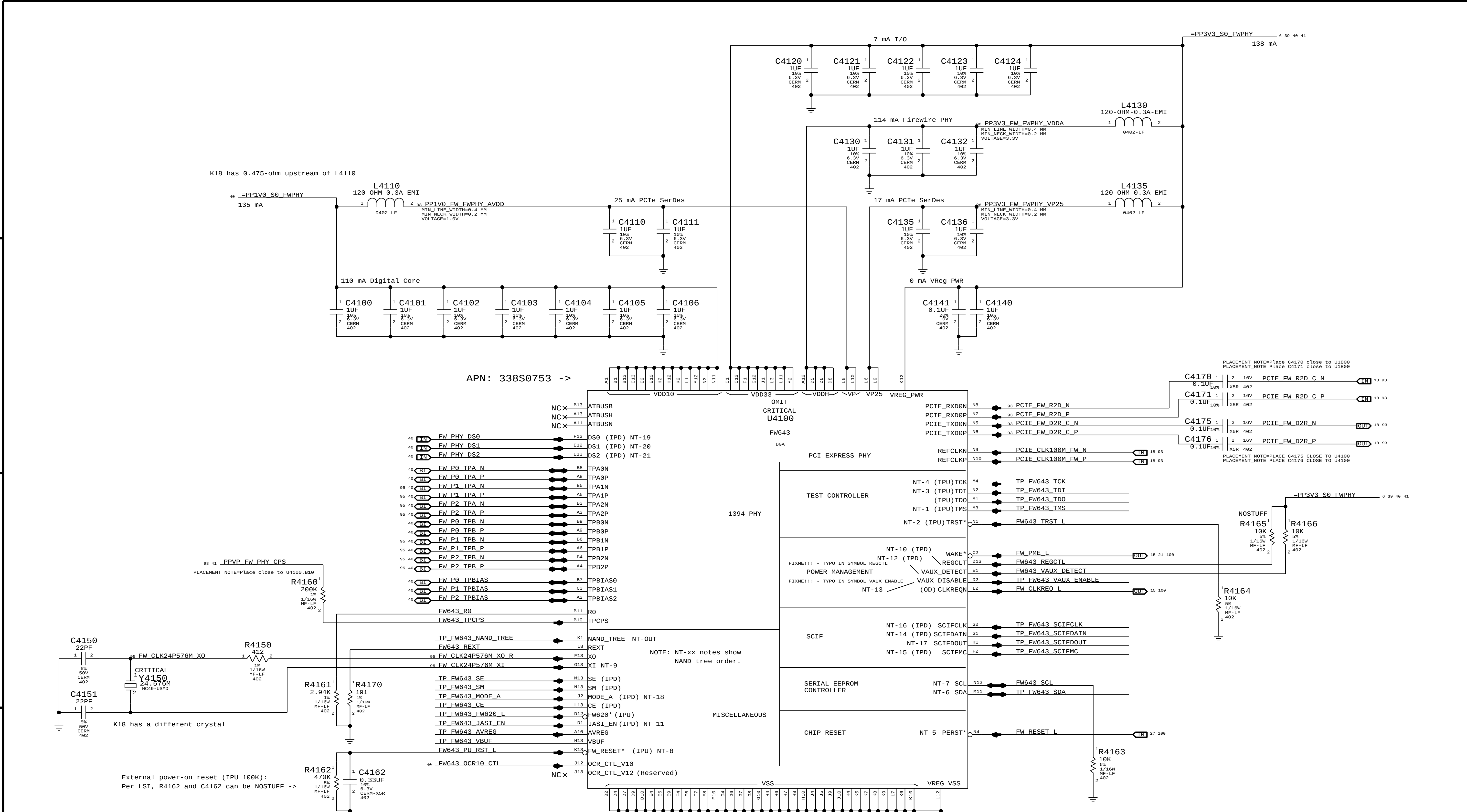
	SHEET: 38 OF 101



SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

FireWire LLC/PHY (FW643)

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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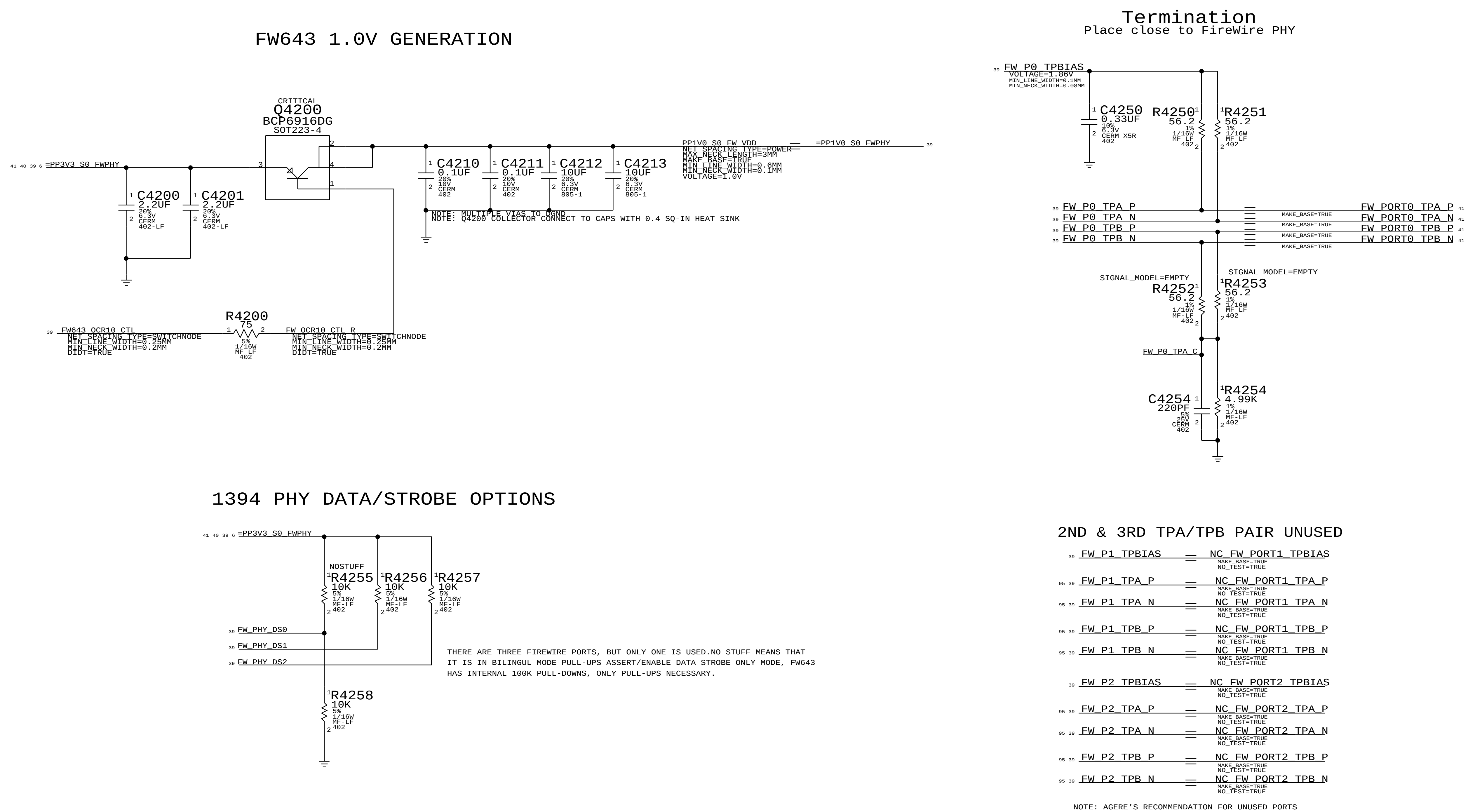


SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

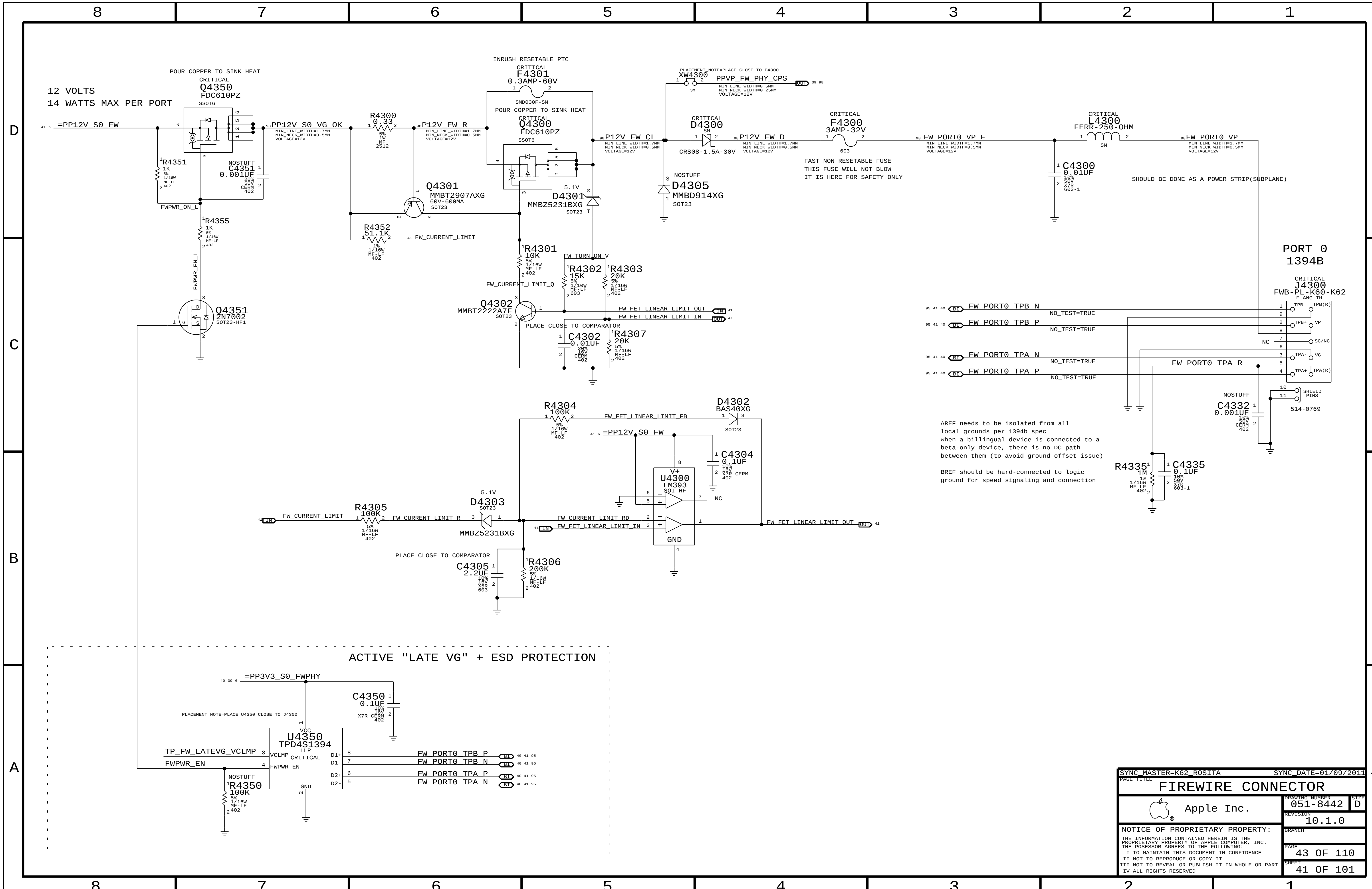
PAGE TITLE: FireWire: 1394B MISC

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

NOTICE OF PROPRIETARY PROPERTY:


12 VOLTS
14 WATTS MAX PER PORT
POUR COPPER TO SINK HEAT
CRITICAL
Q4350
FDC610PZ
SSOT6
=PP12V_S0_FW
PP12V_S0_VG_OK
R4351
1K
NOSTUFF
C4351
0.001UF
FWPWR_ON_L
R4355
1K
FWPWR_EN_L
Q4351
2N7002
SOT23-HF1
R4300
0.33
5%
1W
MF
2512
P12V_FW_R
Q4301
MMBT2907AXG
60V-600MA
SOT23
R4352
51.1K
FW_CURRENT_LIMIT
INRUSH RESETABLE PTC
CRITICAL
F4301
0.3AMP-60V
SMD030F-SM
POUR COPPER TO SINK HEAT
CRITICAL
Q4300
FDC610PZ
SSOT6
5.1V
D4301
MMBZ5231BXG
SOT23
R4301
10K
FW_TURN_ON_V
R4302
15K
R4303
20K
FW_CURRENT_LIMIT_Q
Q4302
MMBT2222A7F
SOT23
FW_FET_LINEAR_LIMIT_OUT
FW_FET_LINEAR_LIMIT_IN
PLACE CLOSE TO COMPARATOR
C4302
0.01UF
R4307
20K
P12V_FW_CL
PLACEMENT_NOTE=PLACE CLOSE TO F4300
XW4300
SM
PPVP_FW_PHY_CPS
CRITICAL
D4300
SM
CRS08-1.5A-30V
P12V_FW_D
NOSTUFF
D4305
MMBD914XG
SOT23
CRITICAL
F4300
3AMP-32V
603
FAST NON-RESETABLE FUSE
THIS FUSE WILL NOT BLOW
IT IS HERE FOR SAFETY ONLY
FW_PORT0_VP_F
C4300
0.01UF
CRITICAL
L4300
FERR-250-OHM
SM
FW_PORT0_VP
SHOULD BE DONE AS A POWER STRIP(SUBPLANE)
PORT 0
1394B
CRITICAL
J4300
FWB-PL-K60-K62
F-ANG-TH
FW_PORT0_TPB_N
FW_PORT0_TPB_P
FW_PORT0_TPA_N
FW_PORT0_TPA_P
FW_PORT0_TPA_R
NO_TEST=TRUE
NC
SHIELD PINS
514-0769
NOSTUFF
C4332
0.001UF
R4335
1M
C4335
0.1UF
AREF needs to be isolated from all
local grounds per 1394b spec
When a billingual device is connected to a
beta-only device, there is no DC path
between them (to avoid ground offset issue)
BREF should be hard-connected to logic
ground for speed signaling and connection
R4304
100K
FW_FET_LINEAR_LIMIT_FB
D4302
BAS40XG
SOT23
=PP12V_S0_FW
C4304
0.1UF
U4300
LM393
SOI-HF
V+
GND
NC
R4305
100K
FW_CURRENT_LIMIT_R
5.1V
D4303
SOT23
MMBZ5231BXG
FW_CURRENT_LIMIT_RD
FW_FET_LINEAR_LIMIT_IN
FW_FET_LINEAR_LIMIT_OUT
PLACE CLOSE TO COMPARATOR
C4305
2.2UF
R4306
200K
ACTIVE "LATE VG" + ESD PROTECTION
=PP3V3_S0_FWPHY
C4350
0.1UF
PLACEMENT_NOTE=PLACE U4350 CLOSE TO J4300
U4350
TPD4S1394
LLP
CRITICAL
TP_FW_LATEVG_VCLMP
FWPWR_EN
NOSTUFF
R4350
100K
FW_PORT0_TPB_P
FW_PORT0_TPB_N
FW_PORT0_TPA_P
FW_PORT0_TPA_N
SYNC_MASTER=K62_ROSITA
SYNC_DATE=01/09/2011
FIREWIRE CONNECTOR
Apple Inc.
DRAWING NUMBER
051-8442
SIZE
D
REVISION
10.1.0
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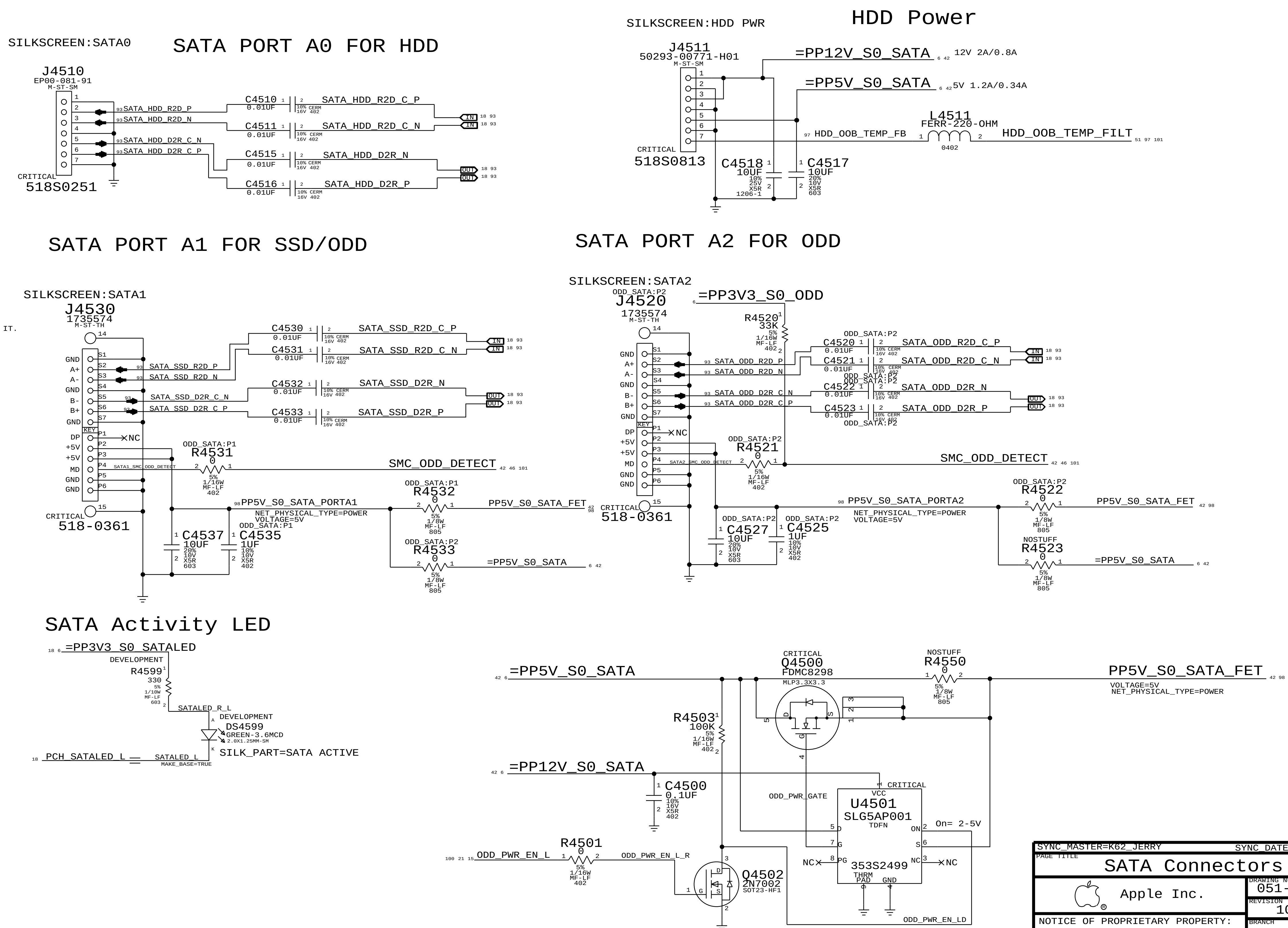
PAGE
43 OF 110
SHEET
41 OF 101

BOMOPTION OPTIONS FOR SATA PORT A1 AND A2

A1	A2	ODD_SATA:P1	ODD_SATA:P2
SSD	ODD		X
ODD		X	

USE OF PORT A2 FOR SSD IS NOT INTENDED VIA BOMOPTION THOUGH MLB SUPPORTS IT.

5V (SSD) 1.4A/0.8A/0.03A
5V (ODD) 1.5A/1A/0.14A



SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE

SATA Connectors

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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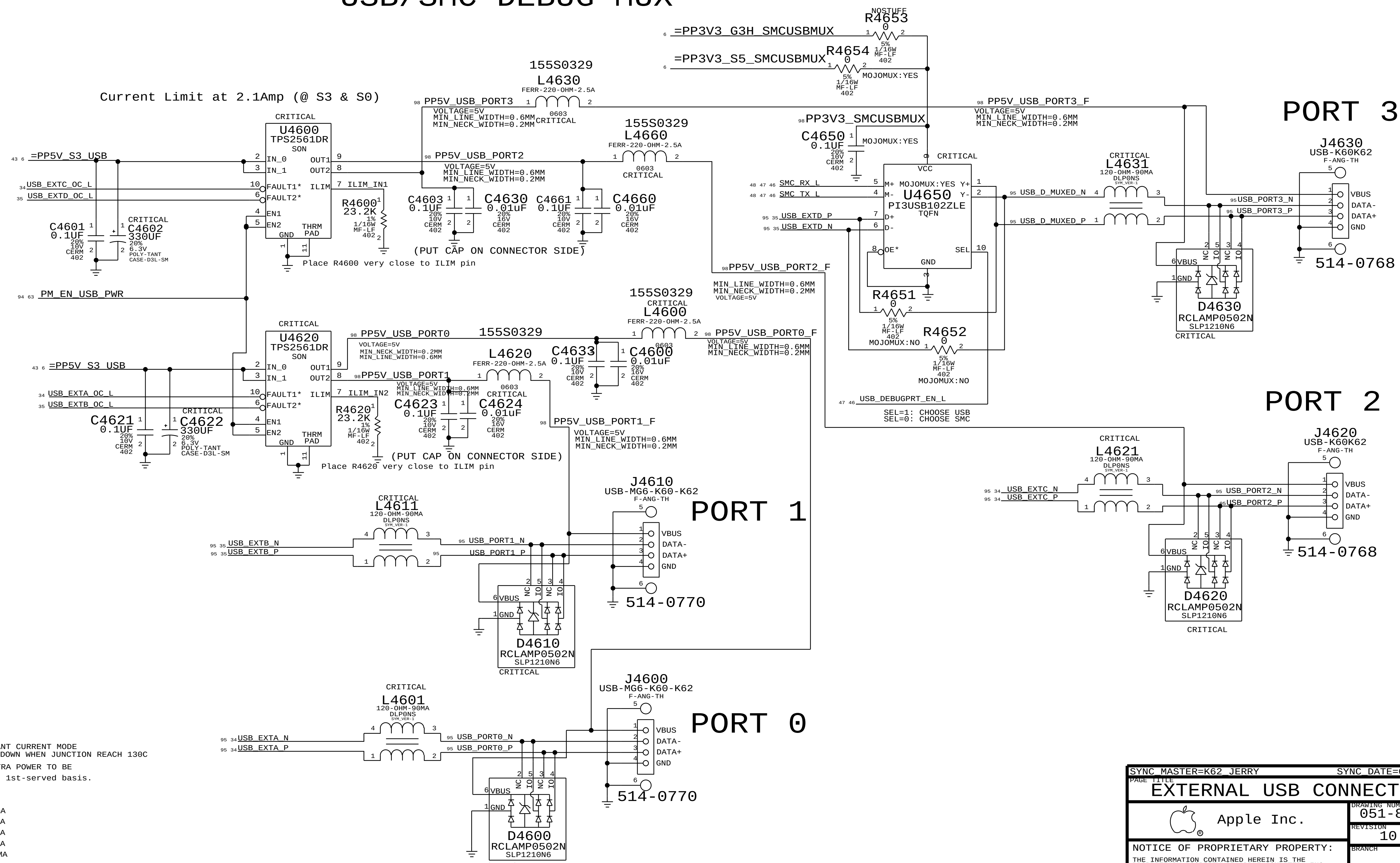
USB/SMC DEBUG MUX

ADDED AT EVT & SWITCH TO S5 RAIL

Current Limit at 2.1Amp (@ S3 & S0)



USB PORT POWER:

EACH PORT IS HARDWARE Capable of :

STATE	MAX	MIN (WITHIN THE TOLERANCE)
S0, S3	2.7A	2.1A -- PER PORT

WHEN CURRENT HITS LIMIT, TPS2561 BECOME CONSTANT CURRENT MODE
AND STAY AT THE LIMIT LEVEL UNTIL THERMAL SHUTDOWN WHEN JUNCTION REACH 130C

SOFTWARE WILL ALLOW 500MA/PORT, PLUS 2700MA EXTRA POWER TO BE
distributed to approved devices on a 1st-come, 1st-served basis.

EXAMPLE: Port 1 - iPad fast charging = 2100mA
Port 2 - Wired Keyboard = 1100mA
Port 3 - iPhone fast charging = 1000mA
Port 4 - USB 2.0 500MA = 500MA
TOTAL: 4700MA

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: EXTERNAL USB CONNECTORS

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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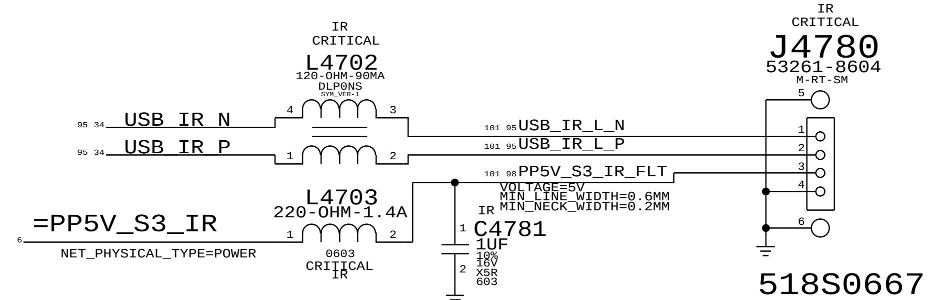
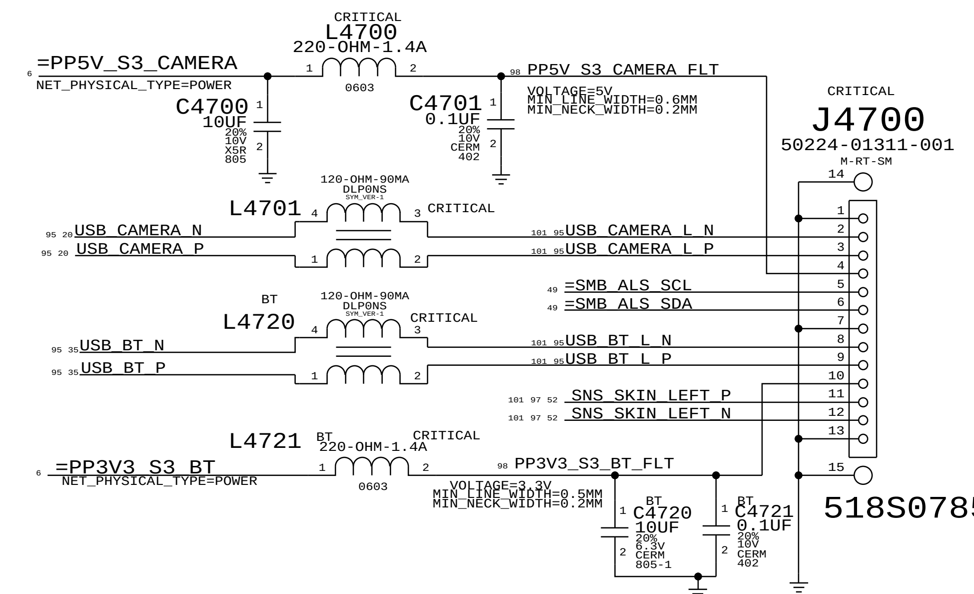
IR RECEIVER CONNECTOR

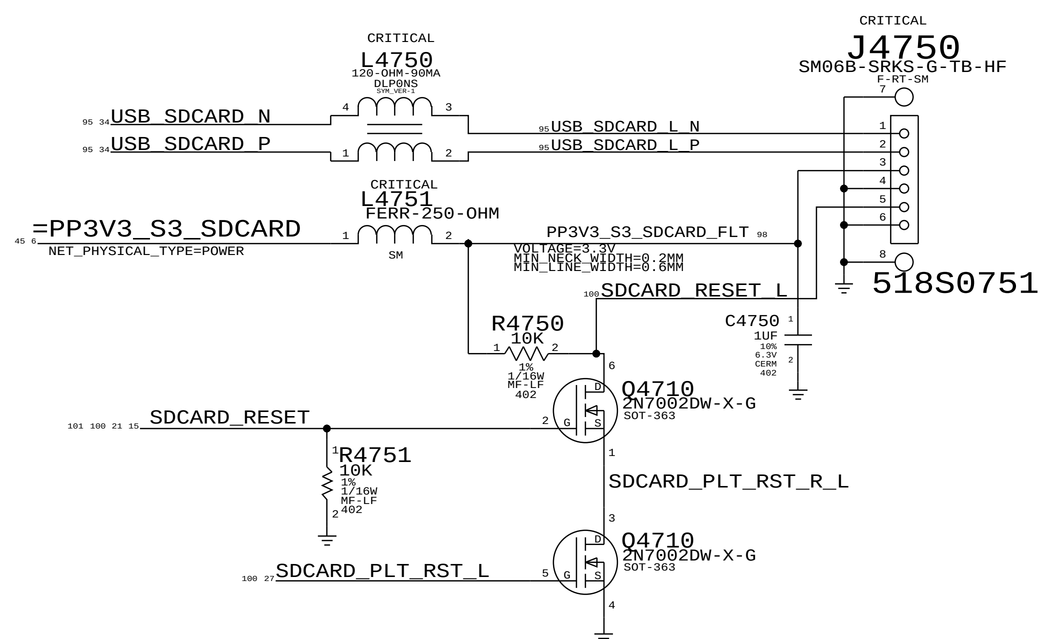


CAMERA/ALS & BLUETOOTH (K37A) CONNECTOR



Skin Temp sense at upper Left Screen corner

SD Card Reader Board (Lazarus)



SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: Internal USB Connections

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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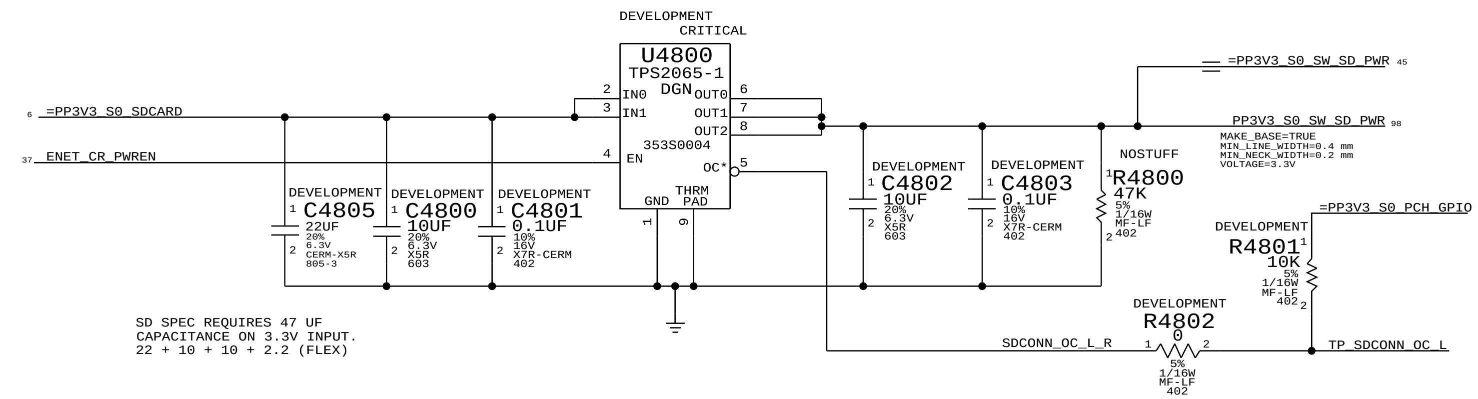
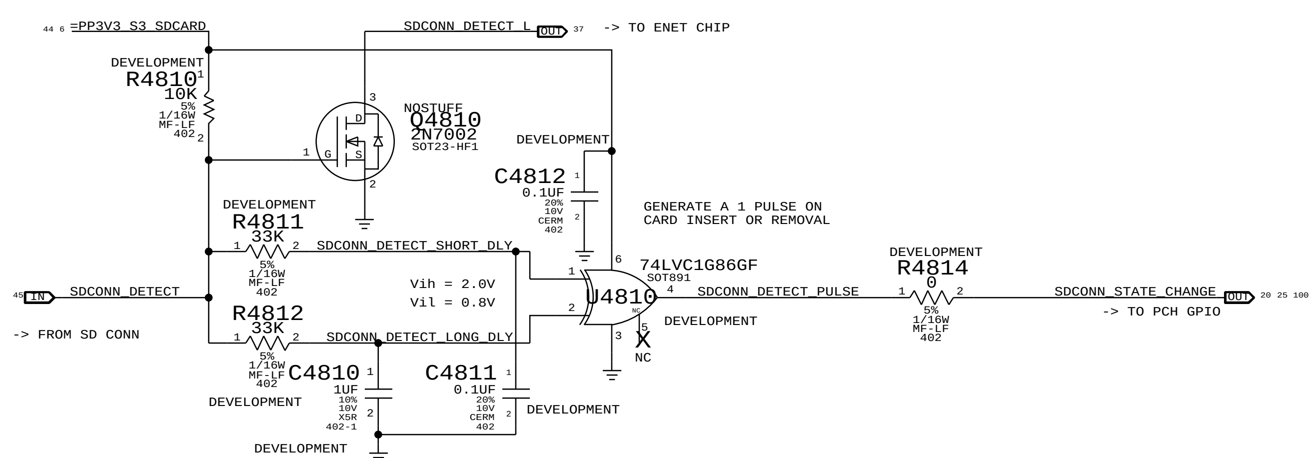
SD CARD 3.3V OVERCURRENT PROTECTION CHIP WITH ACTIVE LOAD DISCHARGE

TPS2065-1 (1.0A LIMIT) HAS ACTIVE LOAD DISCHARGE SO R4800 IS NOSTUFF.

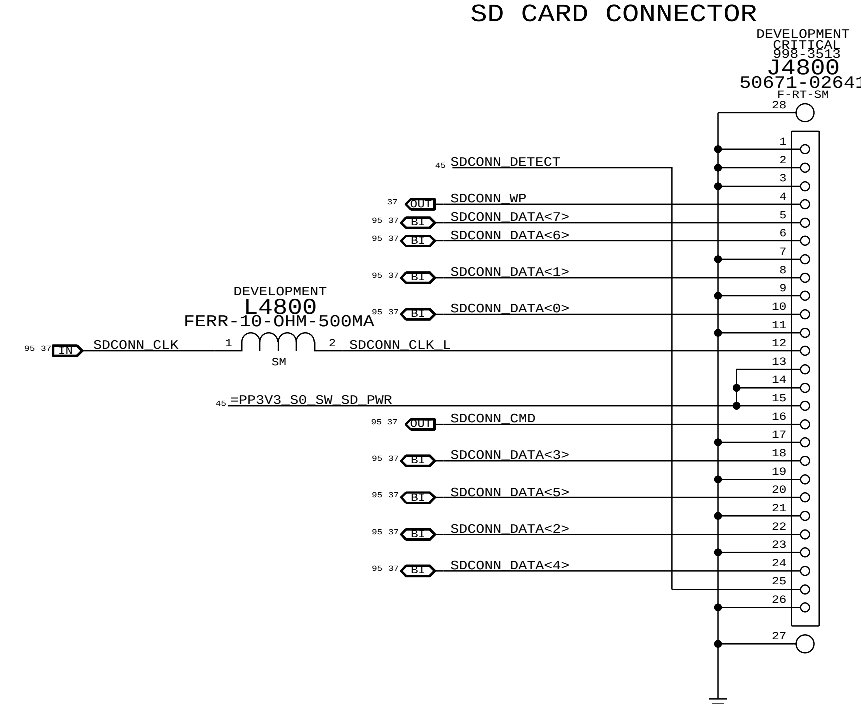


SD SPEC REQUIRES 47 UF
CAPACITANCE ON 3.3V INPUT.
22 + 10 + 10 + 2.2 (FLEX)

SDCONN DETECT DEBOUNCE, INVERSION, AND DETECT-CHANGED PCH GPIO CIRCUIT



SD CARD CONNECTOR



(CARD INSERTED = OPEN)
CAESAR-IV CARD DETECT IS PROGRAMMABLE, BUT A SILICON BUG
MAKES THE ACTIVE-HIGH CASE UNUSABLE.

SYNC_MASTER=K62_MARK	SYNC_DATE=01/09/2011
PAGE TITLE	SD READER CONNECTOR
Apple Inc.	DRAWING NUMBER 051-8442 SIZE D
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	PAGE 48 OF 110
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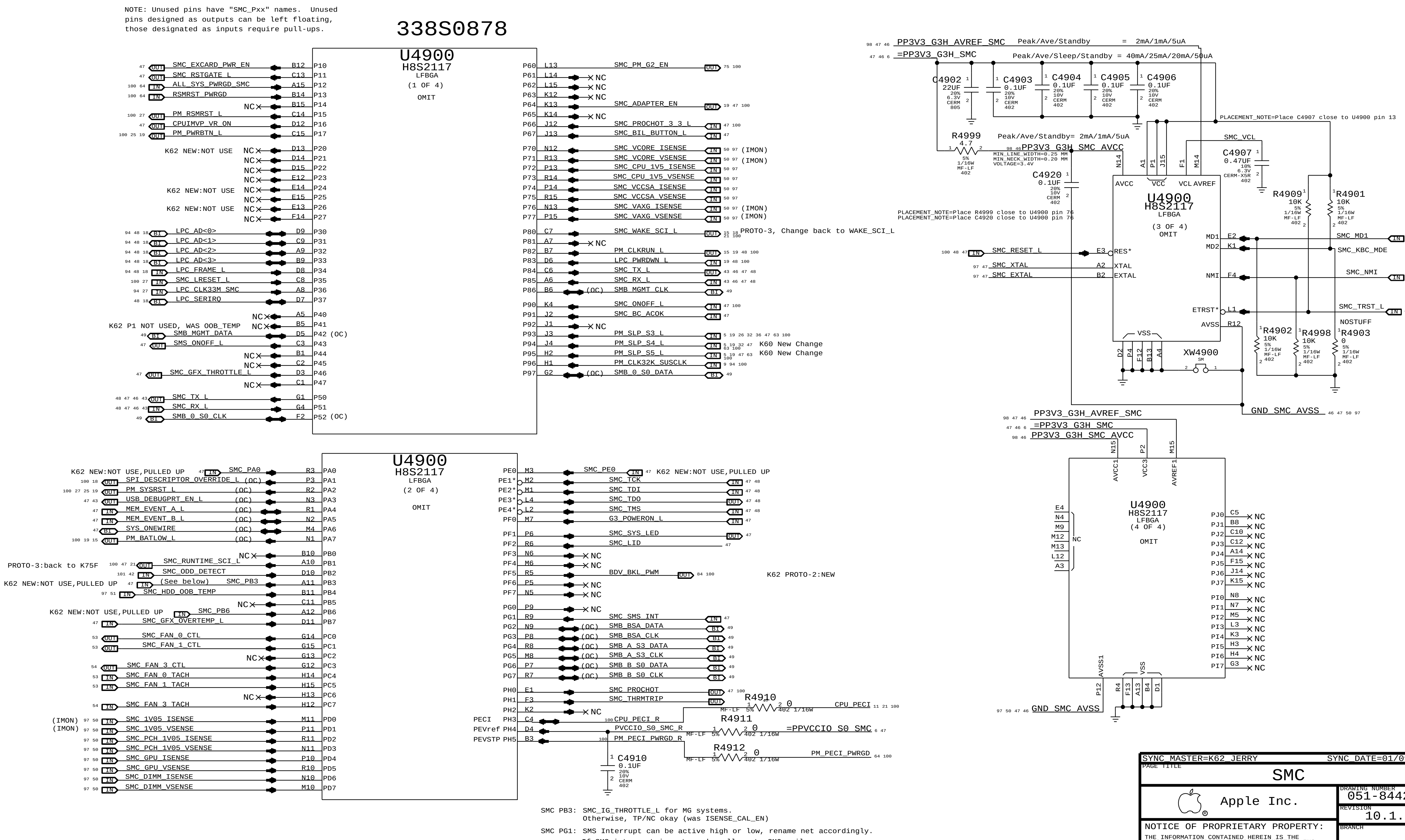

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE

SMC

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

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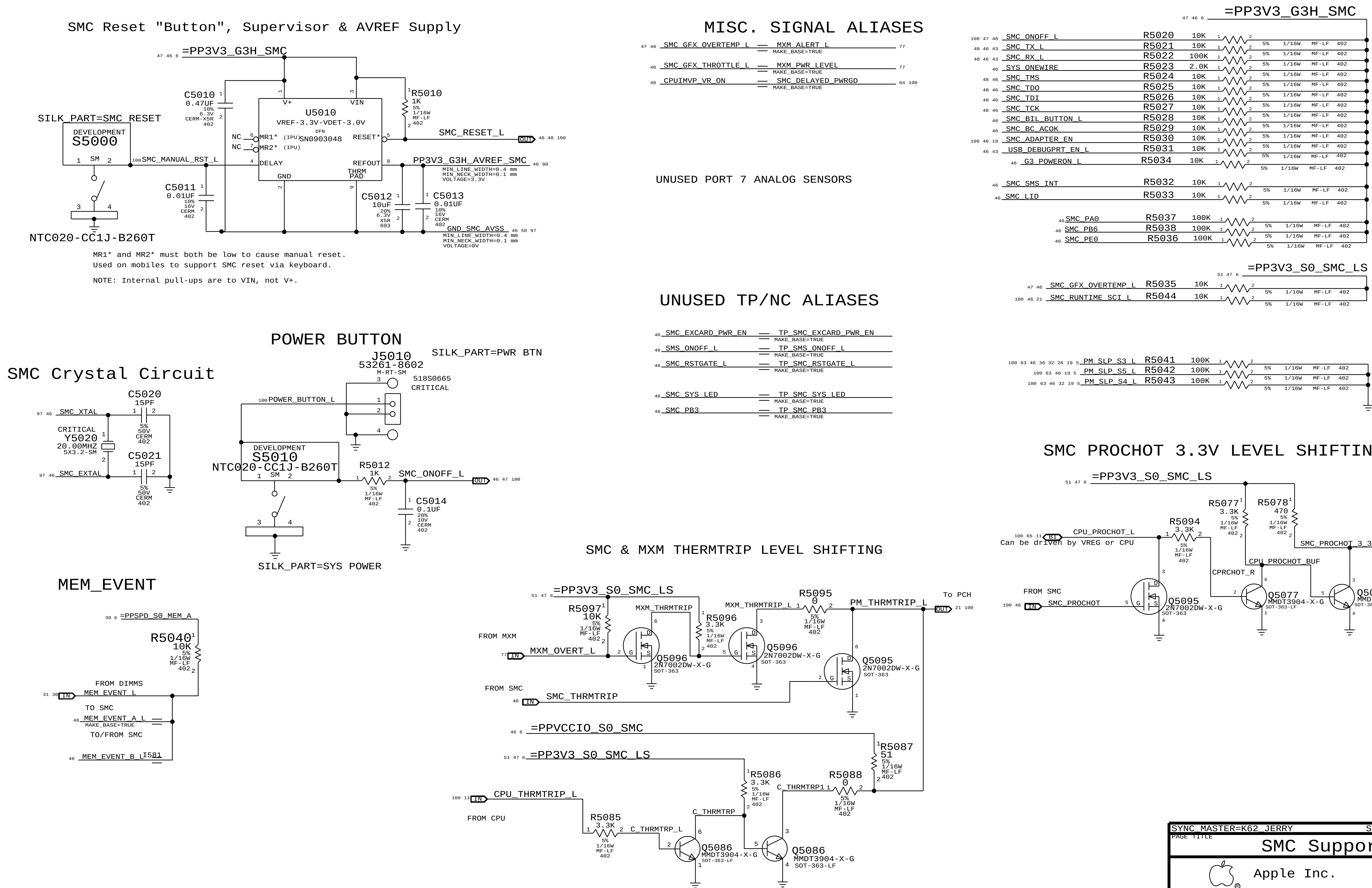

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE

SMC Support

Apple Inc.

DRAWING NUMBER	051-8442
SIZE	D
REVISION	10.1.0
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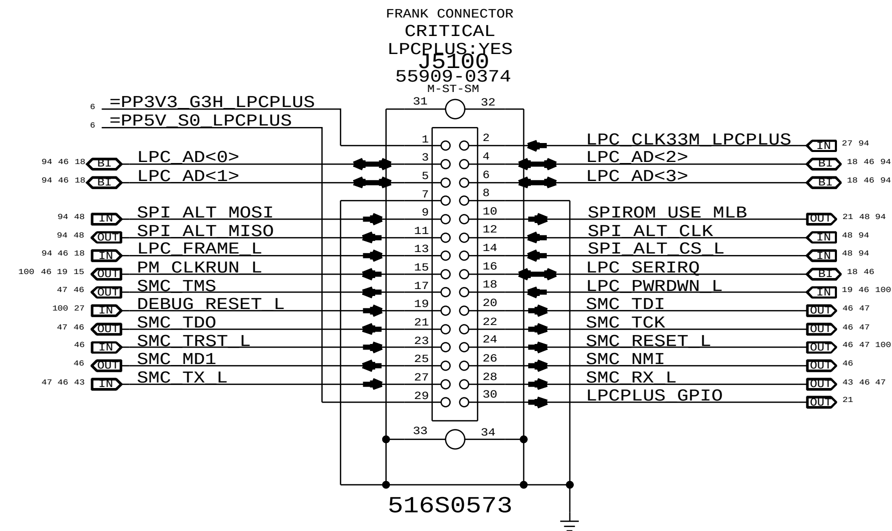
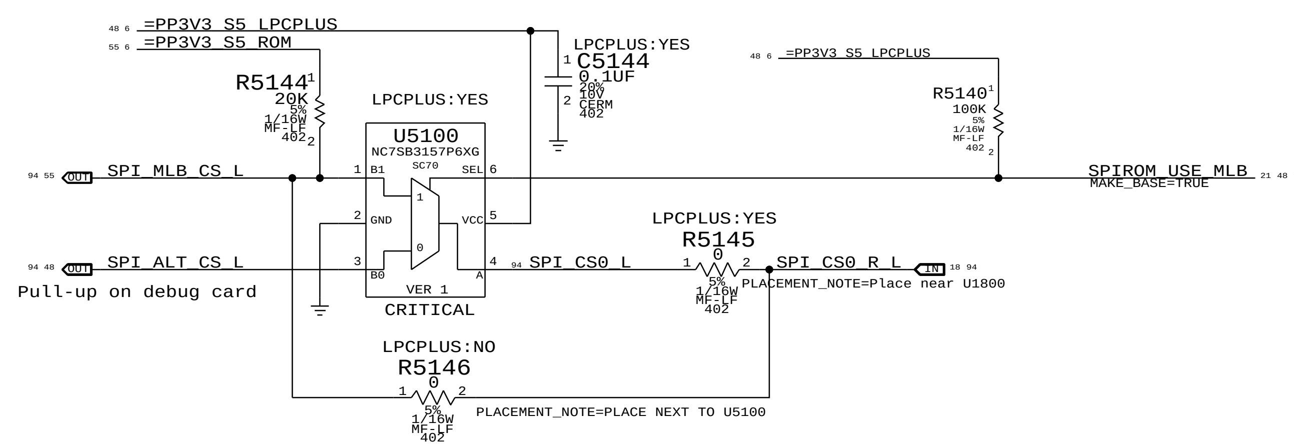
PAGE	50 OF 110
SHEET	47 OF 101

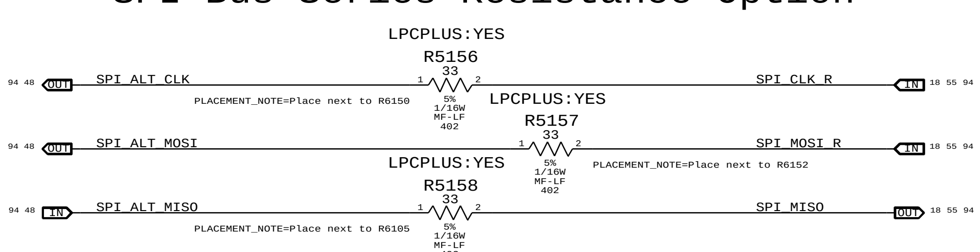
LPC+SPI Connector



Alternate SPI ROM Support



SPI Bus Series Resistance Option



SYNC_MASTER=K62_AARON SYNC_DATE=11/30/2009

PAGE TITLE: LPC+SPI Debug Connector

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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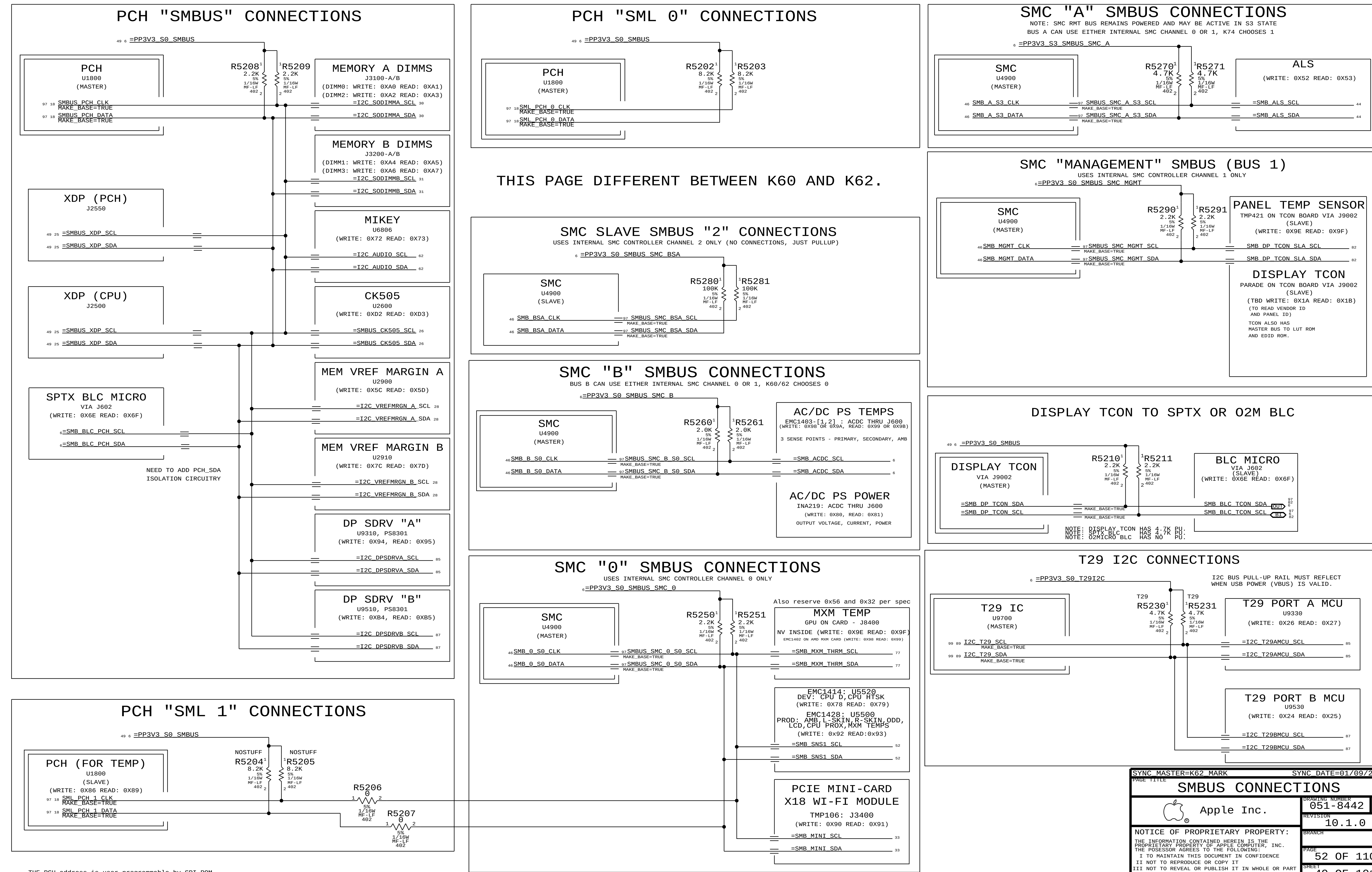

SYNC_MASTER=K62_MARK SYNC_DATE=01/09/2011

PAGE TITLE

SMBUS CONNECTIONS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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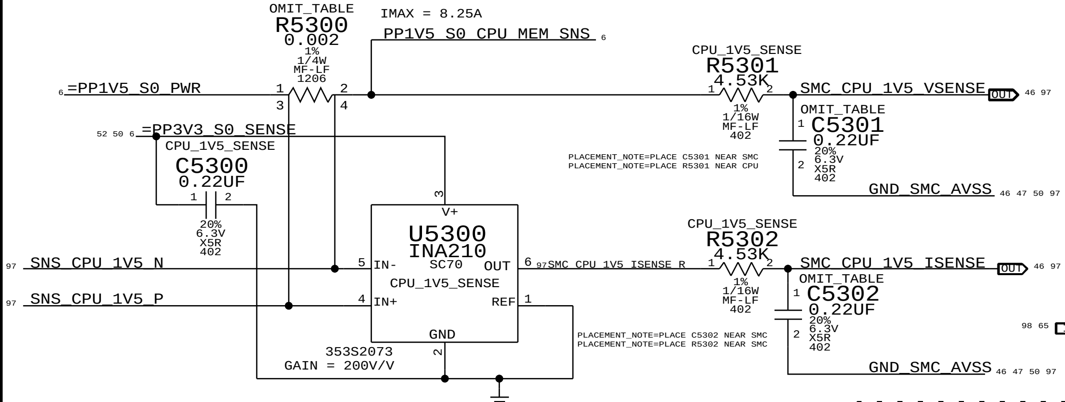
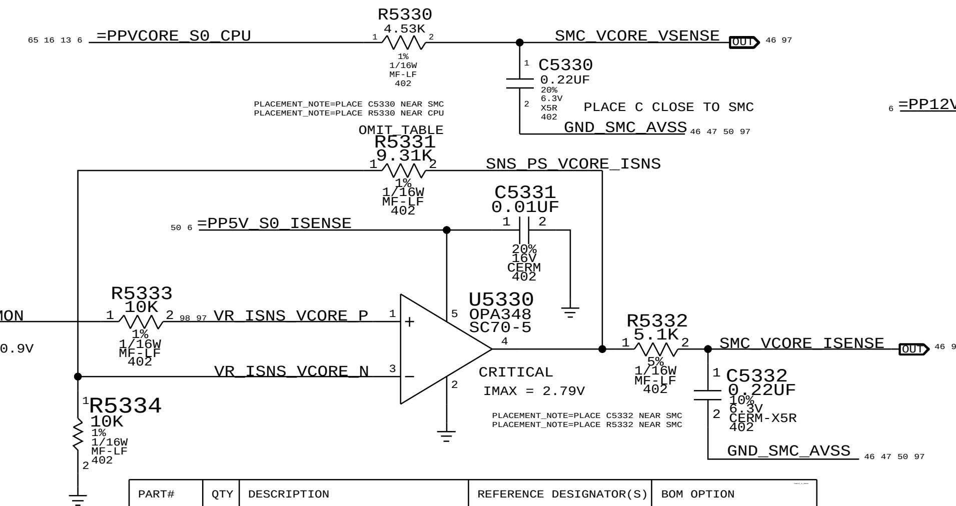
CPU VDD 1.5V
SENSE RESISTOR CURRENT (IC5R) AND VOLTAGE (VC5R) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5300	CPU_1V5_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5300	NO_CPU_1V5_SENSE
132S0080	2	CAP,0.22UF,402	C5301,C5302	CPU_1V5_SENSE
116S0004	2	RES,0 OHM,402	C5301,C5302	NO_CPU_1V5_SENSE

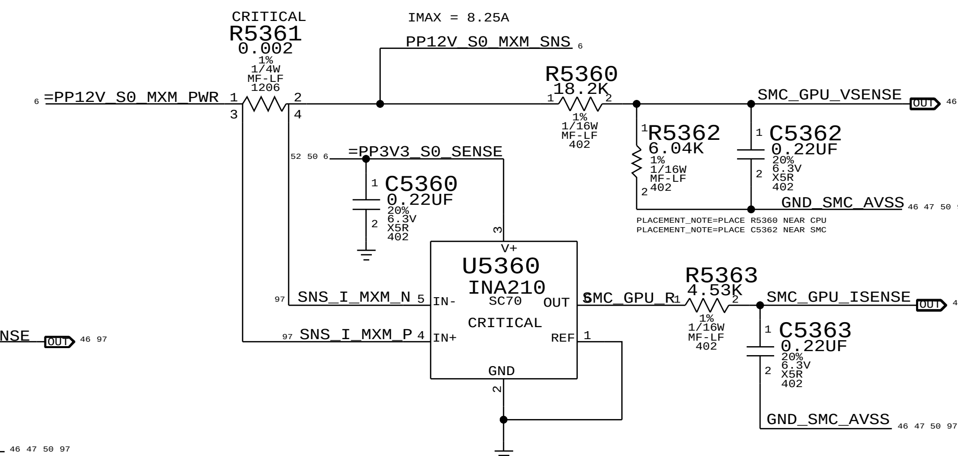
THE NO_1V05_PCH_SENSE, NO_CPU_VCCSA_SENSE, AND NO_CPU_1V5_SENSE BOMOPTIONS SHOULD IDEALLY NEVER BE USED AS TOTAL CPU POWER SENSING REQUIRES ALL 3 SENSORS.

CPU VCC (VCORE)
IMON CURRENT (IC0C) AND VOLTAGE (VC0C) SENSE

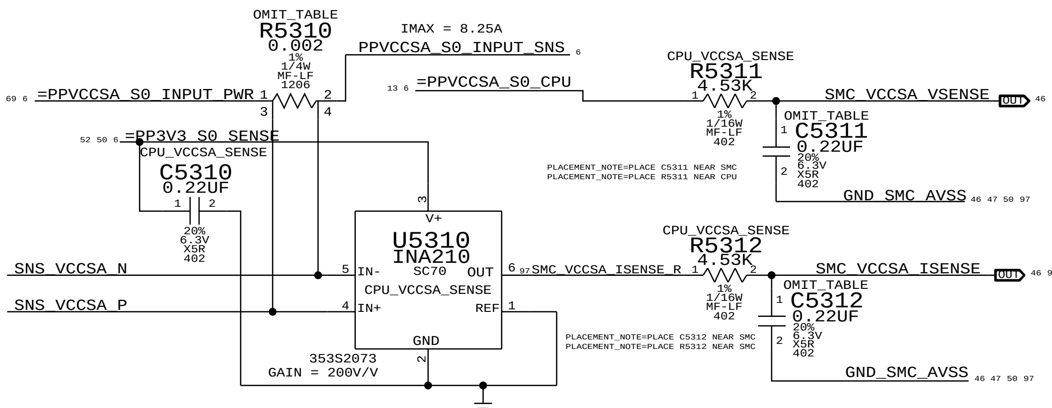


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0312	1	RES,MTL FILM,1/16W,9.31K,0402	R5331	CPUVCORE-3PH
114S0345	1	RES,MTL FILM,1/16W,21K,0402	R5331	CPUVCORE-4PH

GPU MXM
SENSE RESISTOR CURRENT (IG0R) AND VOLTAGE (VG0R) SENSE

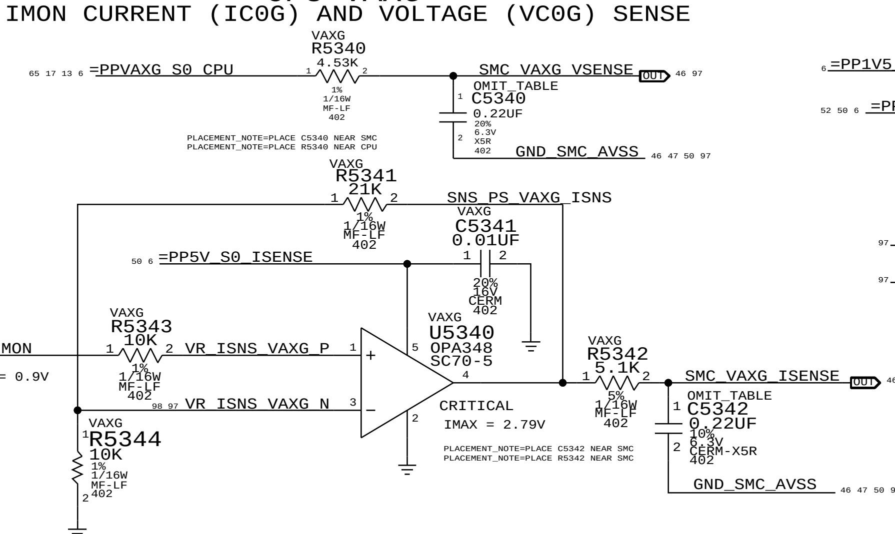


CPU VCCSA
SENSE RESISTOR CURRENT (ICSR) AND VOLTAGE (VCSR) SENSE



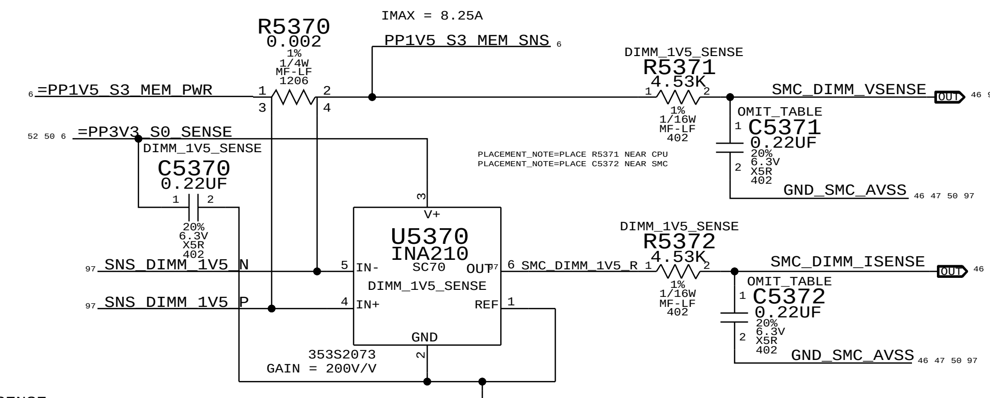
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5310	CPU_VCCSA_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5310	NO_CPU_VCCSA_SENSE
132S0080	2	CAP,0.22UF,402	C5311,C5312	CPU_VCCSA_SENSE
116S0004	2	RES,0 OHM,402	C5311,C5312	NO_CPU_VCCSA_SENSE

CPU VAXG
IMON CURRENT (IC0G) AND VOLTAGE (VC0G) SENSE



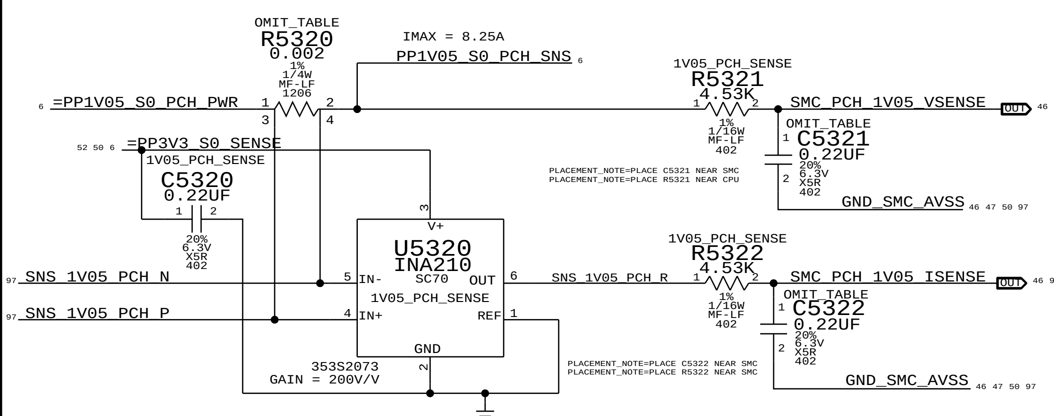
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	2	CAP,0.22UF,402	C5340,C5342	VAXG
116S0004	2	RES,0 OHM,402	C5340,C5342	NO_VAXG

DIMM VDD 1.5V (LIKELY DEVELOPMENT ONLY)
SENSE RESISTOR CURRENT (IM0R) AND VOLTAGE (VM0R) SENSE



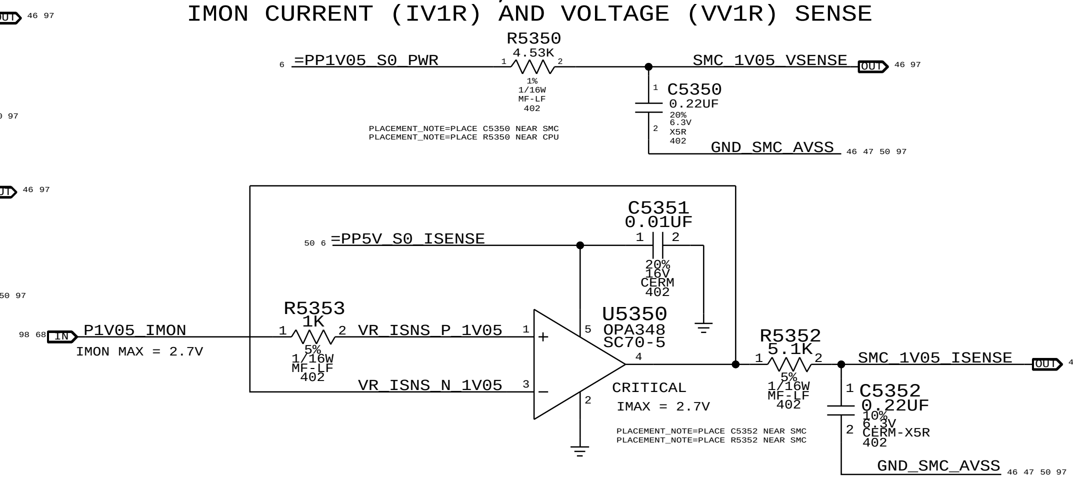
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	2	CAP,0.22UF,402	C5371,C5372	DIMM_1V5_SENSE
116S0004	2	RES,0 OHM,402	C5371,C5372	PRODUCTION

PCH 1.05V
SENSE RESISTOR CURRENT (IN1R) AND VOLTAGE (VN1R) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5320	1V05_PCH_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5320	NO_1V05_PCH_SENSE
132S0080	2	CAP,0.22UF,402	C5321,C5322	1V05_PCH_SENSE
116S0004	2	RES,0 OHM,402	C5321,C5322	NO_1V05_PCH_SENSE

1.05V FOR CPU VCCIO, CPU VCCSA & PCH 1.05V
IMON CURRENT (IV1R) AND VOLTAGE (VV1R) SENSE



NOTE: TOTAL CPU POWER = VC0C*IC0C + VC5R*IC5R + VCSR*ICSR + VC0G*IC0G + VV1R*IC1R

where
IC1R = IV1R - IN1R - ICSR

SYNC_MASTER=K62_MARK SYNC_DATE=01/09/2011

PAGE TITLE
CPU/PCH/GPU POWER SENSE

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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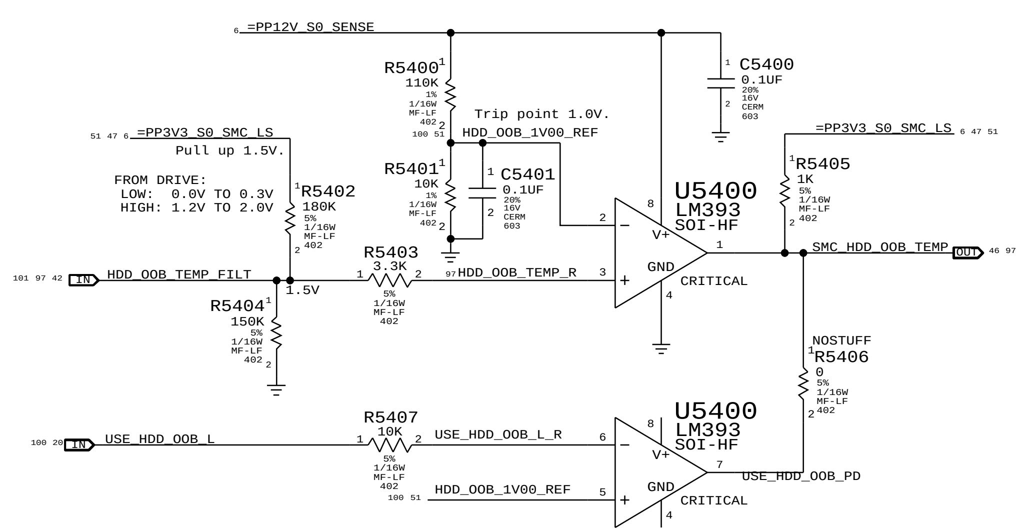
NOTICE OF PROPRIETARY PROPERTY:

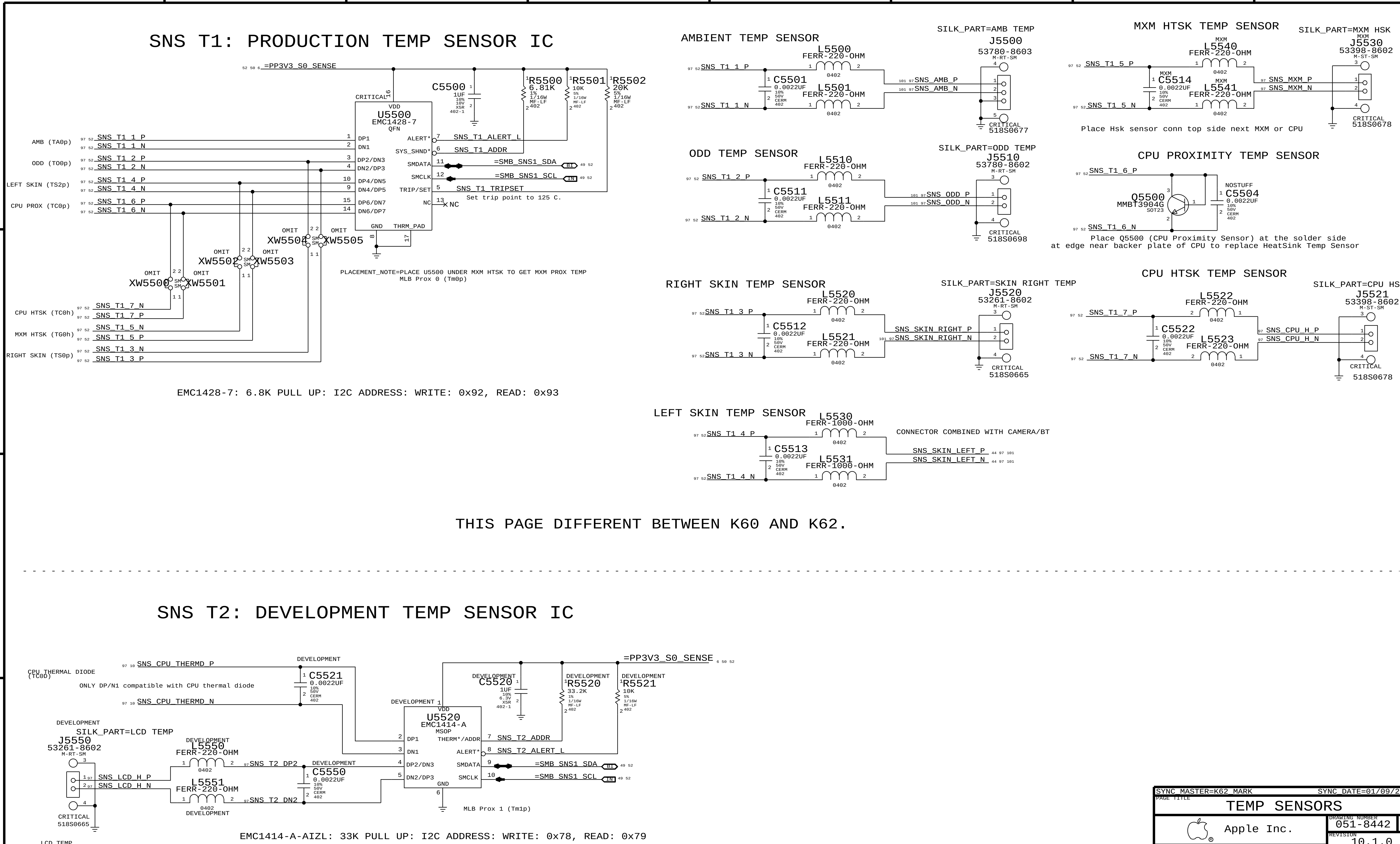
HDD OOB TEMPERATURE SENSING



DRIVE ACTIVE = VALID SIGNAL PROTOCOL BETWEEN 0-2.0V.
DRIVE ASLEEP = HDD DRIVES HDD_OOB_TEMP LOW
DRIVE ABSENT = OOB IS PULLED HIGH UNLESS PCH DETERMINES SSD PRESENT AND DRIVES USE_HDD_OOB_L LOW WHICH THEN PULLS HDD_OOB_TEMP LOW.

NOTE: WILL BE CONNECTED TO SATA PWR CONNECTOR PIN 11
THIS PIN IS ORIGINALLY INTENDED FOR HDD LED OUTPUT,
AND ALSO FOR HDD STAGGERED PIN UP (FLOATING) OR IMMEDIATE SPIN-UP (GROUND).
BOTH FUNCTIONS NOT USED.

SYNC_MASTER=K62_MARK	SYNC_DATE=01/09/2011
PAGE TITLE: HDD OOB SENSE	
Apple Inc.	DRAWING NUMBER: 051-8442 SIZE: D
	REVISION: 10.1.0
	BRANCH
	PAGE: 54 OF 110
	SHEET: 51 OF 101



SYNC_MASTER=K62_MARK SYNC_DATE=01/09/2011

PAGE TITLE

TEMP SENSORS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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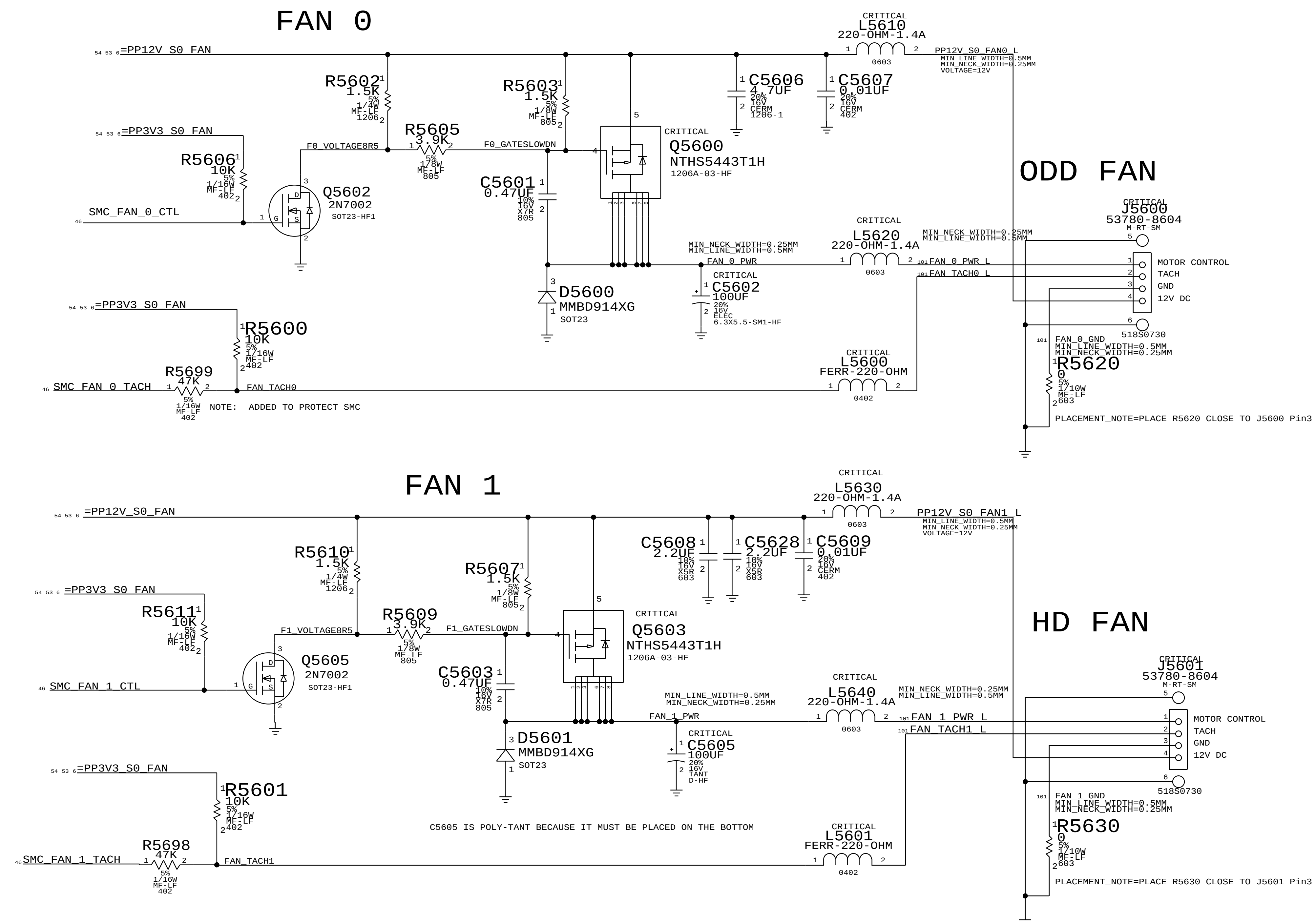

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE

HD AND OD FAN

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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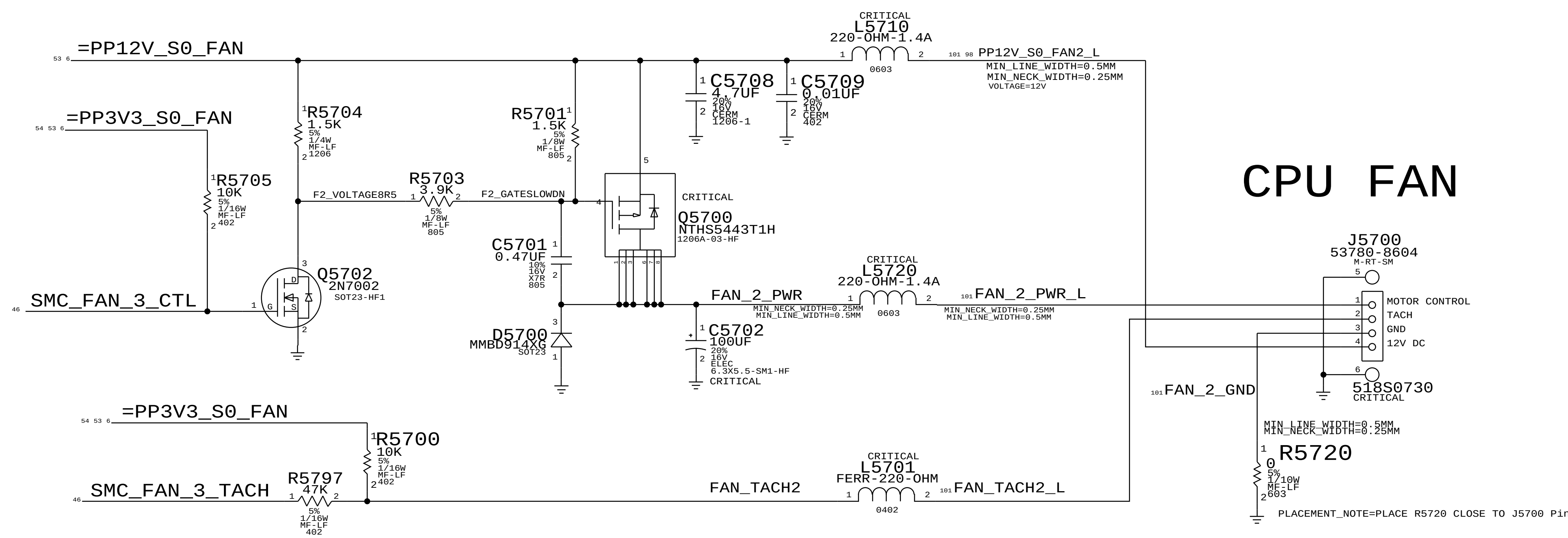
SMC'S FAN3 OUTPUT CONTROL FAN 2



FAN 3 SMC CONTROL (UNUSED)

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: CPU FAN

DRAWING NUMBER: 051-8442 SIZE: D

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REVISION: 10.1.0

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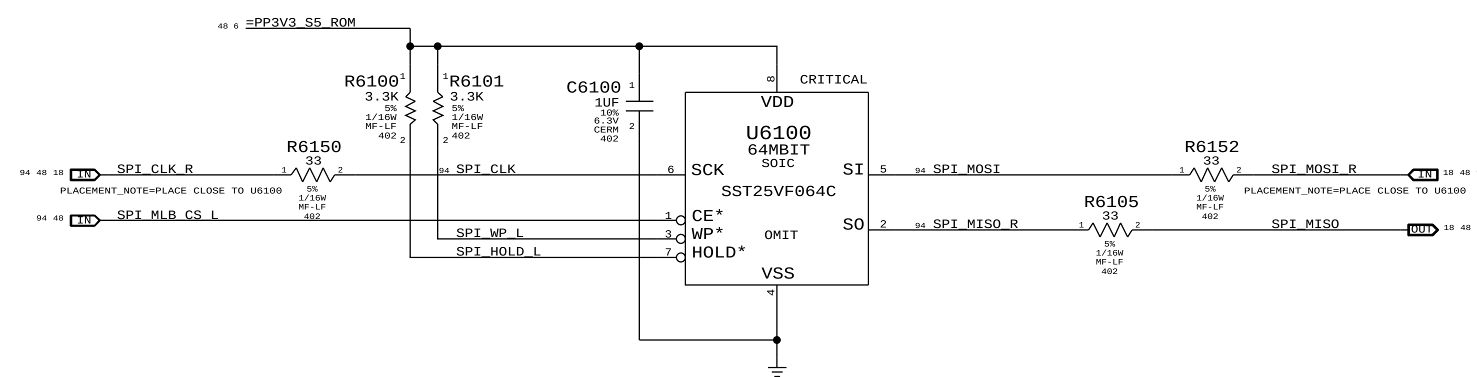

SYNC_MASTER=K62_AARON SYNC_DATE=11/30/2009

PAGE TITLE

SPI ROM

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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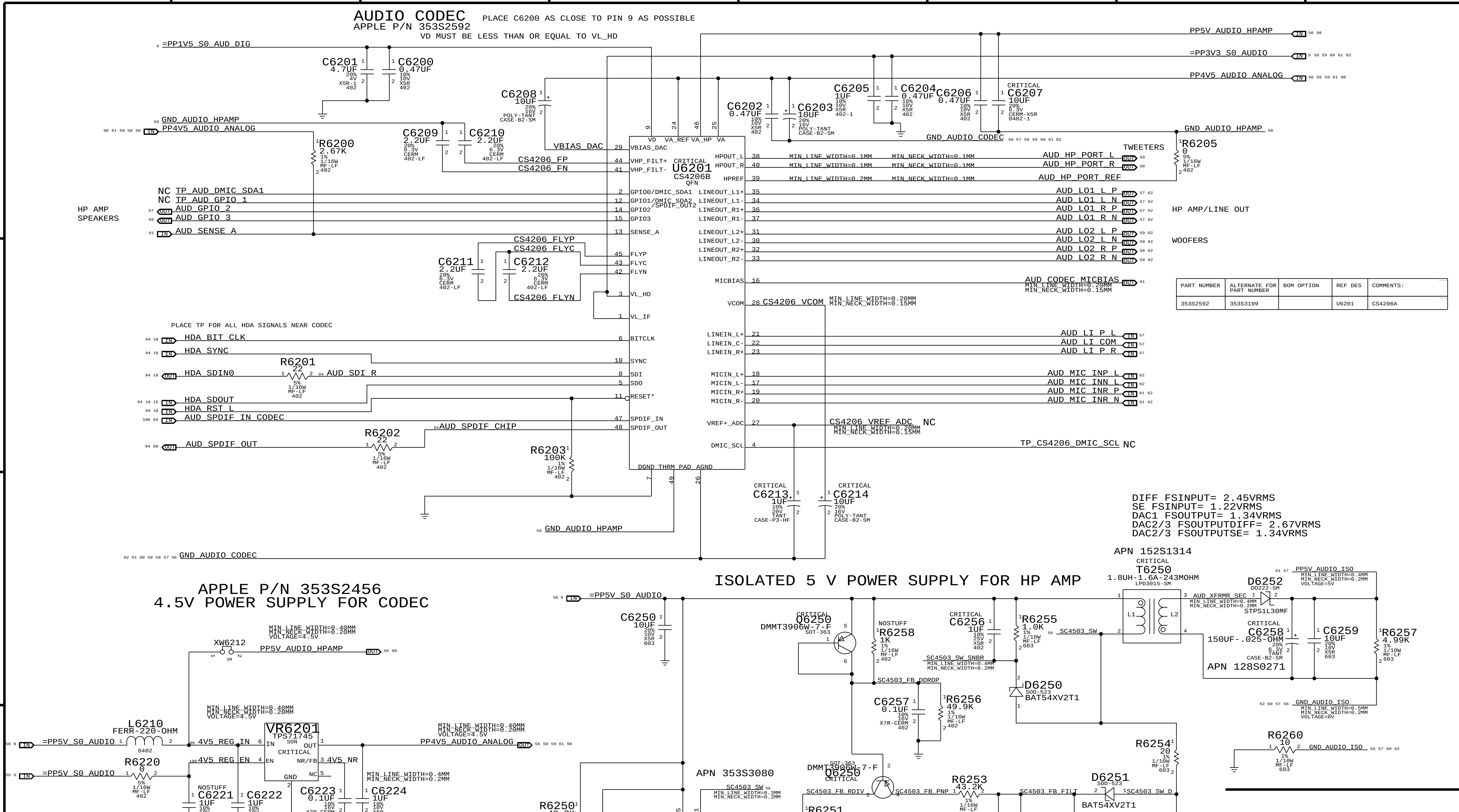
AUDIO: CODEC/REGULATOR



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S2592	353S3199		U6201	CS4206A

SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE: AUDIO: CODEC/REGULATOR

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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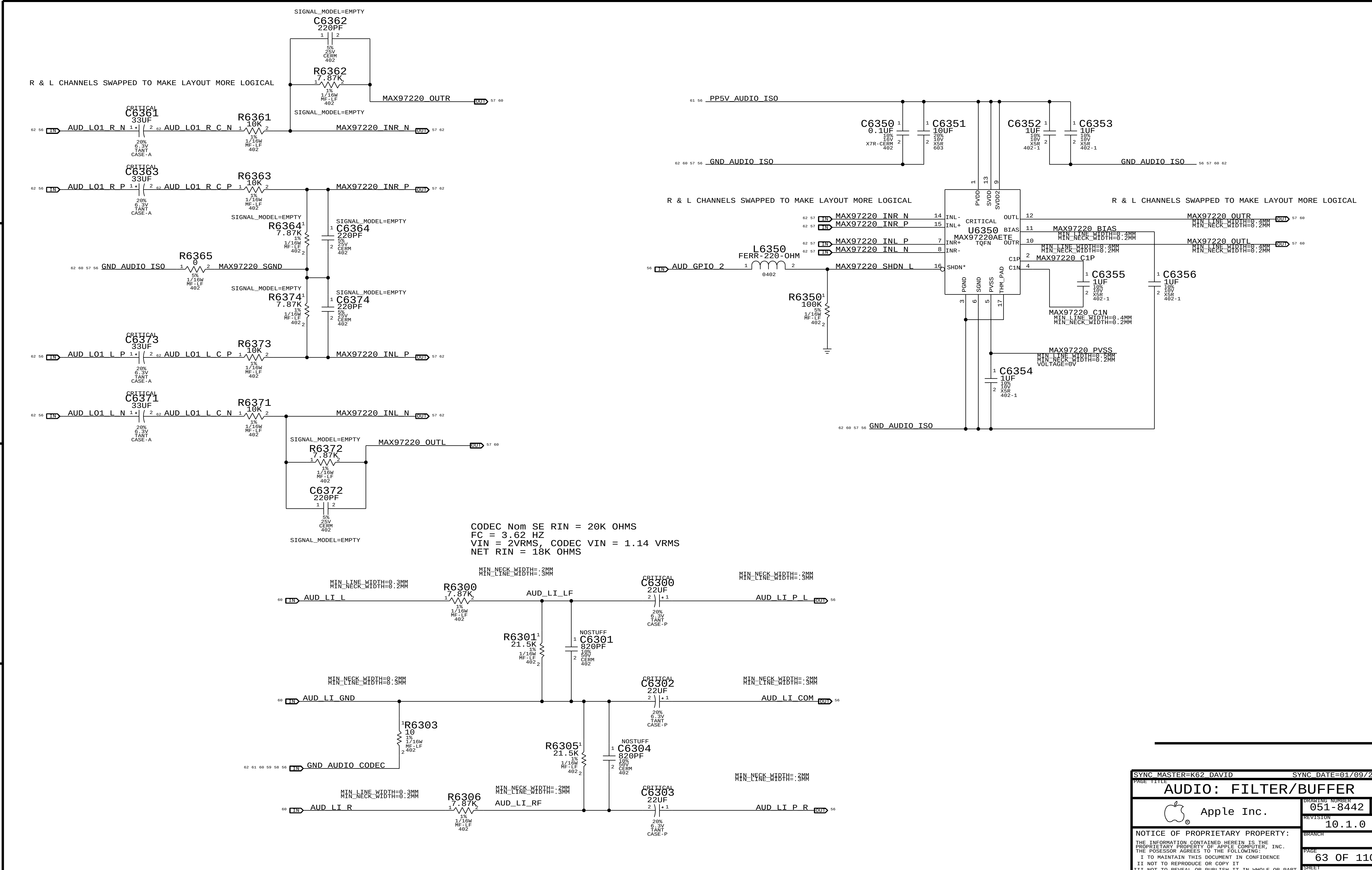

SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE

AUDIO: FILTER/BUFFER

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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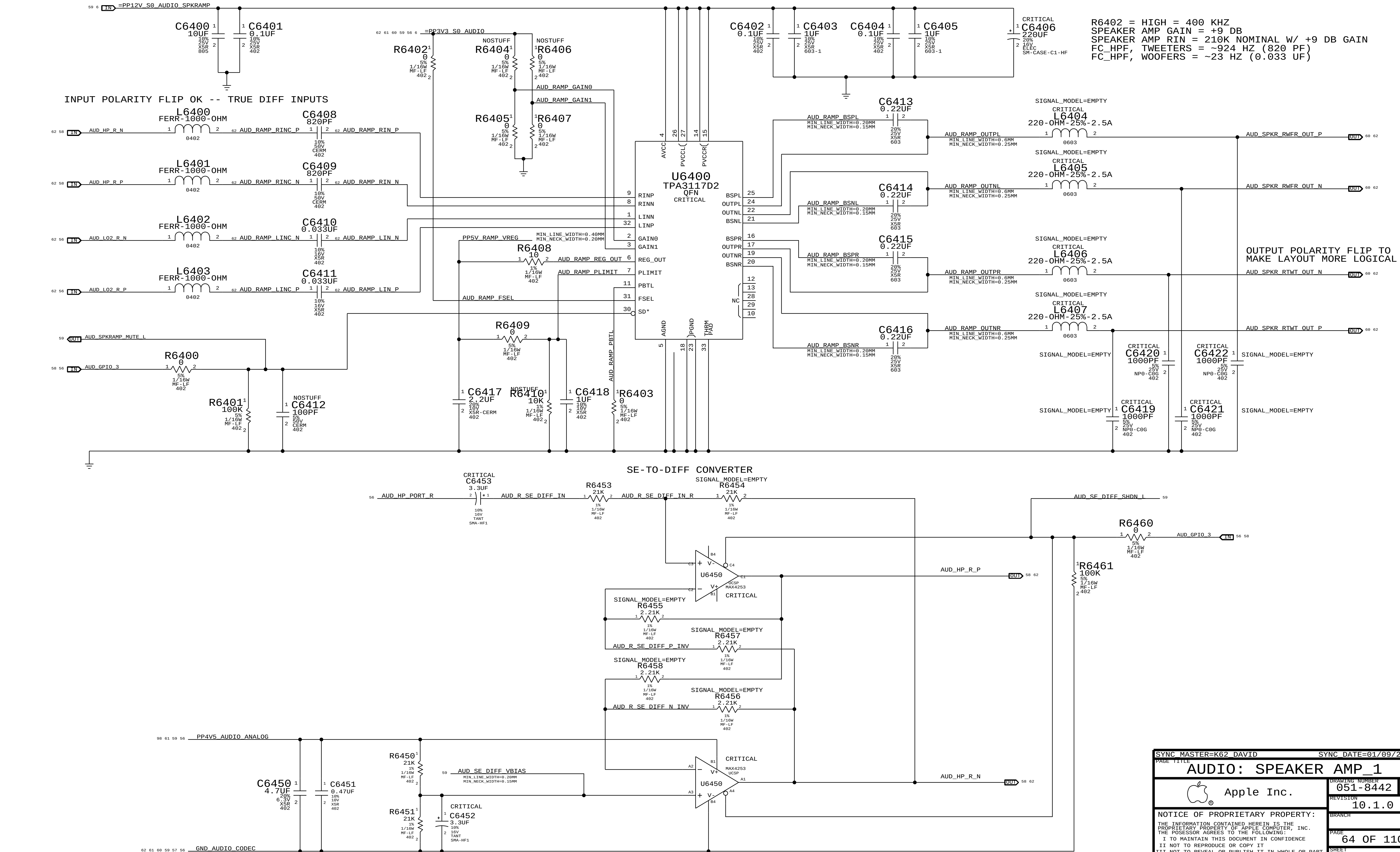
RIGHT CH SPEAKER AMP

APPLE P/N 353S3069



SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE

AUDIO: SPEAKER AMP_1

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DRAWING NUMBER 051-8442 SIZE D

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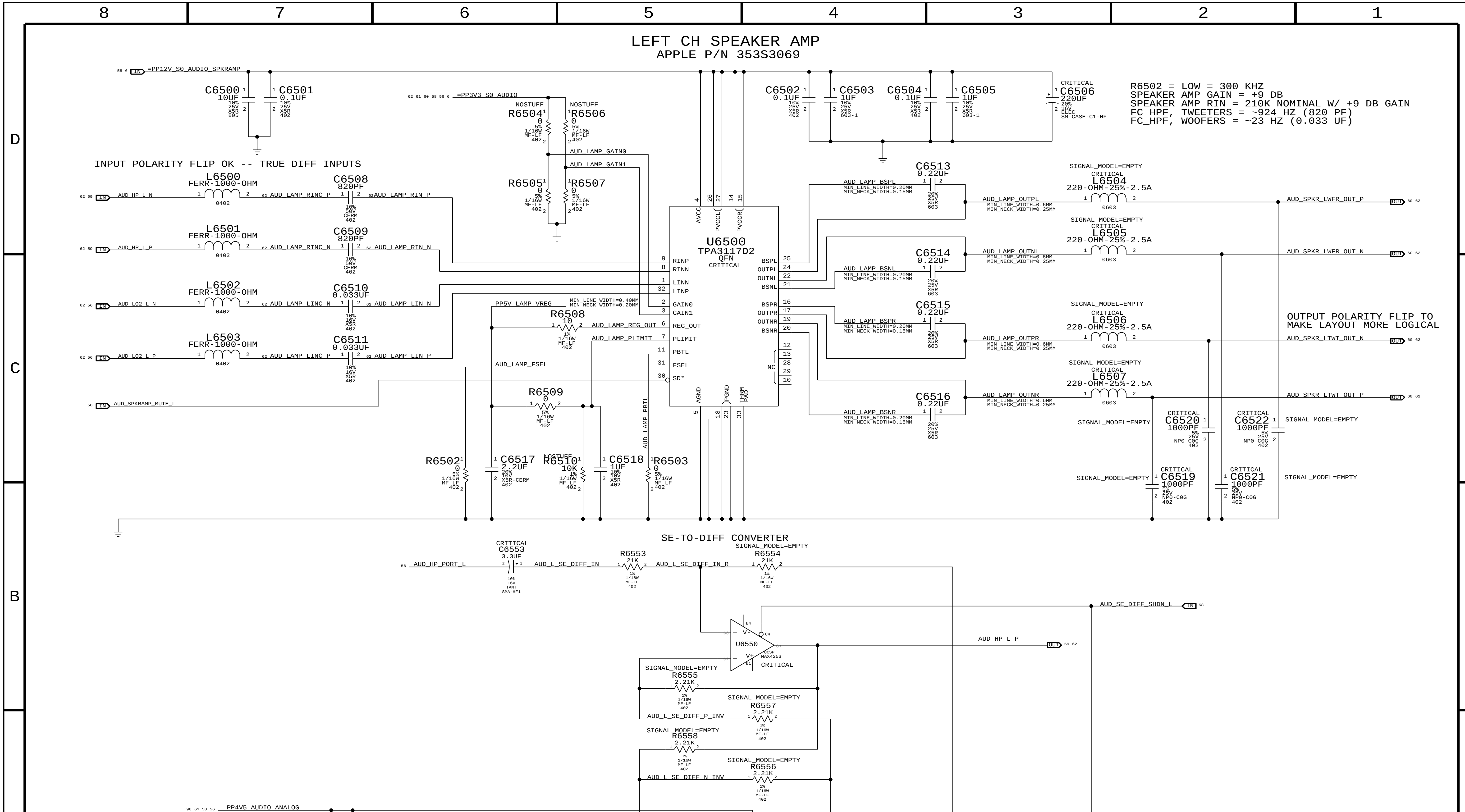

SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE

AUDIO: SPEAKER AMP

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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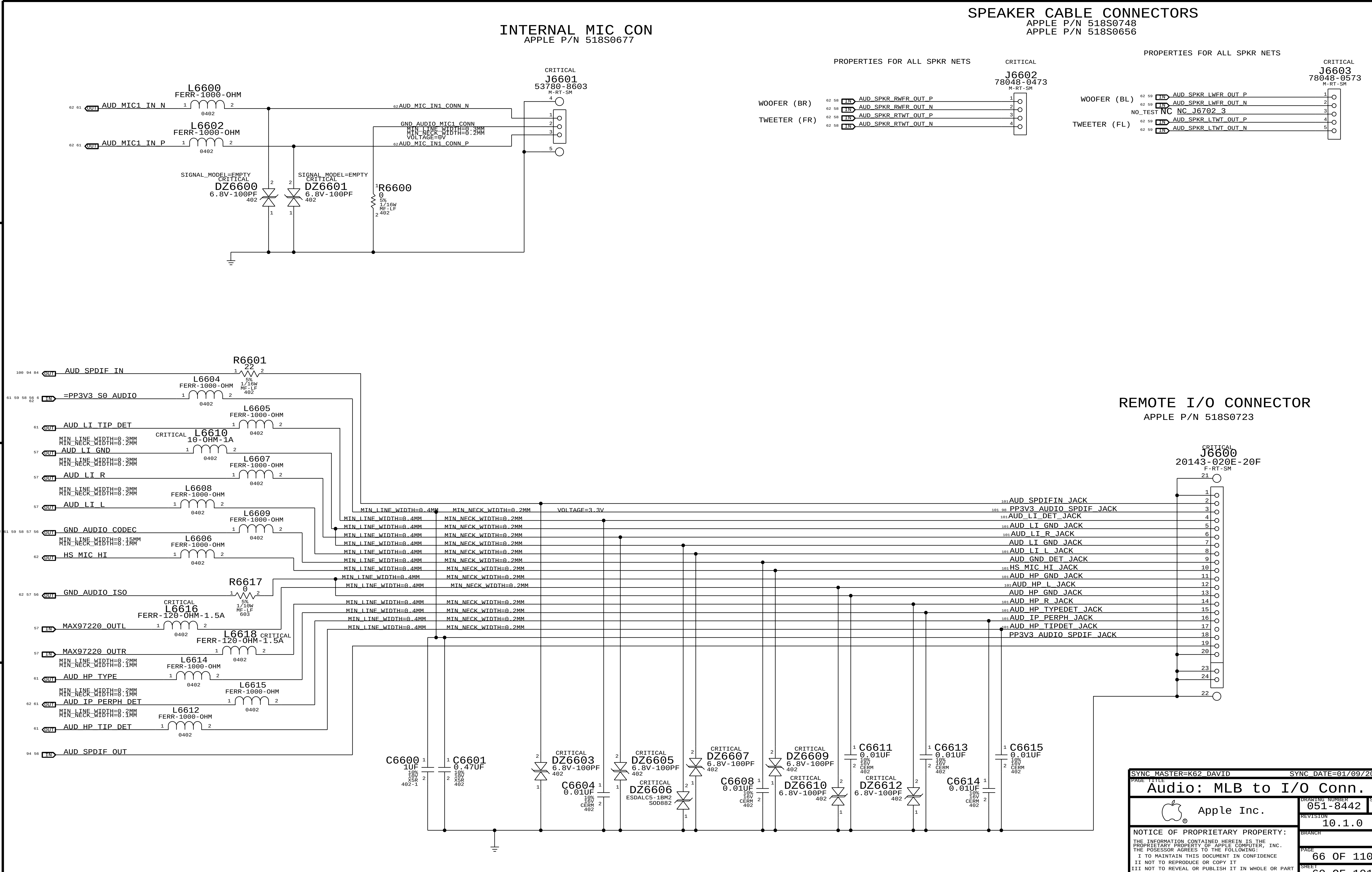

SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE: Audio: MLB to I/O Conn.

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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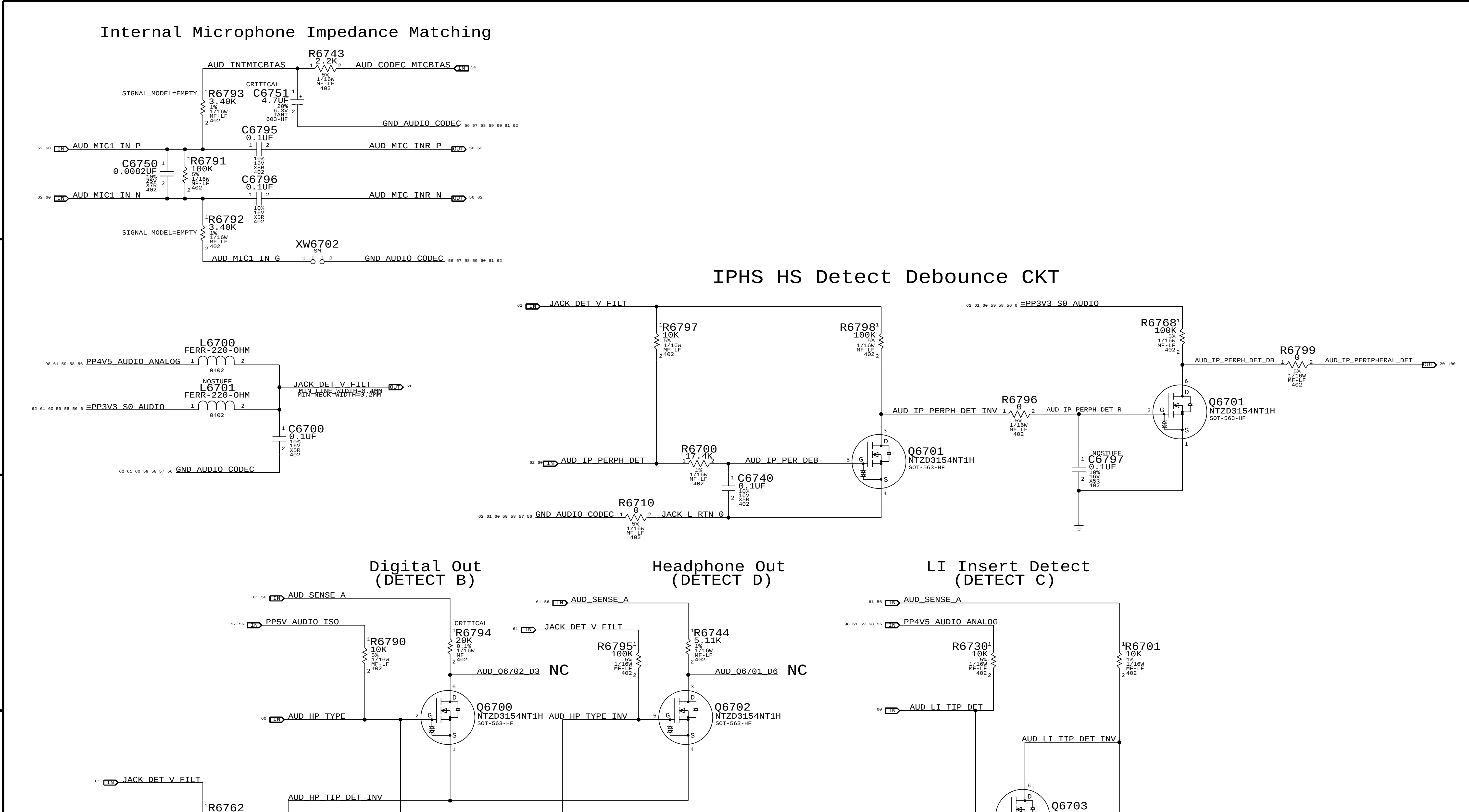
SHEET: 60 OF 101



SYNC_MASTER=K62_DAVID	SYNC_DATE=01/09/2011
PAGE TITLE	AUDIO: Detects/Grounding
Apple Inc.	DRAWING NUMBER 051-8442, SIZE D
	REVISION 10.1.0
	BRANCH
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	SHEET 61 OF 101

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	0X0A (D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X02 (2)	0X02 (2)	0X09 (09)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	0X0D (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
LINE IN	0X05 (5)	0X12 (12,C)	N/A	0X12 (C)
SPDIF IN	0X07 (7)	0x0F (15)	N/A	N/A
INTERNAL MIC	0X06 (6)	0X0E (14,LEFT & RIGHT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	COUGAR POINT GPIO 16	COUGAR POINT GPIO 5 (RCVR INT) COUGAR POINT GPIO 3 (PERIPH DET)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

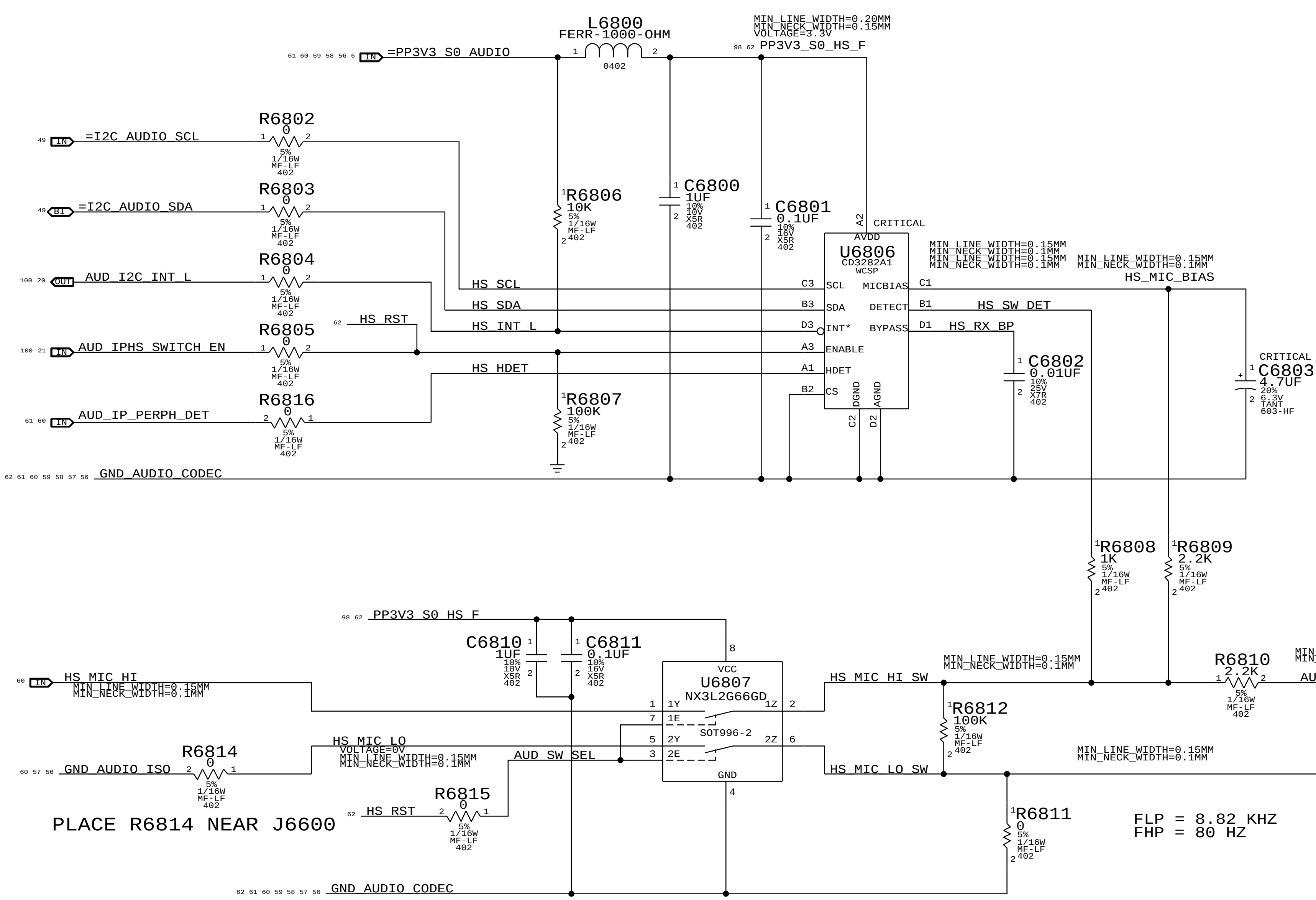
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I213	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_L_P	59
I214	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_L_N	59
I215	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_R_P	58
I216	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_R_N	58
I217	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	56 57
I218	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	56 57
I211	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	56 57
I212	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	56 57
I210	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	56 59
I209	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	56 59
I208	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	56 58
I206	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	56 58
I207	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	58
I204	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	58
I205	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	58
I203	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	58
I220	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	58
I219	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	58
I222	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	58
I221	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	58
I224	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	59
I223	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	59
I226	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	59
I225	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	59
I227	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	59
I229	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	59
I228	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	59
I230	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	59
I232	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	57
I233	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	57
I235	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	57
I234	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	57
I236	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	57
I238	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	57
I237	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	57
I239	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	57
I240	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	58 60
I241	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	58 60
I242	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	58 60
I243	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	58 60
I244	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	59 60
I246	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	59 60
I245	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	59 60
I247	SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	59 60
I252	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC1_IN_P	60 61
I253	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC1_IN_N	60 61
I254	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INR_P	56 61
I255	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INR_N	56 61
I256	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_IN1_CONN_P	60
I257	AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_IN1_CONN_N	60

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640



PLACE R6814 NEAR J6600

FLP = 8.82 KHZ
FHP = 80 HZ

SYNC_MASTER=K62_DAVID SYNC_DATE=01/09/2011

PAGE TITLE
AUDIO: Mikey

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D
REVISION 10.1.0
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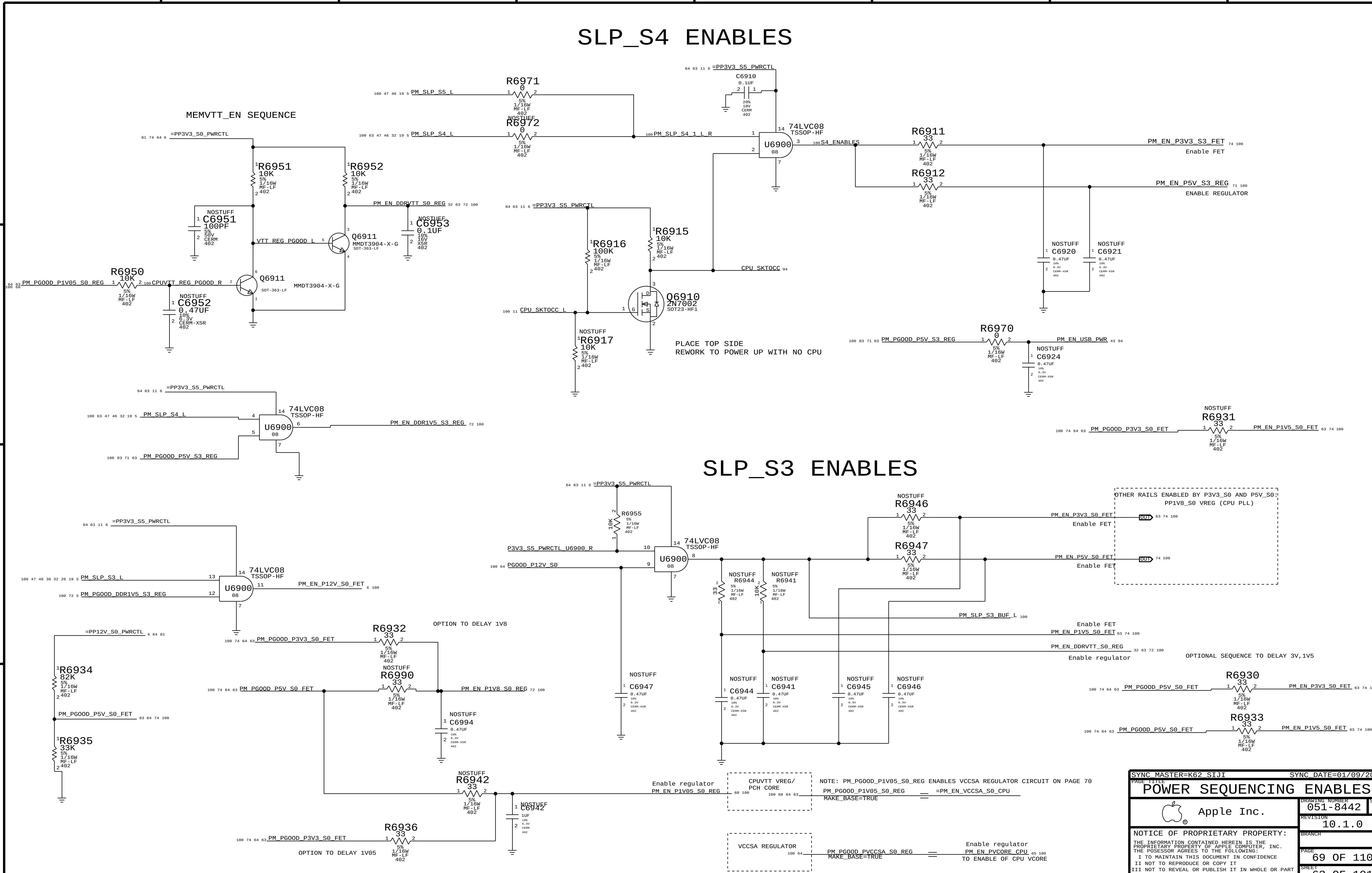

SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

POWER SEQUENCING ENABLES

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DRAWING NUMBER 051-8442 SIZE D

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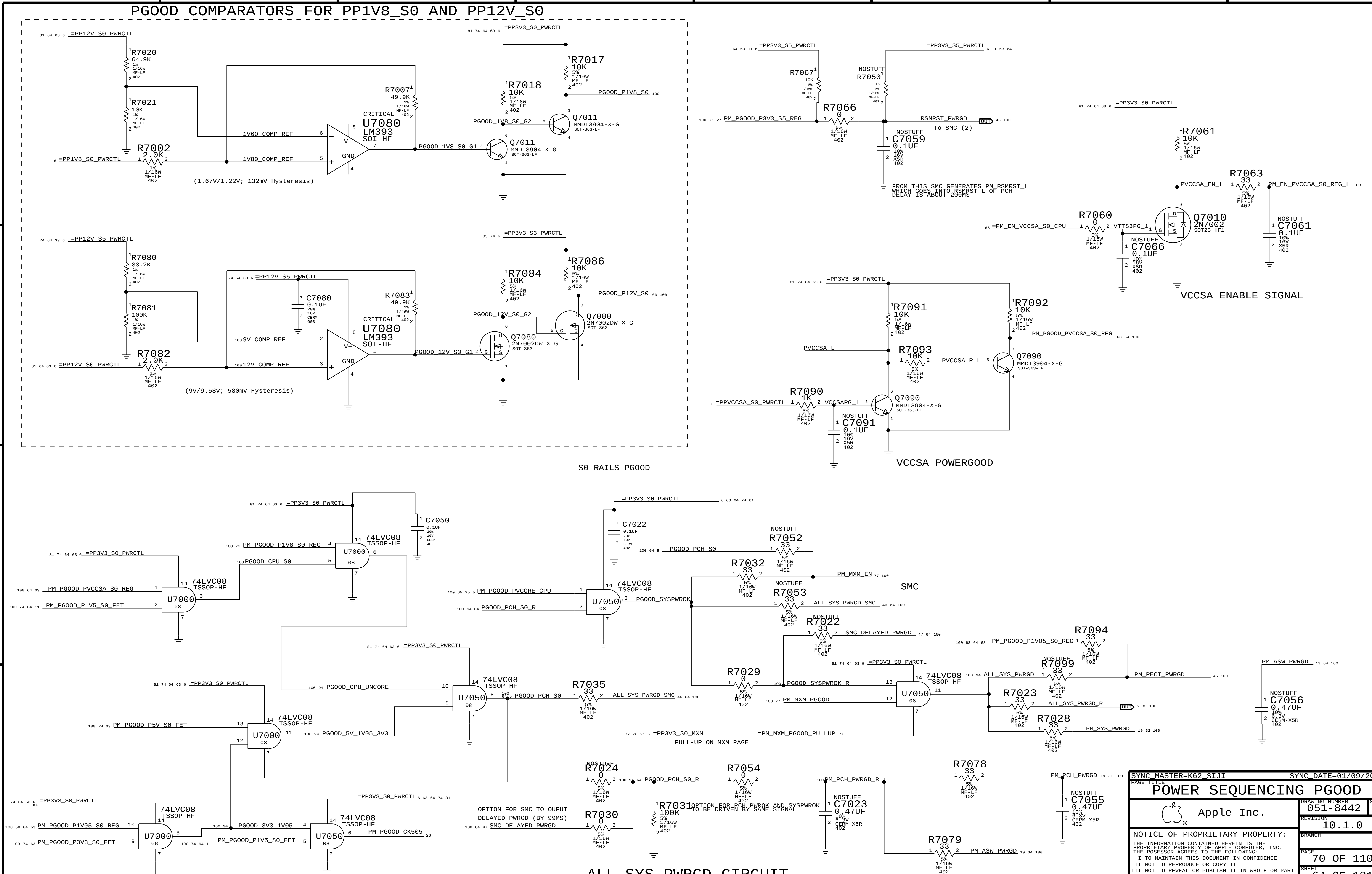

SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE

POWER SEQUENCING PGOOD

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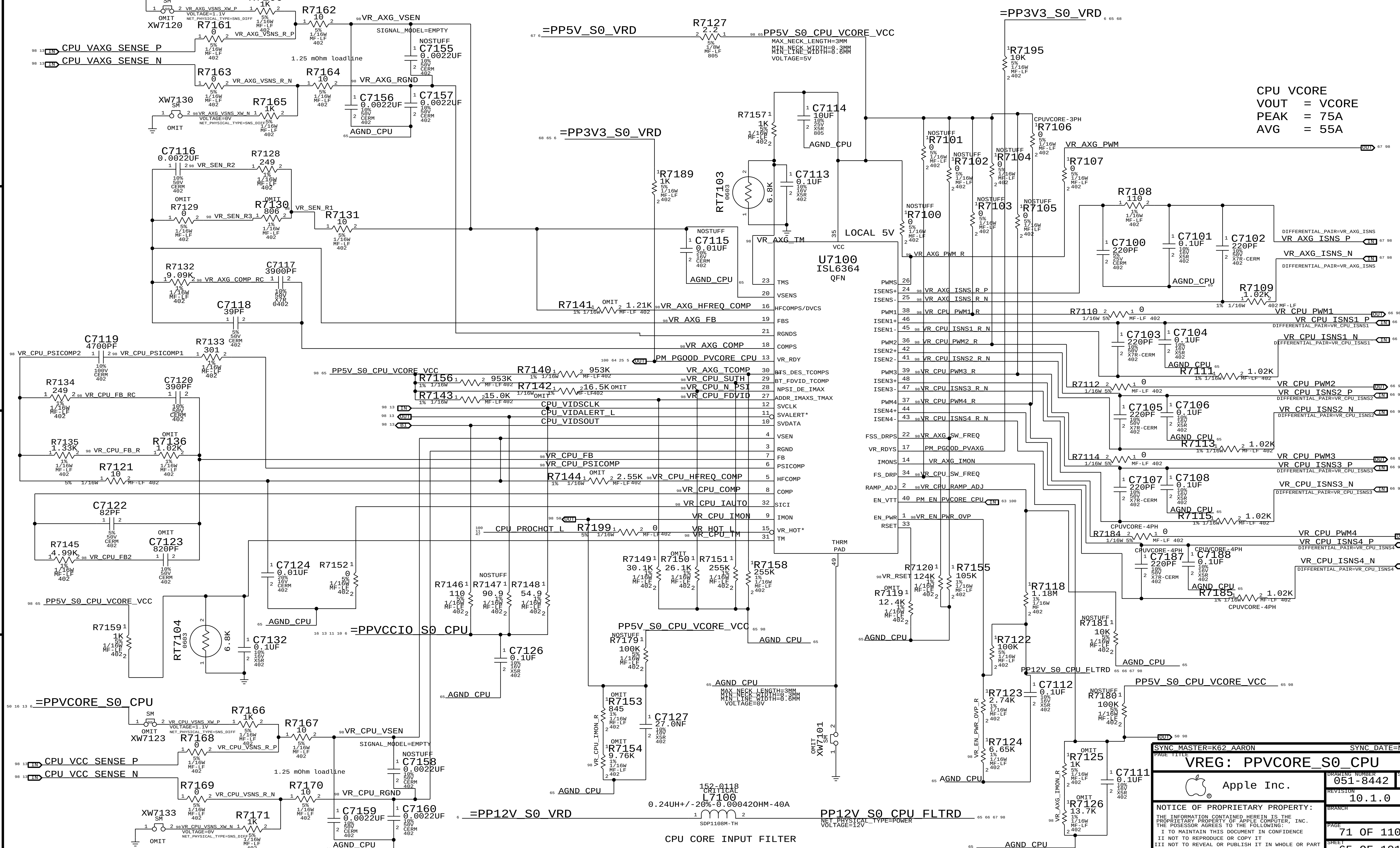
CPU CORE/AXG REG 1.1V/75A O/P= PPVCORE_S0_CPU_REG



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: VREG: PPVCORE_S0_CPU

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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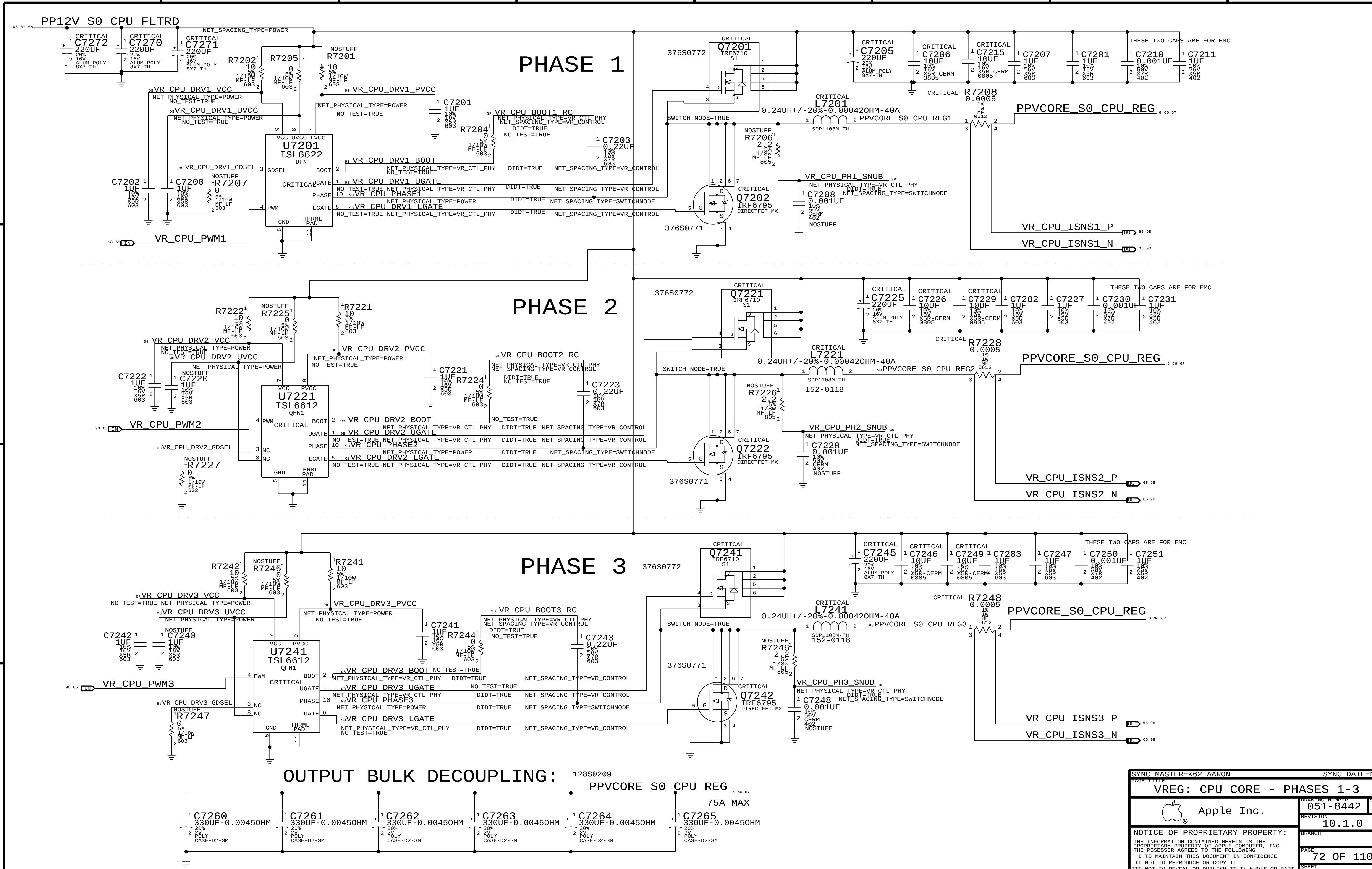

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

VREG: CPU CORE - PHASES 1-3

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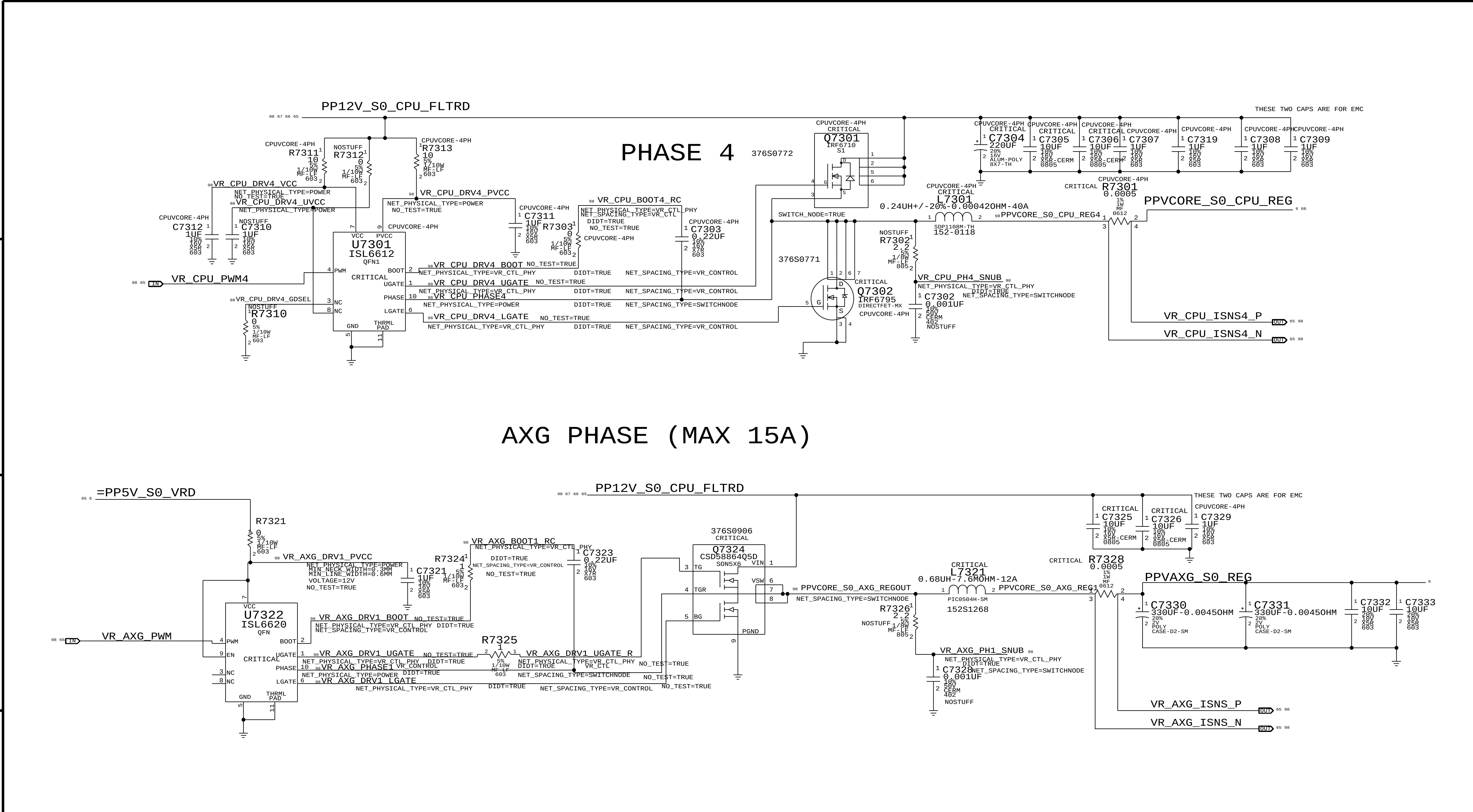

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

VREG:AXG PHASE/CORE - CAPS

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NOTICE OF PROPRIETARY PROPERTY:

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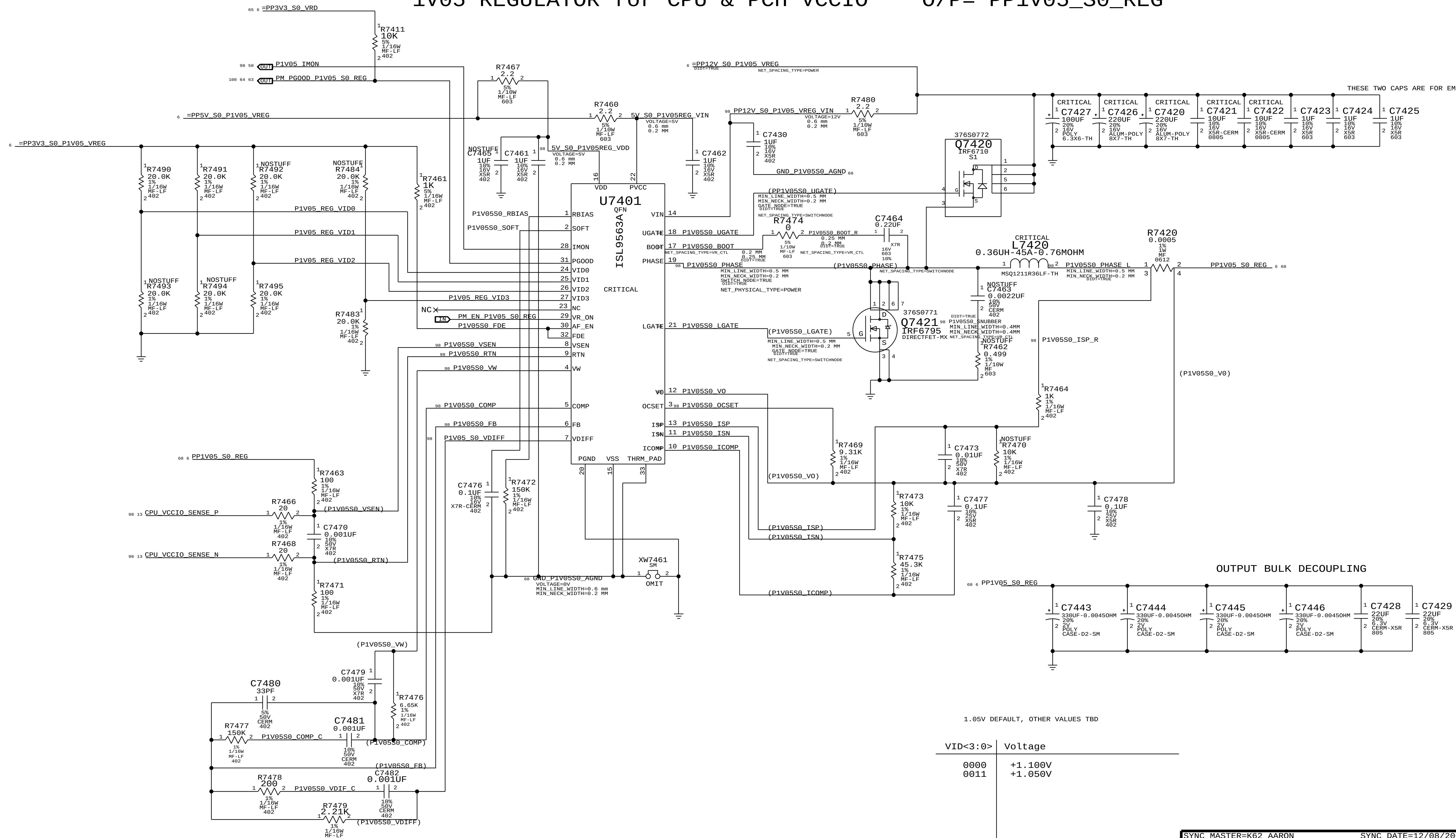
1V05 REGULATOR for CPU & PCH VCCIO O/P= PP1V05_S0_REG



1.05V DEFAULT, OTHER VALUES TBD

VID<3:0>	Voltage
0000	+1.100V
0011	+1.050V

SYNC_MASTER=K62_AARON SYNC_DATE=12/08/2009

PAGE TITLE

1V05 REGULATOR

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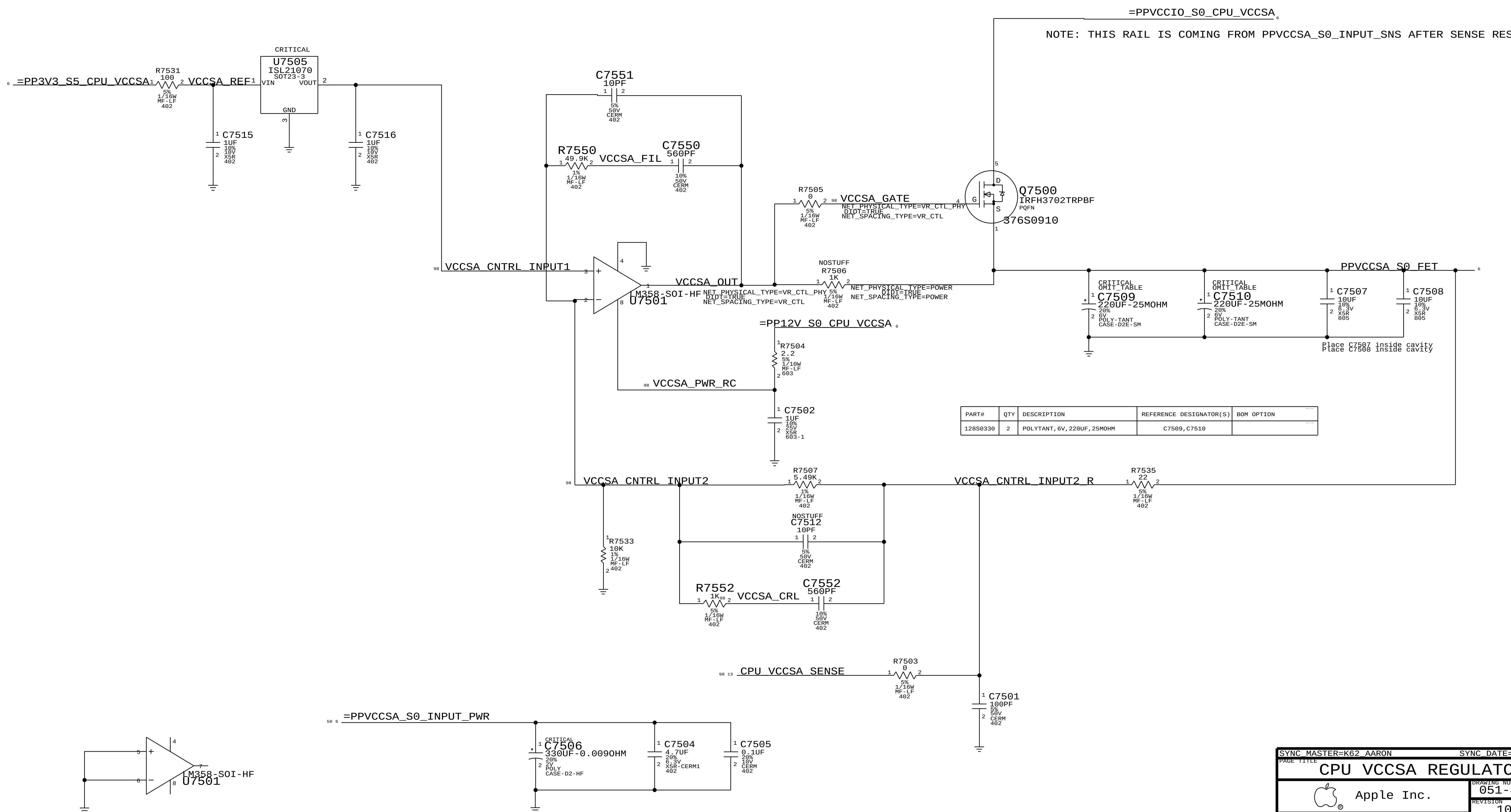
CPU VCCSA 0.925V (8.8A MAX)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0330	2	POLYTANT,6V,220UF,25MOHM	C7509,C7510	

NOTE: THIS POWER RAIL IS BEFORE THE SENSE RES R5310

SYNC_MASTER=K62_AARON SYNC_DATE=12/08/2009

PAGE TITLE: CPU VCCSA REGULATOR

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DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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CPU VCORE 3 PHASE/4 PHASE BOM OPTIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0066	1	RES,1K,5%,0402	R7125	CPUVCORE-4PH
114S0303	1	RES,7.5K,5%,0402	R7125	CPUVCORE-3PH
114S0327	1	RES,13.7K,1%,0402	R7126	CPUVCORE-4PH
114S0316	1	RES,10.2K,1%,0402	R7126	CPUVCORE-3PH
114S0323	1	RES,12.4K,1%,0402	R7119	CPUVCORE-4PH
114S0316	1	RES,10.2K,1%,0402	R7119	CPUVCORE-3PH
114S0355	1	RES,26.1K,1%,0402	R7150	CPUVCORE-4PH
114S0338	1	RES,17.8K,1%,0402	R7150	CPUVCORE-3PH
114S0211	1	RES,845,1%,0402	R7153	CPUVCORE-4PH
116S0004	1	RES,0,1%,0402	R7153	CPUVCORE-3PH
114S0314	1	RES,9.76K,1%,0402	R7154	CPUVCORE-4PH
114S0316	1	RES,10.2K,1%,0402	R7154	CPUVCORE-3PH
114S0225	1	RES,1.21K,1%,0402	R7141	CPUVCORE-4PH
114S0217	1	RES,976,1%,0402	R7141	CPUVCORE-3PH
114S0335	1	RES,16.5K,1%,0402	R7142	CPUVCORE-4PH
114S0349	1	RES,23.2K,1%,0402	R7142	CPUVCORE-3PH
114S0331	1	RES,15K,1%,0402	R7143	CPUVCORE-3PH
114S0257	1	RES,2.55K,1%,0402	R7144	CPUVCORE-4PH
114S0252	1	RES,2.32K,1%,0402	R7144	CPUVCORE-3PH
114S0209	1	RES,806,1%,0402	R7130	CPUVCORE-4PH
114S0188	1	RES,487,1%,0402	R7130	CPUVCORE-3PH
116S0004	1	RES,0R,1%,0402	R7129	CPUVCORE-4PH
114S0189	1	RES,499,1%,0402	R7129	CPUVCORE-3PH
114S0219	1	RES,1.02K,1%,0402	R7136	CPUVCORE-4PH
114S0131	1	RES,130,1%,0402	R7136	CPUVCORE-3PH
132S8221	1	CAP,820PF,10%,0402	C7123	CPUVCORE-4PH
132S1534	1	CAP,0.0012UF,10%,0402	C7123	CPUVCORE-3PH

SYNC_MASTER=K62_AARON SYNC_DATE=12/08/2009

PAGE TITLE
CPU 3P/4P BOM OPTIONS

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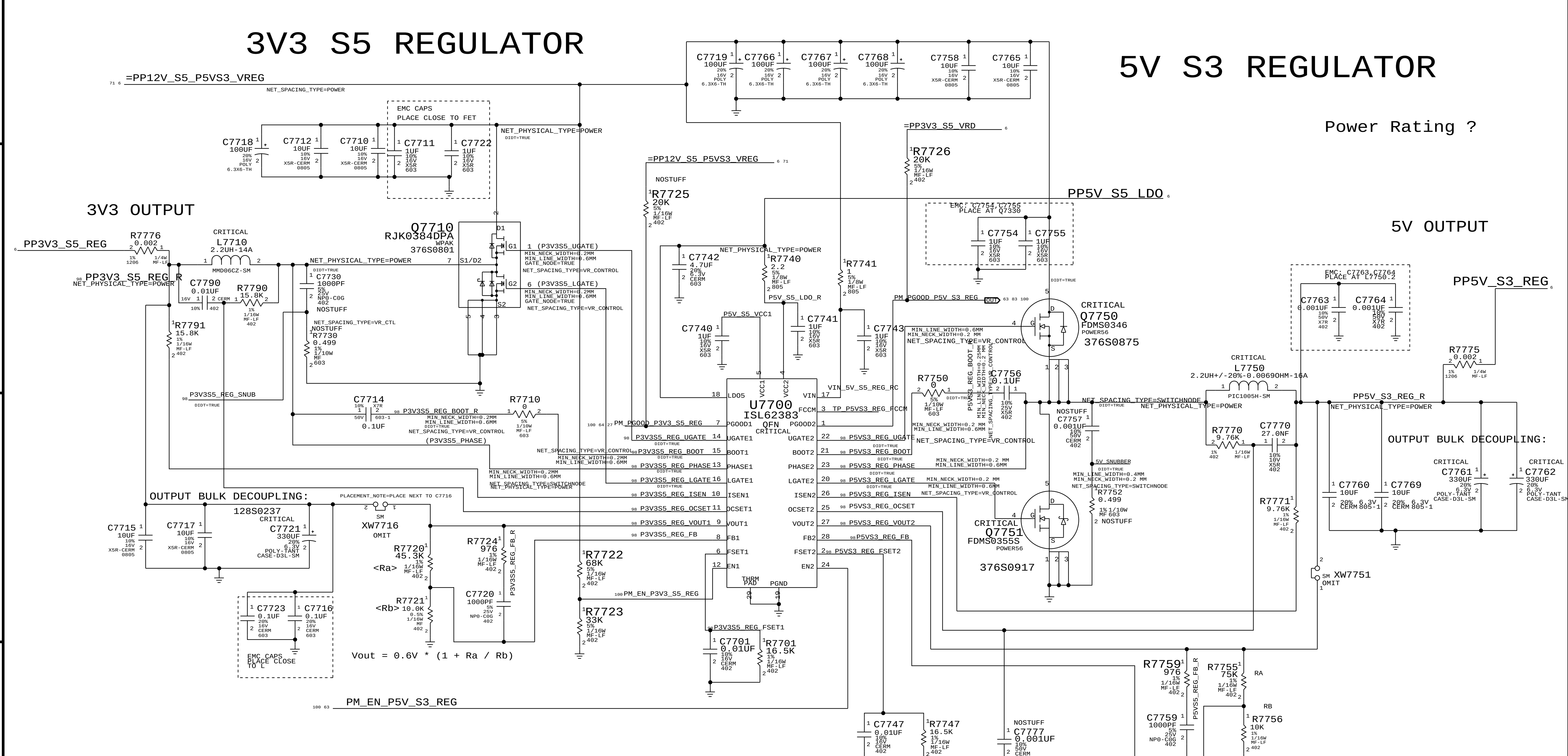

SYNC_MASTER=K62_AARON		SYNC_DATE=12/08/2009
PAGE TITLE	5V_S3 / 3V3_S5 VREGS	
Apple Inc.	DRAWING NUMBER 051-8442	SIZE D
	REVISION 10.1.0	
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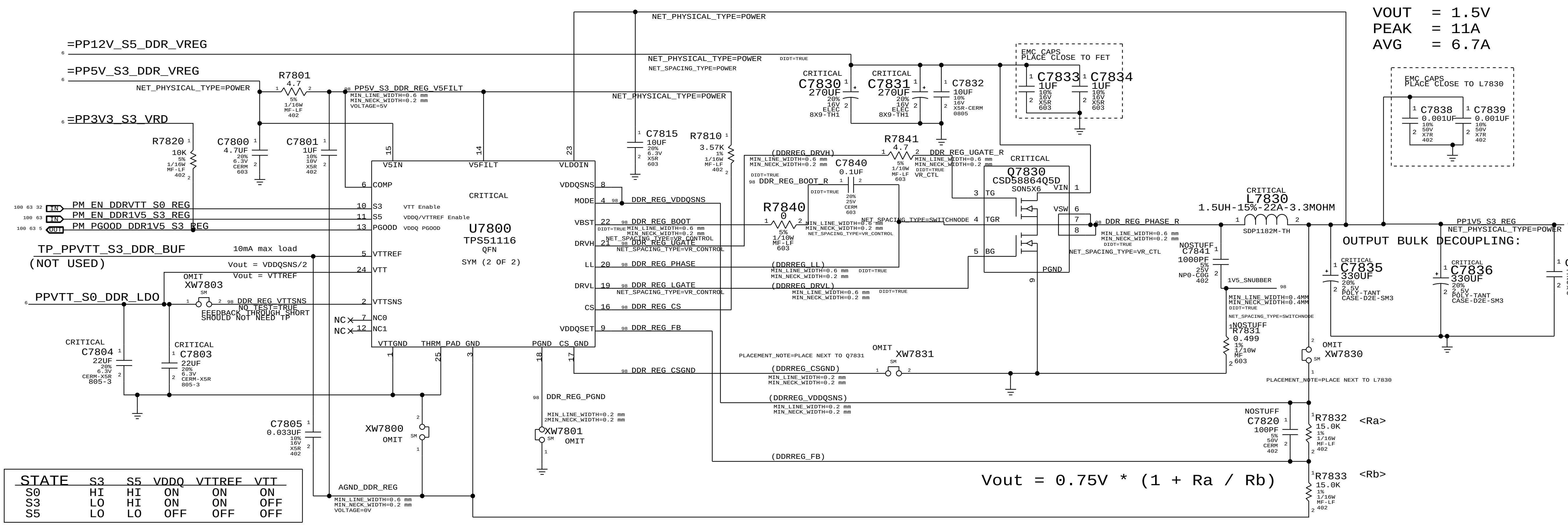
	SHEET 71 OF 101	

1.5 V DDR SUPPLY

PPDDR_S3_REG
VOUT = 1.5V
PEAK = 11A
AVG = 6.7A



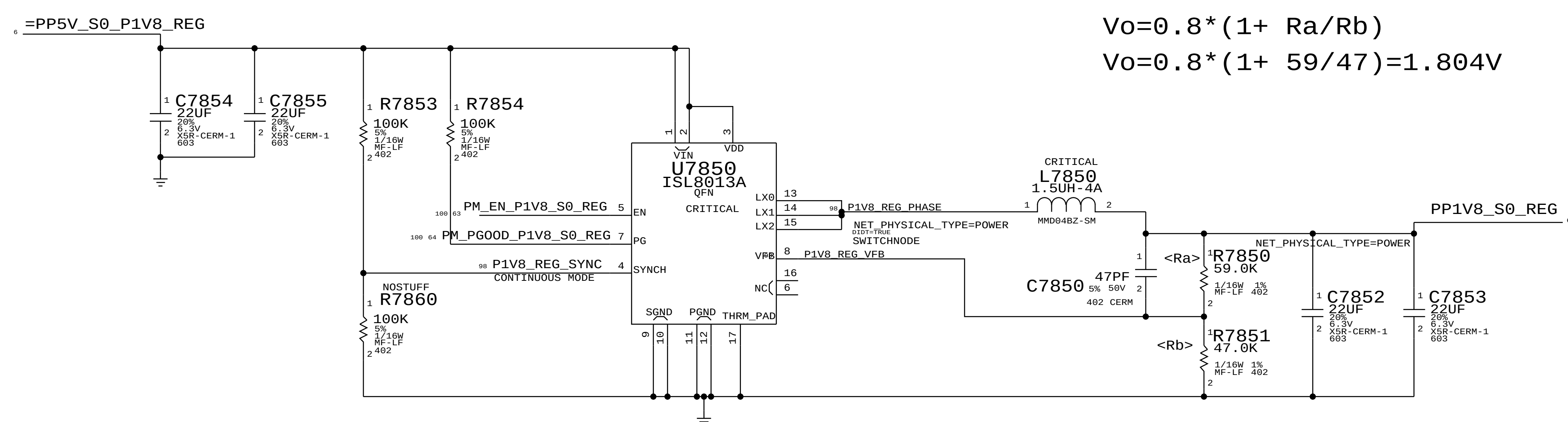
STATE	S3	S5	VDDQ	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

1.8 V SUPPLY

1A Average current

Vo=0.8*(1+ Ra/Rb)

Vo=0.8*(1+ 59/47)=1.804V



SYNC_MASTER=K62_AARON	SYNC_DATE=11/30/2009
PAGE TITLE	1.5V / 1.8V VREGS
Apple Inc.	DRAWING NUMBER 051-8442, SIZE D
	REVISION 10.1.0
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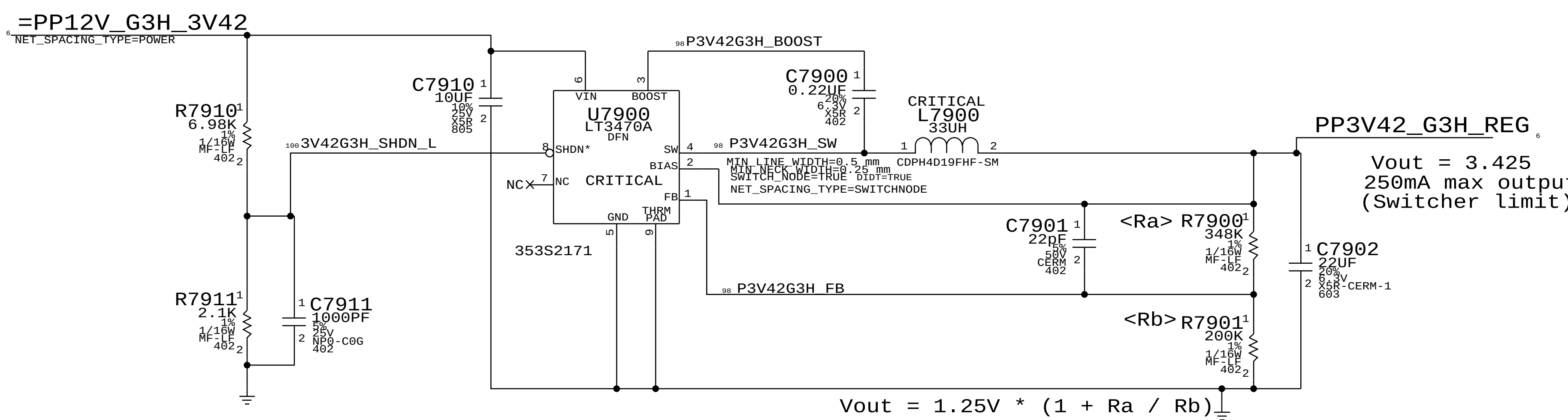
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

3.42 G3HOT SUPPLY

Apple Inc.	DRAWING NUMBER 051-8442 SIZE D
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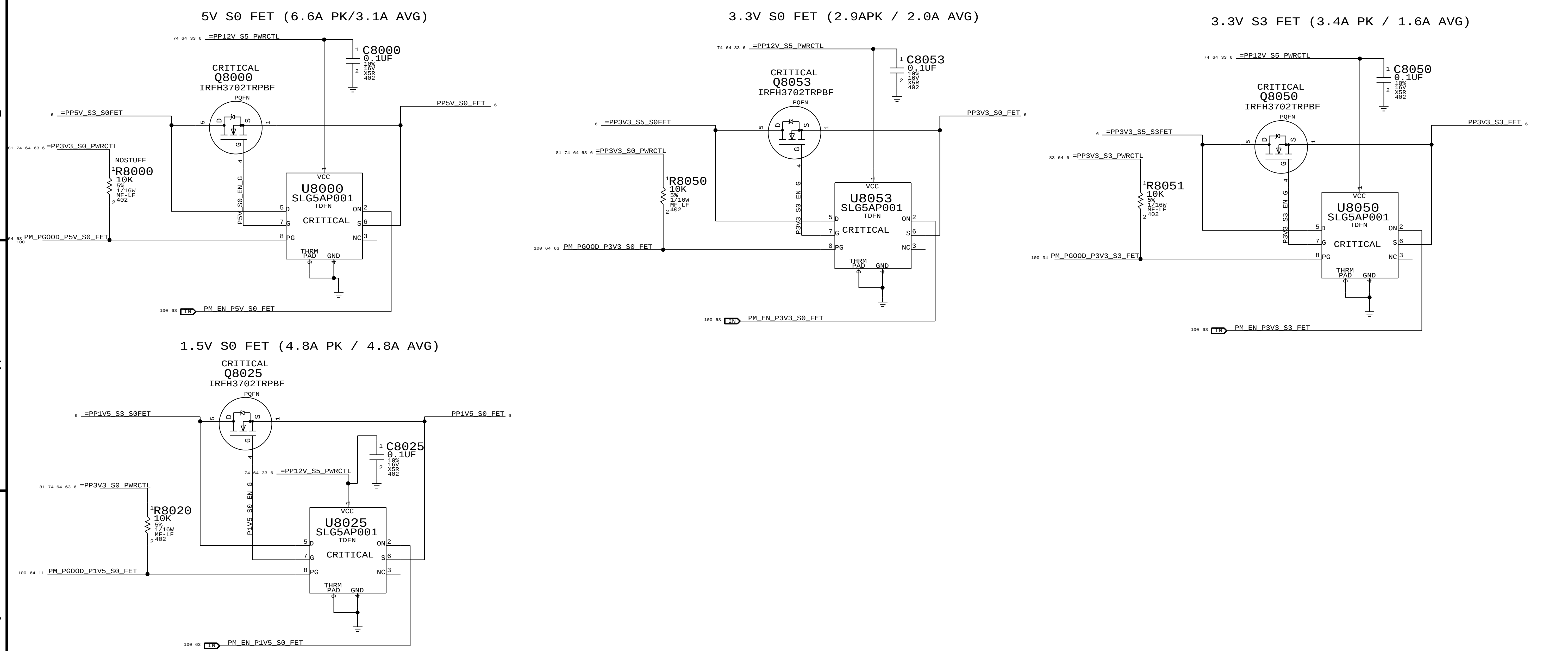

SYNC_MASTER=K62_AARON SYNC_DATE=04/07/2010

PAGE TITLE

S3+S0 FETS

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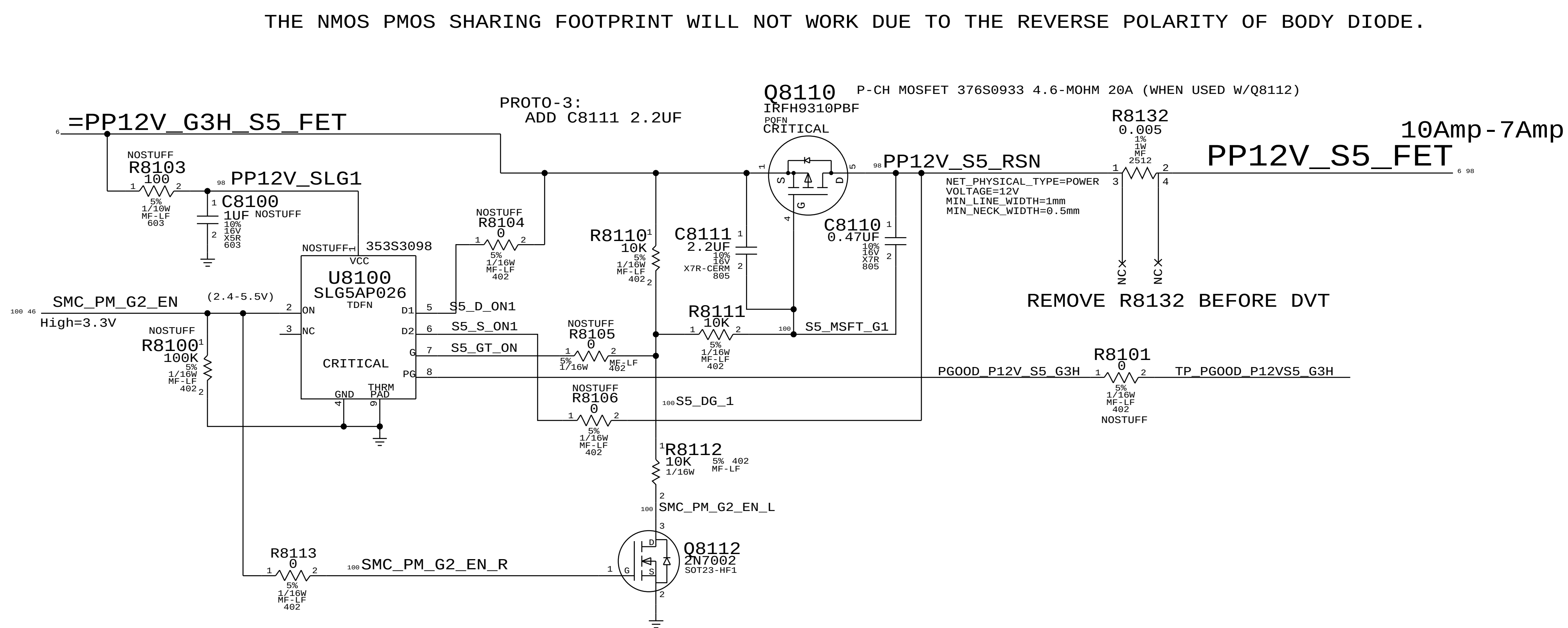

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE

12V_S0 & 12V_S5 switch

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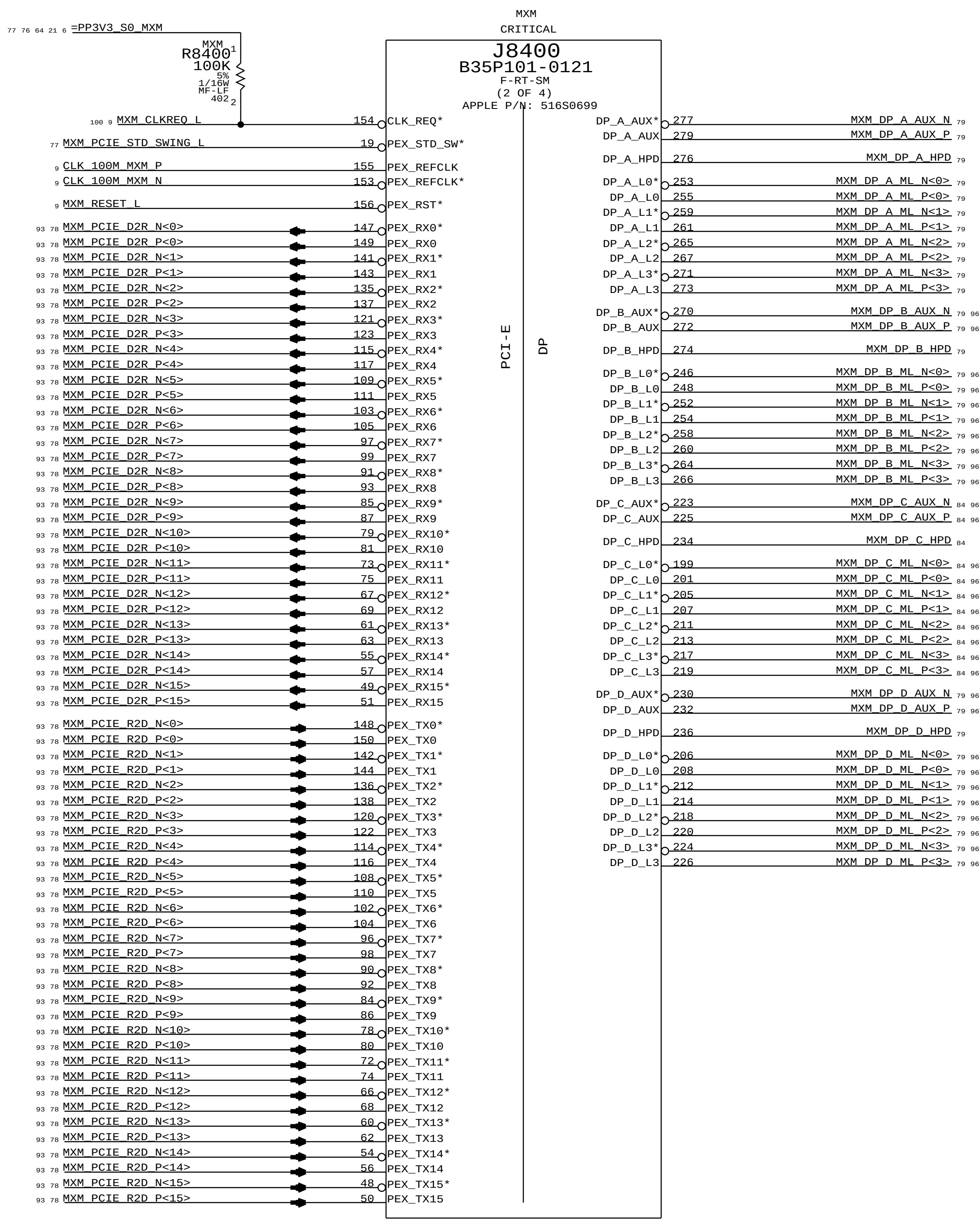
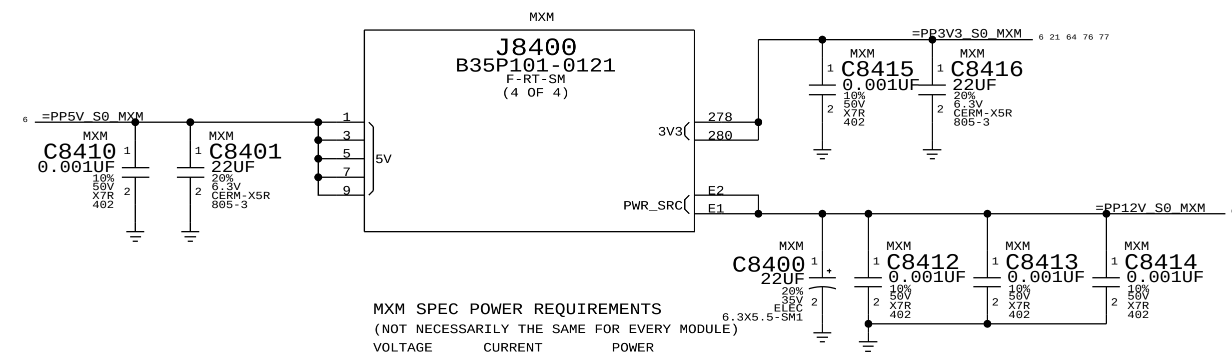
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
- MXM





MXM SPEC POWER REQUIREMENTS
(NOT NECESSARILY THE SAME FOR EVERY MODULE)

VOLTAGE	CURRENT	POWER
3V3	1.0 A	3.3 W
5V	2.5 A	12.5 W
PWR (7-20V)	UP TO 10 A	PLATFORM DEPENDENT

MXM DP PORT ROUTING

	K62	K60
DP A	EXT DP1	EXT DP1
DP B	T29 DP2	T29 DP2
DP C	INT DP	INT DP
DP D	T29 DP1	T29 DP1
DP E	EXT DP2	

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: MXM PCIe, DP & Power

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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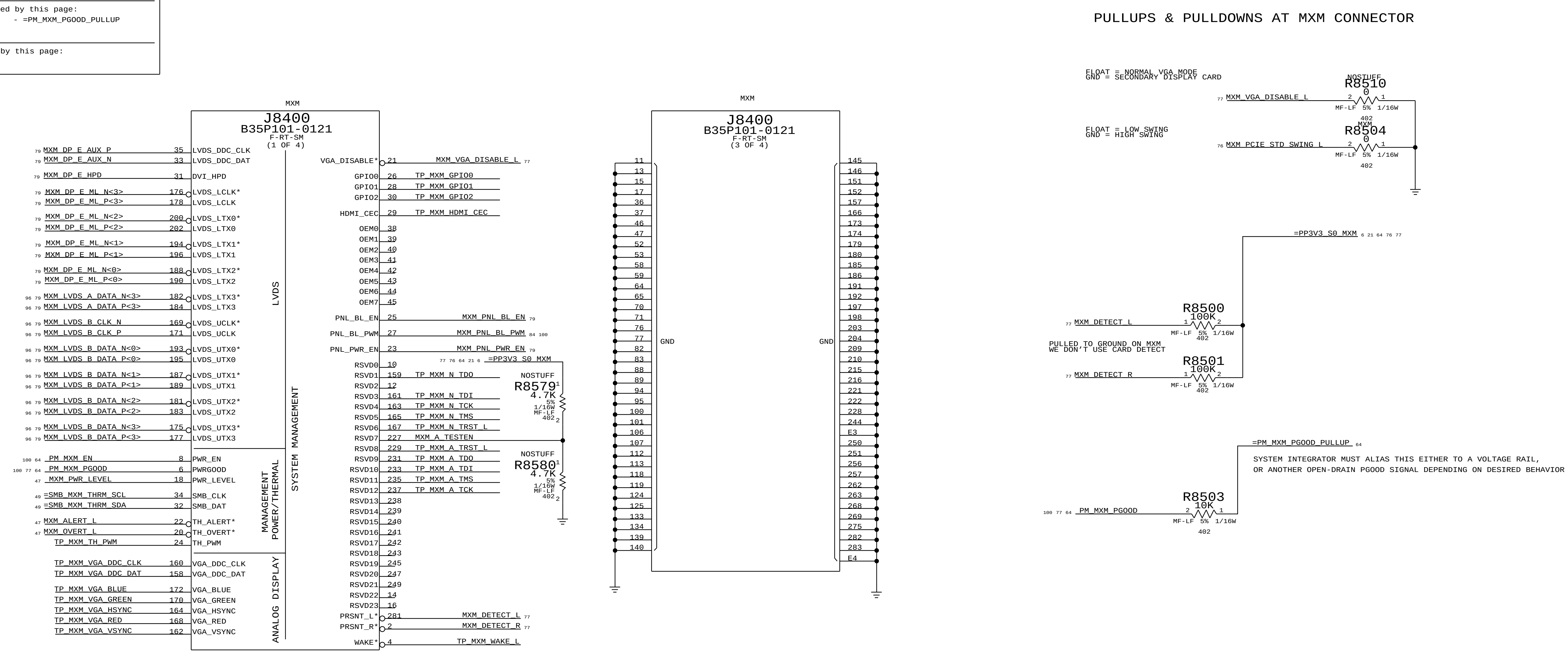
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM

Signal aliases required by this page:
- =SMB_MXM_THRM_DATA
- =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

BOM options provided by this page:



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: MXM I/O

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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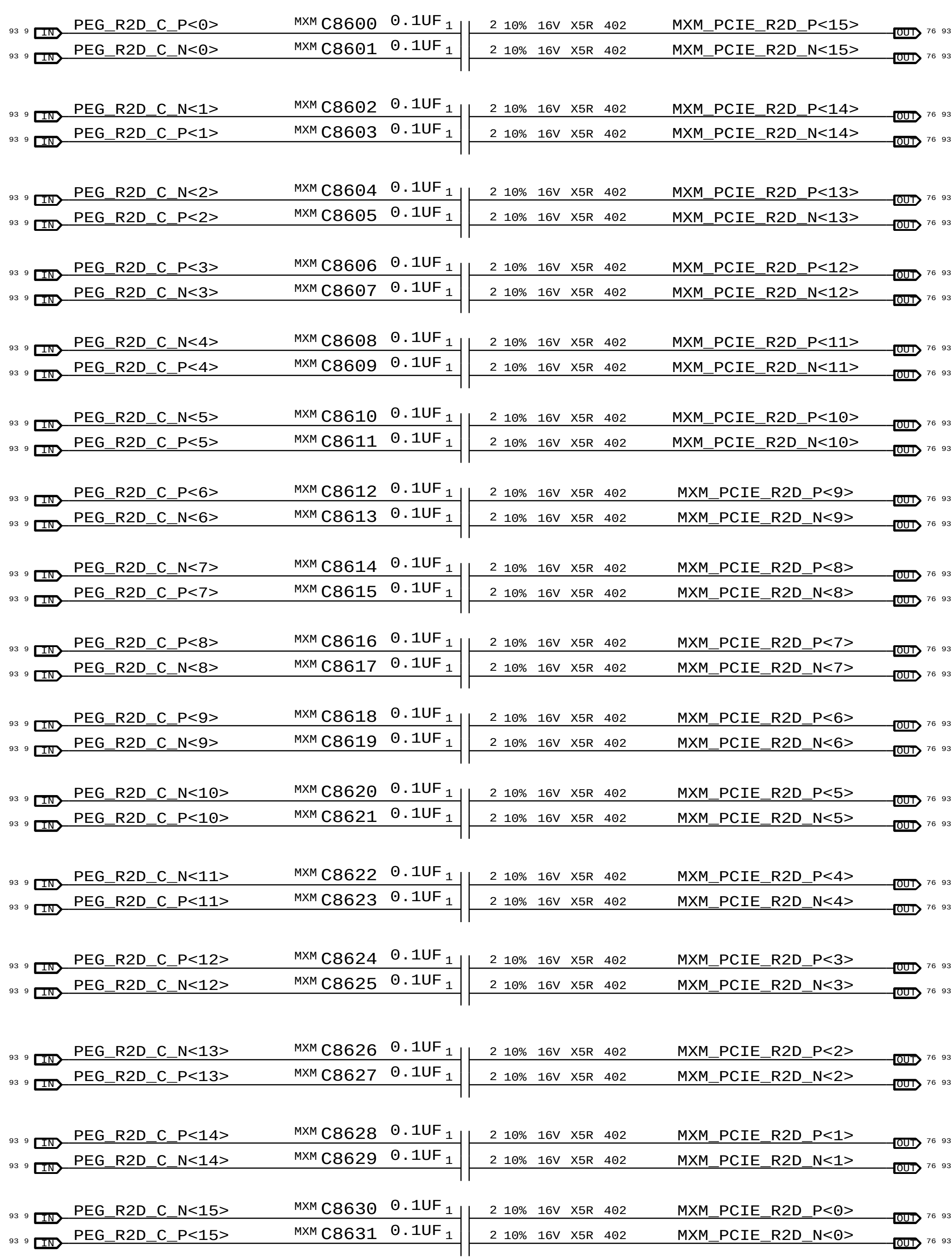
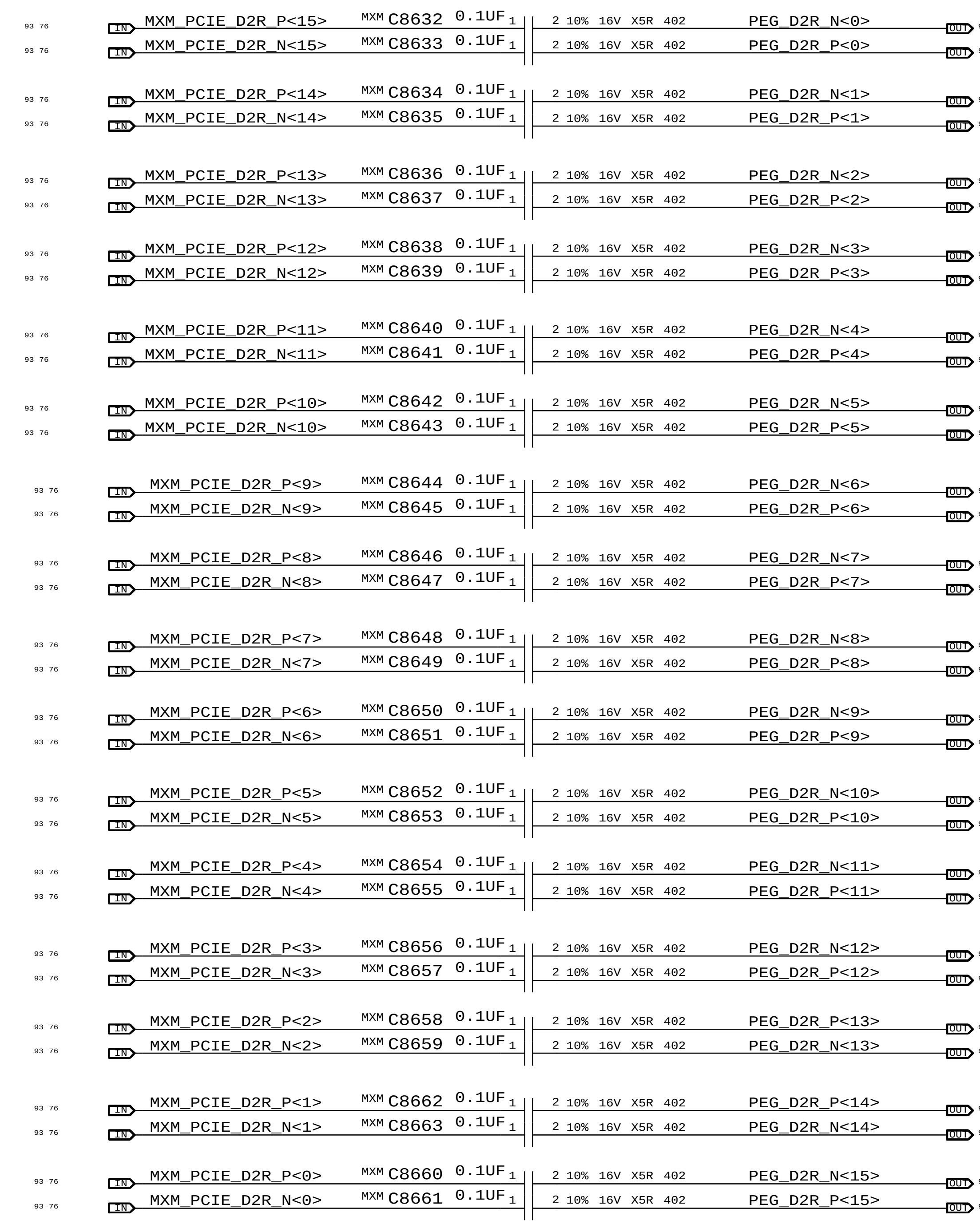
MXM TX CAPS



MXM RX CAPS



SYNC_MASTER=K62 SYNC_DATE=N/A

PAGE TITLE: MXM PCIE CAPS

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

BRANCH

8 7 6 5 4 3 2 1

Page Notes

Power aliases required by this page:

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

MXM ALIAS

76	MXM_DP_A_ML_P<0..3>	DP_EXTA_ML_C_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	85 96
76	MXM_DP_A_ML_N<0..3>	DP_EXTA_ML_C_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	85 96
76	MXM_DP_A_AUX_P	DP_EXTA_AUXCH_C_P	MAKE_BASE=TRUE NO_TEST=TRUE	79 85 96
76	MXM_DP_A_AUX_N	DP_EXTA_AUXCH_C_N	MAKE_BASE=TRUE NO_TEST=TRUE	79 85 96
76	MXM_DP_A_HPD	DP_EXTA_HPD	MAKE_BASE=TRUE NO_TEST=TRUE	85
77	MXM_DP_E_ML_P<0..3>	DP_EXTB_ML_C_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	87 96
77	MXM_DP_E_ML_N<0..3>	DP_EXTB_ML_C_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	87 96
77	MXM_DP_E_AUX_P	DP_EXTB_AUXCH_C_P	MAKE_BASE=TRUE NO_TEST=TRUE	79 87 96
77	MXM_DP_E_AUX_N	DP_EXTB_AUXCH_C_N	MAKE_BASE=TRUE NO_TEST=TRUE	79 87 96
77	MXM_DP_E_HPD	DP_EXTB_HPD	MAKE_BASE=TRUE NO_TEST=TRUE	87

DDC/AUX ALIAS

96 85 79	DP_EXTA_AUXCH_C_P	MAKE_BASE=TRUE	DP_EXTA_DDC_CLK	85
96 85 79	DP_EXTA_AUXCH_C_N	MAKE_BASE=TRUE	DP_EXTA_DDC_DATA	85
96 87 79	DP_EXTB_AUXCH_C_P	MAKE_BASE=TRUE	DP_EXTB_DDC_CLK	87
96 87 79	DP_EXTB_AUXCH_C_N	MAKE_BASE=TRUE	DP_EXTB_DDC_DATA	87

NO_TEST T29 & DP DC BIAS

I81	T29_A_BIAS_R2D_P0 NO_TEST=TRUE	85 86		I99	T29_B_BIAS_R2D_P2 NO_TEST=TRUE	87 88
I82	T29_A_BIAS_R2D_N0 NO_TEST=TRUE	85 86		I100	T29_B_BIAS_R2D_N2 NO_TEST=TRUE	87 88
I83	T29_A_BIAS_R2D_P1 NO_TEST=TRUE	85 86		I101	T29_B_BIAS_R2D_P3 NO_TEST=TRUE	87 88
I84	T29_A_BIAS_R2D_N1 NO_TEST=TRUE	85 86		I102	T29_B_BIAS_R2D_N3 NO_TEST=TRUE	87 88
I93	T29_A_BIAS NO_TEST=TRUE	83 85 86 99		I103	T29_B_BIAS NO_TEST=TRUE	83 87 88 99
I109	T29_A_BIAS_P1 NO_TEST=TRUE	86		I110	T29_B_BIAS_P3 NO_TEST=TRUE	88
I111	T29_A_BIAS_N1 NO_TEST=TRUE	86		I112	T29_B_BIAS_N3 NO_TEST=TRUE	88
I89	DP_A_BIAS_P_0 NO_TEST=TRUE	85 86		I104	DP_B_BIAS_P_0 NO_TEST=TRUE	87 88
I90	DP_A_BIAS_N_0 NO_TEST=TRUE	85 86		I105	DP_B_BIAS_N_0 NO_TEST=TRUE	87 88
I92	DP_A_BIAS_P_2 NO_TEST=TRUE	85 86		I107	DP_B_BIAS_P_2 NO_TEST=TRUE	87 88
I91	DP_A_BIAS_N_2 NO_TEST=TRUE	85 86		I106	DP_B_BIAS_N_2 NO_TEST=TRUE	87 88
I94	DP_A_BIAS NO_TEST=TRUE	85 99		I108	DP_B_BIAS NO_TEST=TRUE	87

T29 CONN POWER AND CONTROL ALIAS

6	=PP3V3_SW_DPAPWR	PP3V3_SW_DPAPWR		85 98
6	=PP3V3_SW_DPBPWR	PP3V3_SW_DPBPWR		87 98
100 36 33 19	PCIE_WAKE_L	=T29_WAKE_L		85 87
96 76	MXM_DP_B_ML_P<0..3>	DP_T29SNK1_ML_C_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_B_ML_N<0..3>	DP_T29SNK1_ML_C_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_B_AUX_P	DP_T29SNK1_AUXCH_C_P	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_B_AUX_N	DP_T29SNK1_AUXCH_C_N	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
76	MXM_DP_B_HPD	DP_T29SNK1_HPD	MAKE_BASE=TRUE NO_TEST=TRUE	89
96 76	MXM_DP_D_ML_P<0..3>	DP_T29SNK0_ML_C_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_D_ML_N<0..3>	DP_T29SNK0_ML_C_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_D_AUX_P	DP_T29SNK0_AUXCH_C_P	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
96 76	MXM_DP_D_AUX_N	DP_T29SNK0_AUXCH_C_N	MAKE_BASE=TRUE NO_TEST=TRUE	89 99
76	MXM_DP_D_HPD	DP_T29SNK0_HPD	MAKE_BASE=TRUE NO_TEST=TRUE	89

UNUSED MXM CONTROL SIGNALS

77	MXM_PNL_BL_EN	NC_MXM_PNL_BL_EN	MAKE_BASE=TRUE NO_TEST=TRUE
77	MXM_PNL_PWR_EN	NC_MXM_PNL_PWR_EN	MAKE_BASE=TRUE NO_TEST=TRUE

Unused MXM Interfaces

96 77	MXM_LVDS_A_DATA_N<3>	NC_MXM_LVDS_A_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_A_DATA_P<3>	NC_MXM_LVDS_A_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_CLK_N	NC_MXM_LVDS_B_CLK_N	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_CLK_P	NC_MXM_LVDS_B_CLK_P	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_N<0>	NC_MXM_LVDS_B_DATA_N<0>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_P<0>	NC_MXM_LVDS_B_DATA_P<0>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_N<1>	NC_MXM_LVDS_B_DATA_N<1>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_P<1>	NC_MXM_LVDS_B_DATA_P<1>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_N<2>	NC_MXM_LVDS_B_DATA_N<2>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_P<2>	NC_MXM_LVDS_B_DATA_P<2>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_N<3>	NC_MXM_LVDS_B_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
96 77	MXM_LVDS_B_DATA_P<3>	NC_MXM_LVDS_B_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE
DP ALIAS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

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8 7 6 5 4 3 2 1

GreenCLK Implementation Notes:

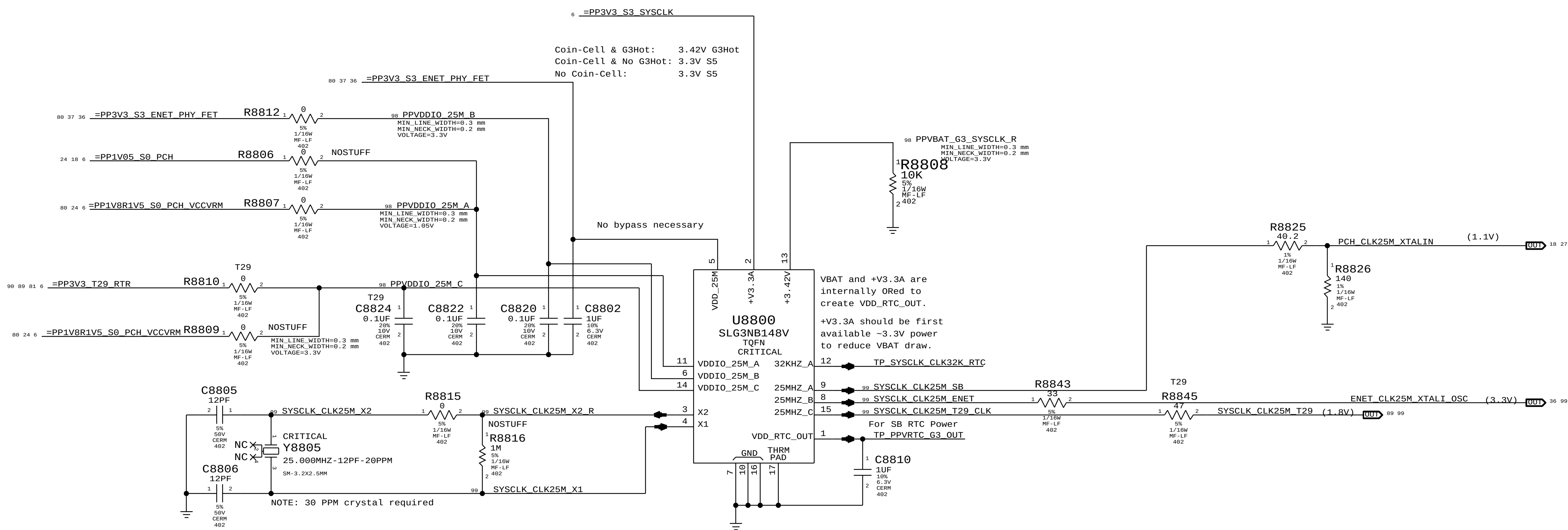
VBAT: Alias as appropriate (see note below & Desktop Example)
+V3.3A: Alias as appropriate (see note below)
VDD_25M: 3.3V matching 'highest' VDDIO power state (ENET)

VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.

For Cougar Point Desktop: VDDIO = VCCVRM (1.8V), Vclk = 1.1V Max, Divider: 604 / 1000
For Cougar Point Mobile: VDDIO = VCCVRM (1.5V), Vclk = 1.1V Max, Divider: 332 / 1000
For Caesar-IV (BCM57765): VDDIO = XTALVDDH (3.3V), Vclk = 3.3V Max. No Divider Necessary

System RTC Power Source & 32kHz / 25MHz Clock Generator



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: GREEN CLOCK

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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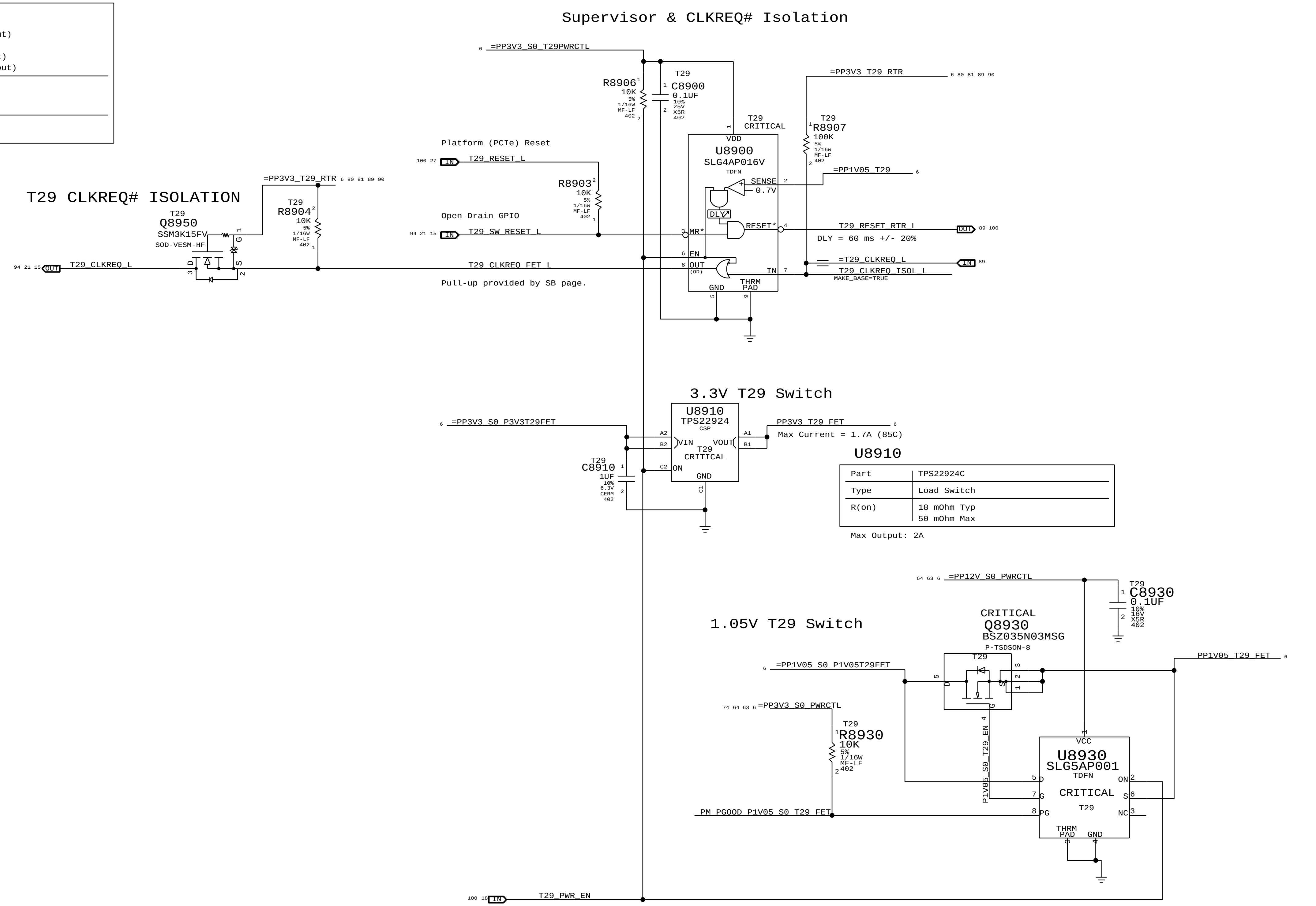
Page Notes

Power aliases required by this page:
- =PP3V3_T29_P3V3T29FET (3.3V FET Input)
- =PP3V3_T29_FET (3.3V FET Output)
- =PP3V3_S0_T29PWRCTL
- =PP1V05_T29_P1V05T29FET (1.05V FET Input)
- =PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:
- =T29_CLKREQ_L
- T29_RESET_L

BOM options provided by this page:
(NONE)



U8910

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

SYNC_MASTER=K62_AARON SYNC_DATE=(MASTER)

PAGE TITLE: T29 POWER

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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Page Notes

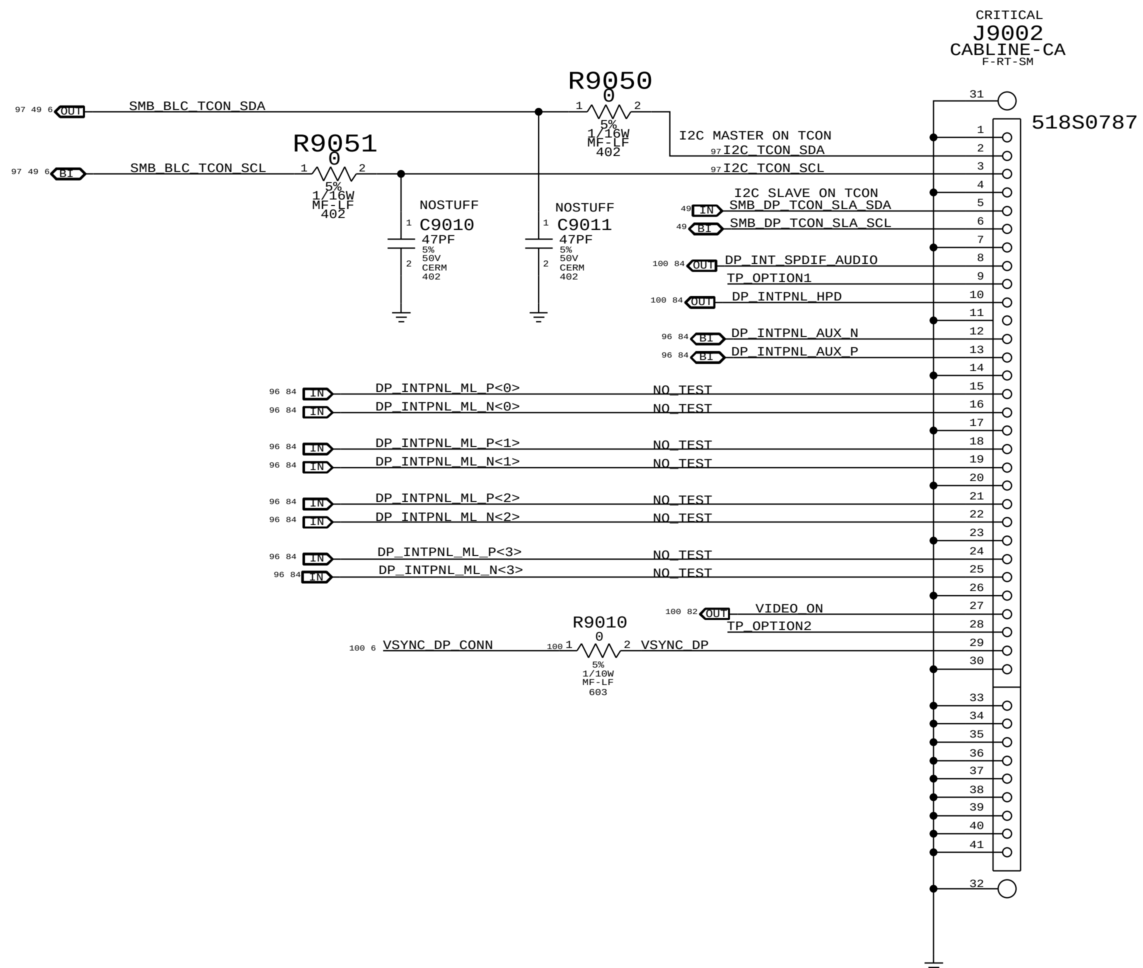
Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

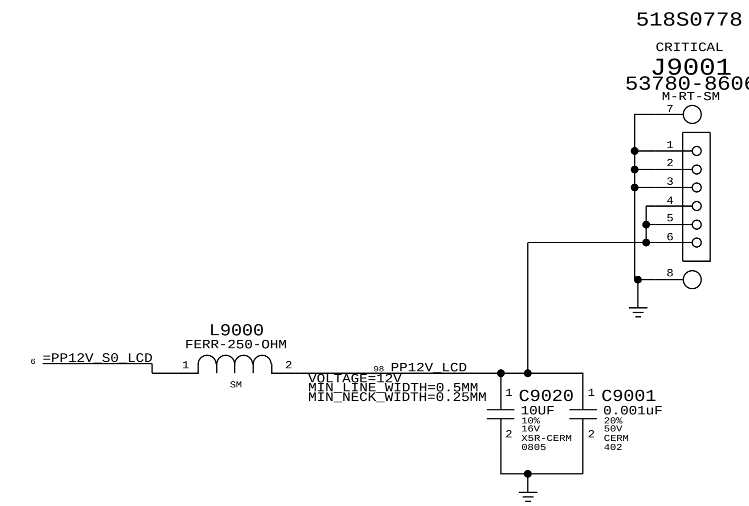
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE

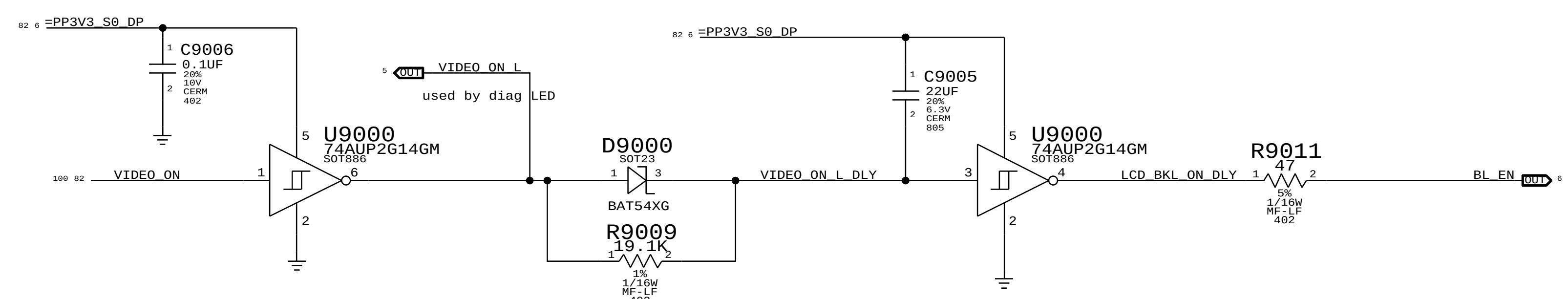


INTERNAL DP POWER



BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE
Display: Int DP Connector

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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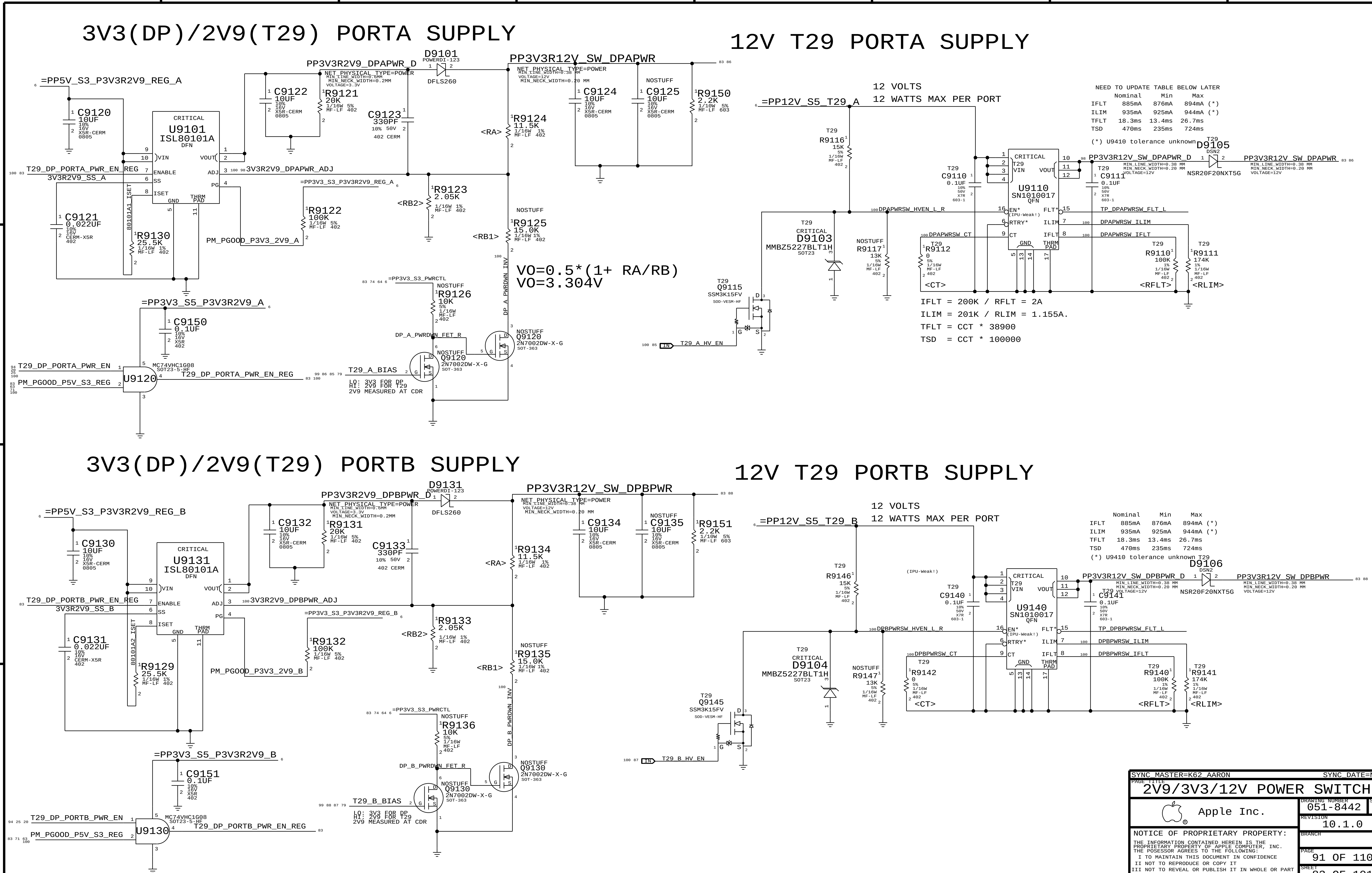

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE: 2V9/3V3/12V POWER SWITCH

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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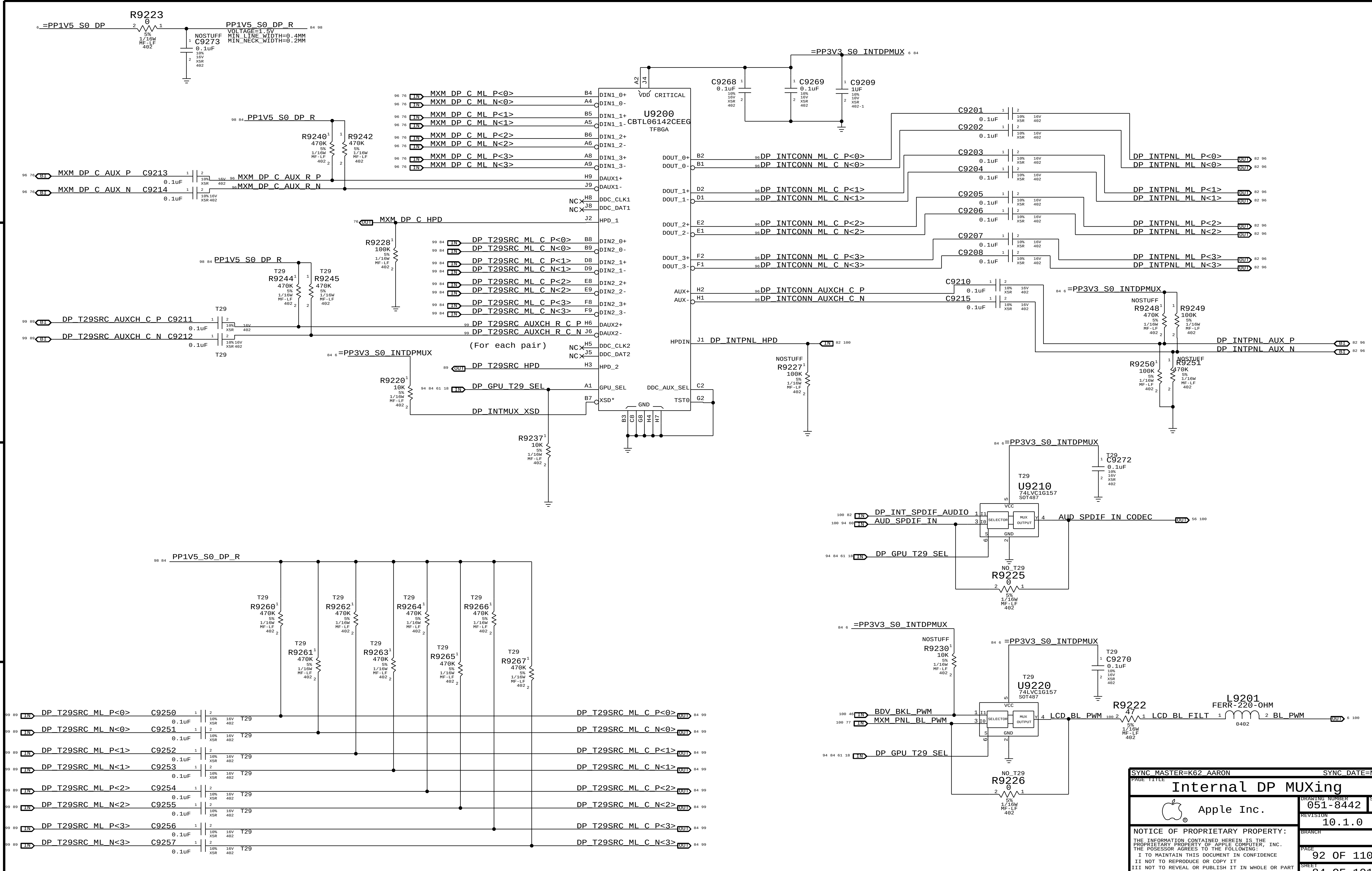
PAGE: 91 OF 110

SHEET: 83 OF 101



SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

Internal DP MUXing

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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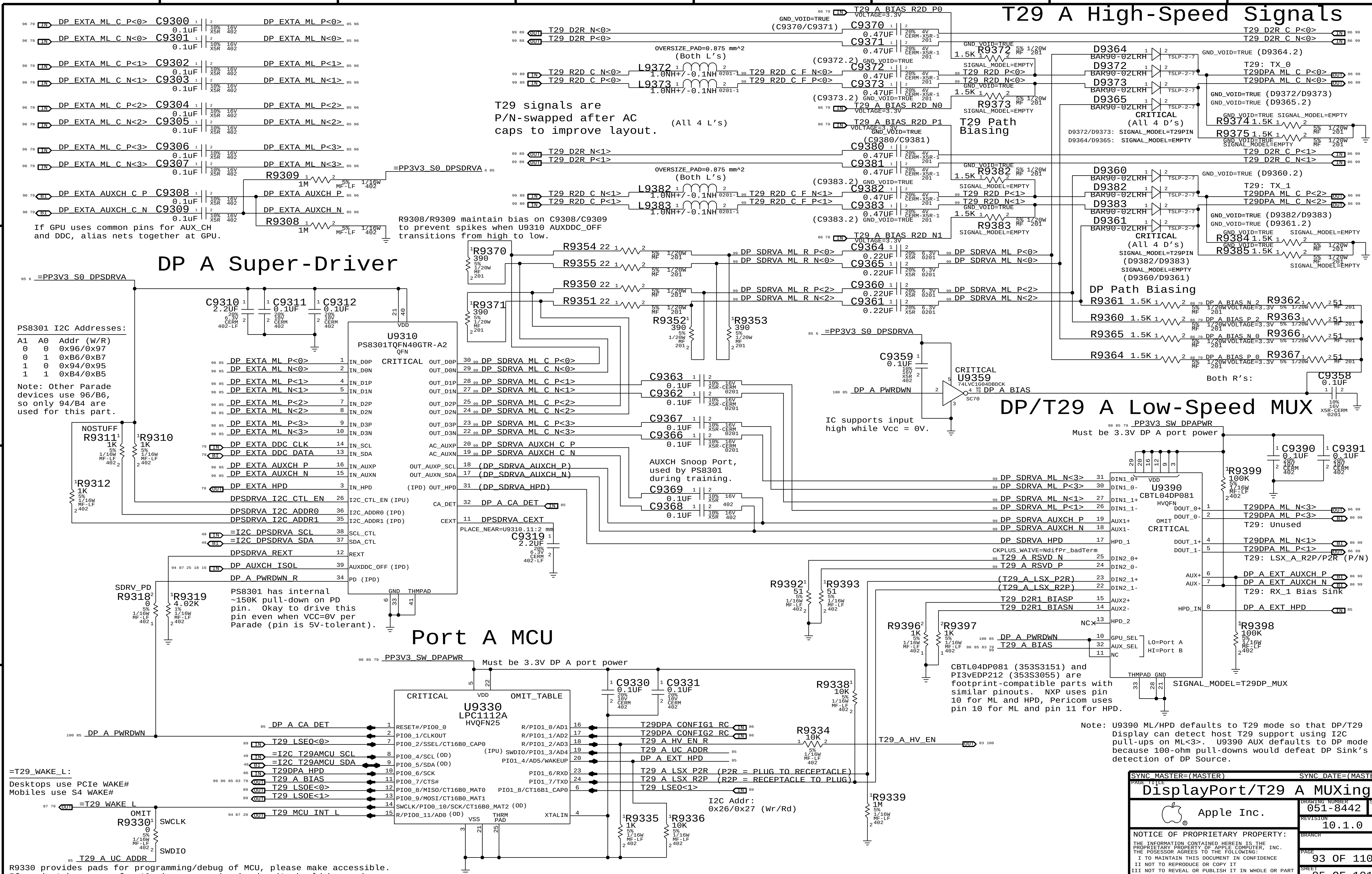
T29 A High-Speed Signals



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE: DisplayPort/T29 A MUXing

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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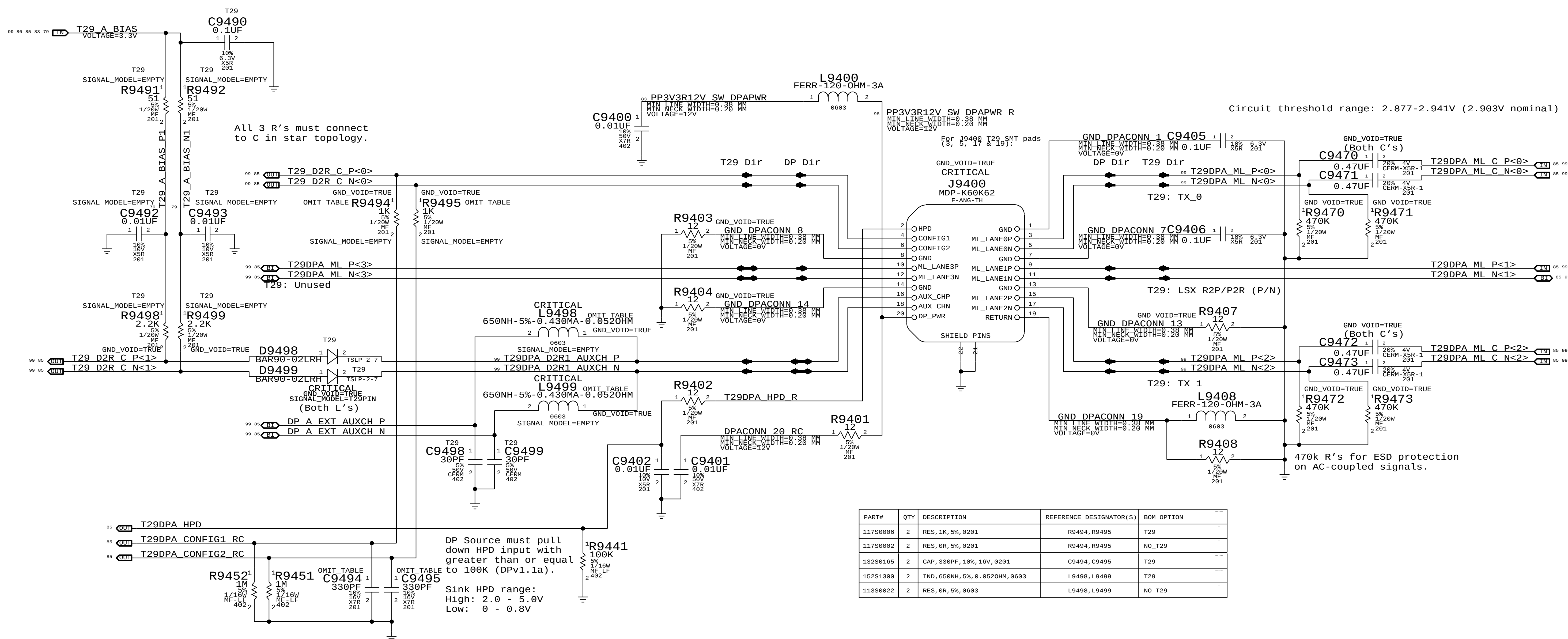
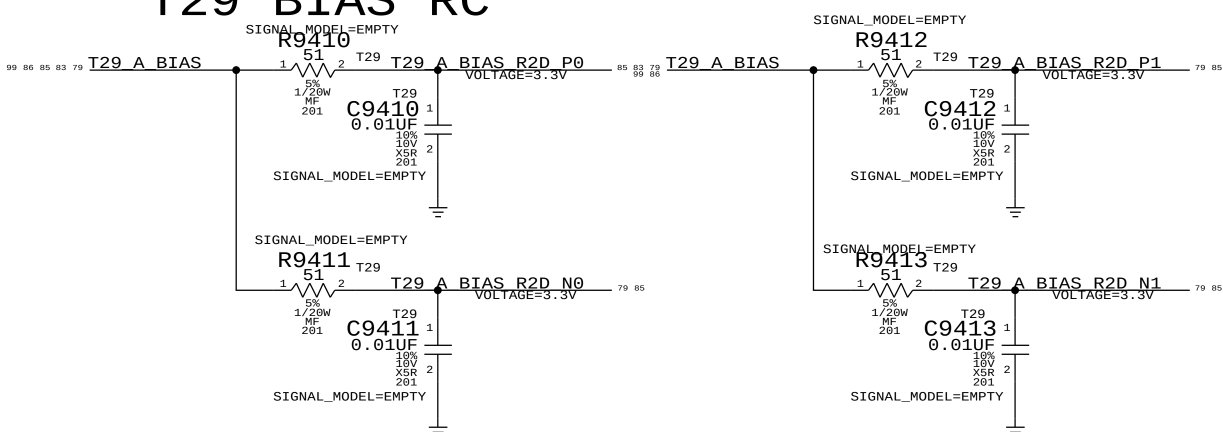
DisplayPort/T29 A Connector

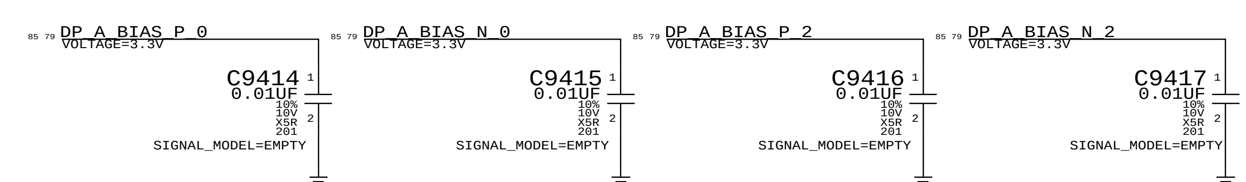


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0006	2	RES,1K,5%,0201	R9494,R9495	T29
117S0002	2	RES,0R,5%,0201	R9494,R9495	NO_T29
132S0165	2	CAP,330PF,10%,16V,0201	C9494,C9495	T29
152S1300	2	IND,650NH,5%,0.052OHM,0603	L9498,L9499	T29
113S0022	2	RES,0R,5%,0603	L9498,L9499	NO_T29

T29 BIAS RC



DP BIAS CAPS



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE
DisplayPort/T29 A Connector

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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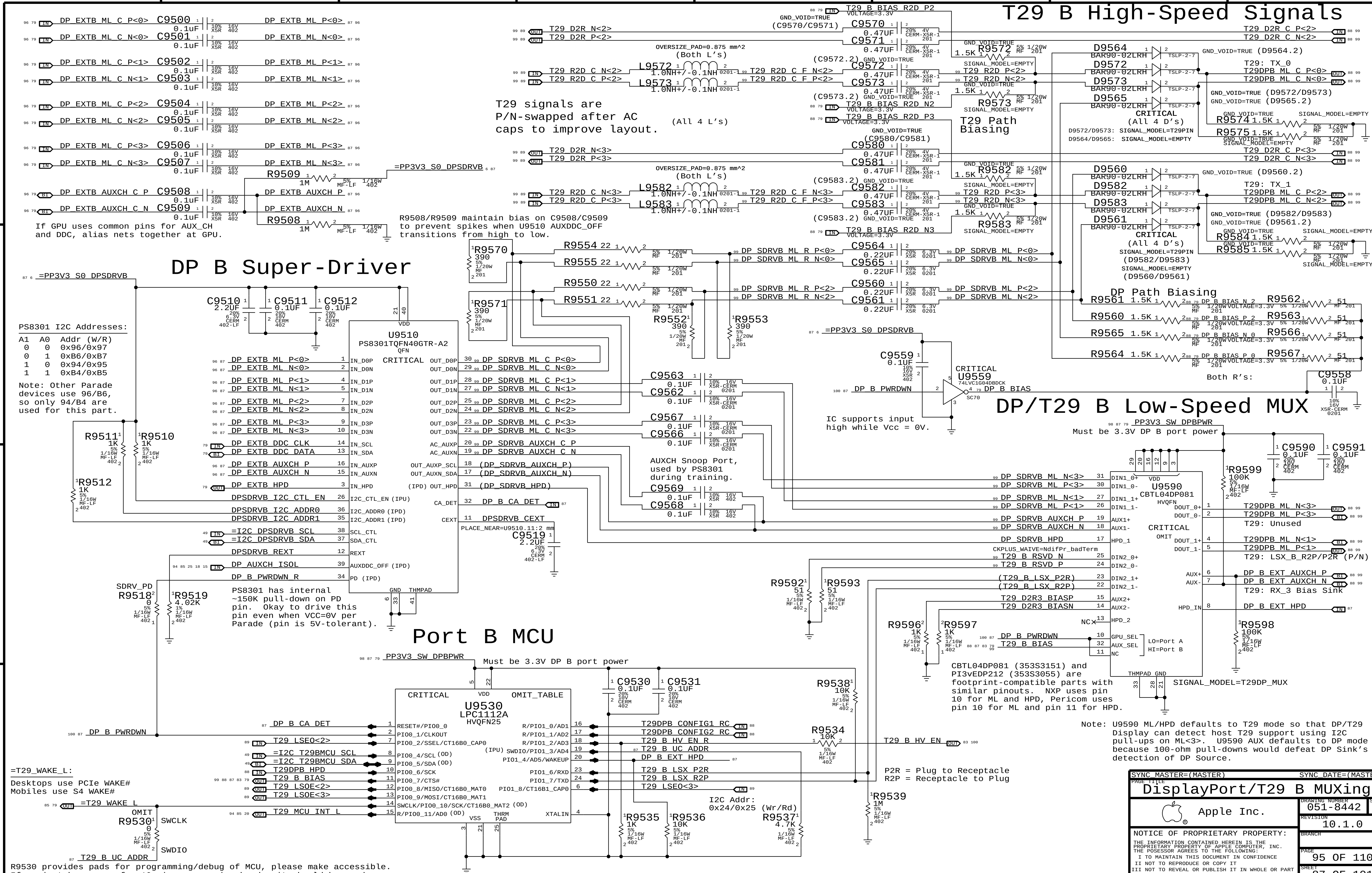

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE: DisplayPort/T29 B MUXing

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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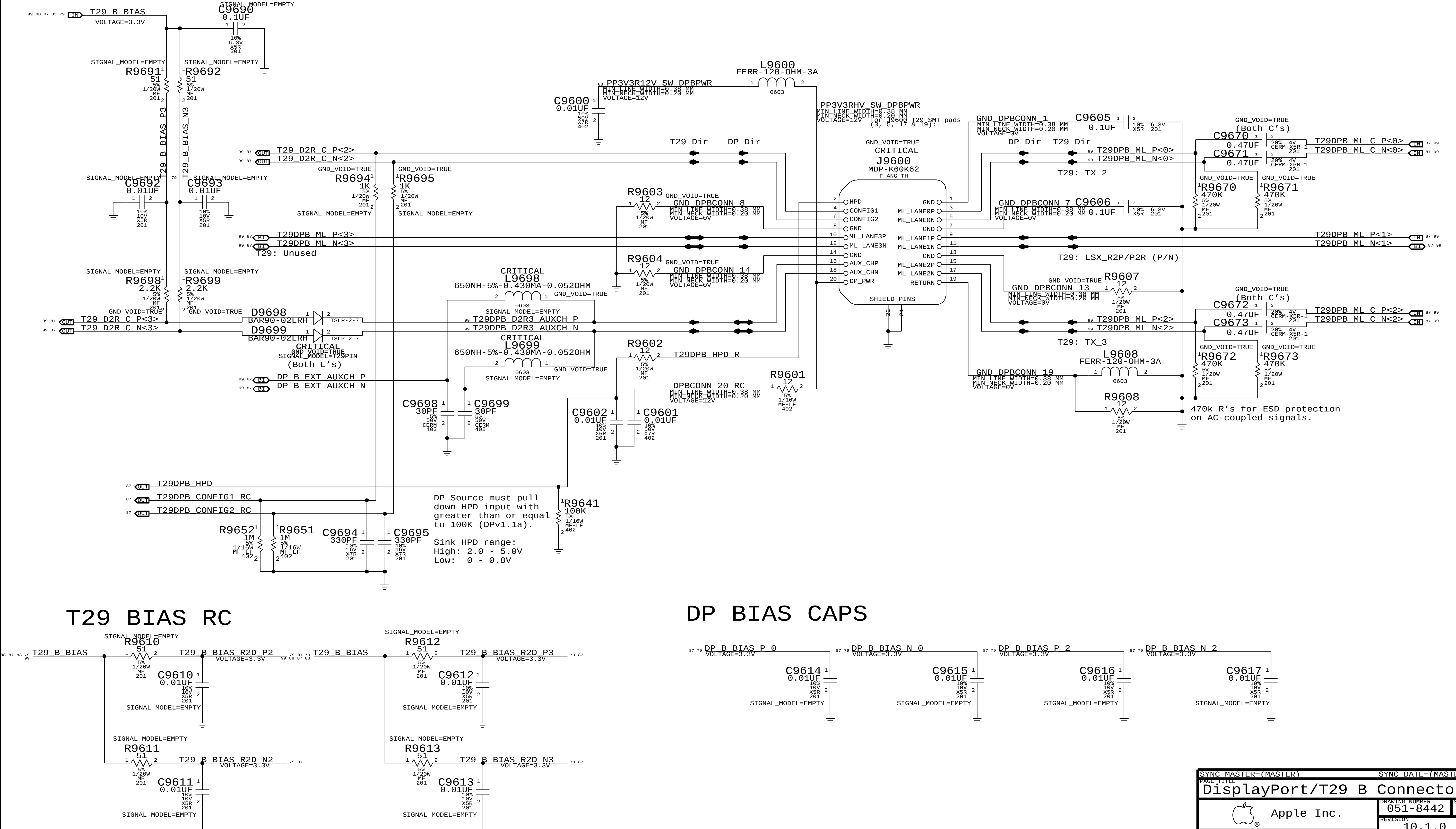
PAGE: 95 OF 110

SHEET: 87 OF 101

DisplayPort/T29 B Connector



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE: DisplayPort/T29 B Connector

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

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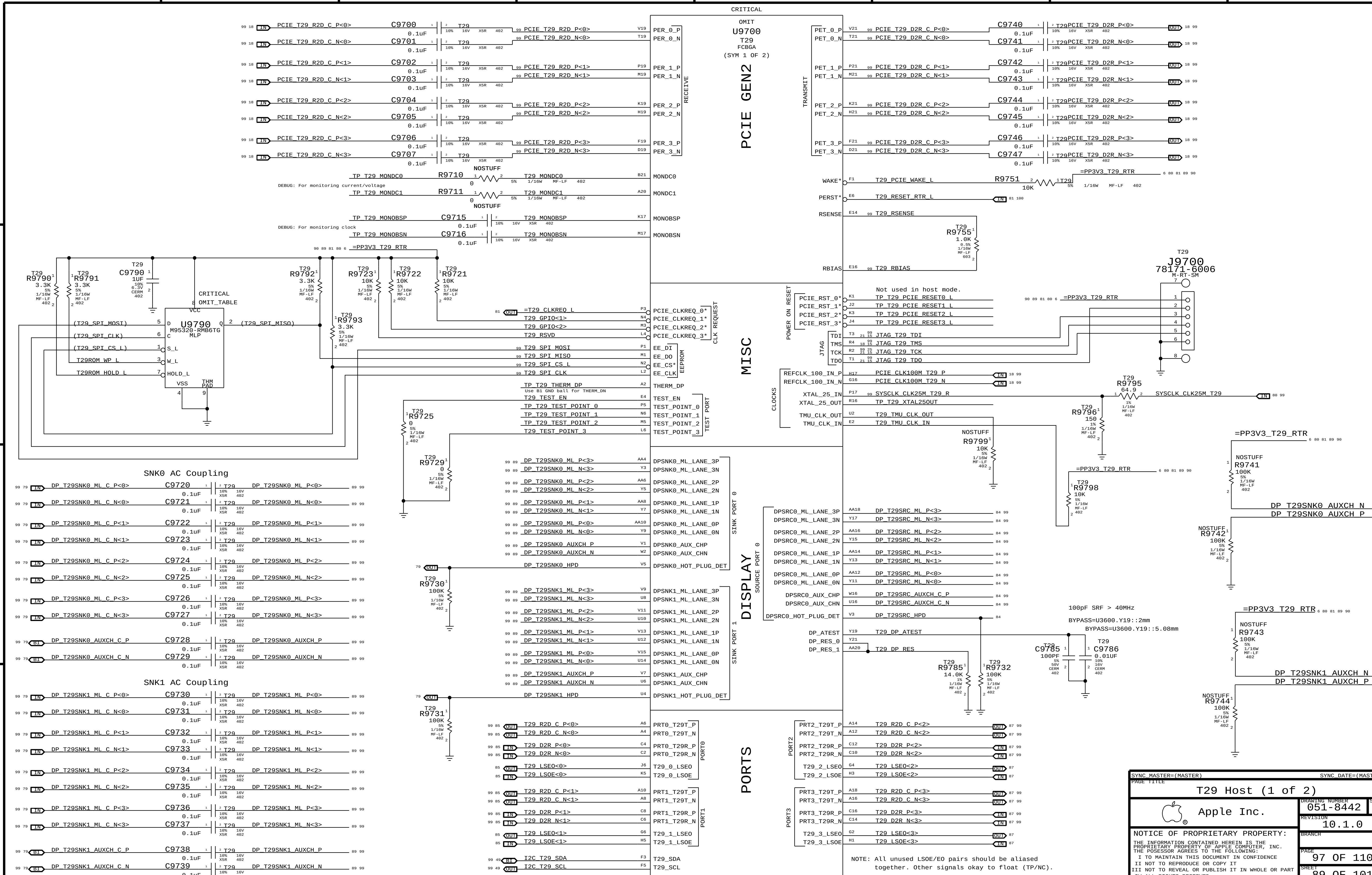
PAGE: 96 OF 110

SHEET: 88 OF 101



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE

T29 Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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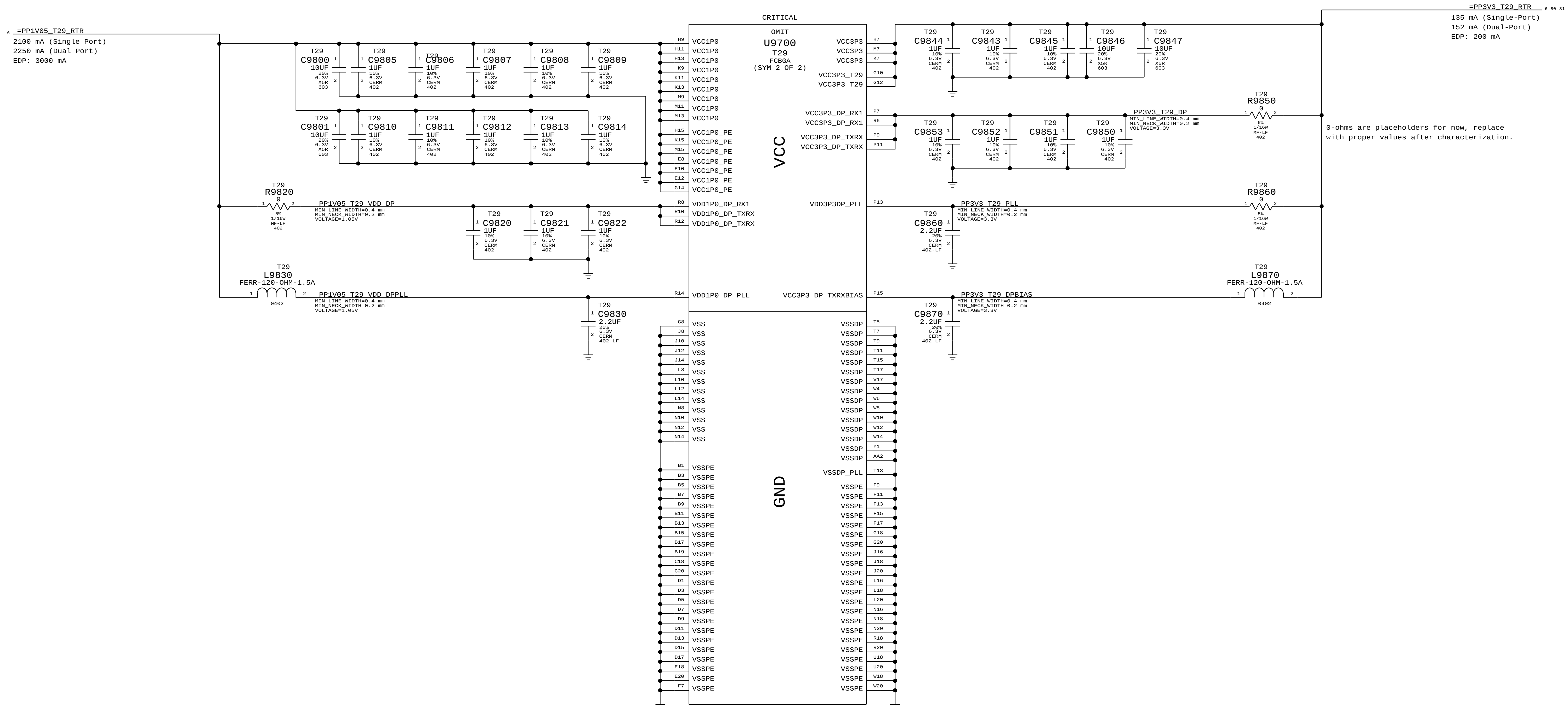
PAGE 97 OF 110

SHEET 89 OF 101



Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC_MASTER=(MASTER)	SYNC_DATE=(MASTER)
PAGE TITLE: T29 Host (2 of 2)	
Apple Inc.	DRAWING NUMBER: 051-8442 SIZE: D
	REVISION: 10.1.0
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	SHEET: 90 OF 101

K60/62 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA,BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	TOP,BOTTOM	Y	0.21 MM	0.090 MM	=STANDARD		
34_OHM_SE	*	Y	0.19 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.090 MM	=STANDARD		
39_OHM_SE	*	Y	0.159 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP,BOTTOM	Y	0.151 MM	0.090 MM	=STANDARD		
42_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
68_OHM_DIFF	ISL3,ISL6	Y	0.16 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL6	Y	0.099 MM	0.085 MM	12 MM	0.2 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL6	Y	0.081 MM	0.081 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3,ISL6	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.075 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
3.5:1_SPACING	*	0.35 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.155 MM	?
2X_DIELECTRIC	TOP,BOTTOM	0.145 MM	?
3X_DIELECTRIC	*	0.230 MM	?
3X_DIELECTRIC	TOP,BOTTOM	0.215 MM	?
4X_DIELECTRIC	*	0.305 MM	?
4X_DIELECTRIC	TOP,BOTTOM	0.285 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP,BOTTOM	0.355 MM	?
7X_DIELECTRIC	*	0.532 MM	?
7X_DIELECTRIC	TOP,BOTTOM	0.497 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P1P5MM	*	0.15 MM	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P1P5MM
CLK_PCIE	*	BGA	BGA_P1MM
CLK_LPC	*	BGA	BGA_P1MM
CLK_PCI	*	BGA	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CONTROL	*	*	SWITCHNODE
VR_CONTROL	VR_CONTROL	*	STANDARD
VR_CONTROL	SWITCHNODE	*	STANDARD
VR_CONTROL	GND	*	STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

BOARD STACK-UP

TOP	HALF OZ	SIGNAL
	0.071	PREPREG
2	TWO OZ	GND
	0.076	PREPREG
3	ONE OZ	SIGNAL
	0.370	PREPREG
4	TWO OZ	POWER
	0.101	CORE
5	TWO OZ	POWER
	0.370	PREPREG
6	ONE OZ	SIGNAL
	0.076	PREPREG
7	TWO OZ	GND
	0.071	PREPREG
BOTTOM	HALF OZ	SIGNAL

BOARD THICKNESS = 62 MIL (1.5748 mm)

SYNC_MASTER=K62_AARON SYNC_DATE=06/08/2010

PAGE TITLE
K60/K62 RULE DEFINITIONS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

MEMORY BUS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
MEM_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
MEM_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=5:1_SPACING	?
MEM_CTRL2CTRL	*	=2.5:1_SPACING	?
MEM_CTRL2MEM	*	=3.5:1_SPACING	?
MEM_CMD2CMD	*	=2:1_SPACING	?
MEM_CMD2MEM	*	=3.5:1_SPACING	?
MEM_DQ_SAMEBYTE	*	=3:1_SPACING	?
MEM_DQ_DIFFBYTE	*	=5:1_SPACING	?
MEM_DATA2MEM	*	=4:1_SPACING	?
MEM_DQS2MEM	*	=4:1_SPACING	?
MEM_2OTHER	*	=5:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE0	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE1	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE2	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE3	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE4	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE5	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE6	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE7	*	MEM_DQS2MEM
MEM_DQS	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE5	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_DQ_BYTE5	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE6	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_DQ_BYTE6	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE6	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE6	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE7	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_DQ_BYTE7	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE7	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE0	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_DQ_BYTE0	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE1	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE1	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_DQ_BYTE1	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE2	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_DQ_BYTE2	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE3	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_DQ_BYTE3	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE4	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_DQ_BYTE4	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	*	*	MEM_2OTHER

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_68D	MEM_CLK	MEM_A_CLK_P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM_A_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_A_CKE<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_CS_L<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_ODT<3..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_A<15..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_BA<2..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_RAS_L	12 30
	MEM_34S	MEM_CMD	MEM_A_CAS_L	12 30
	MEM_34S	MEM_CMD	MEM_A_WE_L	12 30
	MEM_42S	MEM_DQ_BYTE0	MEM_A_DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM_A_DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM_A_DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM_A_DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM_A_DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM_A_DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM_A_DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM_A_DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<7>	12 32
I177	MEM_50S	PM	MEM_RESET_L	30 31 32 100

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_68D	MEM_CLK	MEM_B_CLK_P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM_B_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_B_CKE<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_CS_L<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_ODT<3..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_A<15..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_BA<2..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_RAS_L	12 31
	MEM_34S	MEM_CMD	MEM_B_CAS_L	12 31
	MEM_34S	MEM_CMD	MEM_B_WE_L	12 31
	MEM_42S	MEM_DQ_BYTE0	MEM_B_DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM_B_DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM_B_DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM_B_DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM_B_DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM_B_DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM_B_DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM_B_DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<7>	12 32

MEMORY MISC PROPERTIES

VOLTAGE	NET_TYPE PHYSICAL	NET_TYPE SPACING	VOLTAGE	NET	PAGES
	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_A	
	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_B	
I162	MEM_POWER_PHY	MEM_POWER		CPU_DDR_VREF	11 28
I168	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_DACOUT	28
I169	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_OPFB	28
I170	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_DQ	28
I174	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_A_SW	
I171	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_DACOUT	28
I173	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_OPFB	28
I172	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_DQ	28
I175	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_B_SW	

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_POWER_PHY	*	MEM_POWER_WIDTH

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_POWER	*	=3:1_SPACING	?

SYNC_MASTER=K62_ROSITA SYNC_DATE=01/09/2011

PAGE TITLE

Memory Constraints

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=4X_DIELECTRIC	?
CLK_PCIE	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4:1_SPACING	?

CPU

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_RCOMP_PHY	*	Y	0.254 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ITP	*	0.2 MM	?
CPU_RCOMP	*	0.2 MM	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=6:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=6:1_SPACING	?

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
SATA				
I300	SATA_90D	SATA	SATA_SSD_R2D_C_P	18 42
I299	SATA_90D	SATA	SATA_SSD_R2D_C_N	18 42
I301	SATA_90D	SATA	SATA_SSD_R2D_P	42
I302	SATA_90D	SATA	SATA_SSD_R2D_N	42
I303	SATA_90D	SATA	SATA_SSD_D2R_P	18 42
I304	SATA_90D	SATA	SATA_SSD_D2R_N	18 42
I305	SATA_90D	SATA	SATA_SSD_D2R_C_P	42
I306	SATA_90D	SATA	SATA_SSD_D2R_C_N	42
PCIE				
I319	PCIE_85D	PCIE	PCIE_USB3_1_R2D_P	
I320	PCIE_85D	PCIE	PCIE_USB3_1_R2D_N	
I321	PCIE_85D	PCIE	PCIE_USB3_1_R2D_C_P	
I322	PCIE_85D	PCIE	PCIE_USB3_1_R2D_C_N	
I323	PCIE_85D	PCIE	PCIE_USB3_1_D2R_P	
I324	PCIE_85D	PCIE	PCIE_USB3_1_D2R_N	
I325	PCIE_85D	PCIE	PCIE_USB3_1_D2R_C_P	
I326	PCIE_85D	PCIE	PCIE_USB3_1_D2R_C_N	
I327	PCIE_85D	PCIE	PCIE_USB3_2_R2D_P	
I328	PCIE_85D	PCIE	PCIE_USB3_2_R2D_N	
I330	PCIE_85D	PCIE	PCIE_USB3_2_R2D_C_P	
I329	PCIE_85D	PCIE	PCIE_USB3_2_R2D_C_N	
I332	PCIE_85D	PCIE	PCIE_USB3_2_D2R_P	
I331	PCIE_85D	PCIE	PCIE_USB3_2_D2R_N	
I333	PCIE_85D	PCIE	PCIE_USB3_2_D2R_C_P	
I334	PCIE_85D	PCIE	PCIE_USB3_2_D2R_C_N	
CPU ITP				
	CPU_50S	CPU_ITP	XDP_BPM_L<7..0>	11 25
	CPU_50S	CPU_ITP	CPU_CFG<17..0>	10 15 25
I344	CPU_50S	CPU_ITP	XDP_OBSDATA_B<3..0>	25
I343	CPU_50S	CPU_ITP	XDP_CPU_CFG<0>	25
I345	CPU_50S	CPU_ITP	XDP_CPU_TDO	11 25
I347	CPU_50S	CPU_ITP	XDP_CPU_TDI	11 25
I348	CPU_50S	CPU_ITP	XDP_CPU_TMS	11 25
I349	CPU_50S	CPU_ITP	XDP_CPU_TCK	11 25
I350	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	11 25
CPU_MISC				
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_PEG_COMP	10

ANY OTHER LYNNFIELD CONSTRAINTS NOT COVERED ON PAGES 101 AND 107 SHOULD GO ON THIS PAGE TOO

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
PCIE GRAPHICS				
	PCIE_85D	PCIE	PEG_R2D_C_P<15..0>	9 78
	PCIE_85D	PCIE	PEG_R2D_C_N<15..0>	9 78
	PCIE_85D	PCIE	PEG_D2R_P<15..0>	9 78
	PCIE_85D	PCIE	PEG_D2R_N<15..0>	9 78
	PCIE_85D	PCIE	MXM_PCIE_R2D_P<7..0>	76 78
	PCIE_85D	PCIE	MXM_PCIE_R2D_N<7..0>	76 78
	PCIE_85D	PCIE	MXM_PCIE_D2R_P<7..0>	76 78
	PCIE_85D	PCIE	MXM_PCIE_D2R_N<7..0>	76 78
PCIE I/O				
	PCIE_85D	PCIE	PCIE_MINI_R2D_P	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_N	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_N	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_N	18 33
	PCIE_85D	PCIE	PCIE_FW_R2D_P	39
	PCIE_85D	PCIE	PCIE_FW_R2D_N	39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	18 39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_P	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_N	39
DMI				
I315	CLK_PCIE_90D	CLK_PCIE	DMI_MIDBUS_CLK100M_N	
I318	CLK_PCIE_90D	CLK_PCIE	DMI_MIDBUS_CLK100M_P	
I316	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_N	11 18
I317	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_P	11 18
	PCIE_85D	PCIE	DMI_S2N_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_S2N_N<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_N<3..0>	10 19
PCIE REF CLOCKS				
	CLK_PCIE_90D	CLK_PCIE	GPU_CLK100M_PCIE_P	9
	CLK_PCIE_90D	CLK_PCIE	GPU_CLK100M_PCIE_N	9
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_MINI_P	18 33
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_MINI_N	18 33
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P	18 39
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N	18 39
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_P	18 37
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_N	18 37
SATA				
	SATA_90D	SATA	SATA_HDD_R2D_C_P	18 42
	SATA_90D	SATA	SATA_HDD_R2D_C_N	18 42
	SATA_90D	SATA	SATA_HDD_R2D_P	42
	SATA_90D	SATA	SATA_HDD_R2D_N	42
	SATA_90D	SATA	SATA_HDD_D2R_P	18 42
	SATA_90D	SATA	SATA_HDD_D2R_N	18 42
	SATA_90D	SATA	SATA_HDD_D2R_C_P	42
	SATA_90D	SATA	SATA_HDD_D2R_C_N	42
	SATA_90D	SATA	SATA_ODD_R2D_C_P	18 42
	SATA_90D	SATA	SATA_ODD_R2D_C_N	18 42
	SATA_90D	SATA	SATA_ODD_R2D_P	42
	SATA_90D	SATA	SATA_ODD_R2D_N	42
	SATA_90D	SATA	SATA_ODD_D2R_P	18 42
	SATA_90D	SATA	SATA_ODD_D2R_N	18 42
	SATA_90D	SATA	SATA_ODD_D2R_C_P	42
	SATA_90D	SATA	SATA_ODD_D2R_C_N	42
CLOCKS				
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_DMI_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_DMI_N	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_N	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_N	18 26
	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_N	18 25
	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_P	18 25
	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_N	11 18
	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_P	11 18
I337	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_P	25
I338	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_N	25
UNUSED CLOCKS				
I335	CLK_PCIE_90D	CLK_PCIE	TP_CLK133M_PCH_N	26
I336	CLK_PCIE_90D	CLK_PCIE	TP_CLK133M_PCH_P	26
UNUSED PCIE				
I339	PCIE_85D	PCIE	MXM_PCIE_R2D_P<8..15>	76 78
I340	PCIE_85D	PCIE	MXM_PCIE_R2D_N<8..15>	76 78
I342	PCIE_85D	PCIE	MXM_PCIE_D2R_P<8..15>	76 78
I341	PCIE_85D	PCIE	MXM_PCIE_D2R_N<8..15>	76 78

PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	0.2 MM	?
ITP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

Tag	PHYSICAL	NET_TYPE SPACING	Net	Pages
I329		PM	T29_CLKREQ_L	15 21 81
I330		PM	FW_MINI_CLKREQ_L	15 18
I331		PM	BLC_GPIO	6 15 21
I332		PM	T29_SW_RESET_L	15 21 81
I334		PM	ENET_CLKREQ_L	15 18 36
I337		PM	DP_GPU_T29_SEL	18 61 84
I338		PM	T29_MCU_INT_L	20 85 87
I339		PM	T29_DP_PORTA_PWR_EN	20 25 83 100
I340		PM	T29_DP_PORTB_PWR_EN	20 25 83
I341		PM	DP_AUXCH_ISOL	15 18 25 85 87
I348		PM	PLT_RST_BUF_L	27
I349		PM	XDPCPU_PLTRST_L	25 27
I352		PM	PCH_PEG_CLKREQ_L	21
I357		PM	ENET_SW_RESET_L	15 21 36
I354		PM	CPU_SKTOCC	63
I355		PM	PM_EN_USB_PWR	43 63

Tag	PHYSICAL	NET_TYPE SPACING	Net	Pages
I372		PM	ENET_RESET_LOGIC_L	36
I373		PM	ENET_RESET_FET_L	
I375		PM	ENET_CLKREQ_FET_L	36 37
I374		PM	PGOOD_5V_1V05_3V3	64 100
I376		PM	PGOOD_CPU_UNCORE	64 100
I377		PM	ALL_SYS_PWRGD	64 100
I378		PM	PGOOD_3V3_1V05	64 100
I379		PM	PGOOD_PCH_S0_R	64 100
I380		PM	AUD_IPHS_SWITCH_EN_PCH	21 25

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE SPACING	Net	Pages
	PCI_55S	PCI	PCI_REQ0_L	20
	PCI_55S	PCI	PCI_REQ1_L	20
I381	PCI_55S	PCI	PCI_REQ2_L	20
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	20 27
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	18 27
	LPC_55S	LPC	LPC_AD<3..0>	18 46 48
	LPC_55S	LPC	LPC_FRAME_L	18 46 48
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	20 27
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	27 46
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	27 48
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R	9 19 100
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK	9 46 100
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 27
I325	LPC_55S	LPC	LPC_R_AD<3..0>	18
I326	LPC_55S	LPC	LPC_FRAME_R_L	18
I335	SPI_55S	SPI	SPI_CLK_1_R	18
I336	SPI_55S	SPI	SPI_MOSI_1_R	18
I350	CLK_XTAL	XTAL	USB_HUB2_XTAL1	35
I351	CLK_XTAL	XTAL	USB_HUB2_XTAL2	35
I344	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1_R	27
I345	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	27
I342	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 94
I343	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 94
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 94
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 94
	CLK_XTAL	XTAL	CK505_XTAL_IN	26
	CLK_XTAL	XTAL	CK505_XTAL_OUT	26
I324	CLK_XTAL	XTAL	CK505_XTAL_OUT_R	26
	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK	18 26

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE SPACING	Net	Pages
	SPI_55S	SPI	SPI_CLK_R	18 48 55
	SPI_55S	SPI	SPI_CLK	55
	SPI_55S	SPI	SPI_MOSI_R	18 48 55
	SPI_55S	SPI	SPI_MOSI	55
	SPI_55S	SPI	SPI_MISO	18 48 55
	SPI_55S	SPI	SPI_MISO_R	55
	SPI_55S	SPI	SPI_CS0_R_L	18 48
	SPI_55S	SPI	SPI_CS0_L	48
	SPI_55S	SPI	SPI_MLB_CS_L	48 55
	SPI_55S	SPI	SPI_ALT_CS_L	48
	SPI_55S	SPI	SPIROM_USE_MLB	21 48
	SPI_55S	SPI	SPI_ALT_MOSI	48
	SPI_55S	SPI	SPI_ALT_MISO	48
	SPI_55S	SPI	SPI_ALT_CLK	48
	HDA_55S	HDA	HDA_BIT_CLK	18 56
	HDA_55S	HDA	HDA_BIT_CLK_R	18
	HDA_55S	HDA	HDA_RST_L	18 56
	HDA_55S	HDA	HDA_RST_R_L	18
	HDA_55S	HDA	HDA_SDOUT	15 18 56
	HDA_55S	HDA	HDA_SDOUT_R	18
	HDA_55S	HDA	HDA_SYNC	18 56
	HDA_55S	HDA	HDA_SYNC_R	18
	HDA_55S	HDA	HDA_SDIN0	18 56
	HDA_55S	HDA	AUD_SDI_R	56
		PM	AUD_SPDIF_IN	60 84 100
		HDA	AUD_SPDIF_OUT	56 60
		HDA	AUD_SPDIF_CHIP	56
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_NOUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_POUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_NOUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_POUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_NOUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_POUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_NOUT	101
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_POUT	101
I346	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT_R	27
I347	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN_R	27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT	18 27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN	18 27 80
	PCH_55S	COMP_PCH	PCH_USB_RBIAS	20
	PCH_55S	COMP_PCH	PCH_SATA3COMP	18
	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP	18
	PCH_55S	COMP_PCH	PCH_DMI_COMP	19
I322	PCH_55S	COMP_PCH	PCH_SATAICOMP	18
	CLK_XTAL	XTAL	USB_HUB1_XTAL1	34
	CLK_XTAL	XTAL	USB_HUB1_XTAL2	34
	PCH_55S	COMP_PCH	USB_HUB1_RBIAS	34
	PCH_55S	ITP_PCH	XDP_PCH_TCK	18 25
	PCH_55S	ITP_PCH	XDP_PCH_TMS	18 25
	PCH_55S	ITP_PCH	XDP_PCH_TDI	18 25
I323	PCH_55S	ITP_PCH	XDP_PCH_TDO	18 25
I327	PCH_55S	COMP_PCH	PCH_DMI2RBIAS	19
I328	PCH_55S	COMP_PCH	PCH_SATA3RBIAS	18
I356	PCH_55S	COMP_PCH	USB_HUB2_RBIAS	35

SYNC_MASTER=K62_SIJI SYNC_DATE=01/09/2011

PAGE TITLE
IBEX PEAK CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MII	*	0.3 MM	?
ENET_SE	*	=STANDARD	?

SOURCE:BROADCOM 5764M-DS04-RDS. PAGE 38

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF	*	0.6 MM	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_2OTHER	*	=50MIL_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
50MIL_SPACING	*	1.27 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_DIFF_T	*	*	ENET_2OTHER

CAESAR IV (SD) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD	*	=3:1_SPACING	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=3:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

ELECTRICAL_CONSTRAINT_SET	PHYSICAL (NET_TYPE)	SPACING (NET_TYPE)	NET	PAGES
	ENET_50S	ENET_SE	ENET_RDAC	37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALI	36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALO	36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALO_R	36
	ENET_100D	ENET_DIFF	ENETCONN_MDI_P<3..0>	37 38
	ENET_100D	ENET_DIFF	ENETCONN_MDI_N<3..0>	37 38
	ENET_100D	ENET_DIFF_T	ENETCONN_MDI_T_P<3..0>	38
	ENET_100D	ENET_DIFF_T	ENETCONN_MDI_T_N<3..0>	38
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_P	37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_N	37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_P	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_N	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_C_P	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_C_N	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_C_P	37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_C_N	37
	SD_50S	SD	ENET_SD_CMD	37
	SD_50S	SD	SDCONN_CMD	37 45
	SD_50S	SD	SDCONN_CLK	37 45
	SD_50S	SD	ENET_SD_CLK	37
	SD_50S	SD	SDCONN_DATA<7..0>	37 45
	SD_50S	SD	ENET_CR_DATA<7..0>	37
	SD_50S	PM	ENET_MEDIA_SENSE	15 18 37
	SD_50S	SD	ENET_MEDIA_SENSE_R	
	SD_50S	SD	ENET_SD_DETECT_L	37
	SD_50S	PM	SDCONN_DETECT_BUF_L	100

ELECTRICAL_CONSTRAINT_SET	PHYSICAL (NET_TYPE)	SPACING (NET_TYPE)	NET	PAGES
I252	USB_90D	USB	USB_EXTA_P	34 43
I253	USB_90D	USB	USB_EXTA_N	34 43
I254	USB_90D	USB	USB_PORT0_P	43
I256	USB_90D	USB	USB_PORT0_N	43
I255	USB_90D	USB	USB_EXTB_P	35 43
I257	USB_90D	USB	USB_EXTB_N	35 43
I258	USB_90D	USB	USB_PORT1_P	43
I259	USB_90D	USB	USB_PORT1_N	43
I260	USB_90D	USB	USB_EXTC_P	34 43
I261	USB_90D	USB	USB_EXTC_N	34 43
I262	USB_90D	USB	USB_PORT2_P	43
I264	USB_90D	USB	USB_PORT2_N	43
I263	USB_90D	USB	USB_EXTD_P	35 43
I265	USB_90D	USB	USB_EXTD_N	35 43
I266	USB_90D	USB	USB_D_MUXED_P	43
I267	USB_90D	USB	USB_D_MUXED_N	43
I268	USB_90D	USB	USB_PORT3_P	43
I269	USB_90D	USB	USB_PORT3_N	43
I270	USB_90D	USB	USB_CAMERA_P	20 44
I271	USB_90D	USB	USB_CAMERA_N	20 44
I272	USB_90D	USB	USB_CAMERA_L_P	44 101
I274	USB_90D	USB	USB_CAMERA_L_N	44 101
I273	USB_90D	USB	USB_BT_P	35 44
I275	USB_90D	USB	USB_BT_N	35 44
I276	USB_90D	USB	USB_BT_L_P	44 101
I277	USB_90D	USB	USB_BT_L_N	44 101
I278	USB_90D	USB	USB_IR_P	34 44
I279	USB_90D	USB	USB_IR_N	34 44
I280	USB_90D	USB	USB_IR_L_P	44 101
I281	USB_90D	USB	USB_IR_L_N	44 101
I286	USB_90D	USB	USB_SDCARD_P	34 44
I287	USB_90D	USB	USB_SDCARD_N	34 44
I288	USB_90D	USB	USB_SDCARD_L_P	44
I289	USB_90D	USB	USB_SDCARD_L_N	44
I282	USB_90D	USB	USB_HUB1_UP_P	20 34
I283	USB_90D	USB	USB_HUB1_UP_N	20 34
I284	USB_90D	USB	USB_HUB2_UP_P	20 35
I285	USB_90D	USB	USB_HUB2_UP_N	20 35
I290	USB_90D	USB	USB_HUB2UNUSED_P	35
I291	USB_90D	USB	USB_HUB2UNUSED_N	35

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL (NET_TYPE)	SPACING (NET_TYPE)	NET	PAGES
I251	CLK_PCH_55S	XTAL	FW_CLK24P576M_XO	39
I249	CLK_PCH_55S	XTAL	FW_CLK24P576M_XO_R	39
I250	CLK_PCH_55S	XTAL	FW_CLK24P576M_XI	39
	FW_110D	FW_TP	FW_PORT0_TPA_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPA_N	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_N	40 41

UNUSED FW NETS PHYSICAL PROPERTIES

ELECTRICAL_CONSTRAINT_SET	PHYSICAL (NET_TYPE)	SPACING (NET_TYPE)	NET	PAGES
I243	FW_110D	FW_TP	FW_P1_TPA_P	39 40
I245	FW_110D	FW_TP	FW_P1_TPA_N	39 40
I244	FW_110D	FW_TP	FW_P2_TPA_P	39 40
I246	FW_110D	FW_TP	FW_P2_TPA_N	39 40
	FW_110D	FW_TP	FW_P1_TPB_P	39 40
	FW_110D	FW_TP	FW_P1_TPB_N	39 40
	FW_110D	FW_TP	FW_P2_TPB_P	39 40
	FW_110D	FW_TP	FW_P2_TPB_N	39 40

GRAPHICS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

USE 5X_DIELECTRIC IN K62

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

ELECTRICAL_CONSTRAINT_SET
ASSINGED IN CONT. MGR. NET_TYPE

	PHYSICAL	SPACING		
I169	DP_85D	DISPLAYPORT	DP_INTCONN_ML_C_P<3..0>	84
I170	DP_85D	DISPLAYPORT	DP_INTCONN_ML_C_N<3..0>	84
	DP_85D	DISPLAYPORT	DP_INTCONN_AUXCH_C_P	84
	DP_85D	DISPLAYPORT	DP_INTCONN_AUXCH_C_N	84
I153	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	82 84
I154	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	82 84
	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P	82 84
	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N	82 84
I185	DP_85D	DISPLAYPORT	DP_EXTA_ML_P<3..0>	85
I186	DP_85D	DISPLAYPORT	DP_EXTA_ML_N<3..0>	85
I188	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_P	85
I187	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_N	85
I171	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_P<3..0>	79 85
I174	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_N<3..0>	79 85
I172	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_P	79 85
I173	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_N	79 85
I191	DP_85D	DISPLAYPORT	DP_EXTB_ML_P<3..0>	87
I192	DP_85D	DISPLAYPORT	DP_EXTB_ML_N<3..0>	87
I193	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_P	87
I194	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_N	87
I195	DP_85D	DISPLAYPORT	DP_EXTB_ML_C_P<3..0>	79 87
I196	DP_85D	DISPLAYPORT	DP_EXTB_ML_C_N<3..0>	79 87
I198	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_C_P	79 87
I197	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_C_N	79 87
I199	DP_85D	DISPLAYPORT	MXM_DP_B_ML_P<3..0>	76 79
I200	DP_85D	DISPLAYPORT	MXM_DP_B_ML_N<3..0>	76 79
I201	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_P	76 79
I202	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_N	76 79
I203	DP_85D	DISPLAYPORT	MXM_DP_C_ML_P<3..0>	76 84
I204	DP_85D	DISPLAYPORT	MXM_DP_C_ML_N<3..0>	76 84
I205	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_P	76 84
I206	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_N	76 84
I211	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_R_P	84
I212	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_R_N	84
I207	DP_85D	DISPLAYPORT	MXM_DP_D_ML_P<3..0>	76 79
I208	DP_85D	DISPLAYPORT	MXM_DP_D_ML_N<3..0>	76 79
I209	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_P	76 79
I210	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_N	76 79

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_P	
DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_N	
DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_P	77 79
DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_N	77 79
DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_P<3..0>	77 79
DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_N<3..0>	77 79
DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_P<3..0>	77 79
DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_N<3..0>	77 79

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PAGE TITLE
GRAPHICS CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D
REVISION 10.1.0
BRANCH

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
THERMAL	POWER	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	49
	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	49
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 97
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 97
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 97
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 97
	SMB_55S	SMB	SMBUS_PCH_CLK	18 49
	SMB_55S	SMB	SMBUS_PCH_DATA	18 49
	SMB_55S	SMB	SML_PCH_0_CLK	18 49
	SMB_55S	SMB	SML_PCH_0_DATA	18 49
	SMB_55S	SMB	SML_PCH_1_CLK	18 49
	SMB_55S	SMB	SML_PCH_1_DATA	18 49
	CLK_XTAL	XTAL	SMC_EXTAL	46 47
	CLK_XTAL	XTAL	SMC_XTAL	46 47
I254	SMB_55S	SMB	I2C_VREFMRGN_DIMMA_SCL	28
I253	SMB_55S	SMB	I2C_VREFMRGN_DIMMA_SDA	28
I255	SMB_55S	SMB	I2C_VREFMRGN_DIMMB_SCL	28
I256	SMB_55S	SMB	I2C_VREFMRGN_DIMMB_SDA	28
I265	SMB_55S	SMB	SMB_BLC_TCON_SCL	6 49 82
I264	SMB_55S	SMB	SMB_BLC_TCON_SDA	6 49 82
I268	SMB_55S	SMB	I2C_TCON_SCL	82
I269	SMB_55S	SMB	I2C_TCON_SDA	82
I266	SMB_55S	SMB	SMB_BLC_PCH_SCL_R	6
I267	SMB_55S	SMB	SMB_BLC_PCH_SDA_R	6

SMC VOLTAGE/CURRENT NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	THERM_DIFF	THERMAL	SNS_I_MXM_P	50
	THERM_DIFF	THERMAL	SNS_I_MXM_N	50
	THERM_DIFF	THERMAL	SNS_DIMM_1V5_P	50
	THERM_DIFF	THERMAL	SNS_DIMM_1V5_N	50
	SNS_DIFF	THERMAL	VR_ISNS_VCORE_P	50 98
	SNS_DIFF	THERMAL	VR_ISNS_VCORE_N	50 98
	SNS_DIFF	THERMAL	VR_ISNS_VAXG_P	50 98
	SNS_DIFF	THERMAL	VR_ISNS_VAXG_N	50 98
I243	SNS_DIFF	THERMAL	VR_ISNS_1V05_P	98
I244	SNS_DIFF	THERMAL	VR_ISNS_1V05_N	98
I245	THERM_DIFF	THERMAL	SNS_CPU_1V5_P	50
I246	THERM_DIFF	THERMAL	SNS_CPU_1V5_N	50
I247	THERM_DIFF	THERMAL	SNS_VCCSA_P	50
I248	THERM_DIFF	THERMAL	SNS_VCCSA_N	50
I251	THERM_DIFF	THERMAL	SNS_1V05_PCH_P	50
I252	THERM_DIFF	THERMAL	SNS_1V05_PCH_N	50
I239		THERMAL	GND_SMC_AVSS	46 47 50 97
I241		THERMAL	SMC_CPU_1V5_ISENSE	46 50
I240		THERMAL	SMC_CPU_1V5_ISENSE_R	50
I242		THERMAL	SMC_CPU_1V5_VSENSE	46 50
I229		THERMAL	GND_SMC_AVSS	46 47 50 97
I230		THERMAL	SMC_DIMM_ISENSE	46 50
I228		THERMAL	SMC_DIMM_1V5_R	50
I231		THERMAL	SMC_DIMM_VSENSE	46 50
I225		THERMAL	GND_SMC_AVSS	46 47 50 97
I226		THERMAL	SMC_VCCSA_ISENSE	46 50
I224		THERMAL	SMC_VCCSA_ISENSE_R	50
I227		THERMAL	SMC_VCCSA_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47 50 97
		THERMAL	SMC_PCH_1V05_ISENSE	46 50
		THERMAL	SMC_VAXG_VSENSE	46 50
		THERMAL	SMC_PCH_1V05_VSENSE	46 50
I233		THERMAL	GND_SMC_AVSS	46 47 50 97
I234		THERMAL	SMC_1V05_ISENSE	46 50
I232		THERMAL	SMC_VAXG_ISENSE	46 50
I235		THERMAL	SMC_1V05_VSENSE	46 50
I236		THERMAL	SMC_GPU_ISENSE	46 50
		THERMAL	SMC_GPU_VSENSE	46 50
		THERMAL	SMC_VCORE_ISENSE	46 50
I237		THERMAL	SMC_VCORE_VSENSE	46 50
I238		THERMAL	SMC_CPU_VSENSE	

SMC THERMAL NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	THERM_DIFF	THERMAL	SNS_T1_1_P	52
	THERM_DIFF	THERMAL	SNS_T1_1_N	52
	THERM_DIFF	THERMAL	SNS_T2_DP2	52
	THERM_DIFF	THERMAL	SNS_T2_DN2	52
I206	THERM_DIFF	THERMAL	SNS_T1_2_P	52
I208	THERM_DIFF	THERMAL	SNS_T1_2_N	52
I207	THERM_DIFF	THERMAL	SNS_T1_3_P	52
I209	THERM_DIFF	THERMAL	SNS_T1_3_N	52
I210	THERM_DIFF	THERMAL	SNS_T1_4_P	52
I212	THERM_DIFF	THERMAL	SNS_T1_4_N	52
I211	THERM_DIFF	THERMAL	SNS_T1_5_P	52
I213	THERM_DIFF	THERMAL	SNS_T1_5_N	52
I214	THERM_DIFF	THERMAL	SNS_T1_6_P	52
I216	THERM_DIFF	THERMAL	SNS_T1_6_N	52
I215	THERM_DIFF	THERMAL	SNS_T1_7_P	52
I217	THERM_DIFF	THERMAL	SNS_T1_7_N	52
I221	THERM_DIFF	THERMAL	SNS_CPU_THERMD_P	10 52
I220	THERM_DIFF	THERMAL	SNS_CPU_THERMD_N	10 52
I219	THERM_DIFF	THERMAL	SNS_LCD_H_P	52
I218	THERM_DIFF	THERMAL	SNS_LCD_H_N	52
	THERM_DIFF	THERMAL	SNS_ODD_P	52 101
	THERM_DIFF	THERMAL	SNS_ODD_N	52 101
	THERM_DIFF	THERMAL	SNS_CPU_H_P	52
	THERM_DIFF	THERMAL	SNS_CPU_H_N	52
I222	THERM_DIFF	THERMAL	SNS_SKIN_RIGHT_P	52 101
I223	THERM_DIFF	THERMAL	SNS_SKIN_RIGHT_N	52 101
	THERM_DIFF	THERMAL	SNS_SKIN_LEFT_P	44 52 101
	THERM_DIFF	THERMAL	SNS_SKIN_LEFT_N	44 52 101
	THERM_DIFF	THERMAL	SNS_AMB_P	52 101
	THERM_DIFF	THERMAL	SNS_AMB_N	52 101
	THERM_DIFF	THERMAL	SNS_MXM_P	52
	THERM_DIFF	THERMAL	SNS_MXM_N	52
		THERMAL	HDD_OOB_TEMP_FILT	42 51 101
		THERMAL	HDD_OOB_TEMP_FB	42
		THERMAL	HDD_OOB_TEMP_R	51
		THERMAL	SMC_HDD_OOB_TEMP	46 51

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PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

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NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SWITCHNODE	SWITCHNODE	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	BGA_P1MM	BGA_P2MM
SWITCHNODE	GND	BGA_P1MM	BGA_P2MM
SWITCHNODE	*	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	*	6:1_SPACING
SWITCHNODE	GND	*	6:1_SPACING
SWITCHNODE	*	*	SWITCHNODE

POWER NET PROPERTIES

PHYSICAL	SPACING	VOLTAGE	NET
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE1
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE2
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE3
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE4
POWER	SWITCHNODE	1.5V	VR_AXG_PHASE1
POWER	SWITCHNODE	3.3V	P3V3S5_REG_PHASE
POWER	SWITCHNODE	5V	P5VS3_REG_PHASE
POWER	SWITCHNODE	1.05V	P1V05S0_PHASE
POWER	SWITCHNODE	1.05V	P1V05_REG_PHASE
POWER	SWITCHNODE	3.4V	P3V42G3H_SW
POWER	SWITCHNODE	3.3V	ENET_3V3_S3_SR_IN
POWER	SWITCHNODE	1.2V	ENET_SR_LX
POWER	SWITCHNODE	1.5V	DDR_REG_PHASE
POWER	SWITCHNODE	1.8V	P1V8_REG_PHASE
POWER	POWER	0.75V	PP0V75_S3_MEM_VREFCA_A
POWER	POWER	0.75V	PP0V75_S3_MEM_VREFCA_B
POWER	POWER	0.75V	PP0V75_S3_MEM_VREFDQ_A
POWER	POWER	0.75V	PP0V75_S3_MEM_VREFDQ_B
POWER	POWER	12V	PP12V_S0
POWER	POWER	12V	PP12V_S0_CPU_FLTRD
POWER	POWER	12V	PP12V_S0_FAN0_L
POWER	POWER	12V	PP12V_S0_FAN1_L
POWER	POWER	12V	PP12V_S0_FAN2_L
POWER	POWER	12V	VR_CPU_DRV1_PVCC
POWER	POWER	12V	VR_CPU_DRV1_UVCC
POWER	POWER	12V	VR_CPU_DRV2_PVCC
POWER	POWER	12V	VR_CPU_DRV2_UVCC
POWER	POWER	12V	VR_CPU_DRV3_PVCC
POWER	POWER	12V	VR_CPU_DRV3_UVCC
POWER	POWER	12V	VR_CPU_DRV4_PVCC
POWER	POWER	12V	VR_CPU_DRV4_UVCC
POWER	POWER	12V	VR_AXG_DRV1_PVCC
POWER	POWER	12V	VR_AXG_DRV1_UVCC
POWER	POWER	12V	VR_CPU_DRV1_VCC
POWER	POWER	12V	VR_CPU_DRV2_VCC
POWER	POWER	12V	VR_CPU_DRV3_VCC
POWER	POWER	12V	VR_CPU_DRV4_VCC
POWER	POWER	12V	PP12V_S0_VG_OK
POWER	POWER	12V	P12V_FW_R
POWER	POWER	12V	P12V_FW_CL
POWER	POWER	12V	P12V_FW_D
POWER	POWER	12V	FW_PORT0_VP
POWER	POWER	12V	FW_PORT0_VP_F
POWER	POWER	12V	PPVP_FW_PHY_CPS
POWER	POWER	12V	PP12V_S0_P1V05_VREG_VIN
POWER	POWER	12V	VCCSA_PWR_RC
POWER	POWER	12V	PP12V_S0_BL_R
POWER	POWER	12V	PP12V_S5_FET
POWER	POWER	12V	PP12V_LCD
POWER	POWER	12V	PP12V_G3H
POWER	POWER	12V	PP12V_S5
POWER	POWER	12V	PP12V_S5_RSN
POWER	POWER	0.75V	PPVTT_S0_DDR
POWER	POWER	0.75V	PP0V75_S0
POWER	POWER	0.925V	PPVCCSA_S0_CPU
POWER	POWER	1.0V	PP1V0_FW_FWPHY_AVDD
POWER	POWER	1.0V	PP1V0_S0_FW_VDD
POWER	POWER	1.05V	PP1V05_S0_T29
POWER	POWER	1.05V	PP1V05_S0_INPUT_VCCSA
POWER	POWER	1.05V	PPVAXG_S0
POWER	POWER	1.05V	PP1V05_S0
POWER	POWER	1.05V	PP1V05_S0_PCH
POWER	POWER	1.05V	PP1V05_S0_CK505_F
POWER	POWER	1.05V	PPVOUT_S5_PCH_DCPSUS
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_EXP_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLLDMI2_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_SATA_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCCLKDMI_F
POWER	POWER	1.1V	PPVCORE_S0_AXG_REG1
POWER	POWER	1.1V	PPVCORE_S0_AXG_REGOUT
POWER	POWER	1.1V	PPVCORE_S0_CPU
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG1
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG2
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG3
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG4
VR_CTL_PHY	VR_CTL	1.2V	PP1V2_S3_ENET_INTREG
POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_AVDDL
POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_PCIEPLL
POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_GPHYPLL
POWER	POWER	1.5V	PP1V5_S0
POWER	POWER	1.5V	PP1V5_S0_CK505_F
POWER	POWER	1.5V	PP1V5_S0_CK505_R
POWER	POWER	1.5V	PP1V5_S0_CPU_MEM
POWER	POWER	1.5V	PP1V5_S0_CPU_SNS
POWER	POWER	1.5V	PP1V5_S3
POWER	POWER	1.5V	PP1V5_S3_MEM

POWER NET PROPERTIES

PHYSICAL	SPACING	VOLTAGE	NET
POWER	POWER	12V	PP12V_S0_MXM
POWER	POWER	12V	PP12V_SLG1
POWER	POWER	12V	PP3V3R12V_SW_DPAPWR_1
POWER	POWER	12V	PP3V3R12V_SW_DPBPWR_1
POWER	POWER	1.8V	USB_HUB2_VDD1V8
POWER	POWER	1.8V	USB_HUB2_VDD1V8PLL
POWER	POWER	1.8V	PP1V8_S0
POWER	POWER	1.8V	PP1V8R1V5_S0_PCH_VCCVRM_F
POWER	POWER	1.8V	USB_HUB1_VDD1V8
POWER	POWER	1.8V	USB_HUB1_VDD1V8PLL
POWER	POWER	1.8V	PP3V3R1V8_ENET_LR_OUT_REG
POWER	POWER	3.3V	PPVBAT_G3_SYSCLK_R
POWER	POWER	3.3V	USB_HUB1_VDDPLL3V3
POWER	POWER	3.3V	USB_HUB1_VDDA3V3
POWER	POWER	3.3V	USB_HUB2_VDDPLL3V3
POWER	POWER	3.3V	USB_HUB2_VDDA3V3
POWER	POWER	3.3V	PP3V3R1V5_PCH_VCCSUSHDA
POWER	POWER	3.3V	PP3V3_S0_CK505_F
POWER	POWER	3.3V	PP3V3_S0_HS_F
POWER	POWER	3.3V	PP3V3_S0_SW_SD_PWR
POWER	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC_F
POWER	POWER	3.3V	PP3V3_S0
POWER	POWER	3.3V	PP3V3_SMCUSBMUX
POWER	POWER	3.3V	PP3V3_S3
POWER	POWER	3.3V	PP3V3_S3_BT_FLT
POWER	POWER	3.3V	PP3V3_S3_MINI_CONN
POWER	POWER	3.3V	PP3V3_S3_SDCARD_FLT
POWER	SWITCHNODE	3.3V	PP3V3_S3_ENET_PHY_FET
POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_XTALVDDH
POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_BIASVDDH
POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_AVDDH
POWER	POWER	3.3V	ENET_XTALVDDH
POWER	POWER	3.3V	PP3V3_S5
POWER	POWER	3.3V	PP3V3_S5_REG_R
POWER	POWER	3.3V	PPVOUT_S0_PCH_DCPSST
POWER	POWER	3.3V	PPVOUT_PCH_DCPSUSBYP
POWER	POWER	3.3V	PPVOUT_G3_PCH_DCPRTC
POWER	POWER	3.3V	PPVBATT_G3_RTC
POWER	POWER	3.3V	PPVBATT_G3_RTC_R
POWER	POWER	3.3V	PP3V3_AUDIO_SPDIF_JACK
POWER	POWER	3.3V	PP3V3_FW_FWPHY_VDDA
POWER	POWER	3.3V	PP3V3_FW_FWPHY_VP25
POWER	POWER	3.3V	PP3V3_S0_T29
POWER	POWER	3.3V	PP3V3_SW_DPAPWR
POWER	POWER	3.3V	PP3V3_SW_DPBPWR
POWER	POWER	3.3V	PP3V3_G3H_RTC
POWER	POWER	3.3V	PP3V3_G3H_AVREF_SMC
POWER	POWER	3.4V	PP3V3_G3H_SMC_AVCC
POWER	POWER	3.42V	PP3V42_G3H
POWER	POWER	4.5V	4V5_REG_IN
POWER	POWER	4.5V	PP4V5_AUDIO_ANALOG
POWER	POWER	4.5V	PP5V_AUDIO_HPAMP
POWER	POWER	5V	PP5V_LAMP_VREG
POWER	POWER	5V	PP5V_RAMP_VREG
POWER	POWER	5V	PP5V_S0
POWER	POWER	5V	PP5V_S0_CPU_VCORE_VCC
POWER	POWER	5V	PP5V_S0_PCH_V5REF
POWER	POWER	5V	PP5V_S0_SATA_FET
POWER	POWER	5V	5V_S0_P1V05REG_VIN
POWER	POWER	5V	5V_S0_P1V05REG_VDD
POWER	POWER	5V	PP5V_S3
POWER	POWER	5V	PP5V_S3_VREFMRGN_A
POWER	POWER	5V	PP5V_S3_VREFMRGN_B
POWER	POWER	5V	PP5V_S3_DDR_REG_V5FILT
POWER	POWER	5V	PP5V_S3_CAMERA_FLT
POWER	POWER	5V	PP5V_S3_REG_R
POWER	POWER	5V	PP5V_S3_IR_FLT
POWER	POWER	5V	PP5V_S5
POWER	POWER	5V	PP5V_S5_PCH_V5REFSUS
POWER	POWER	5V	PP5V_USB_PORT0
POWER	POWER	5V	PP5V_USB_PORT0_F
POWER	POWER	5V	PP5V_USB_PORT1
POWER	POWER	5V	PP5V_USB_PORT1_F
POWER	POWER	5V	PP5V_USB_PORT2
POWER	POWER	5V	PP5V_USB_PORT2_F
POWER	POWER	5V	PP5V_USB_PORT3
POWER	POWER	5V	PP5V_USB_PORT3_F
POWER	POWER		DDR_REG_PGND
POWER	POWER		DDR_REG_CSGND
	POWER	3.3V	PPVDDIO_25M_B
	POWER	1.05V	PPVDDIO_25M_A
	POWER	3.3V	PPVDDIO_25M_C
	POWER	12V	PP3V3R12V_SW_DPAPWR_R
	POWER	12V	PP3V3R12V_SW_DPAPWR_D
	POWER	1.5V	PP1V5_S0_DP_R
POWER	POWER	5V	PP5V_S0_SATA_PORTA1
POWER	POWER	5V	PP5V_S0_SATA_PORTA2

SENSING NET PROPERTIES

PHYSICAL	SPACING	NET
SNS_DIFF	THERMAL	VR_CPU_ISNS1_P
SNS_DIFF	THERMAL	VR_CPU_ISNS1_N
SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N
SNS_DIFF	THERMAL	VR_CPU_ISNS2_P
SNS_DIFF	THERMAL	VR_CPU_ISNS2_N
SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N
SNS_DIFF	THERMAL	VR_CPU_ISNS3_P
SNS_DIFF	THERMAL	VR_CPU_ISNS3_N
SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N
SNS_DIFF	THERMAL	VR_CPU_ISNS4_P
SNS_DIFF	THERMAL	VR_CPU_ISNS4_N
SNS_DIFF	THERMAL	VR_CPU_ISNS4_R_N
SNS_DIFF	THERMAL	P1V05S0_ISN
SNS_DIFF	THERMAL	P1V05S0_ISP
SNS_DIFF	THERMAL	VR_AXG_ISNS_R_P
SNS_DIFF	THERMAL	VR_AXG_ISNS_R_N
SNS_DIFF	THERMAL	VR_AXG_ISNS_N
SNS_DIFF	THERMAL	VR_AXG_ISNS_P
SNS_DIFF	THERMAL	VR_ISNS_1V05_N
SNS_DIFF	THERMAL	VR_ISNS_1V05_P
SNS_DIFF	THERMAL	VR_ISNS_VAXG_N
SNS_DIFF	THERMAL	VR_ISNS_VAXG_P
SNS_DIFF	THERMAL	VR_ISNS_VCORE_N
SNS_DIFF	THERMAL	VR_ISNS_VCORE_P
SNS_DIFF		CPU_VAXG_SENSE_P
SNS_DIFF		CPU_VAXG_SENSE_N
SNS_DIFF		VR_AXG_VSNS_R_N
SNS_DIFF		VR_AXG_VSNS_R_P
SNS_DIFF		VR_AXG_VSEN
SNS_DIFF		VR_AXG_RGND
SNS_DIFF		VR_AXG_VSNS_XW_N
SNS_DIFF		VR_AXG_VSNS_XW_P
SNS_DIFF		CPU_VCC_SENSE_N
SNS_DIFF		CPU_VCC_SENSE_P
SNS_DIFF		VR_CPU_VSEN
SNS_DIFF		VR_CPU_RGND
SNS_DIFF		VR_CPU_VSNS_R_N
SNS_DIFF		VR_CPU_VSNS_R_P
SNS_DIFF		VR_CPU_VSNS_XW_P
SNS_DIFF		VR_CPU_VSNS_XW_N
SNS_DIFF		CPU_VCCIO_SENSE_P
SNS_DIFF		CPU_VCCIO_SENSE_N
SNS_DIFF		P1V05S0_VSEN
SNS_DIFF		P1V05S0_RTN

VR CTRL NET PROPERTIES

PHYSICAL	SPACING	NET
VR_CTL_PHY	SWITCHNODE	VR_CPU_PH1_SNUB
VR_CTL_PHY	SWITCHNODE	VR_CPU_PH2_SNUB
VR_CTL_PHY	SWITCHNODE	VR_CPU_PH3_SNUB
VR_CTL_PHY	SWITCHNODE	VR_CPU_PH4_SNUB
VR_CTL_PHY	VR_CTL	VR_CPU_COMP
VR_CTL_PHY	VR_CTL	VR_AXG_TM
VR_CTL_PHY	VR_CTL	VR_AXG_PWM_R
VR_CTL_PHY	VR_CTL	VR_AXG_PWM
VR_CTL_PHY	VR_CTL	VR_CPU_PWM2
VR_CTL_PHY	VR_CTL	VR_CPU_PWM1
VR_CTL_PHY	VR_CTL	VR_CPU_PWM1_R
VR_CTL_PHY	VR_CTL	VR_CPU_PWM2_R
VR_CTL_PHY	VR_CTL	VR_CPU_PWM3
VR_CTL_PHY	VR_CTL	VR_CPU_PWM3_R
VR_CTL_PHY	VR_CTL	VR_CPU_PWM4
VR_CTL_PHY	VR_CTL	VR_CPU_PWM4_R
VR_CTL_PHY	VR_CTL	VR_AXG_SW_FREQ
VR_CTL_PHY	VR_CTL	VR_EN_PWR_OVP
VR_CTL_PHY	VR_CTL	VR_RSET
VR_CTL_PHY	VR_CTL	VR_EN_PWR_OVP_R
VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT1_RC
VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT2_RC
VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT3_RC
VR_CTL_PHY	VR_CONTROL	VR_AXG_BOOT1_RC
VR_CTL_PHY	VR_CTL	VR_CPU_BOOT4_RC
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_GDSEL
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_LGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_UGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_GDSEL
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_LGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_UGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_GDSEL
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_LGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_UGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV4_GDSEL
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_LGATE
VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_UGATE
VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_BOOT
VR_CTL_PHY	VR_CTL	VR_AXG_DRV1_UGATE_R
VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_LGATE
VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_UGATE
VR_CTL_PHY	VR_CTL	VR_CPU_SW_FREQ
VR_CTL_PHY	VR_CTL	VR_CPU_FB
VR_CTL_PHY	VR_CTL	VR_CPU_FB_R
VR_CTL_PHY	VR_CTL	VR_CPU_RAMP_ADJ
VR_CTL_PHY	VR_CTL	VR_CPU_IMON
VR_CTL_PHY	VR_CTL	VR_CPU_SUTH
VR_CTL_PHY	VR_CTL	VR_CPU_N_PSI
VR_CTL_PHY	VR_CTL	VR_CPU_FDVID
VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP
VR_CTL_PHY	VR_CTL	VR_CPU_IAUTO
VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP1
VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP2
VR_CTL_PHY	VR_CTL	VR_CPU_FB_RC
VR_CTL_PHY	VR_CTL	VR_CPU_FB2
VR_CTL_PHY	VR_CTL	VR_CPU_IMON_R
VR_CTL_PHY	VR_CTL	VR_CPU_HFREQ_COMP
VR_CTL_PHY	SWITCHNODE	VR_AXG_PH1_SNUB
VR_CTL_PHY	VR_CTL	VR_AXG_IMON
VR_CTL_PHY	VR_CTL	VR_AXG_IMON_R
VR_CTL_PHY	VR_CTL	VR_AXG_HFREQ_COMP
VR_CTL_PHY	VR_CTL	VR_AXG_FB
VR_CTL_PHY	VR_CTL	VR_SEN_R1
VR_CTL_PHY	VR_CTL	VR_SEN_R2
VR_CTL_PHY	VR_CTL	VR_SEN_R3
VR_CTL_PHY	VR_CTL	VR_AXG_COMP_RC
VR_CTL_PHY	VR_CTL	VR_AXG_COMP
VR_CTL_PHY	VR_CTL	VR_AXG_TCOMP
VR_CTL_PHY	SWITCHNODE	P1V05S0_UGATE
VR_CTL_PHY	SWITCHNODE	P1V05S0_LGATE
VR_CTL_PHY	VR_CTL	P1V05S0_BOOT
VR_CTL_PHY	VR_CTL	P1V05S0_BOOT_R
VR_CTL_PHY	VR_CTL	P1V05_IMON
VR_CTL_PHY	VR_CTL	P1V05S0_FB
VR_CTL_PHY	VR_CTL	P1V05S0_PHASE_L
VR_CTL_PHY	VR_CTL	P1V05S0_VW
VR_CTL_PHY	VR_CTL	P1V05S0_COMP
VR_CTL_PHY	VR_CTL	P1V05S0_ICOMP
VR_CTL_PHY	VR_CTL	P1V05_S0_VDIFF
VR_CTL_PHY	VR_CTL	P1V05S0_COMP_C
VR_CTL_PHY	VR_CTL	P1V05S0_VDIF_C
VR_CTL_PHY	VR_CTL	P1V05S0_VO
VR_CTL_PHY	VR_CTL	P1V05S0_OCSET
VR_CTL_PHY	VR_CTL	P1V05S0_SNUBBER
VR_CTL_PHY	VR_CTL	P1V05_REG_VFB
VR_CTL_PHY	VR_CTL	ENET_SR_VFB
VR_CTL_PHY	VR_CTL	P1V05S0_ISP_R
VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2_R
VR_CTL_PHY	VR_CTL	T29BST_FBX
VR_CTL_PHY	VR_CTL	P3V42G3H_BOOST
VR_CTL_PHY	VR_CTL	P3V42G3H_FB

VR CTRL NET PROPERTIES

PHYSICAL	SPACING	NET
VR_CTL_PHY	VR_CTL	DDR_REG_CS
VR_CTL_PHY	VR_CTL	DDR_REG_FB
VR_CTL_PHY	VR_CONTROL	DDR_REG_LGATE
VR_CTL_PHY	VR_CONTROL	DDR_REG_UGATE
VR_CTL_PHY	VR_CONTROL	DDR_REG_BOOT
VR_CTL_PHY	VR_CONTROL	DDR_REG_BOOT_R
VR_CTL_PHY	VR_CTL	DDR_REG_VDDQSNS
VR_CTL_PHY	VR_CTL	DDR_REG_VTTSNS
VR_CTL_PHY	VR_CTL	DDR_REG_PHASE_R
VR_CTL_PHY	VR_CTL	DDR_REG_UGATE_R
VR_CTL_PHY	SWITCHNODE	1V5_SNUBBER
VR_CTL_PHY	VR_CTL	P1V8_REG_VFB
VR_CTL_PHY	VR_CTL	P1V8_REG_SYNC
VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_BOOT
VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_BOOT_R
VR_CTL_PHY	VR_CTL	P3V3S5_REG_FB
VR_CTL_PHY	VR_CTL	P3V3S5_REG_ISEN
VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_LGATE
VR_CTL_PHY	VR_CTL	P3V3S5_REG_OCSET
VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_VOUT1
VR_CTL_PHY	VR_CTL	P3V3S5_REG_FSET1
VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_UGATE
VR_CTL_PHY	VR_CTL	P3V3S5_REG_SNUB
VR_CTL_PHY	VR_CONTROL	P5VS3_REG_BOOT
VR_CTL_PHY	VR_CTL	P5VS3_REG_FB
VR_CTL_PHY	VR_CTL	P5VS3_REG_ISEN
VR_CTL_PHY	VR_CONTROL	P5VS3_REG_LGATE
VR_CTL_PHY	VR_CTL	P5VS3_REG_OCSET
VR_CTL_PHY	VR_CTL	P5VS3_REG_FSET2
VR_CTL_PHY	VR_CONTROL	P5VS3_REG_UGATE
VR_CTL_PHY	VR_CONTROL	P5VS3_REG_VOUT2
VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT1
VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2
VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2_R
VR_CTL_PHY	VR_CTL	VCCSA_OUT
VR_CTL_PHY	VR_CTL	VCCSA_GATE
VR_CTL_PHY	VR_CTL	VCCSA_REF
VR_CTL_PHY	VR_CTL	VCCSA_CRL
VR_CTL_PHY	VR_CTL	CPU_VCCSA_SENSE
VR_CTL_PHY	VR_CTL	VR_CPU_TM
VR_CTL_PHY	PM	3V3R2V9_DPAPWR_ADJ

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.35MM	?

VR VID NET PROPERTIES

PULL-UP STUB < 1-INCH
VID LENGTH SKEW < 1-INCH
VID LENGTH RANGE< 1 TO 15-INCH

PHYSICAL	SPACING	NET
VID_PHY	VR_CTL_VIDS	CPU_VIDSCLK_R
VID_PHY	VR_CTL_VIDS	CPU_VIDALERT_L_R
VID_PHY	VR_CTL_VIDS	CPU_VIDSOUT_R
VID_PHY	VR_CTL_VIDS	CPU_VIDSCLK
VID_PHY	VR_CTL_VIDS	CPU_VIDALERT_L
VID_PHY	VR_CTL_VIDS	CPU_VIDSOUT

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_VIDS	*	=4X_DIELECTRIC	?

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: POWER CONSTRAINTS

Apple Inc.

DRAWING NUMBER: 051-8442 SIZE: D

REVISION: 10.1.0

T29 ELECTRICAL ROUTES

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29	*	=5X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29	TOP,BOTTOM	=7X_DIELECTRIC	?

T29 PCI-EXPRESS (SAME RULE AS PCIE)

T29 SPI INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	0.2 MM	?

T29 XTAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_XTAL_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_XTAL	*	=4X_DIELECTRIC	?

T29 SMBUS INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SMB	*	=2x_DIELECTRIC	?

GREEN CLOCK CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

T29 BIAS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_COMP	*	0.2 MM	?

T29 NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_P<3..0>	85 87 89
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_N<3..0>	85 87 89
NO_TEST=TRUE	T29_90D	T29	T29_D2R_C_P<3..0>	85 86 87 88
NO_TEST=TRUE	T29_90D	T29	T29_D2R_C_N<3..0>	85 86 87 88
NO_TEST=TRUE	T29_90D	T29	T29_R2D_P<3..0>	85 87
NO_TEST=TRUE	T29_90D	T29	T29_R2D_N<3..0>	85 87
NO_TEST=TRUE	T29_90D	T29	T29_D2R_P<3..0>	85 87 89
NO_TEST=TRUE	T29_90D	T29	T29_D2R_N<3..0>	85 87 89
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_F_P<3..0>	85 87
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_F_N<3..0>	85 87
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_P<3..0>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_N<3..0>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_P<3..0>	87 88
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_N<3..0>	87 88
NO_TEST=TRUE	T29_90D	T29	DP_A_EXT_AUXCH_P	85 86
NO_TEST=TRUE	T29_90D	T29	DP_A_EXT_AUXCH_N	85 86
NO_TEST=TRUE	T29_90D	T29	DP_SDRVA_AUXCH_C_P	85
NO_TEST=TRUE	T29_90D	T29	DP_SDRVA_AUXCH_C_N	85
NO_TEST=TRUE	T29_90D	T29	DP_SDRVB_AUXCH_C_P	87
NO_TEST=TRUE	T29_90D	T29	DP_SDRVB_AUXCH_C_N	87
NO_TEST=TRUE	T29_90D	T29	DP_B_EXT_AUXCH_P	87 88
NO_TEST=TRUE	T29_90D	T29	DP_B_EXT_AUXCH_N	87 88
NO_TEST=TRUE	T29_90D	T29	T29DPA_D2R1_AUXCH_P	86
NO_TEST=TRUE	T29_90D	T29	T29DPA_D2R1_AUXCH_N	86
NO_TEST=TRUE	T29_90D	T29	T29DPB_D2R3_AUXCH_P	88
NO_TEST=TRUE	T29_90D	T29	T29DPB_D2R3_AUXCH_N	88
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_N<0>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_P<0>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_N<2>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_P<2>	85 86
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_N<0>	87 88
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_P<0>	87 88
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_N<2>	87 88
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_P<2>	87 88
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_P<3..0>	89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_N<3..0>	89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_P<3..0>	18 89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_N<3..0>	18 89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_P<3..0>	18 89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_N<3..0>	18 89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_C_P<3..0>	89
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_C_N<3..0>	89
NO_TEST=TRUE	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_P	18 89
NO_TEST=TRUE	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_N	18 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_P<3..0>	84
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_N<3..0>	84
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_P<3..0>	84 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_N<3..0>	84 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_P<3..0>	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_N<3..0>	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_P	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_N	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_P<3..0>	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_N<3..0>	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_P	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_N	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_P<3..0>	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_N<3..0>	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_P	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_N	89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_P<3..0>	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_N<3..0>	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_P	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_N	79 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_R_C_P	84
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_R_C_N	84
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_C_P	84 89
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_C_N	84 89
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_AUXCH_P	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_AUXCH_N	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_AUXCH_P	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_AUXCH_N	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_C_P<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_C_N<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_P<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_N<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_R_P<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_R_N<3..0>	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_C_P<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_C_N<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_P<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_N<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_R_P<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_R_N<3..0>	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_A_RSVD_N	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_A_RSVD_P	85
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_B_RSVD_N	87
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_B_RSVD_P	87

T29 NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	T29_SPI_55S	T29_SPI	JTAG_T29_TDI	15 21 89
	T29_SPI_55S	T29_SPI	JTAG_T29_TMS	15 18 89
	T29_SPI_55S	T29_SPI	JTAG_T29_TCK	15 21 25 89
	T29_SPI_55S	T29_SPI	JTAG_T29_TDO	15 21 89
I437	T29_SPI_55S	T29_SPI	T29_SPI_MOSI	89
I438	T29_SPI_55S	T29_SPI	T29_SPI_MISO	89
I439	T29_SPI_55S	T29_SPI	T29_SPI_CS_L	89
I440	T29_SPI_55S	T29_SPI	T29_SPI_CLK	89
	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	80 89 99
	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	89 99
	T29_SMB_55S	T29_SMB	I2C_T29_SDA	49 89
	T29_SMB_55S	T29_SMB	I2C_T29_SCL	49 89
I421	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB	80
I422	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET	80
I423	CLK_25M_55S	CLK_25M	ENET_CLK25M_XTALI_OSC	36 80
I424	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_CLK	80
I433	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	80 89 99
I425	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	89 99
I426	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X2	80
I428	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X2_R	80
I427	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X1	80
I431	T29_55S	T29_COMP	T29_RSENSE	89
I432	T29_55S	T29_COMP	T29_RBIAS	89
I434		T29_COMP	T29_A_BIAS	79 83 85 86
I435		T29_COMP	T29_B_BIAS	79 83 87 88
I436		T29_COMP	DP_A_BIAS	79 85

SYNC_MASTER=K62_AARON SYNC_DATE=06/11/2010

PAGE TITLE

T29 CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8442 SIZE D

REVISION 10.1.0

BRANCH

PM NET PROPERTIES
(PM, RESET, EN, PGOOD)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

Port	PHYSICAL / SPACING (NET_TYPE)	Net	Pages
I357	PM	4V5_REG_EN	56
I358	PM	3V42G3H_SHDN_L	73
I359	PM	ALL_SYS_PWRGD_R	5 32 64
I361	PM	ALL_SYS_PWRGD_SMC	46 64
I360	PM	AP_PWR_EN	20 25 33
I362	PM	AP_MINI_RESET_L	33
I364	PM	AUD_I2C_INT_L	20 62
I363	PM	AUD_IP_PERIPHERAL_DET	20 61
I365	PM	AUD_IPHS_SWITCH_EN	21 62
I406	PM	AUD_SPDIF_IN	60 84 94
I407	PM	AUD_SPDIF_IN_CODEC	56 84
I404	PM	BDV_BKL_PWM	46 84 100
I401	PM	BL_PWM	6 84
I410	PM	BL_EN	6 82
I411	PM	BDV_BKL_PWM	46 84 100
I366	PM	CK505_27MHZ_EN	26
I367	PM	CPUVTT_REG_EN	
I369	PM_VTT	CPUVTT_REG_PGOOD_R	63
I368	PM	CPU_MEM_RESET_L	11 32
I372	PM	CPU_PECI_R	46
I370	PM_VTT	CPU_PWRGD	11 21 25
I371	PM	CPU_RESET_L	11 27
I374	PM	CPU_SKTOCC_L	11 63
I373	PM	CPU_CATERR_L	11
I375	PM	CPU_PECI	11 21 46
I376	PM	CPU_PROCHOT_L	11 47 65
I377	PM	CPU_THRMTRIP_L	11 47
I378	PM	CPU_PROC_SEL	11 19
I379	PM	DEBUG_RESET_L	27 48
I380	PM	DDRVTT_EN	
I382	PM	DP_INT_SPDIF_AUDIO	82 84
I381	PM	DP_INTPNL_HPD	82 84
I385	PM	3V3R2V9_DPAPWR_ADJ	83 98
I384	PM	DP_A_PWRDWN	85
I383	PM	DP_A_PWRDWN_FET_R	83
I386	PM	DP_A_PWRDWN_INV	83
I388	PM	DPAPWRSW_HVEN_L_R	83
I387	PM	DPAPWRSW_CT	83
I389	PM	DPAPWRSW_ILIM	83
I391	PM	DPAPWRSW_IFLT	83
I390	PM	T29_A_HV_EN	83 85
I392	PM	3V3R2V9_DPBPWR_ADJ	83
I395	PM	DP_B_PWRDWN	87
I393	PM	DP_B_PWRDWN_FET_R	83
I394	PM	DP_B_PWRDWN_INV	83
I398	PM	DPBPWRSW_HVEN_L_R	83
I396	PM	DPBPWRSW_CT	83
I397	PM	DPBPWRSW_ILIM	83
I399	PM	DPBPWRSW_IFLT	83
I400	PM	T29_B_HV_EN	83 87
I420	PM	T29_PWR_EN	18 81 100
I421	PM	T29_RESET_RTR_L	81 89
I412	PM	LCD_BL_FILT	84
I402	PM	LCD_BLK_ON_DLY	
I403	PM	LCD_BL_PWM	84
I405	PM	MXM_PNL_BL_PWM	77 84

Port	PHYSICAL / SPACING (NET_TYPE)	Net	Pages
I250	PM	ENET_PWR_EN	20 25 36
I227	PM	ENET_LOW_PWR	15 21 37
I229	PM	FW_RESET_L	27 39
I231	PM	ENET_RESET_L	27 36
I236	PM	FW_PME_L	15 21 39
I237	PM	FW_PWR_EN	15 21
I238	PM	FW_CLKREQ_L	15 39
I248	PM	ISOLATE_CPU_MEM_L	21 25 32
I266	PM	LPC_PWRDWN_L	19 46 48
I265	PM	MEM_RESET_L	30 31 32 92
I269	PM	MINI_CLKREQ_L	15 33
I268	PM	MINI_RESET_L	27 33
I267	PM	MXM_CLKREQ_L	9 76
I270	PM	MXM_GOOD	5 21 25
I271	PM	ODD_PWR_EN_L	15 21 42
I68	PM	RTC_RESET_L	18 27 100
I48	PM	RSMRST_PWRGD	46 64
I47	PM	RTC_RESET_L	18 27 100
I319	PM	S4_ENABLES	63
I280	PM	SDCONN_STATE_RST_L	
I281	PM	SDCONN_DETECT_BUF_L	95
I282	PM	SDCONN_STATE_CHANGE	20 25 45
I283	PM	SDCARD_RESET	15 21 44 101
I315	PM	SDCARD_RESET_L	44
I284	PM	SDCARD_PLT_RST_L	27 44
I350	PM	SDCARD_PLT_RST_L_R	
I351	PM	SMC_PM_G2_EN	46 75
I352	PM	SMC_PM_G2_EN_R	75
I353	PM	SMC_PM_G2_EN_L	75
I354	PM	S5_DG_1	75
I355	PM	S5_MSFT_G1	75
I415	PM	USE_HDD_OOB_L	20 51
I414	PM	HDD_OOB_1V00_REF	51
I285	PM	SMC_ADAPTER_EN	19 46 47
I286	PM	SMC_RUNTIME_SCI_L	21 46 47
I287	PM	SMC_WAKE_SCI_L	15 18 21 46
I288	PM	SMC_DELAYED_PWRGD	47 64
I289	PM	SMC_LRESET_L	27 46
I290	PM	SMC_RESET_L	46 47 48
I416	PM	SMC_PROCHOT	46 47
I417	PM	SMC_PROCHOT_3_3_L	46 47
I419	PM	SMC_ONOFF_L	46 47
I418	PM	SMC_MANUAL_RST_L	47
I291	PM	SPI_DESCRIPTOR_OVERRIDE_L	18 46
I413	PM	T29_PWR_EN	18 81 100
I260	PM	T29_RESET_L	27 81
I408	PM	T29_DP_PORTA_PWR_EN	20 25 83 94
I409	PM	T29_DP_PORTA_PWR_EN_REG	83
I259	PM_VTT	XDP_CPUPWRGD	
I261	PM_VTT	XDP_DBRESET_L	11 25
I263	PM_VTT	XDP_PWRGD	
I262	PM	XDPPCH_PLTRST_L	25 27
I264	PM	USB_HUB_SOFT_RESET_L	20 25 34
I323	PM	VSYNC_DP_CONN	6 82
I324	PM	VSYNC_DP	82
I325	PM	VIDEO_ON	82
I344	PM	VTT_REG_PGOOD_L	63

PHYSICAL / SPACING (NET_TYPE)	Net	Pages
PM	PLT_RESET_L	20 27
PM_VTT	PLT_RESET_LS1V05_L	11
PM	PM_BATLOW_L	35 19 46
PM	PM_CLK32K_SUSCLK	9 46 94
PM	PM_CLK32K_SUSCLK_R	9 19 94
PM	PM_CLKRUN_L	15 19 46 48
PM	PM_PWRBTN_L	19 25 46
PM	PM_RSMRST_L	27 46
PM	PM_RSMRST_PCH_L	19 27
PM	PCH_SRTCRST_L	18
PM	PCH_INTVRMEN_L	18
PM	PCH_DSWVRMEN	19
PM	PCH_DF_TVS	19
PM	PCH_PROCPWRGD	21
PM	PCIE_WAKE_L	19 33 36 79
PM	PM_DSW_PWRGD	19
PM	PM_ASW_PWRGD	19 64
PM	PM_MEM_PWRGD_R	11
PM	PM_EN_DDR1V5_S3_REG	63 72
PM	PM_EN_DDRVTT_S0_REG	32 63 72
PM	PM_EN_P12V_S0_FET	6 63
PM	PM_EN_P1V05_S0_REG	63 68
PM	PM_EN_P1V05_S3_REG	
PM	PM_EN_P1V5_S0_FET	63 74
PM	PM_EN_P1V8_S0_REG	63 72
PM	PM_EN_P3V3_S0_FET	63 74
PM	PM_EN_P3V3_S3_FET	63 74
PM	PM_EN_P3V3_S5_REG	71
PM	PM_EN_P5V_S0_FET	63 74
PM	PM_EN_P5V_S3_REG	63 71
PM	PM_EN_PVCCSA_S0_REG_L	64
PM	PM_EN_VCCSA_S0_CPU	
PM	PM_EN_PVCORE_CPU	63 65
PM_VTT	PM_MEM_PWRGD	11 19 100
PM	PM_MXM_EN	64 77
PM	PM_PCH_PWRGD_R	64
PM	PM_PECI_PWRGD	46 64
PM	PM_PECI_PWRGD_R	46
PM	PM_PGOOD_DDR1V5_S3_REG	5 63 72
PM	PM_PGOOD_P1V05_S0_REG	63 64 68
PM	PM_PGOOD_P1V5_S0_FET	11 64 74
PM	PM_PGOOD_P1V8_S0_REG	64 72
PM	PM_PGOOD_P3V3_S0_FET	63 64 74
PM	PM_PGOOD_P3V3_S3_FET	34 74
PM	PM_PGOOD_P3V3_S5_REG	27 64 71
PM	PM_PGOOD_P5V_S0_FET	63 64 74
PM	PM_PGOOD_MINI	33
PM	PM_PGOOD_PVCORE_CPU	5 25 64 65
PM	PM_PGOOD_PVCCSA_S0_REG	63 64
PM	PM_PGOOD_P5V_S3_REG	63 71 83
PM	PM_PGOOD_PVAXG	5 65
PM_VTT	PM_MEM_PWRGD	11 19 100
PM	PM_MEM_PWRGD_L	11
PM	PM_MXM_PGOOD	64 77
PM	PM_PCH_PWRGD	19 21 64
PM	PM_SLP_S3_5V	32
PM	PM_SLP_S3_5V_L	32
PM	PM_SLP_S3_5V_R2	32
PM	PM_SLP_S3_L	5 19 26 32 36 46 47 63
PM	PM_SLP_S4_L	5 19 32 46 47 63 100
PM	PM_SLP_S5_L	5 19 46 47 63
PM_VTT	PM_SYNC	11 19
PM	PM_SYSRST_L	19 25 27 46
PM	PM_SYS_PWRGD	19 32 64
PM_VTT	PM_THRMTRIP_L	21 47
PM	PM_SLP_S3_BUF_L	63
PM	PM_SLP_S4_1_L_R	63
PM	PM_SLP_S4_D_L	32
PM	PM_SLP_S4_L	5 19 32 46 47 63 100
PM	PGOOD_P1V5_S0_DLY	11
PM	PGOOD_1V8_S0_G1	64
PM	PGOOD_1V8_S0_G2	64
PM	PGOOD_P12V_S0	63 64
PM	PGOOD_P1V8_S0	64
PM	PGOOD_PCH_S0	5 64
PM	PGOOD_PCH_S0_R	64 94
PM	PGOOD_SYSPWROK	64
PM	PGOOD_SYSPWROK_R	64
PM	POWER_BUTTON_L	47
PM	PEG_RESET_L	9 27
PM	PGOOD_CPU_S0	64
PM	PGOOD_CPU_UNCORE	64 94
PM	PGOOD_5V_1V05_3V3	64 94
PM	PGOOD_3V3_1V05	64 94
PM	PGOOD_12V_S0_G1	64
PM	PGOOD_12V_S0_G2	64
PM	9V_COMP_REF	64
PM	12V_COMP_REF	64
PM	ALL_SYS_PWRGD	64 94

SYNC_MASTER=K62_JERRY SYNC_DATE=01/09/2011

PAGE TITLE: PM RESETS ENABLES PGOOD CONST

Apple Inc.

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BRANCH

FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT

J4700 USB CAMERA

- 98 6 IN PP5V_S3 FUNC_TEST=TRUE MIN_ALLOWED_TPS=1
- 95 44 IN USB_CAMERA_L_P FUNC_TEST=TRUE
- 95 44 IN USB_CAMERA_L_N FUNC_TEST=TRUE
- 95 44 IN USB_BT_L_P FUNC_TEST=TRUE
- 95 44 IN USB_BT_L_N FUNC_TEST=TRUE

1 PP5V_S3_REG Testpoint near J4700
1 PP3V3_S3 TESTPOINT NEAR J4700
6 GROUND TESTPOINTS NEAR J4700

J4750 USB CARD READER

- 100 44 21 15 IN SDCARD_RESET FUNC_TEST=TRUE

1 PP3V3_S3 Testpoint near J4750
2 Ground Testpoints near J4750

J4780 IR BOARD

- 95 44 IN USB_IR_L_P FUNC_TEST=TRUE
- 95 44 IN USB_IR_L_N FUNC_TEST=TRUE
- 98 44 IN PP5V_S3_IR_FLT FUNC_TEST=TRUE

1 GROUND TESTPOINT NEAR J4780

J4520 SATA ODD (HIGH SPEED)

- 46 42 IN SMC_ODD_DETECT FUNC_TEST=TRUE

1 PP5V_S0 Testpoint near J4520
1 GROUND TESTPOINTS NEAR J4520

J5551 ODD TEMP SENSOR

- 97 52 IN SNS_ODD_P FUNC_TEST=TRUE
- 97 52 IN SNS_ODD_N FUNC_TEST=TRUE

J5600 ODD FAN

- 53 IN FAN_0_PWR_L FUNC_TEST=TRUE
- 53 IN FAN_TACH0_L FUNC_TEST=TRUE
- 98 53 IN PP12V_S0_FAN0_L FUNC_TEST=TRUE
- 53 IN FAN_0_GND FUNC_TEST=TRUE

J5700 CPU FAN

- 54 IN FAN_2_PWR_L FUNC_TEST=TRUE
- 54 IN FAN_TACH2_L FUNC_TEST=TRUE
- 98 54 IN PP12V_S0_FAN2_L FUNC_TEST=TRUE
- 54 IN FAN_2_GND FUNC_TEST=TRUE
- 97 52 IN SNS_AMB_P FUNC_TEST=TRUE
- 97 52 IN SNS_AMB_N FUNC_TEST=TRUE

1 GROUND TESTPOINT NEAR J5700

J5601 HD FAN

- 53 IN FAN_1_PWR_L FUNC_TEST=TRUE
- 53 IN FAN_TACH1_L FUNC_TEST=TRUE
- 98 53 IN PP12V_S0_FAN1_L FUNC_TEST=TRUE
- 53 IN FAN_1_GND FUNC_TEST=TRUE

J5400 HDD TEMP SENSOR

- 97 51 42 IN HDD_OOB_TEMP_FILT FUNC_TEST=TRUE

1 GROUND TESTPOINTS NEAR J5400

J5560 SKIN TEMP SENSOR

- 97 52 44 IN SNS_SKIN_LEFT_P FUNC_TEST=TRUE
- 97 52 44 IN SNS_SKIN_LEFT_N FUNC_TEST=TRUE
- 97 52 IN SNS_SKIN_RIGHT_P FUNC_TEST=TRUE
- 97 52 IN SNS_SKIN_RIGHT_N FUNC_TEST=TRUE

J6602 AUDIO RIGHT SPEAKER

- 94 IN AUD_SPKR_OUTLO2R_POUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO2R_NOUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO1R_POUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO1R_NOUT FUNC_TEST=TRUE

J6603 AUDIO LEFT SPEAKER

- 94 IN AUD_SPKR_OUTLO2L_POUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO2L_NOUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO1L_POUT FUNC_TEST=TRUE
- 94 IN AUD_SPKR_OUTLO1L_NOUT FUNC_TEST=TRUE

J6600 AUDIO AUXILIARY CONNECTOR

- 98 60 IN PP3V3_AUDIO_SPDIF_JACK 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=2
- 60 IN AUD_LI_DET_JACK FUNC_TEST=TRUE
- 60 IN AUD_LI_R_JACK FUNC_TEST=TRUE
- 60 IN AUD_LI_GND_JACK FUNC_TEST=TRUE
- 60 IN AUD_LI_L_JACK FUNC_TEST=TRUE
- 60 IN HS_MIC_HI_JACK FUNC_TEST=TRUE
- 60 IN AUD_HP_L_JACK FUNC_TEST=TRUE
- 60 IN AUD_HP_GND_JACK FUNC_TEST=TRUE
- 60 IN AUD_HP_R_JACK FUNC_TEST=TRUE
- 60 IN AUD_HP_TYPEDET_JACK FUNC_TEST=TRUE
- 60 IN AUD_IP_PERPH_JACK FUNC_TEST=TRUE
- 60 IN AUD_HP_TIPDET_JACK FUNC_TEST=TRUE
- 60 IN AUD_SPDIFIN_JACK FUNC_TEST=TRUE

4 GROUND TESTPOINTS NEAR J6600

Power / Ground

- GND 17 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=17
- 98 6 IN PP3V3_S3 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=2
- 98 6 IN PP5V_S0 FUNC_TEST=TRUE MIN_ALLOWED_TPS=1

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